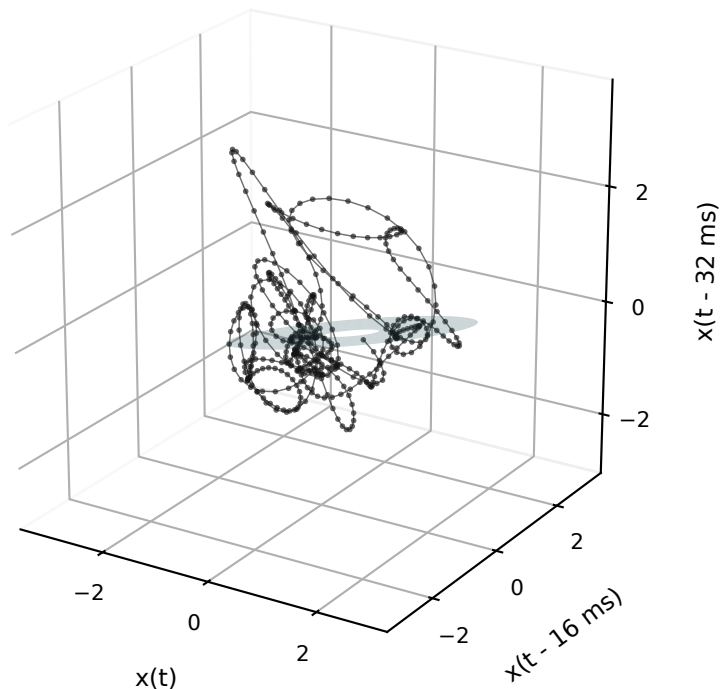


# Direction 1 | original trial 159

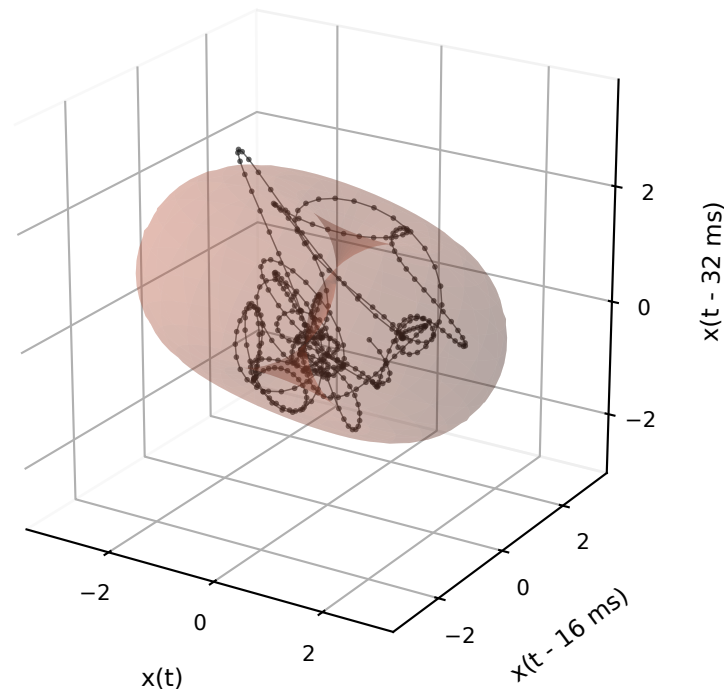
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.3180  
Native coverage: 0.908 | nfev: 82  
Flags: none

2-torus: donut volume



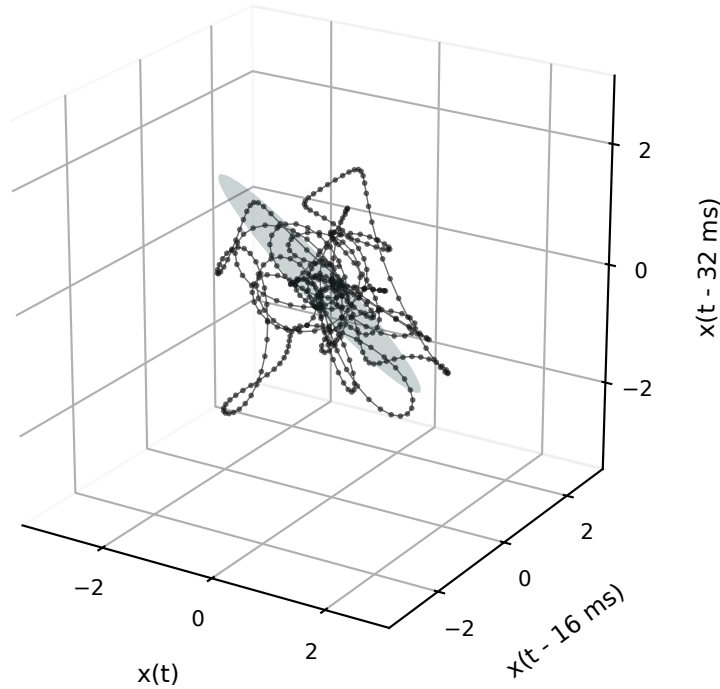
Normalized 3D set error: 0.0016  
Native coverage: 0.908 | nfev: 29  
Flags: collapsed hole; active optimizer bound; near optimizer bound; nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 1 | original trial 16

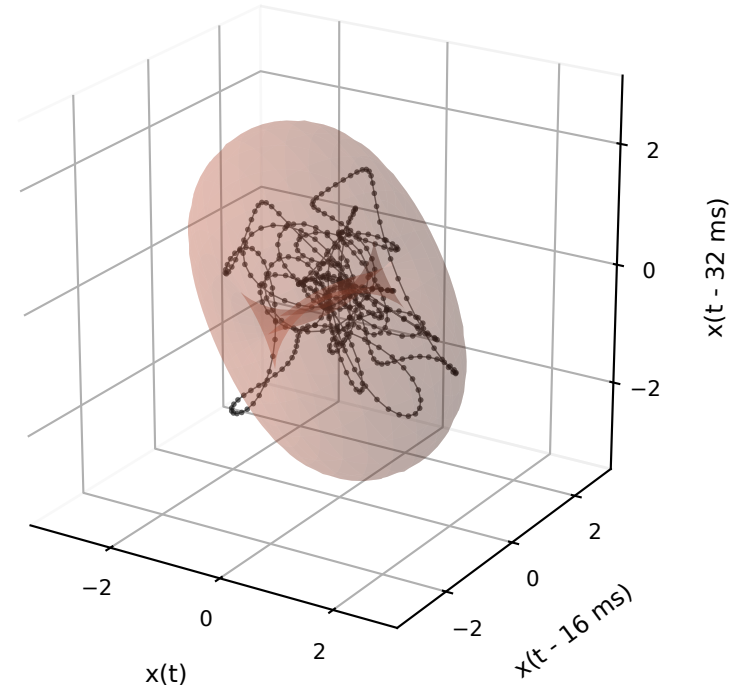
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2092  
Native coverage: 0.902 | nfev: 40  
Flags: near optimizer bound

2-torus: donut volume



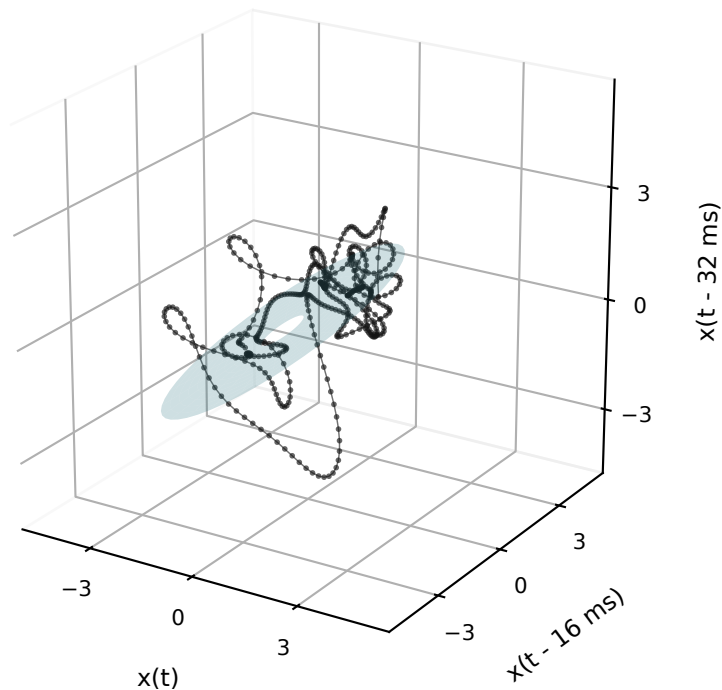
Normalized 3D set error: 0.0016  
Native coverage: 0.874 | nfev: 29  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

# Direction 1 | original trial 566

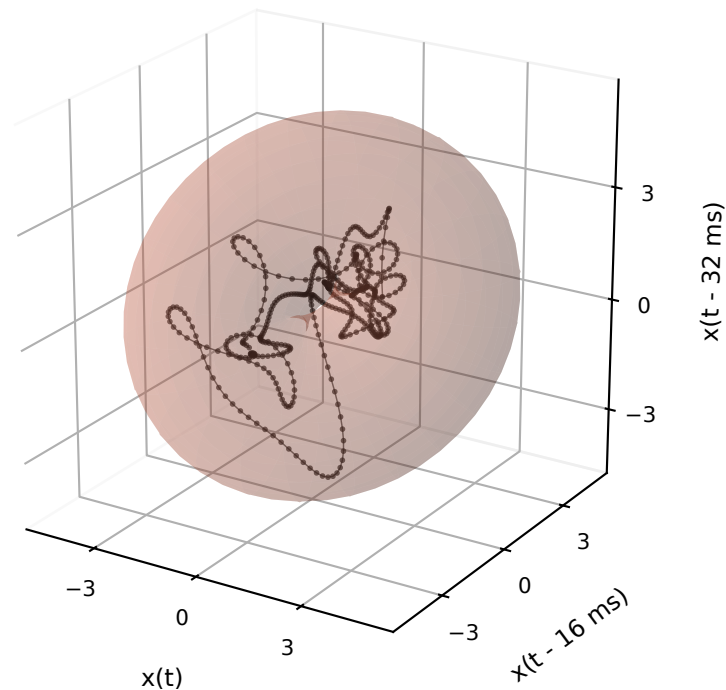
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

**1-torus: planar band**



Normalized 3D set error: 0.2379  
Native coverage: 0.940 | nfev: 121  
Flags: active optimizer bound; near optimizer bound

**2-torus: donut volume**



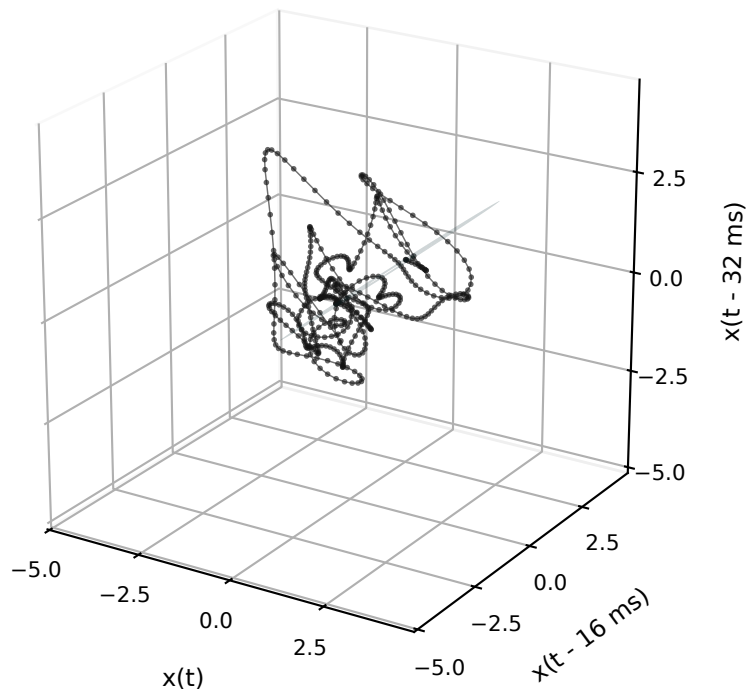
Normalized 3D set error: 0.0006  
Native coverage: 0.908 | nfev: 61  
Flags: none

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

# Direction 1 | original trial 673

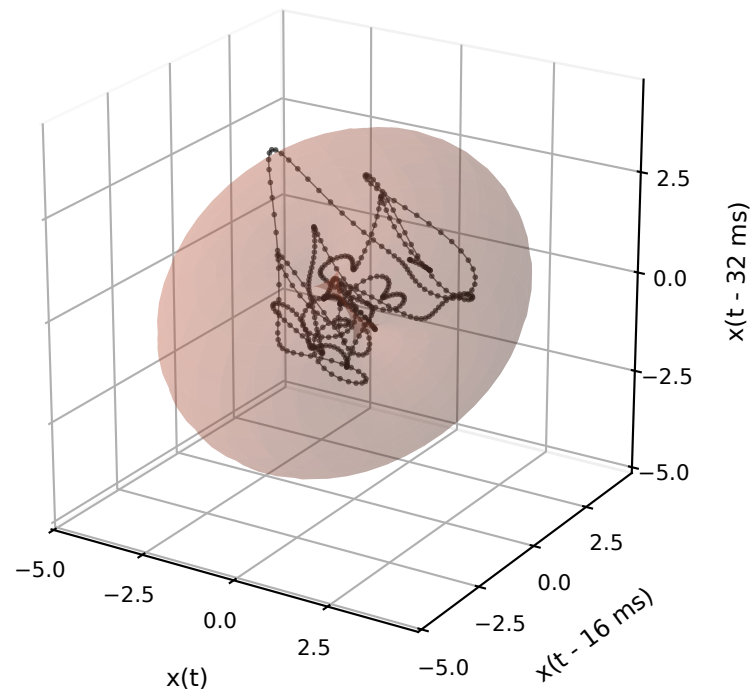
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2980  
Native coverage: 0.923 | nfev: 97  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

2-torus: donut volume



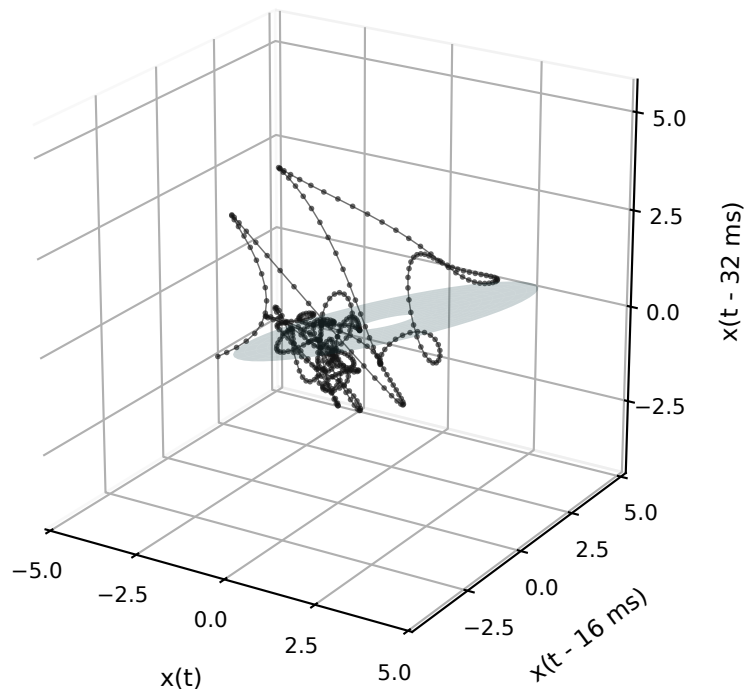
Normalized 3D set error: 0.0009  
Native coverage: 0.923 | nfev: 14  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

# Direction 1 | original trial 263

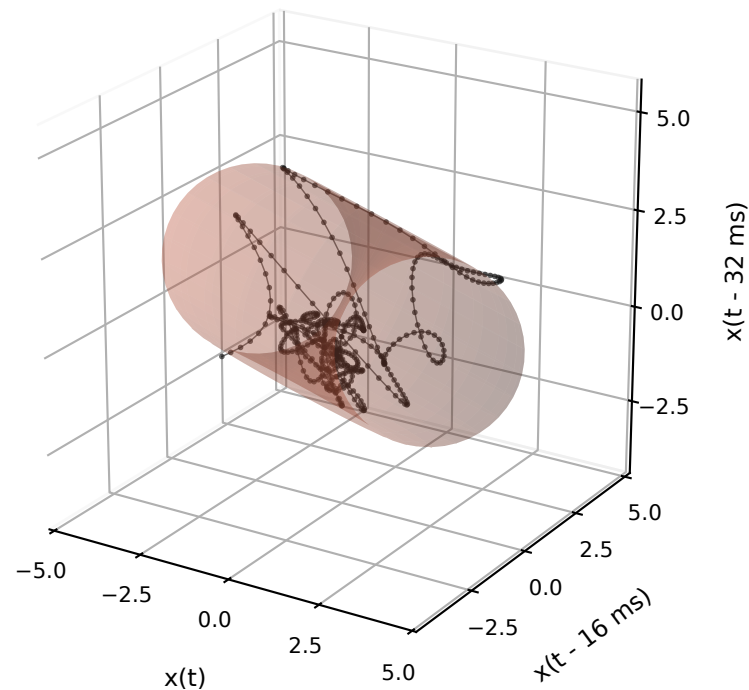
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.3140  
Native coverage: 0.944 | nfev: 106  
Flags: nearly circular orientation

2-torus: donut volume



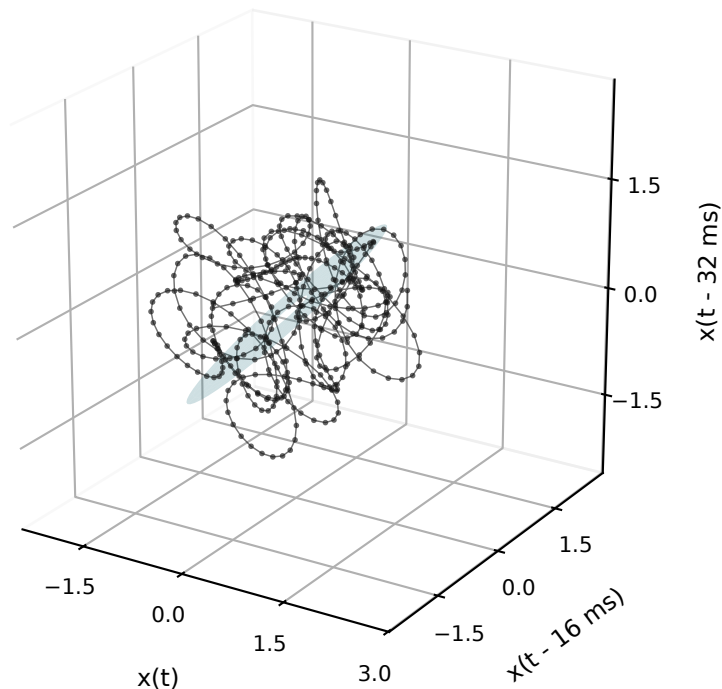
Normalized 3D set error: 0.0004  
Native coverage: 0.878 | nfev: 20  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

# Direction 1 | original trial 389

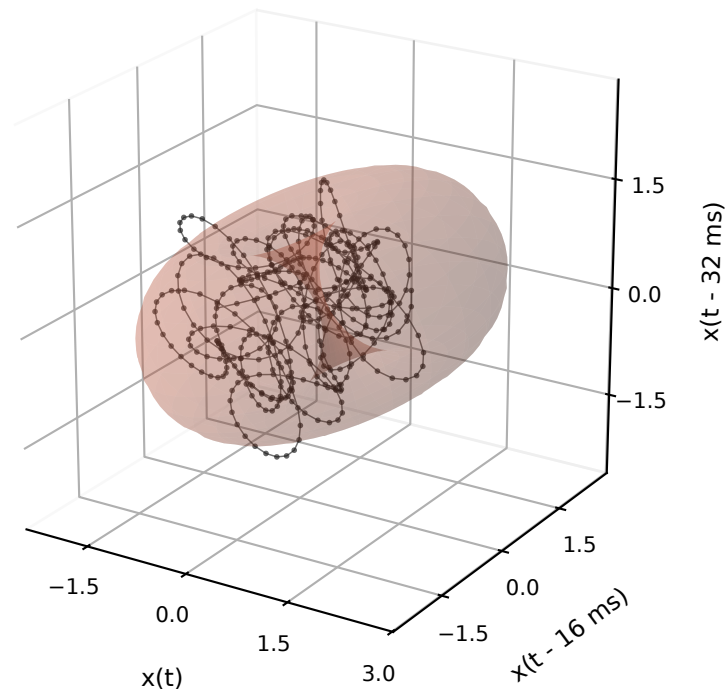
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2906  
Native coverage: 0.870 | nfev: 33  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

2-torus: donut volume



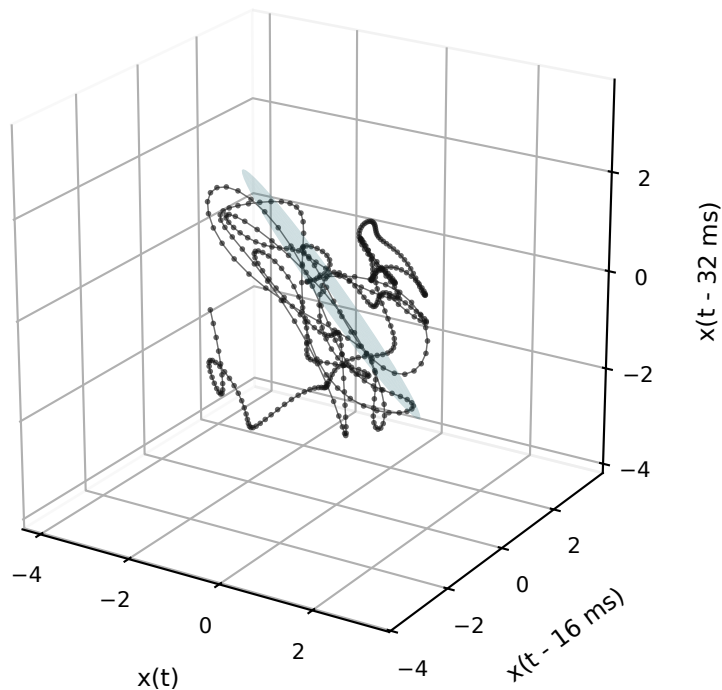
Normalized 3D set error: 0.0028  
Native coverage: 0.861 | nfev: 20  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 1 | original trial 684

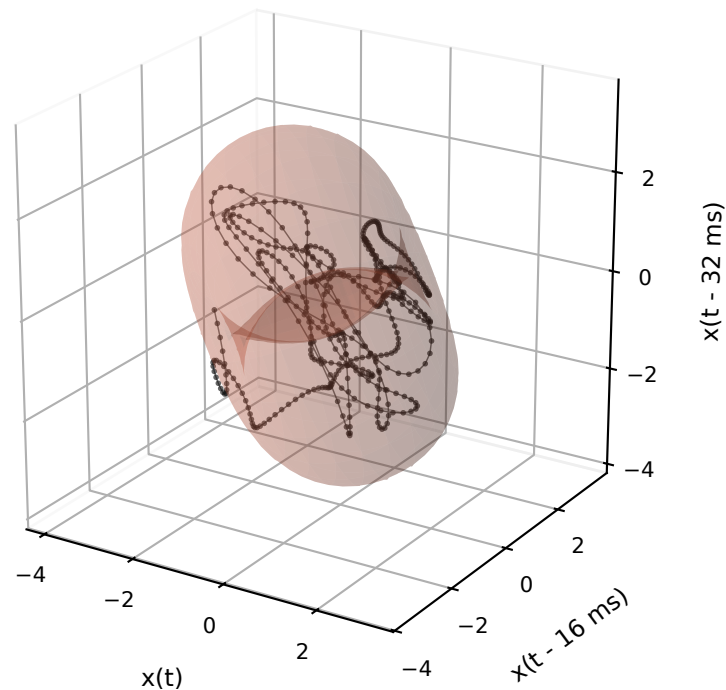
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2868  
Native coverage: 0.923 | nfev: 83  
Flags: none

2-torus: donut volume



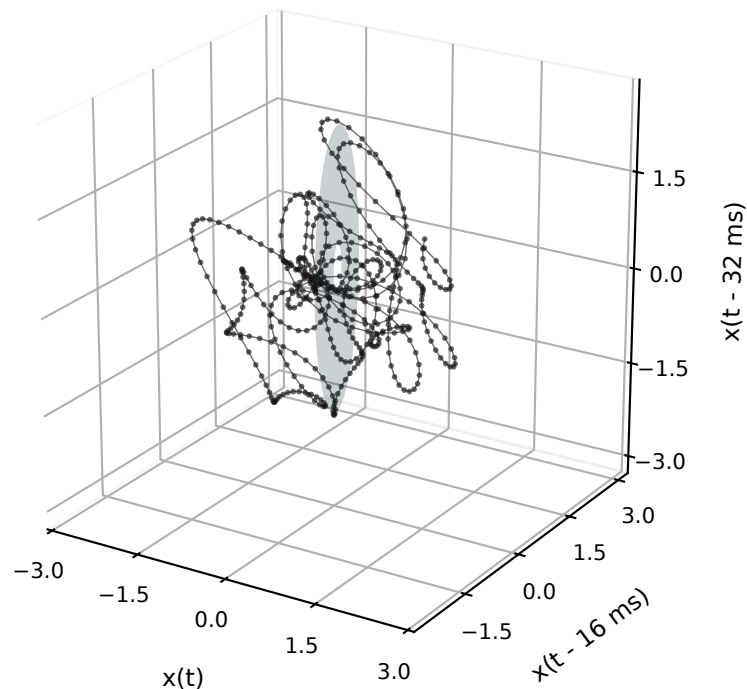
Normalized 3D set error: 0.0011  
Native coverage: 0.876 | nfev: 57  
Flags: collapsed hole; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

# Direction 1 | original trial 437

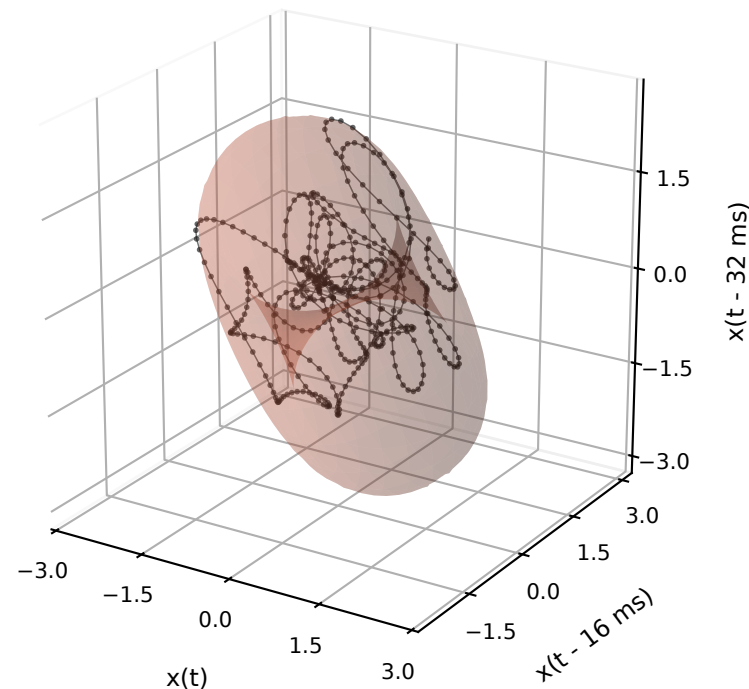
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2907  
Native coverage: 0.857 | nfev: 75  
Flags: near optimizer bound

2-torus: donut volume



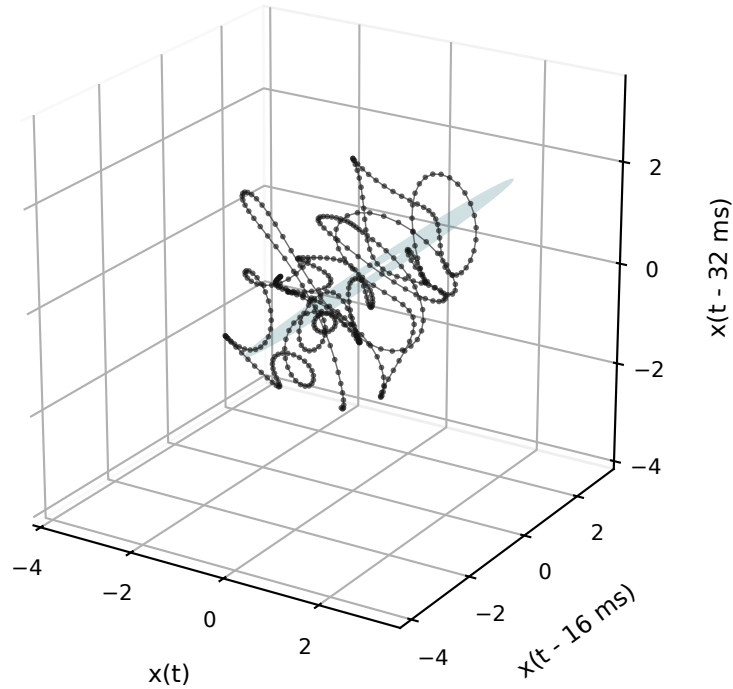
Normalized 3D set error: 0.0014  
Native coverage: 0.868 | nfev: 18  
Flags: active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 1 | original trial 56

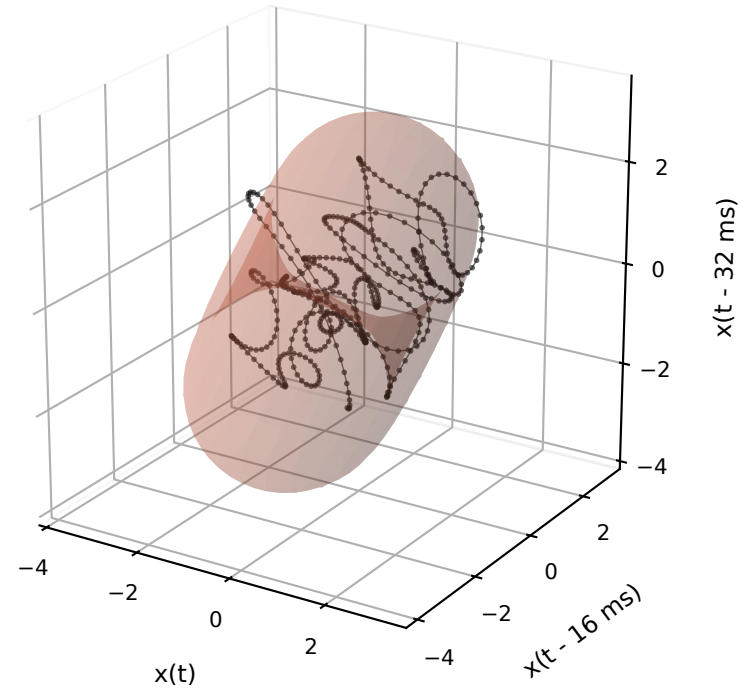
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2414  
Native coverage: 0.906 | nfev: 39  
Flags: none

2-torus: donut volume



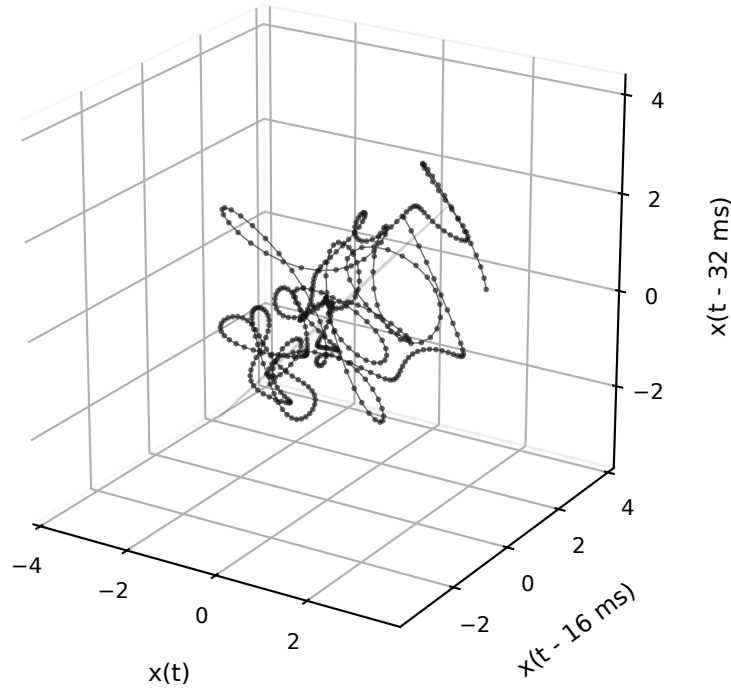
Normalized 3D set error: 0.0010  
Native coverage: 0.893 | nfev: 26  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 1 | original trial 82

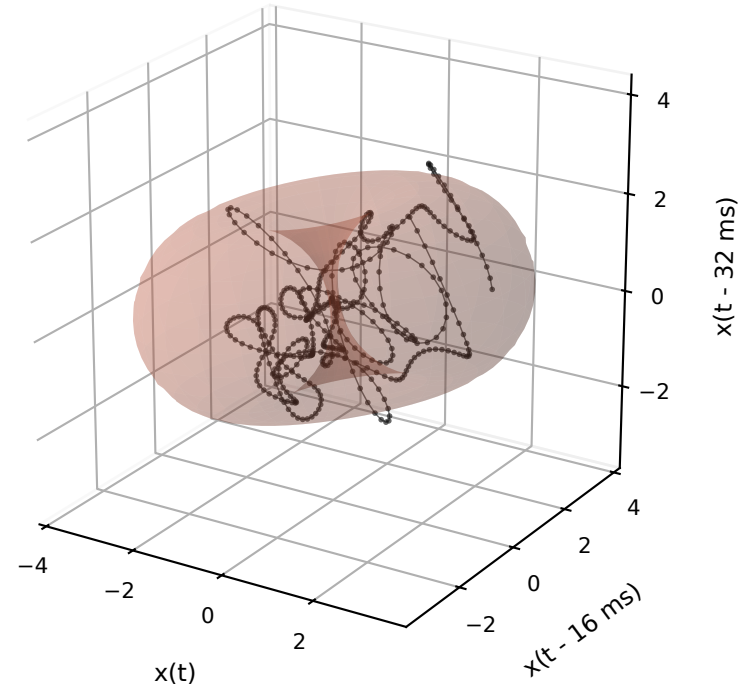
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2317  
Native coverage: 0.908 | nfev: 42  
Flags: none

2-torus: donut volume



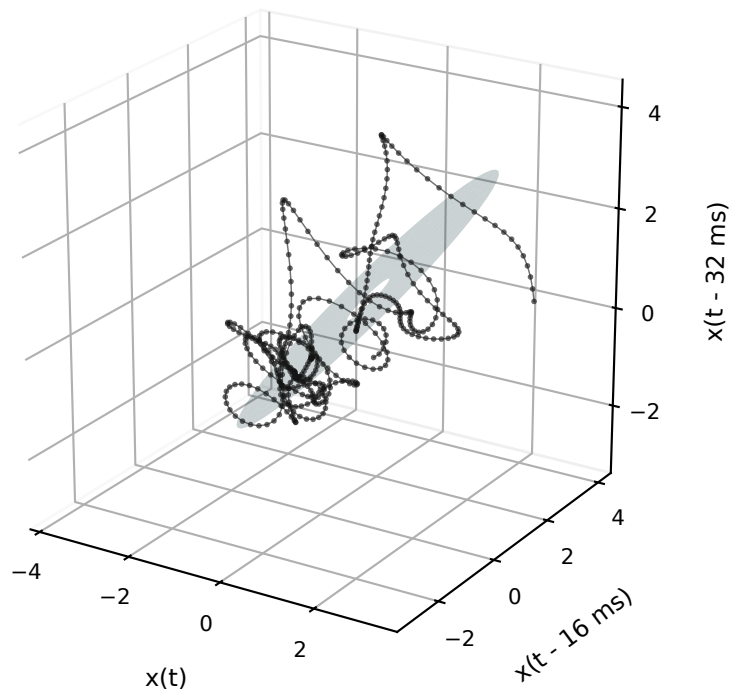
Normalized 3D set error: 0.0010  
Native coverage: 0.906 | nfev: 23  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

# Direction 1 | original trial 517

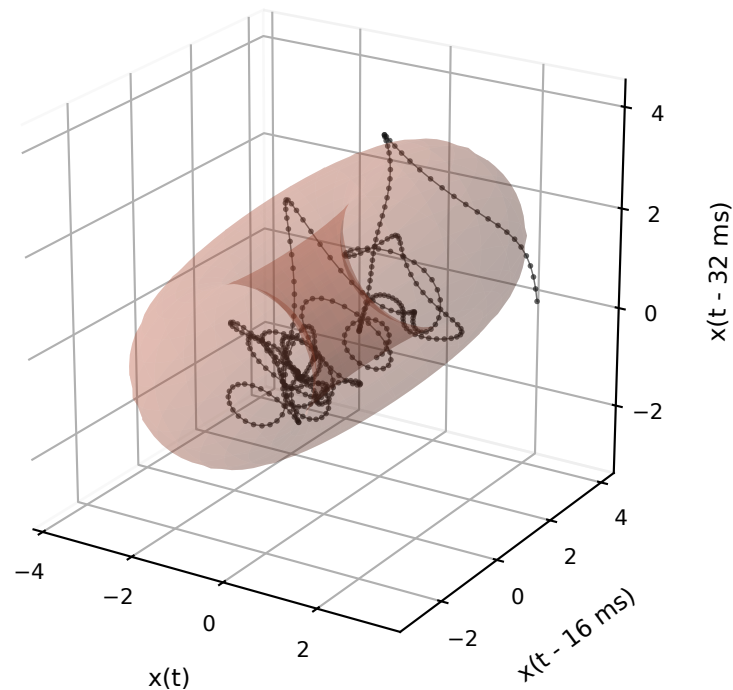
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1832  
Native coverage: 0.927 | nfev: 41  
Flags: none

2-torus: donut volume



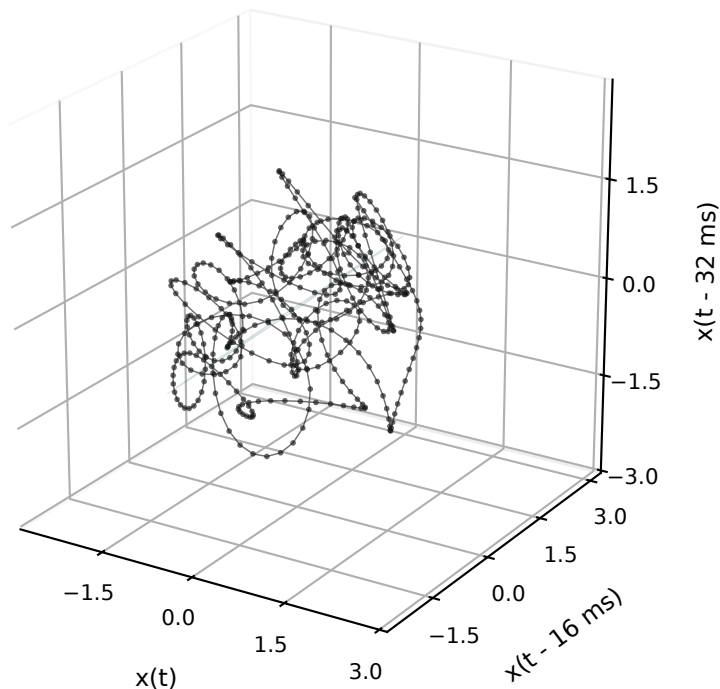
Normalized 3D set error: 0.0013  
Native coverage: 0.850 | nfev: 37  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

# Direction 1 | original trial 124

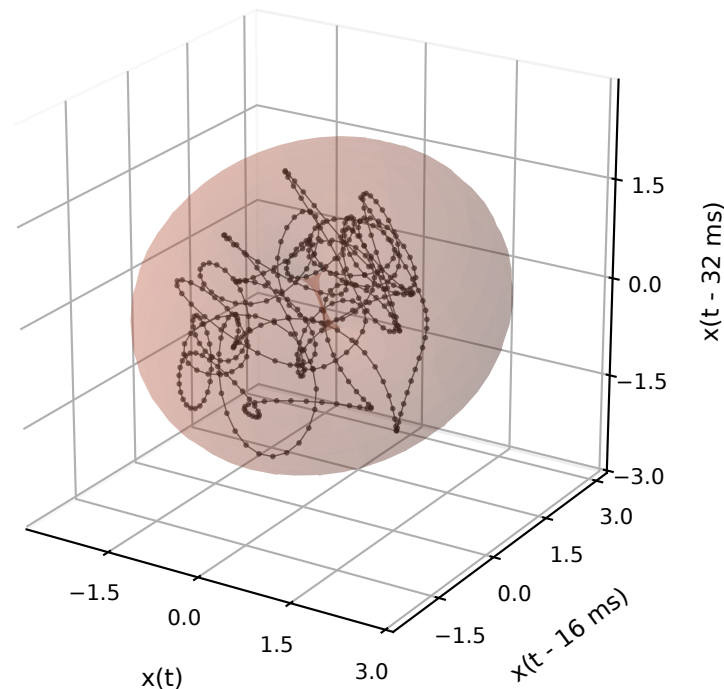
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2653  
Native coverage: 0.853 | nfev: 96  
Flags: none

2-torus: donut volume



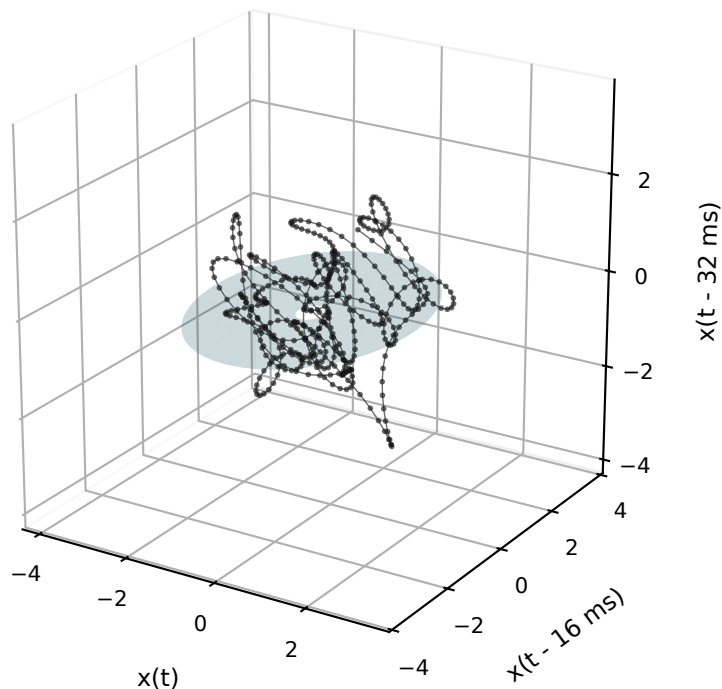
Normalized 3D set error: 0.0019  
Native coverage: 0.878 | nfev: 51  
Flags: collapsed hole; active optimizer bound; near optimizer bound; nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

# Direction 1 | original trial 526

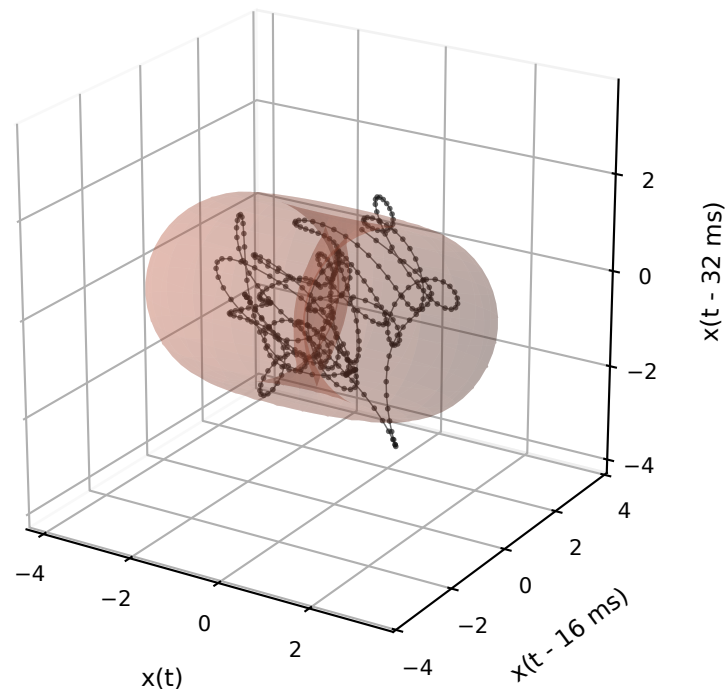
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2897  
Native coverage: 0.902 | nfev: 39  
Flags: active optimizer bound; near optimizer bound

2-torus: donut volume



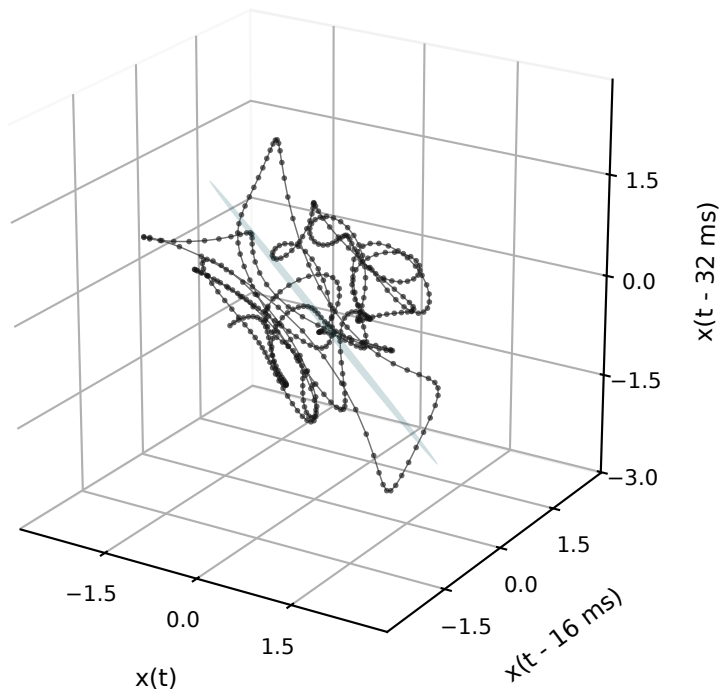
Normalized 3D set error: 0.0017  
Native coverage: 0.887 | nfev: 35  
Flags: collapsed hole

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

# Direction 1 | original trial 382

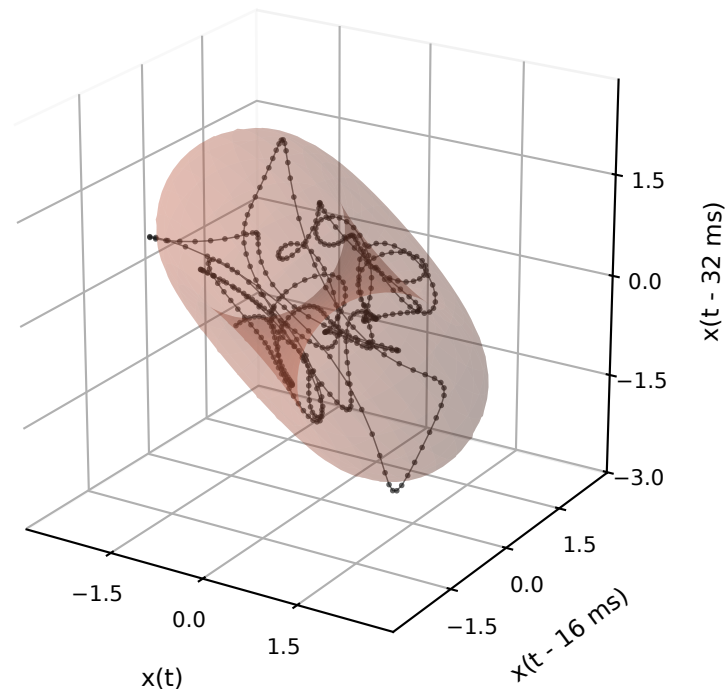
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

**1-torus: planar band**



Normalized 3D set error: 0.2273  
Native coverage: 0.902 | nfev: 33  
Flags: nearly circular orientation

**2-torus: donut volume**



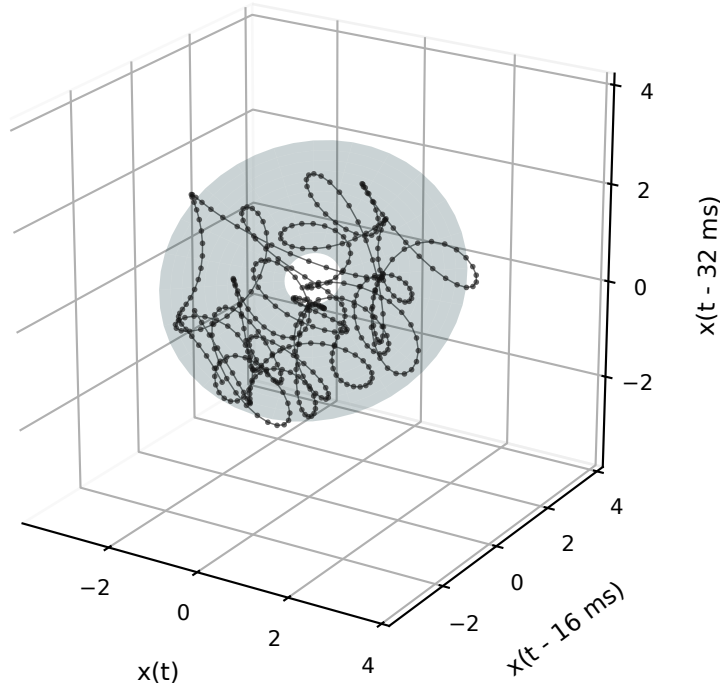
Normalized 3D set error: 0.0020  
Native coverage: 0.861 | nfev: 54  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 1 | original trial 80

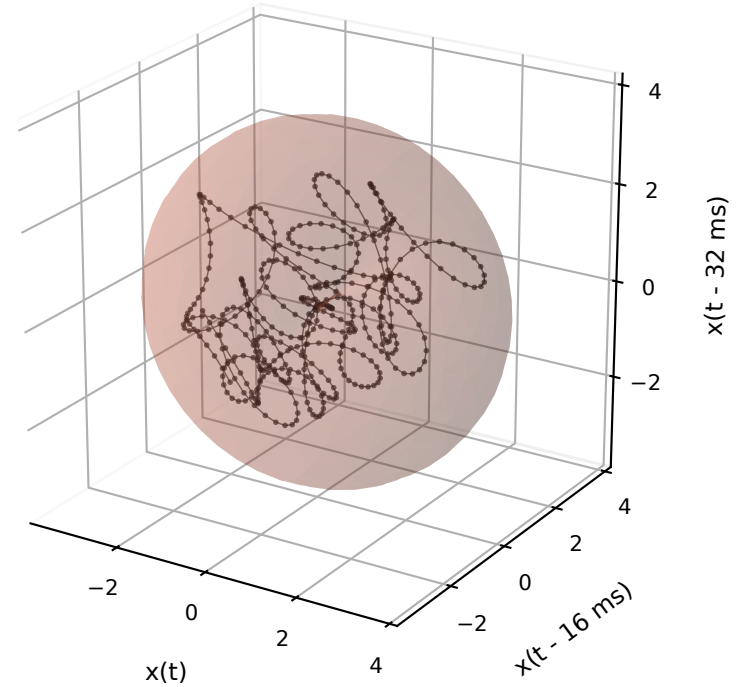
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2207  
Native coverage: 0.925 | nfev: 99  
Flags: none

2-torus: donut volume



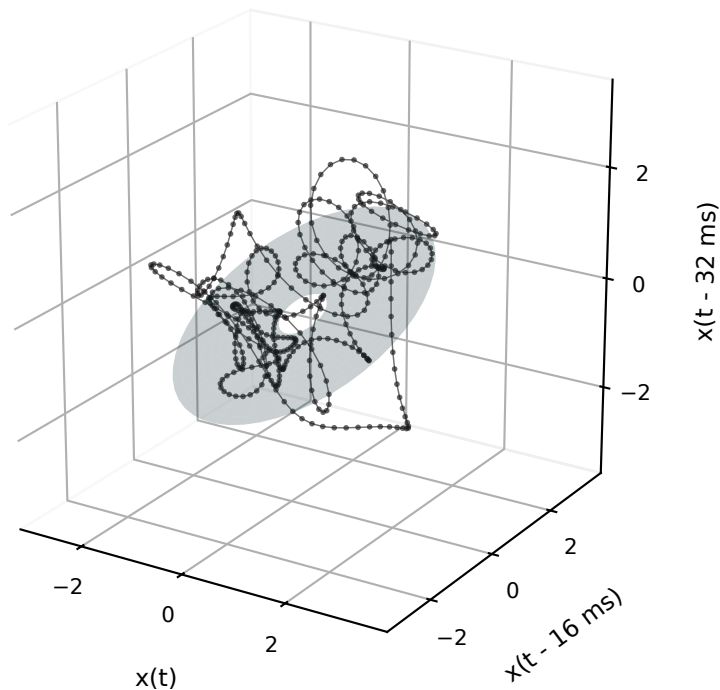
Normalized 3D set error: 0.0009  
Native coverage: 0.880 | nfev: 33  
Flags: near optimizer bound; nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 1 | original trial 366

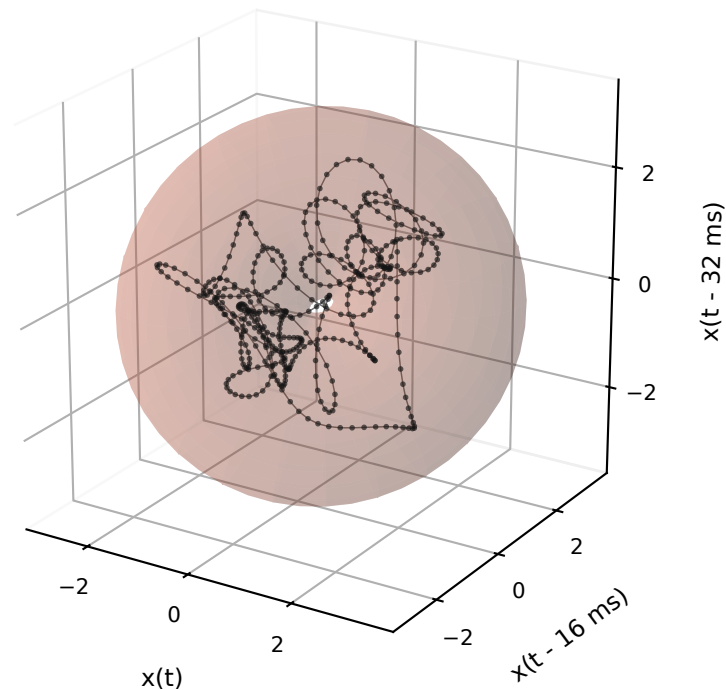
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2637  
Native coverage: 0.893 | nfev: 188  
Flags: nearly circular orientation

2-torus: donut volume



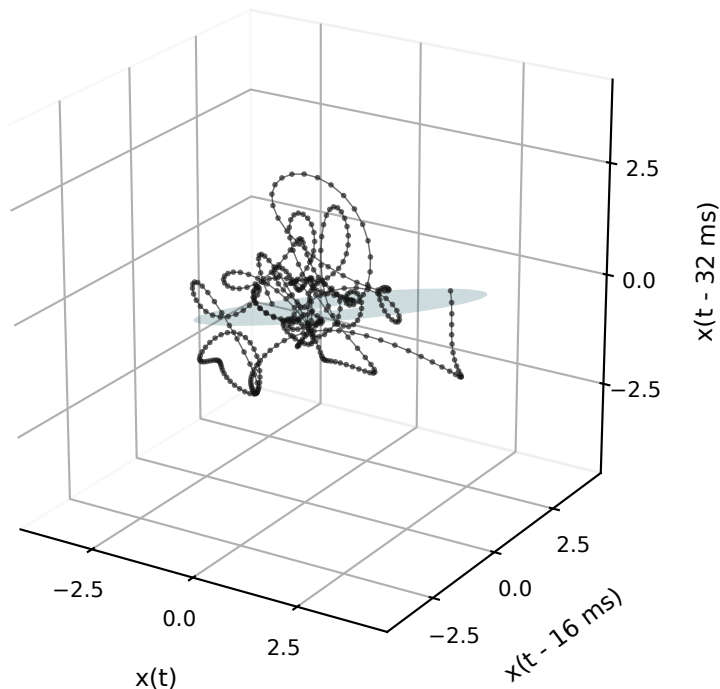
Normalized 3D set error: 0.0014  
Native coverage: 0.868 | nfev: 28  
Flags: none

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

# Direction 1 | original trial 156

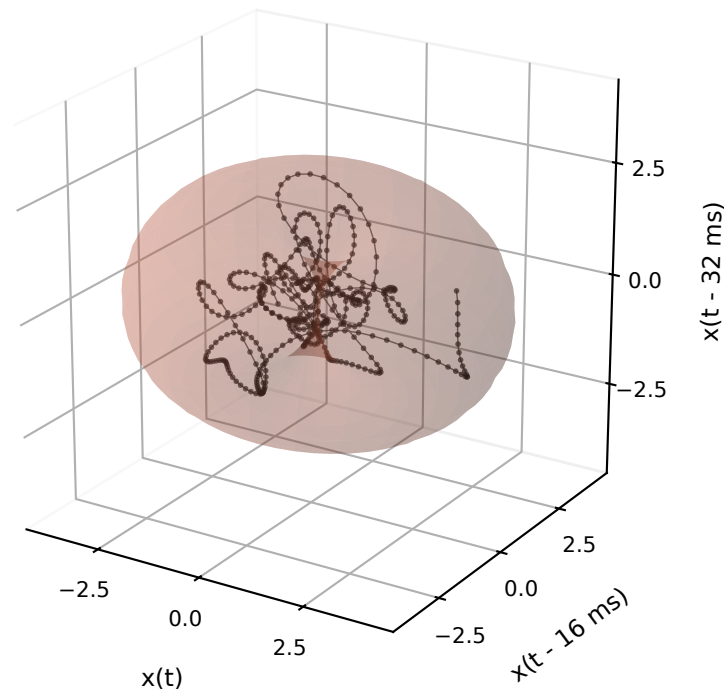
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2431  
Native coverage: 0.919 | nfev: 36  
Flags: none

2-torus: donut volume



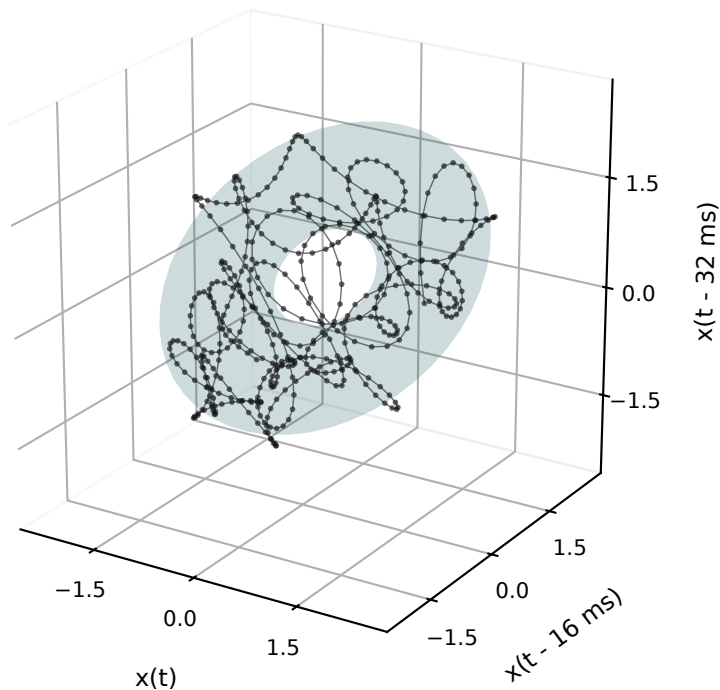
Normalized 3D set error: 0.0010  
Native coverage: 0.891 | nfev: 38  
Flags: collapsed hole; active optimizer bound; near optimizer bound; nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

# Direction 1 | original trial 690

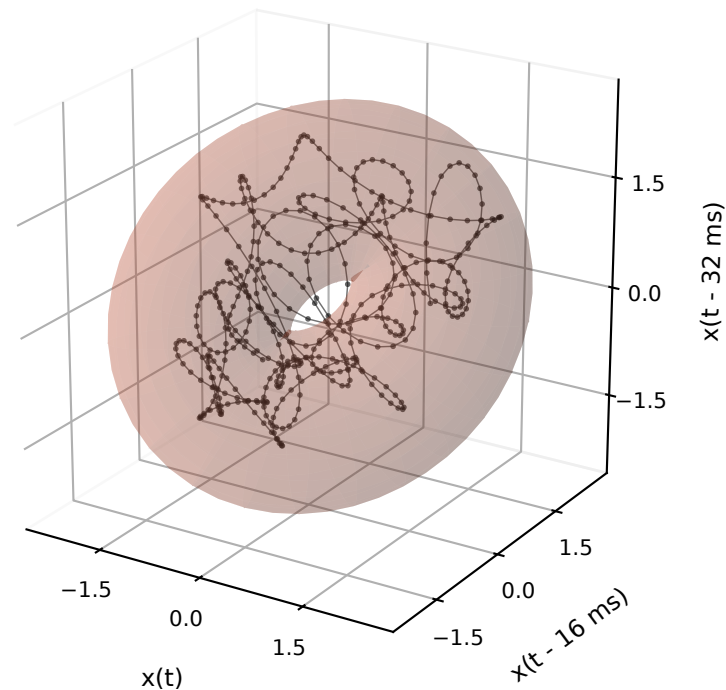
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

**1-torus: planar band**



Normalized 3D set error: 0.1463  
Native coverage: 0.889 | nfev: 52  
Flags: none

**2-torus: donut volume**



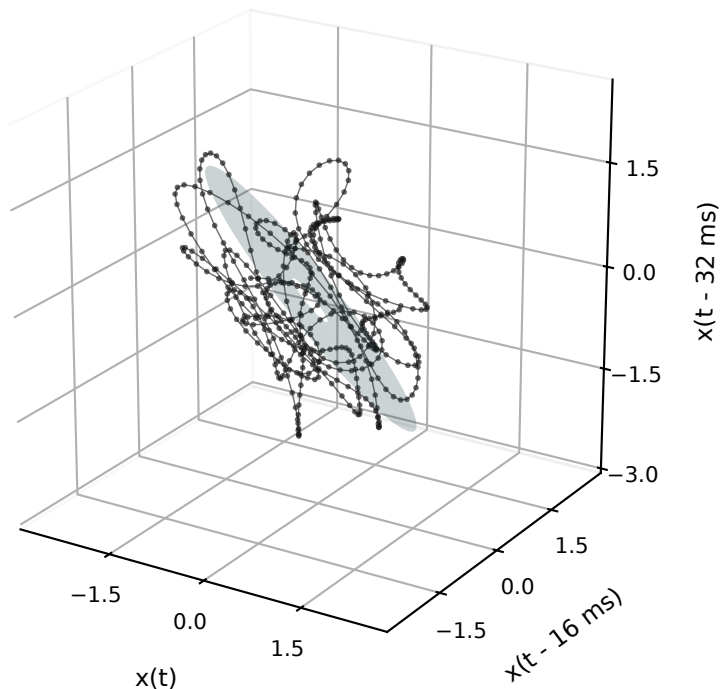
Normalized 3D set error: 0.0011  
Native coverage: 0.878 | nfev: 17  
Flags: active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 1 | original trial 265

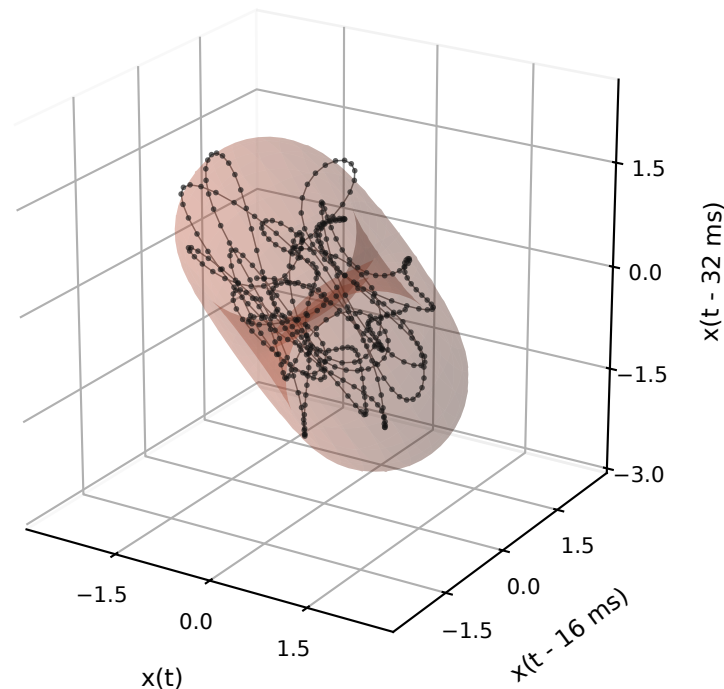
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1448  
Native coverage: 0.915 | nfev: 19  
Flags: active optimizer bound; near optimizer bound

2-torus: donut volume



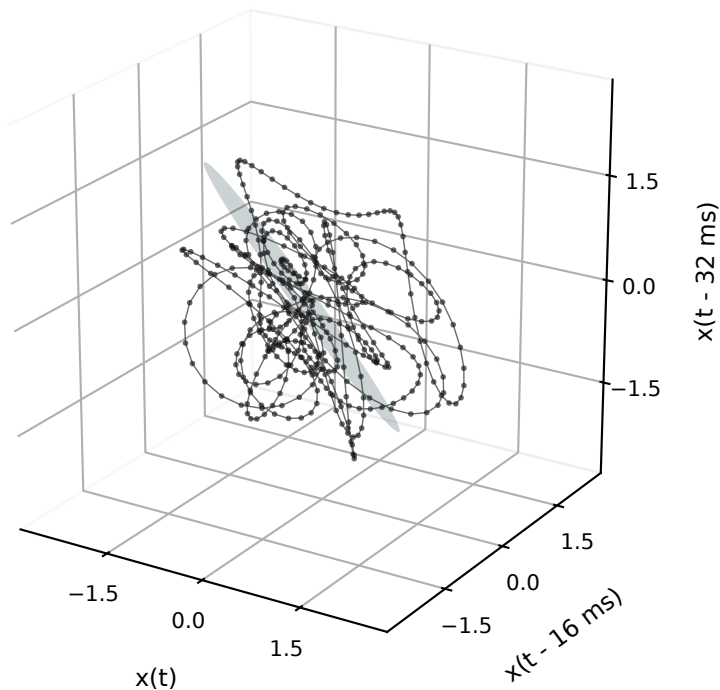
Normalized 3D set error: 0.0020  
Native coverage: 0.874 | nfev: 122  
Flags: near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

# Direction 1 | original trial 445

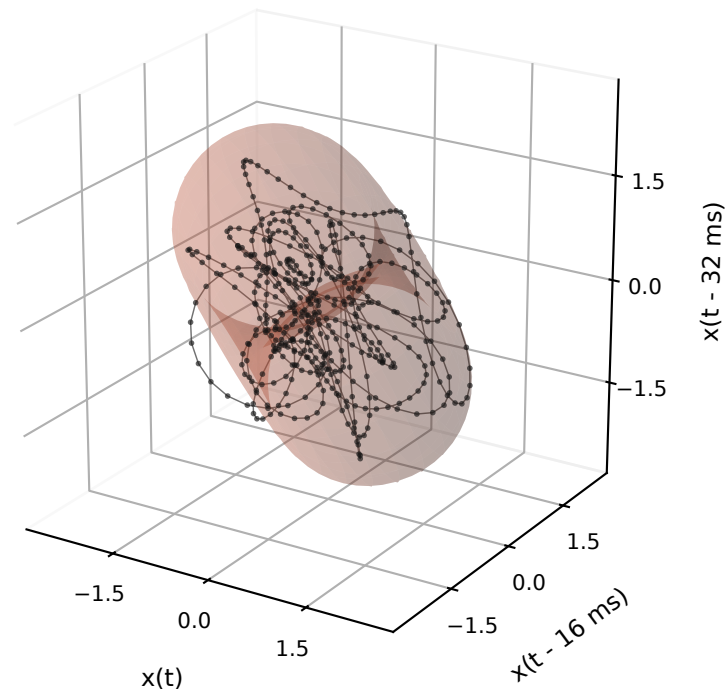
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2065  
Native coverage: 0.912 | nfev: 57  
Flags: none

2-torus: donut volume



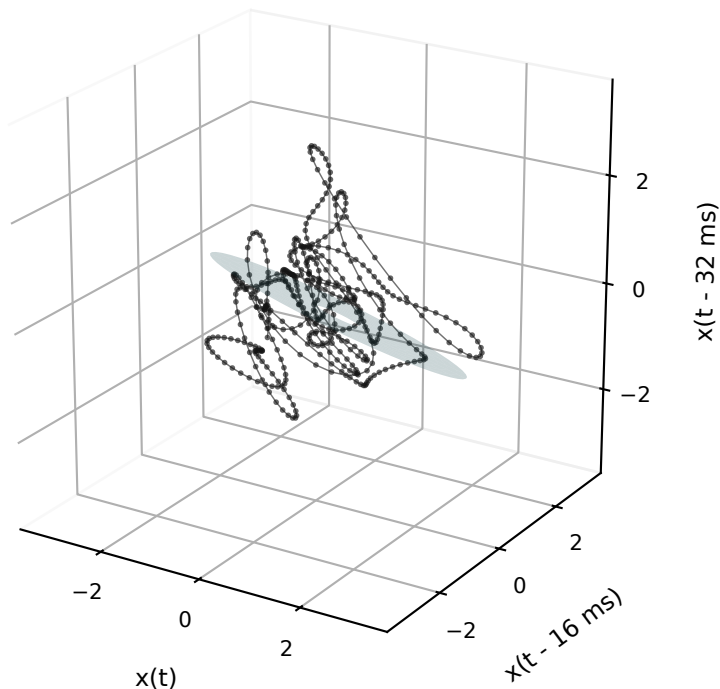
Normalized 3D set error: 0.0019  
Native coverage: 0.857 | nfev: 34  
Flags: collapsed hole; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 1 | original trial 462

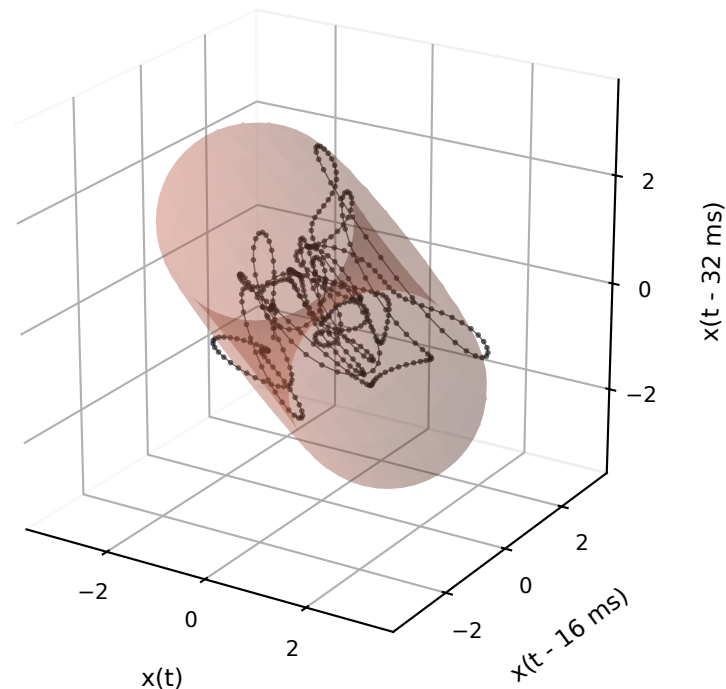
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2759  
Native coverage: 0.902 | nfev: 54  
Flags: none

2-torus: donut volume



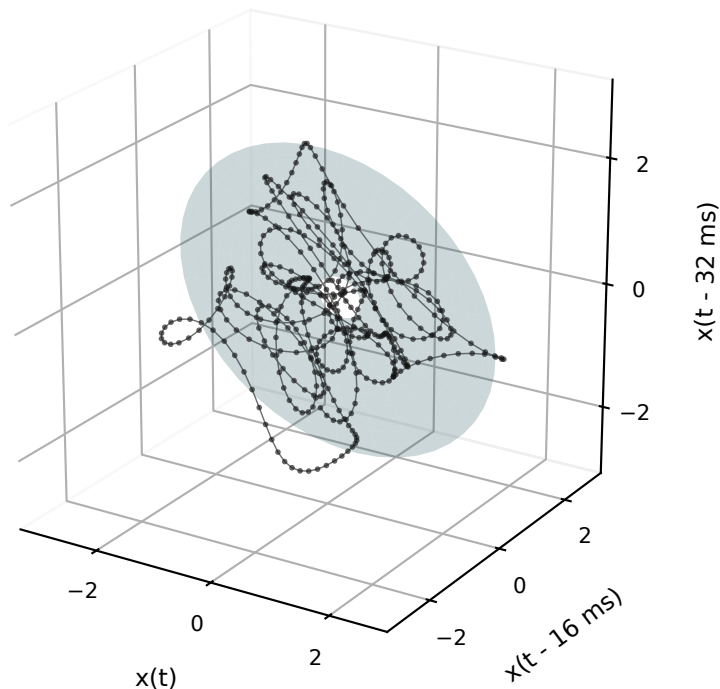
Normalized 3D set error: 0.0009  
Native coverage: 0.891 | nfev: 21  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

# Direction 1 | original trial 152

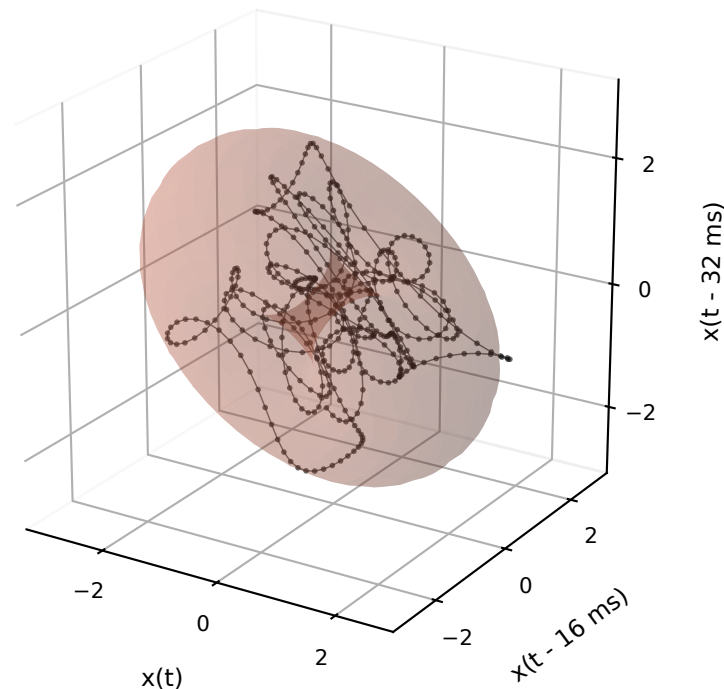
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2572  
Native coverage: 0.900 | nfev: 63  
Flags: active optimizer bound; near optimizer bound

2-torus: donut volume



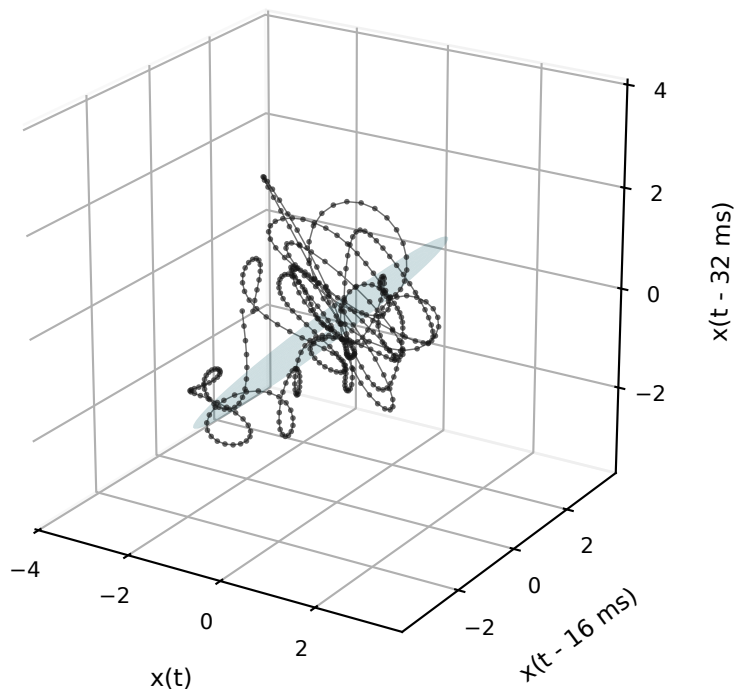
Normalized 3D set error: 0.0010  
Native coverage: 0.857 | nfev: 30  
Flags: active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

# Direction 1 | original trial 188

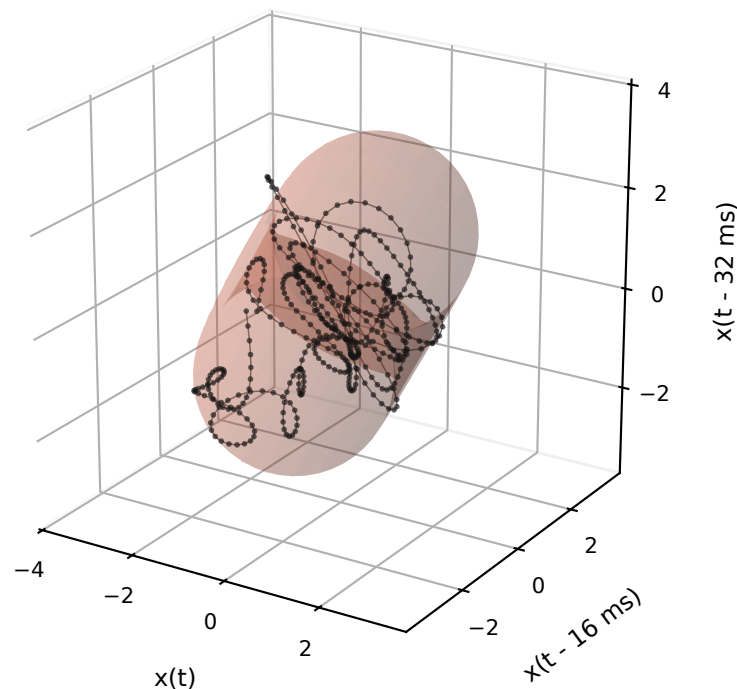
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2298  
Native coverage: 0.915 | nfev: 34  
Flags: none

2-torus: donut volume



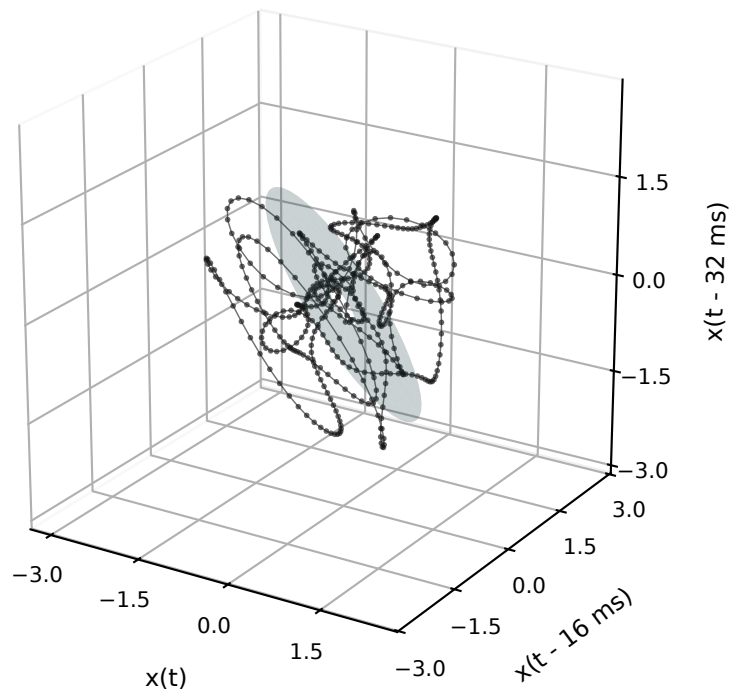
Normalized 3D set error: 0.0019  
Native coverage: 0.880 | nfev: 31  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

# Direction 1 | original trial 195

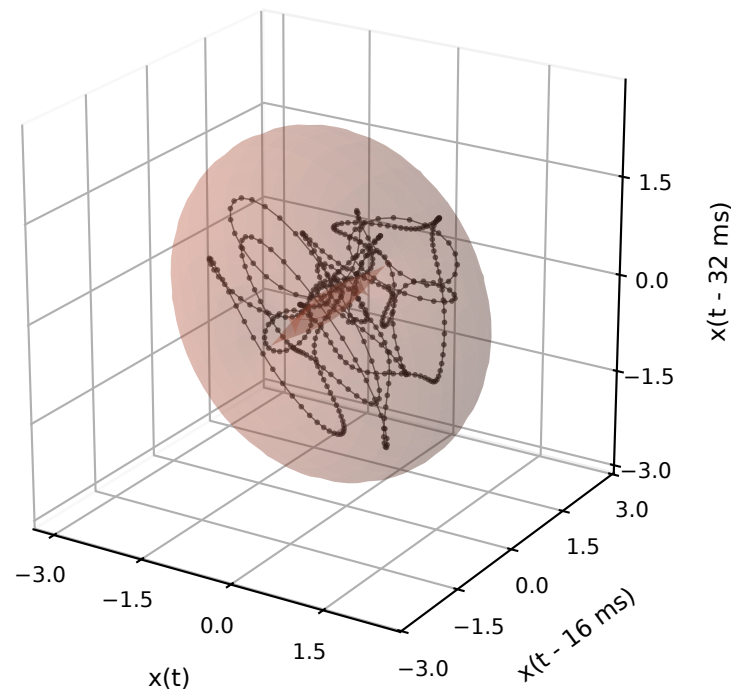
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2977  
Native coverage: 0.876 | nfev: 105  
Flags: none

2-torus: donut volume



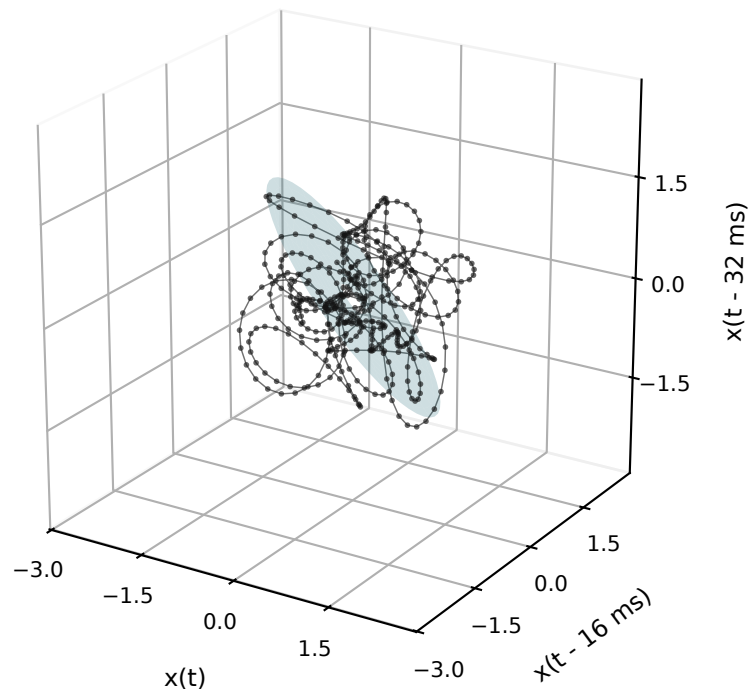
Normalized 3D set error: 0.0022  
Native coverage: 0.865 | nfev: 37  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

# Direction 1 | original trial 205

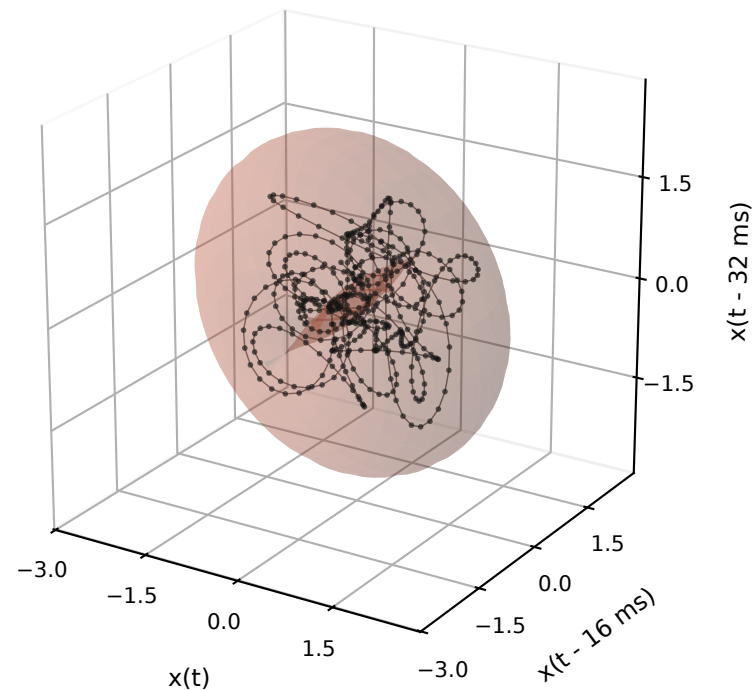
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2802  
Native coverage: 0.880 | nfev: 97  
Flags: none

2-torus: donut volume



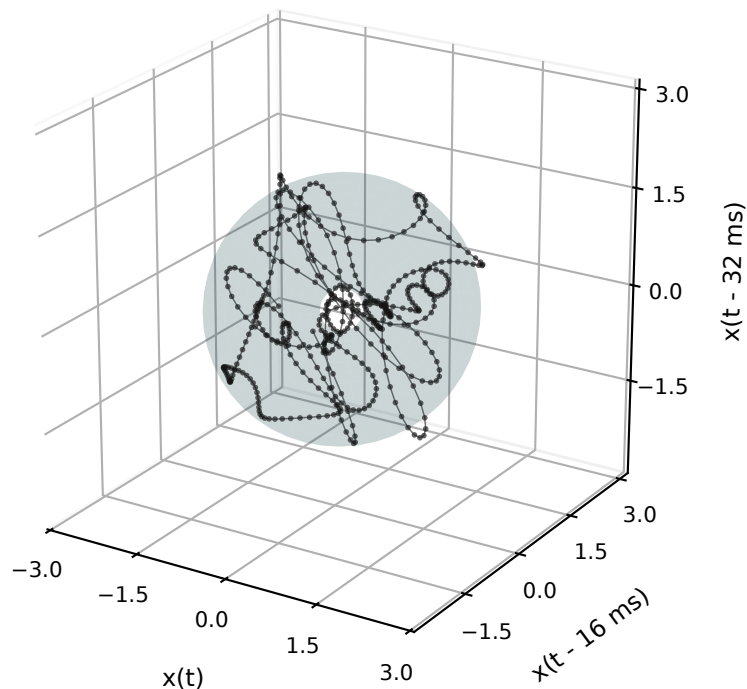
Normalized 3D set error: 0.0042  
Native coverage: 0.859 | nfev: 36  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

# Direction 1 | original trial 262

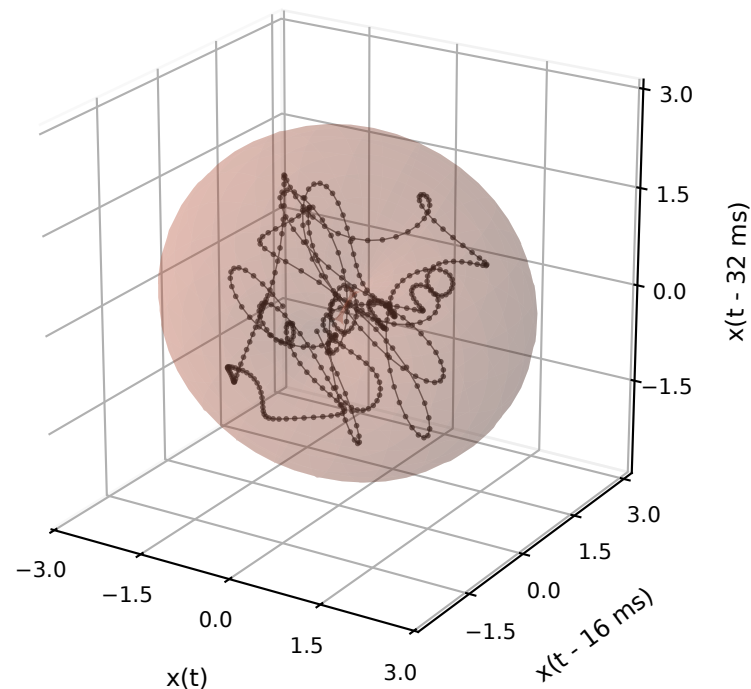
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2959  
Native coverage: 0.889 | nfev: 63  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

2-torus: donut volume



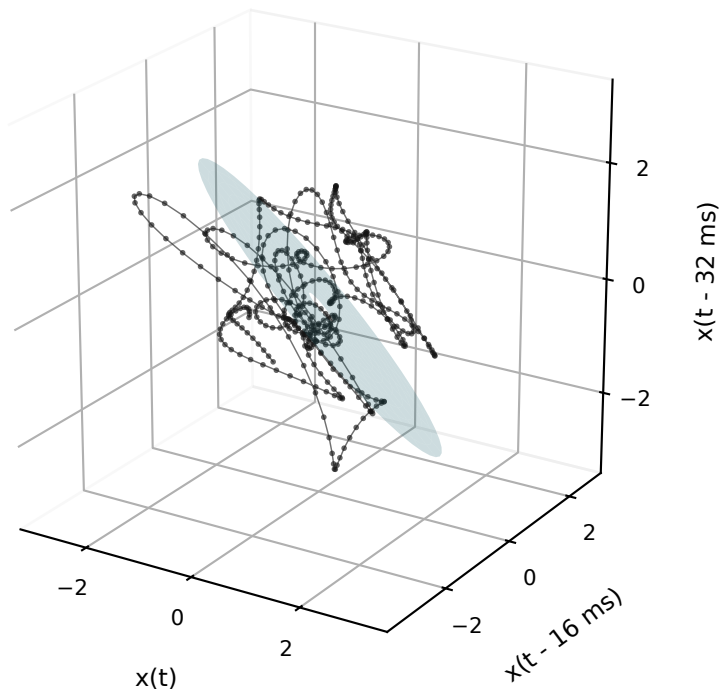
Normalized 3D set error: 0.0013  
Native coverage: 0.863 | nfev: 27  
Flags: collapsed hole; active optimizer bound; near  
optimizer bound; nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

# Direction 1 | original trial 571

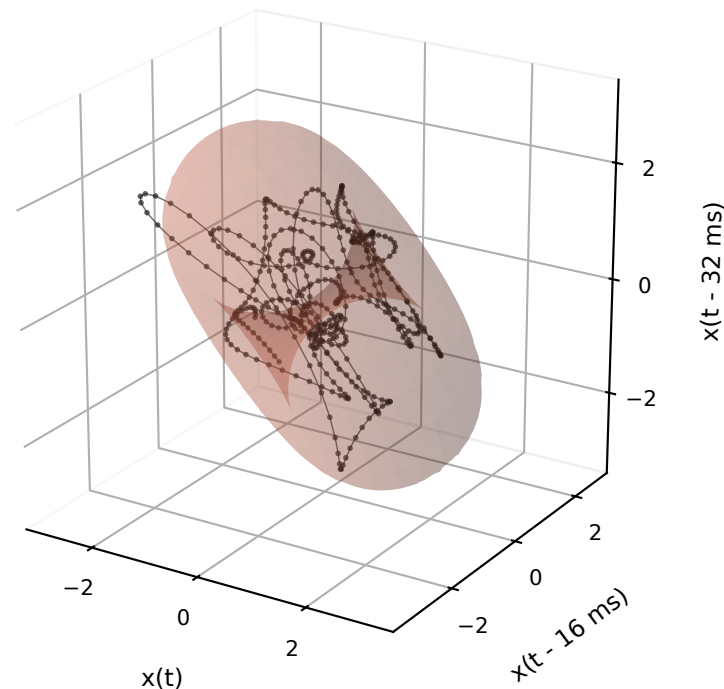
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2386  
Native coverage: 0.902 | nfev: 52  
Flags: none

2-torus: donut volume



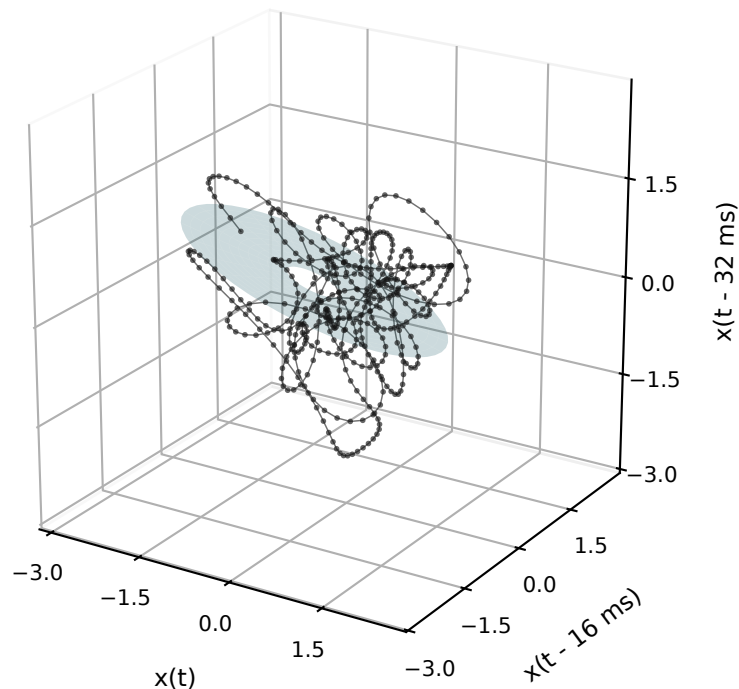
Normalized 3D set error: 0.0013  
Native coverage: 0.855 | nfev: 16  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

# Direction 1 | original trial 601

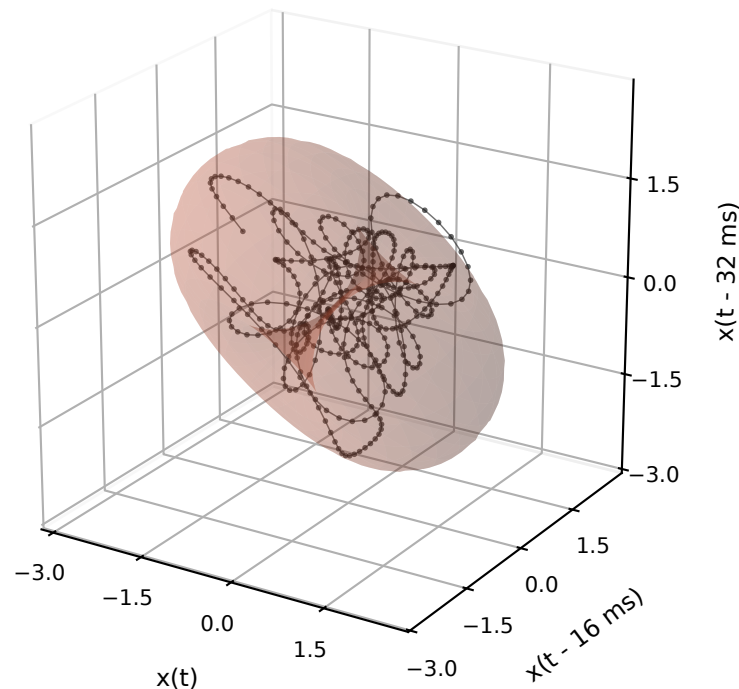
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2358  
Native coverage: 0.910 | nfev: 38  
Flags: none

2-torus: donut volume



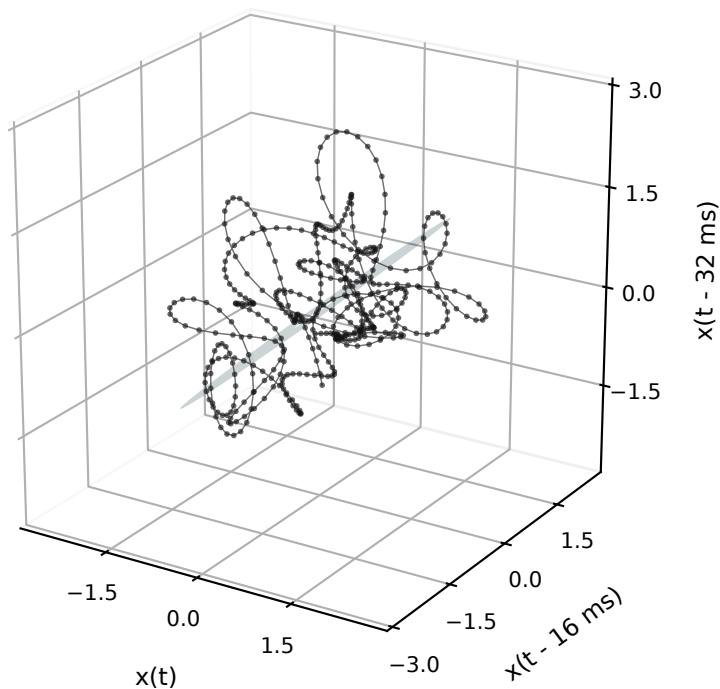
Normalized 3D set error: 0.0016  
Native coverage: 0.876 | nfev: 275  
Flags: none

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 1 | original trial 349

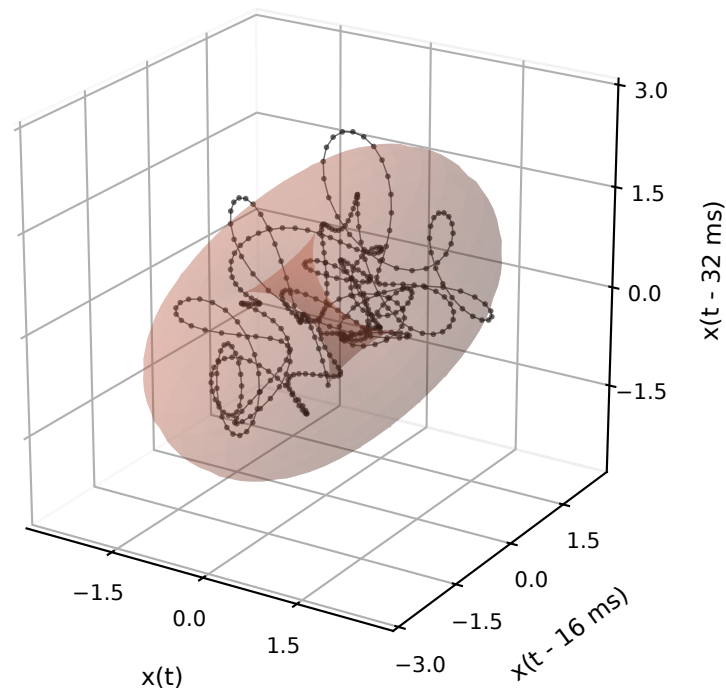
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

**1-torus: planar band**



Normalized 3D set error: 0.2288  
Native coverage: 0.908 | nfev: 145  
Flags: nearly circular orientation

**2-torus: donut volume**



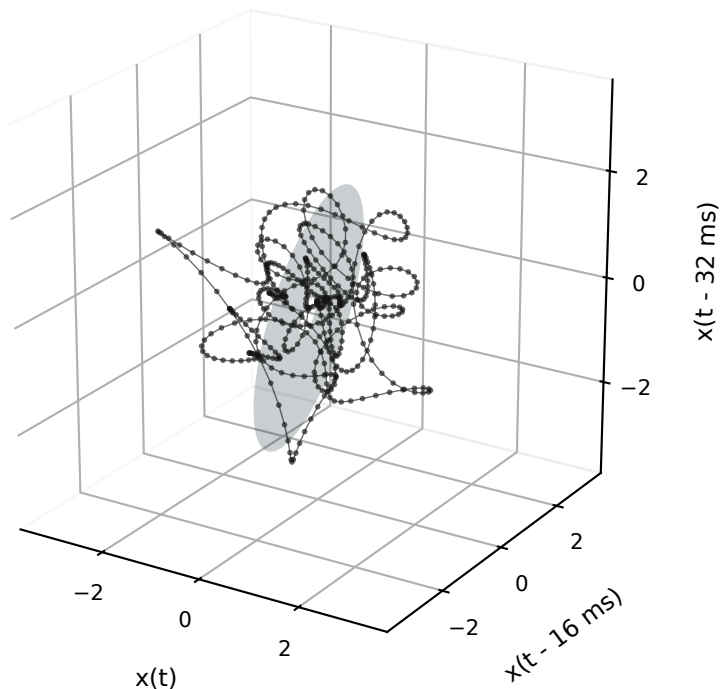
Normalized 3D set error: 0.0014  
Native coverage: 0.863 | nfev: 30  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

# Direction 1 | original trial 236

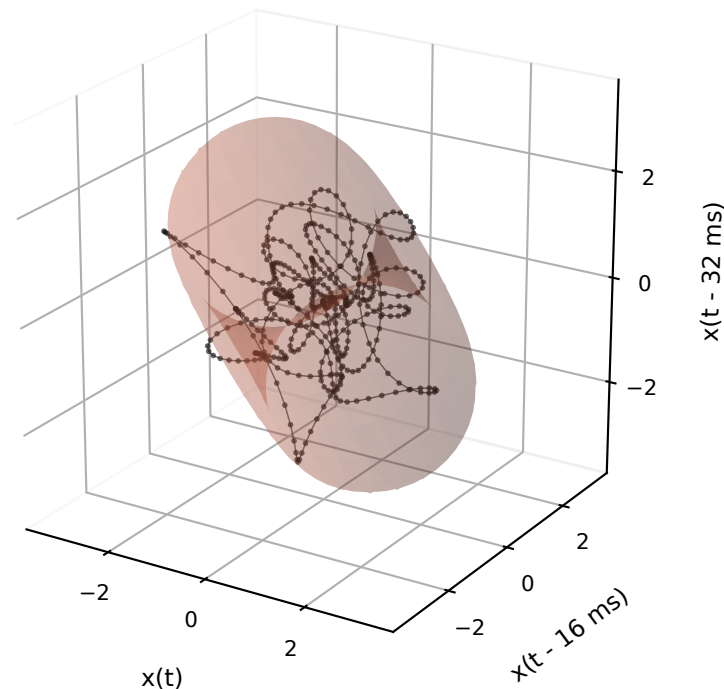
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.3155  
Native coverage: 0.910 | nfev: 54  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

2-torus: donut volume



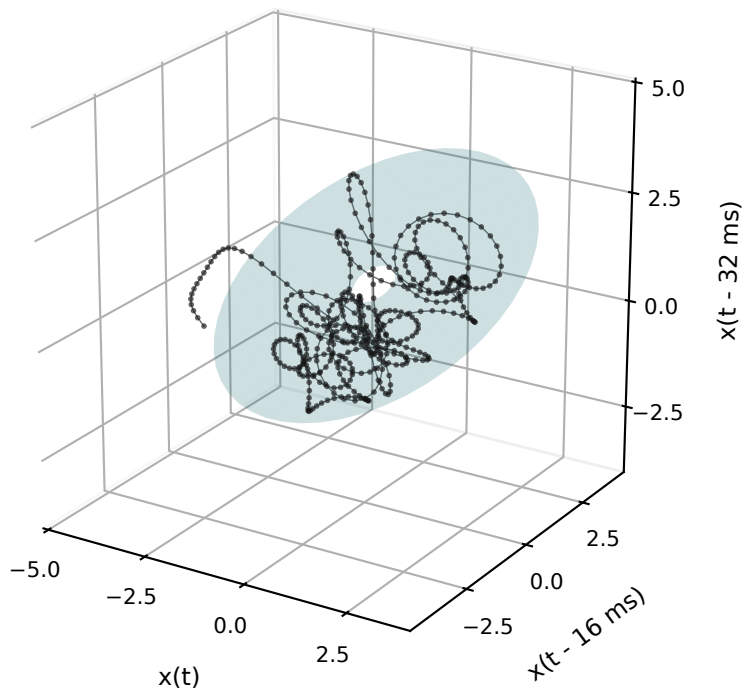
Normalized 3D set error: 0.0014  
Native coverage: 0.908 | nfev: 18  
Flags: collapsed hole; active optimizer bound; near  
optimizer bound; nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

# Direction 1 | original trial 569

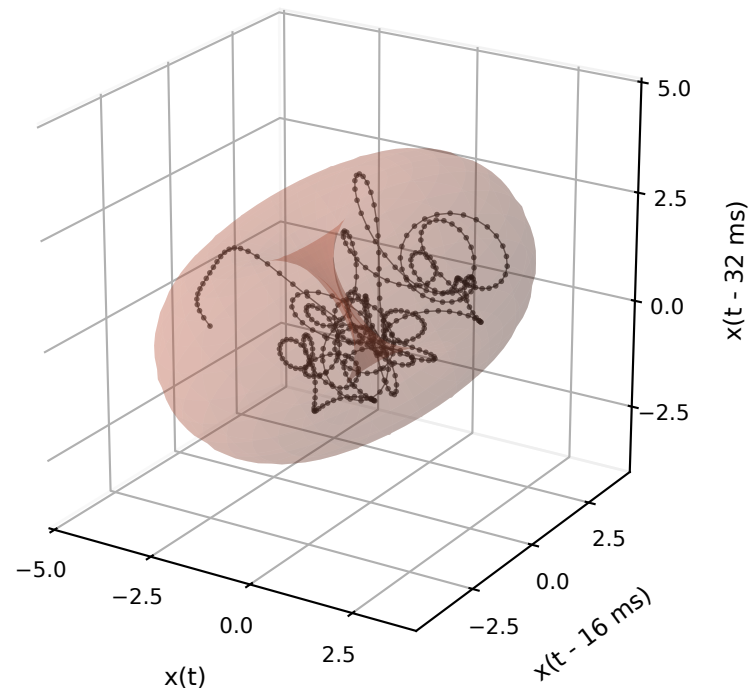
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2435  
Native coverage: 0.949 | nfev: 55  
Flags: none

2-torus: donut volume



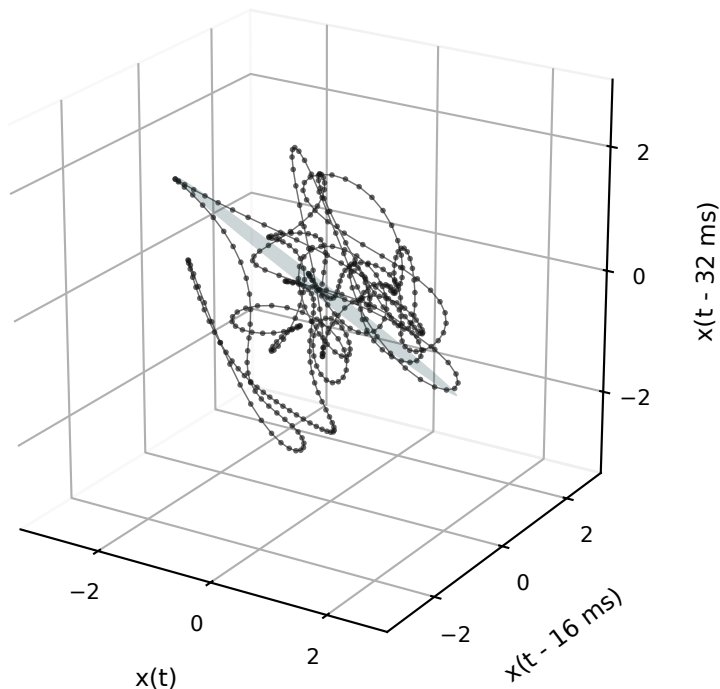
Normalized 3D set error: 0.0013  
Native coverage: 0.906 | nfev: 72  
Flags: collapsed hole

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

# Direction 1 | original trial 276

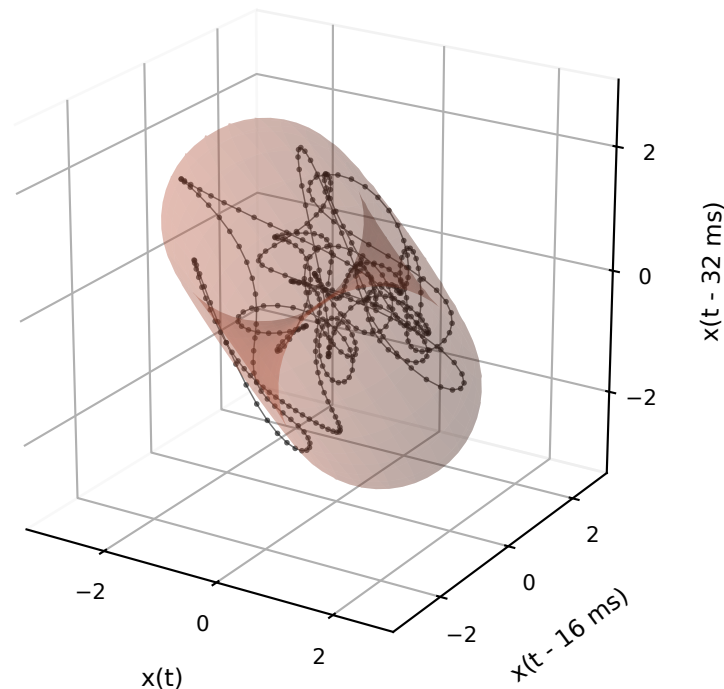
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

**1-torus: planar band**



Normalized 3D set error: 0.2618  
Native coverage: 0.915 | nfev: 52  
Flags: active optimizer bound; near optimizer bound

**2-torus: donut volume**



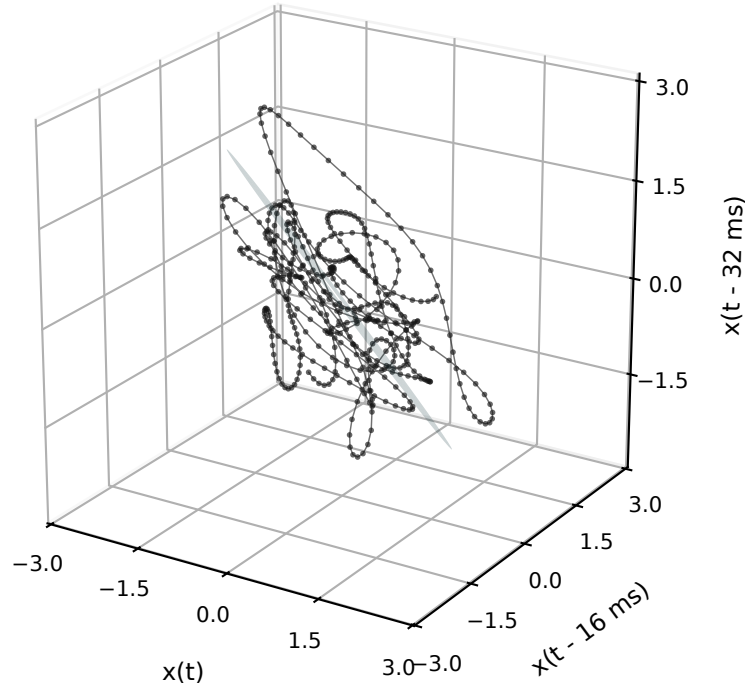
Normalized 3D set error: 0.0010  
Native coverage: 0.859 | nfev: 16  
Flags: active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

# Direction 1 | original trial 40

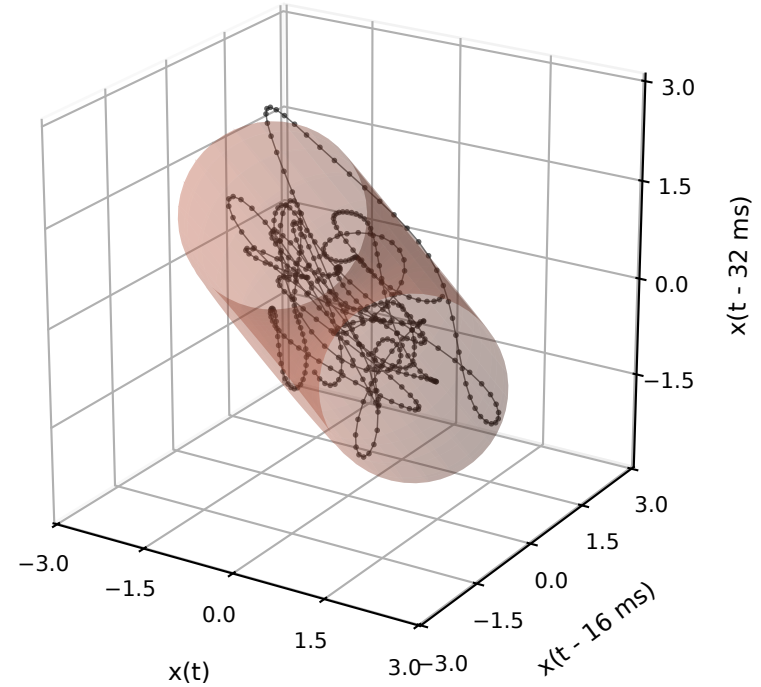
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1185  
Native coverage: 0.893 | nfev: 191  
Flags: nearly circular orientation

2-torus: donut volume



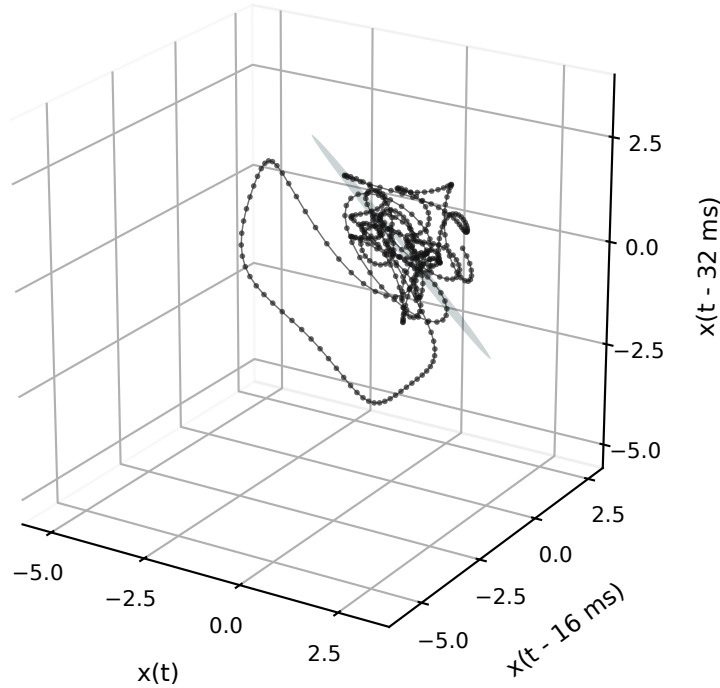
Normalized 3D set error: 0.0007  
Native coverage: 0.893 | nfev: 12  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 2 | original trial 98

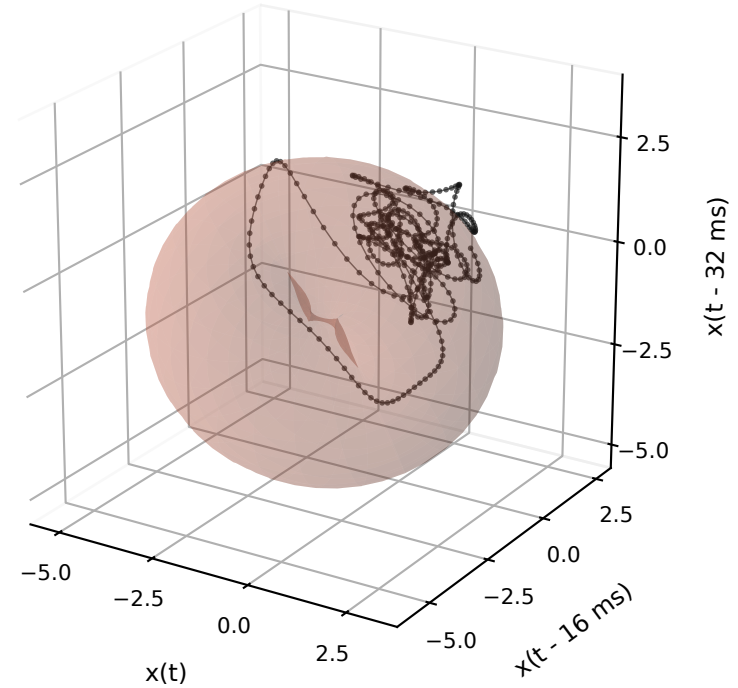
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2976  
Native coverage: 0.921 | nfev: 70  
Flags: none

2-torus: donut volume



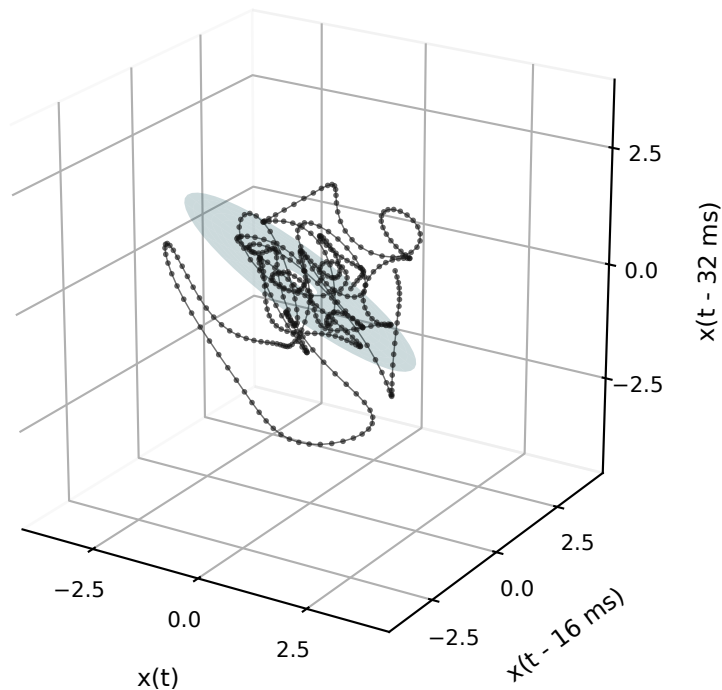
Normalized 3D set error: 0.0045  
Native coverage: 0.793 | nfev: 28  
Flags: active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 2 | original trial 260

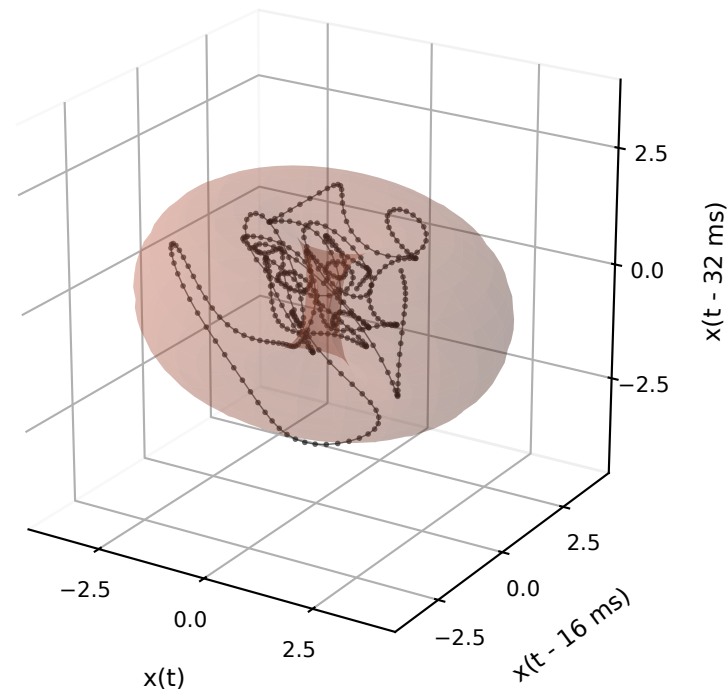
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

**1-torus: planar band**



Normalized 3D set error: 0.2863  
Native coverage: 0.882 | nfev: 146  
Flags: none

**2-torus: donut volume**



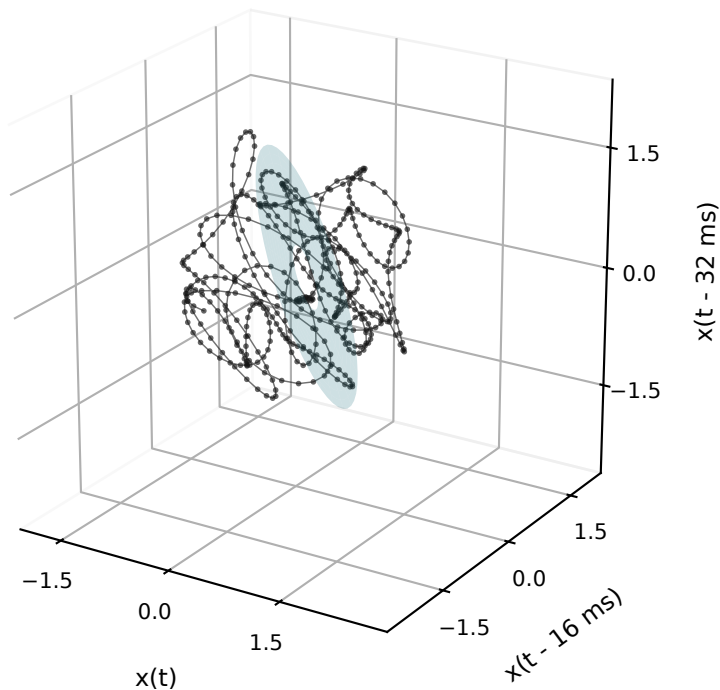
Normalized 3D set error: 0.0011  
Native coverage: 0.876 | nfev: 33  
Flags: collapsed hole; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 2 | original trial 622

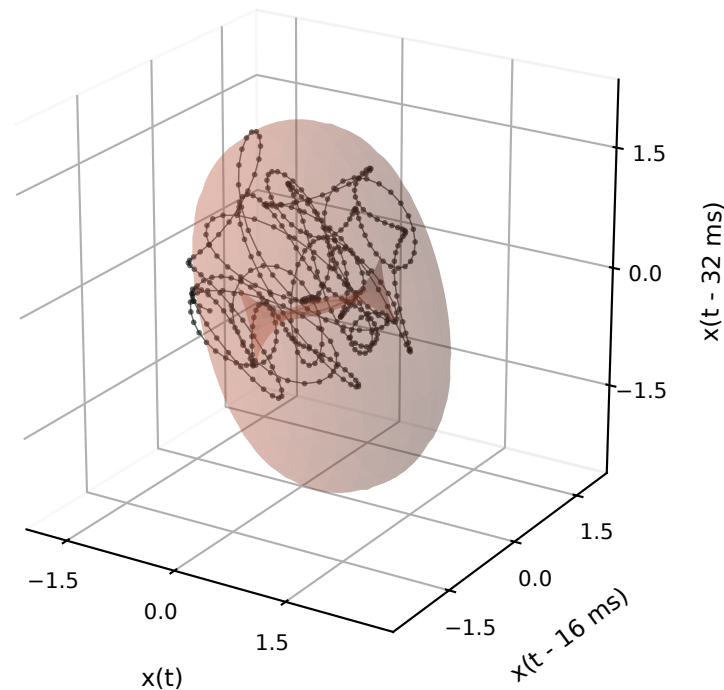
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2876  
Native coverage: 0.865 | nfev: 128  
Flags: none

2-torus: donut volume



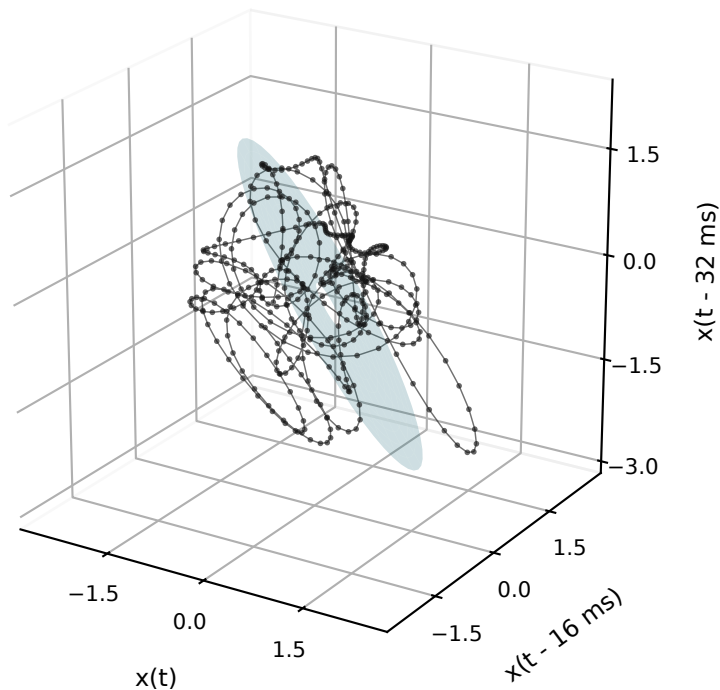
Normalized 3D set error: 0.0029  
Native coverage: 0.833 | nfev: 29  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 2 | original trial 198

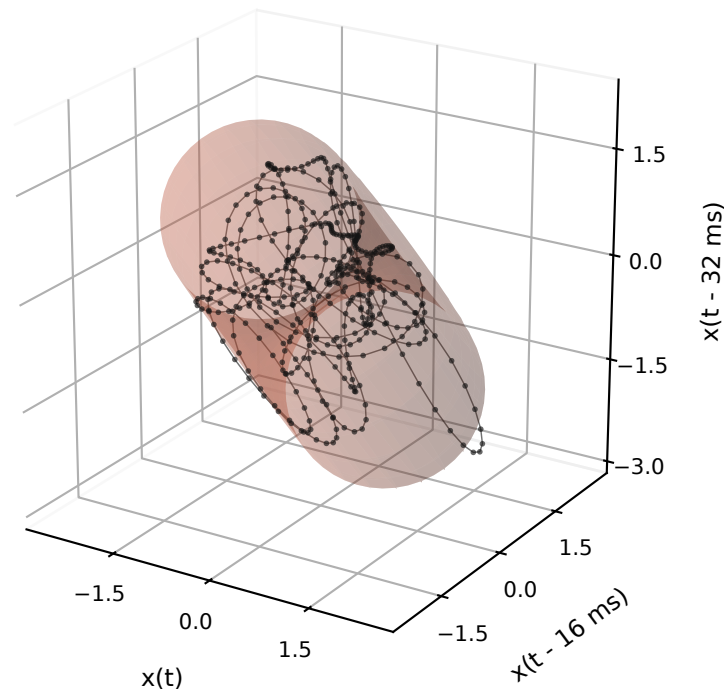
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2025  
Native coverage: 0.919 | nfev: 38  
Flags: none

2-torus: donut volume



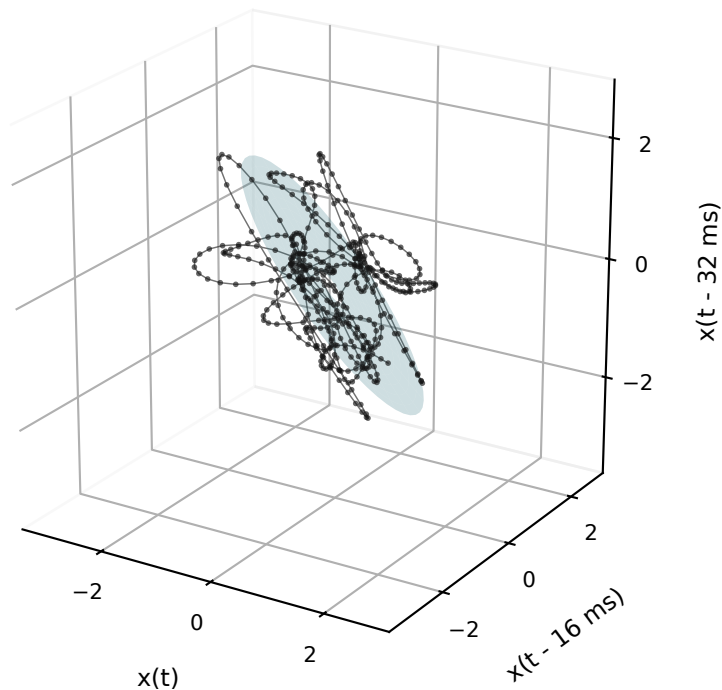
Normalized 3D set error: 0.0017  
Native coverage: 0.850 | nfev: 37  
Flags: none

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 2 | original trial 628

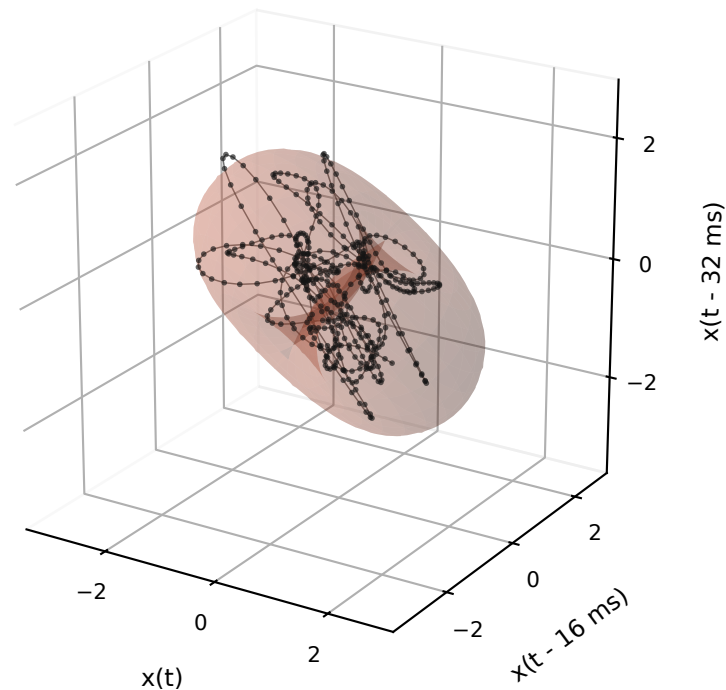
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1174  
Native coverage: 0.925 | nfev: 28  
Flags: none

2-torus: donut volume



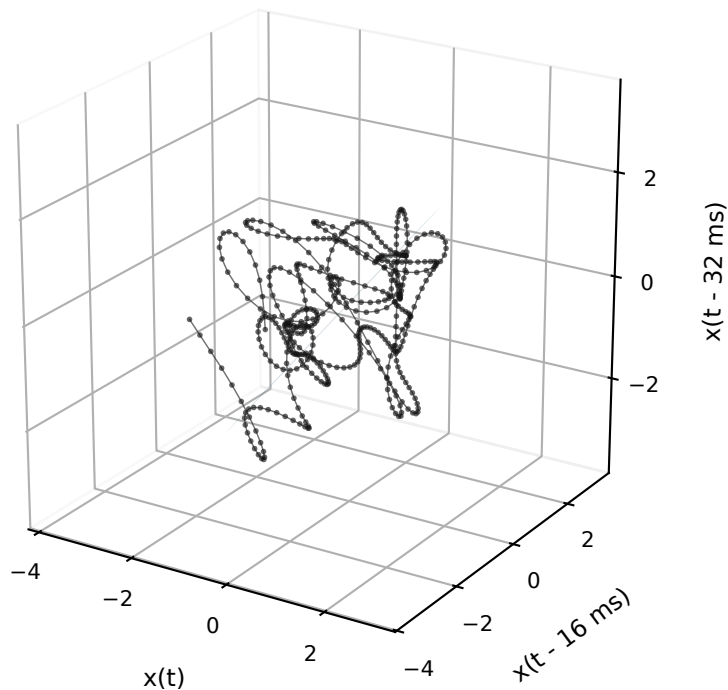
Normalized 3D set error: 0.0050  
Native coverage: 0.842 | nfev: 31  
Flags: near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 2 | original trial 647

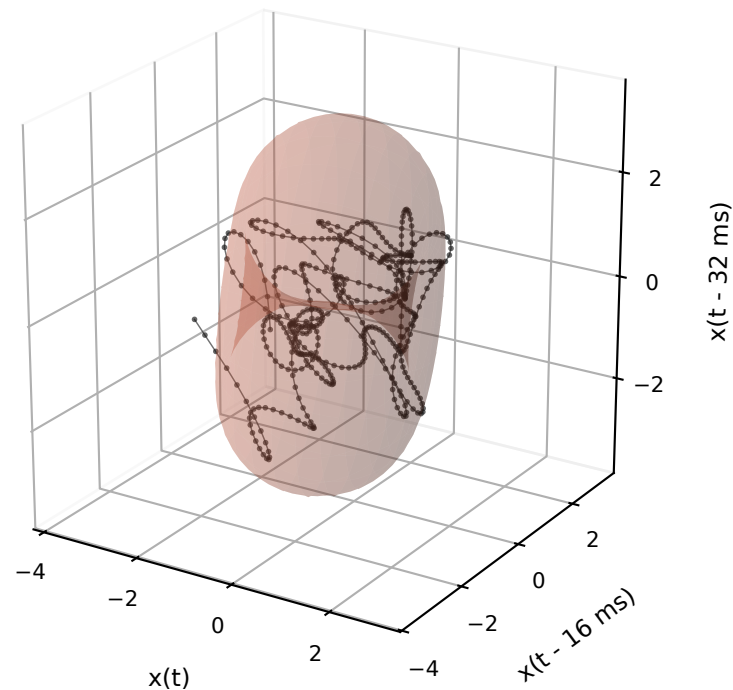
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2648  
Native coverage: 0.904 | nfev: 29  
Flags: none

2-torus: donut volume



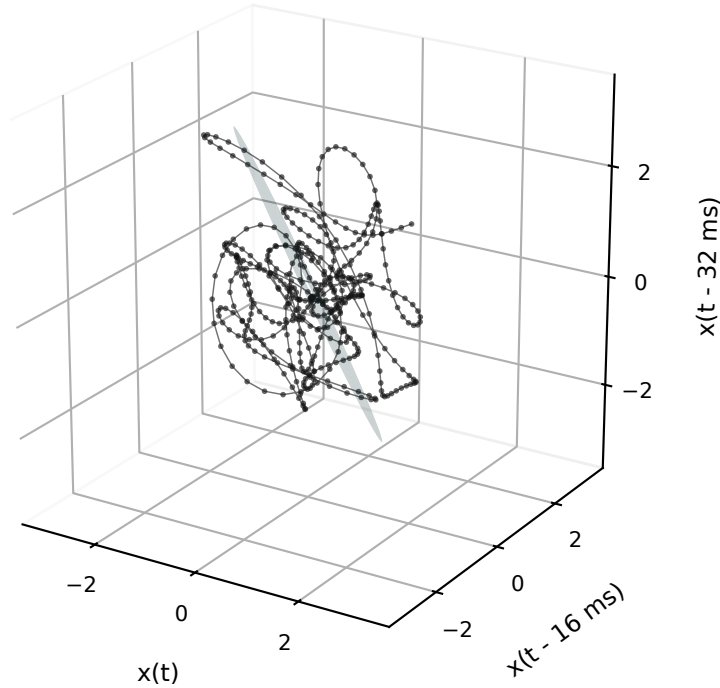
Normalized 3D set error: 0.0014  
Native coverage: 0.891 | nfev: 37  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 2 | original trial 10

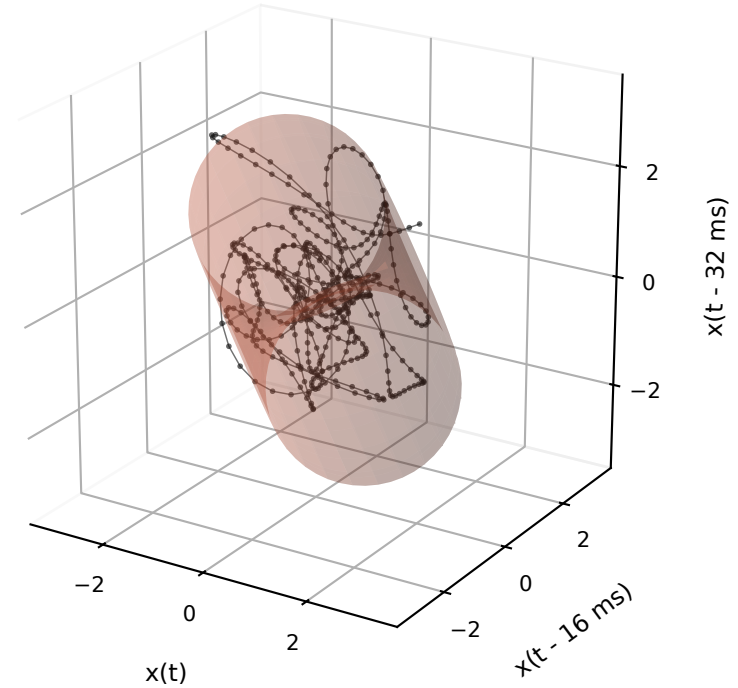
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1980  
Native coverage: 0.906 | nfev: 44  
Flags: none

2-torus: donut volume



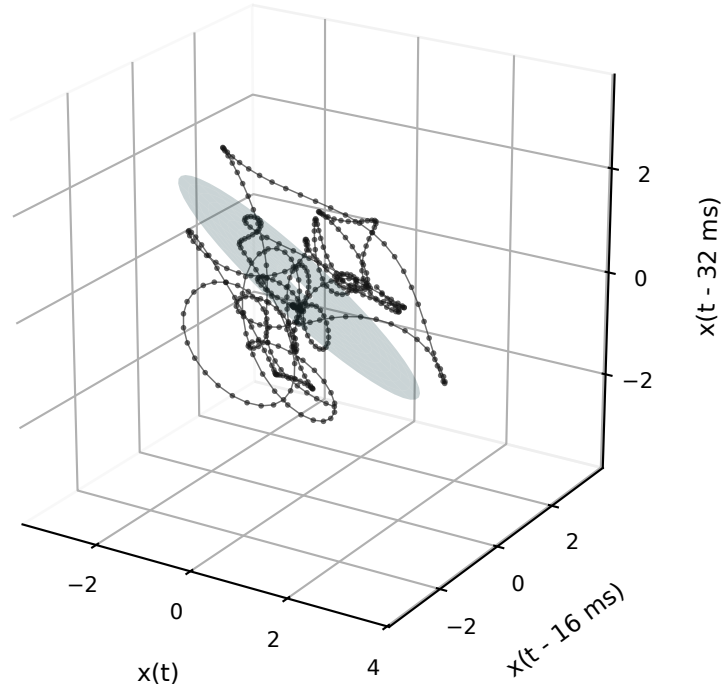
Normalized 3D set error: 0.0009  
Native coverage: 0.882 | nfev: 34  
Flags: collapsed hole; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 2 | original trial 412

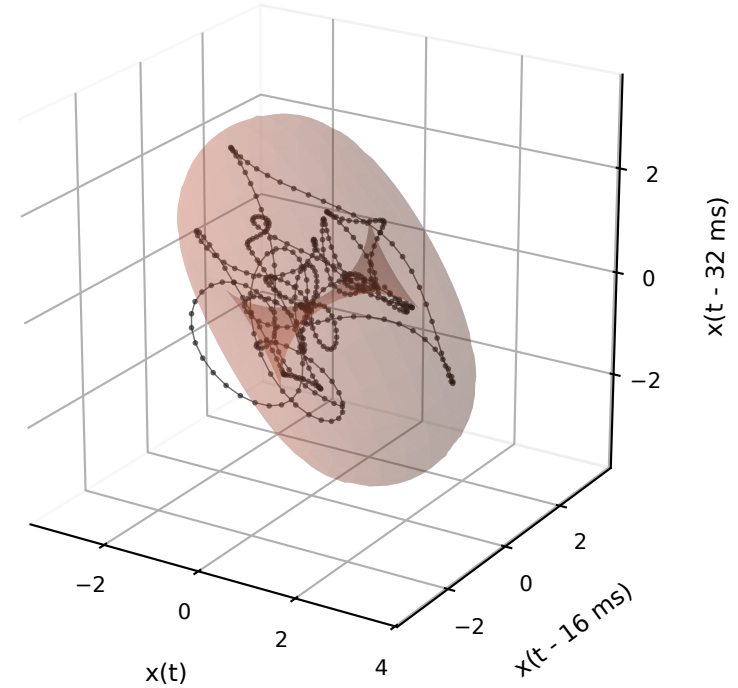
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2680  
Native coverage: 0.910 | nfev: 106  
Flags: nearly circular orientation

2-torus: donut volume



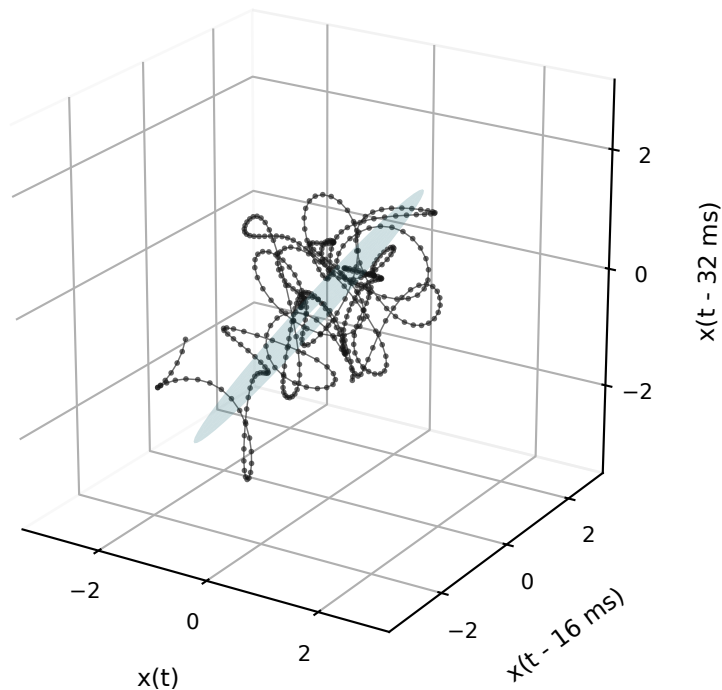
Normalized 3D set error: 0.0011  
Native coverage: 0.917 | nfev: 19  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 2 | original trial 670

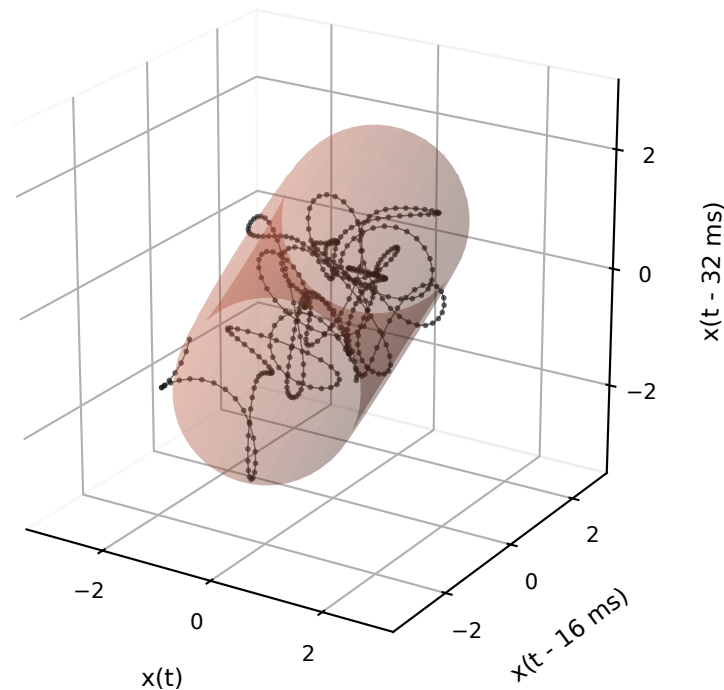
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1698  
Native coverage: 0.921 | nfev: 54  
Flags: none

2-torus: donut volume



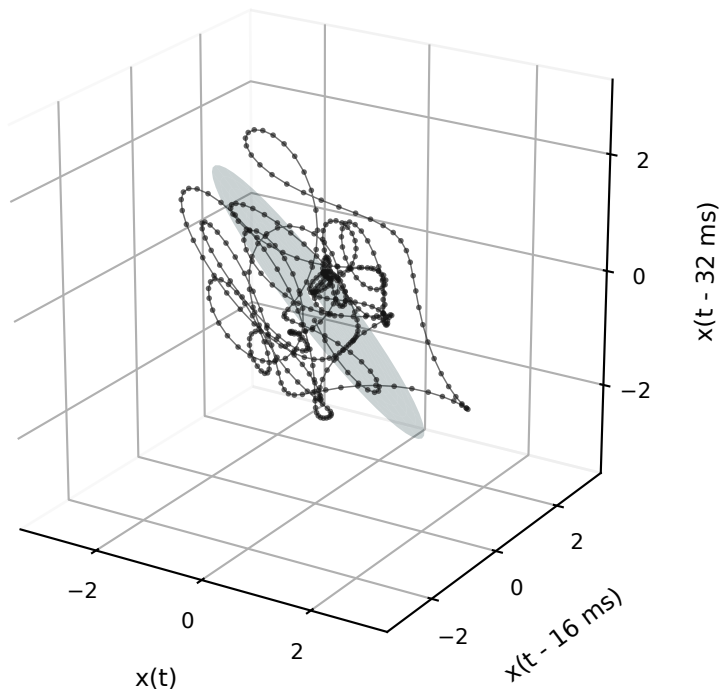
Normalized 3D set error: 0.0016  
Native coverage: 0.885 | nfev: 25  
Flags: near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 2 | original trial 150

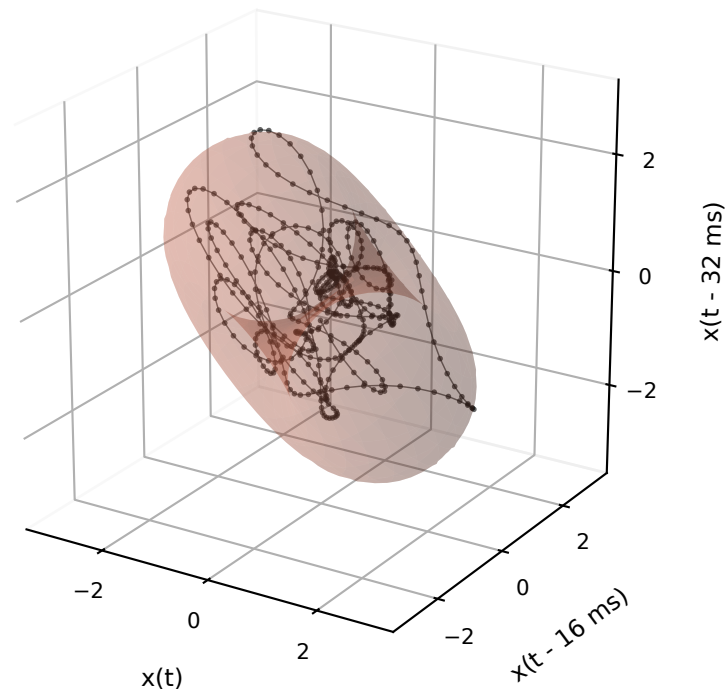
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1829  
Native coverage: 0.889 | nfev: 64  
Flags: none

2-torus: donut volume



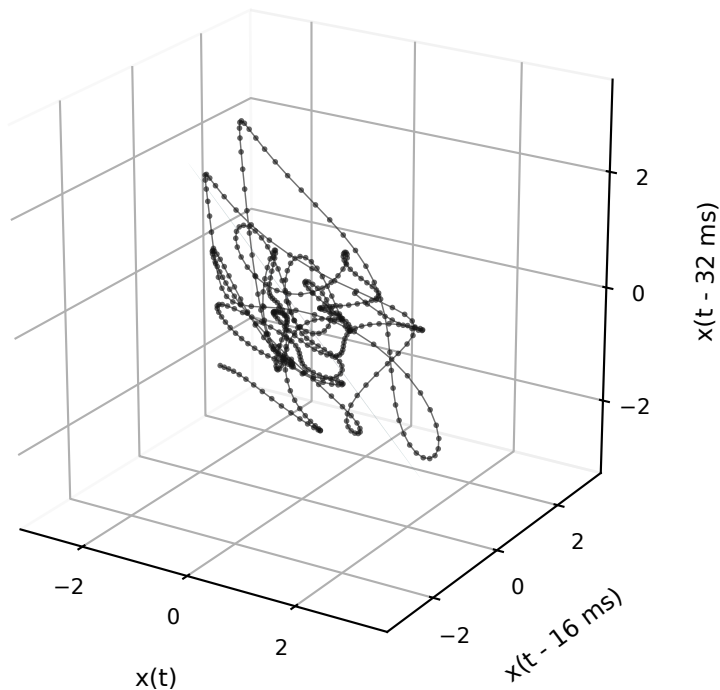
Normalized 3D set error: 0.0009  
Native coverage: 0.878 | nfev: 35  
Flags: near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 2 | original trial 521

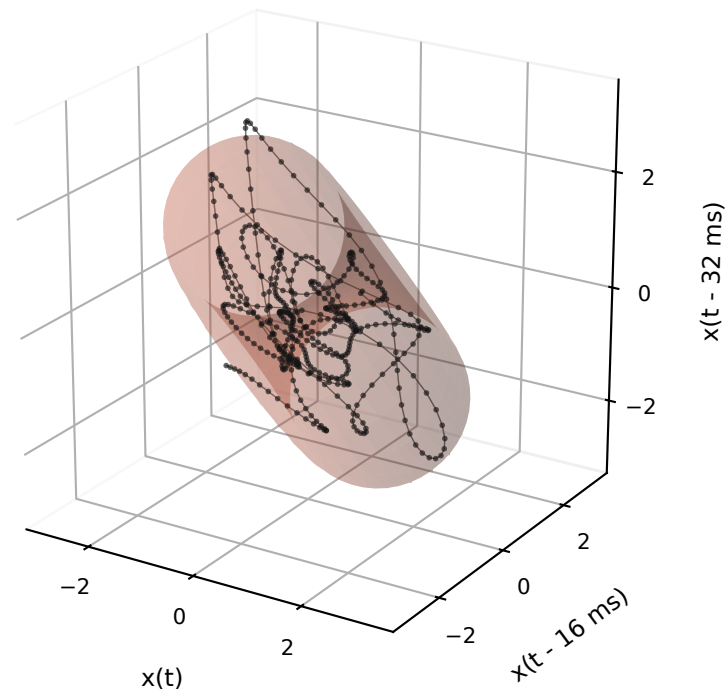
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1503  
Native coverage: 0.915 | nfev: 78  
Flags: near optimizer bound

2-torus: donut volume



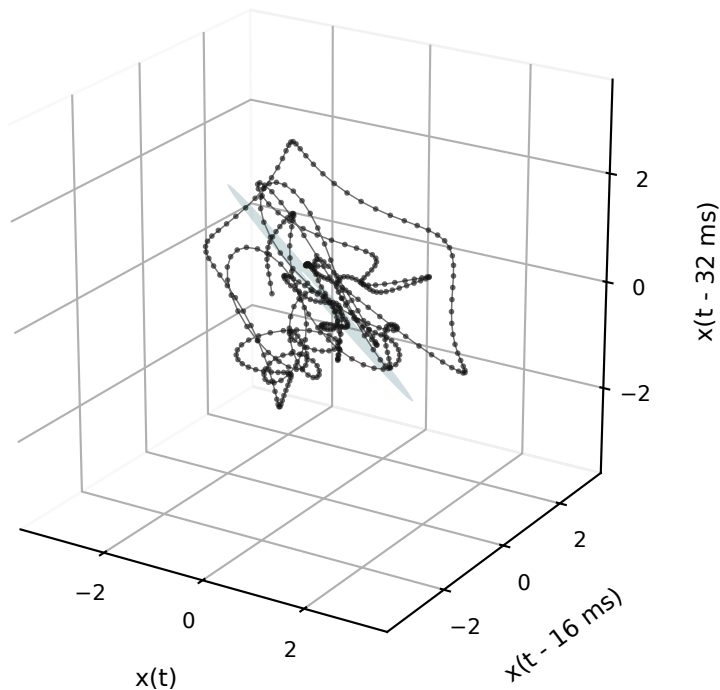
Normalized 3D set error: 0.0019  
Native coverage: 0.812 | nfev: 27  
Flags: near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 2 | original trial 131

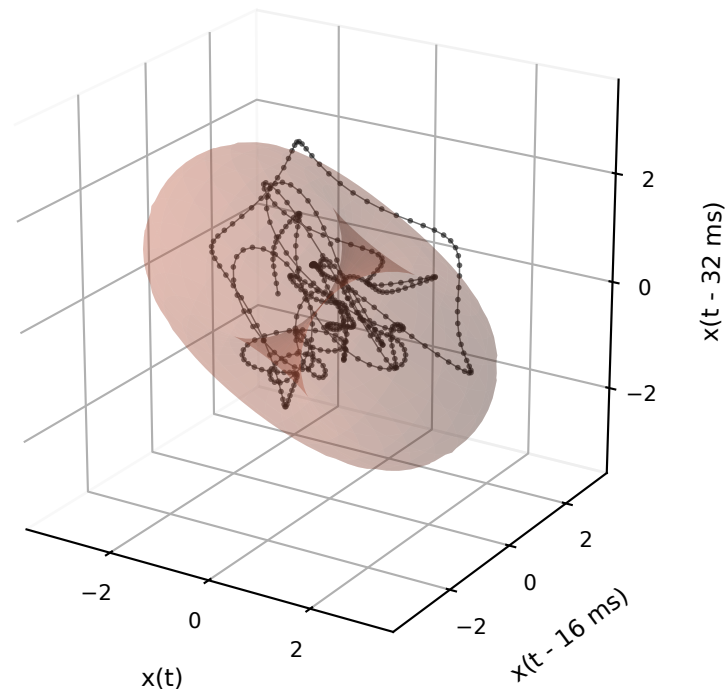
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2454  
Native coverage: 0.893 | nfev: 91  
Flags: nearly circular orientation

2-torus: donut volume



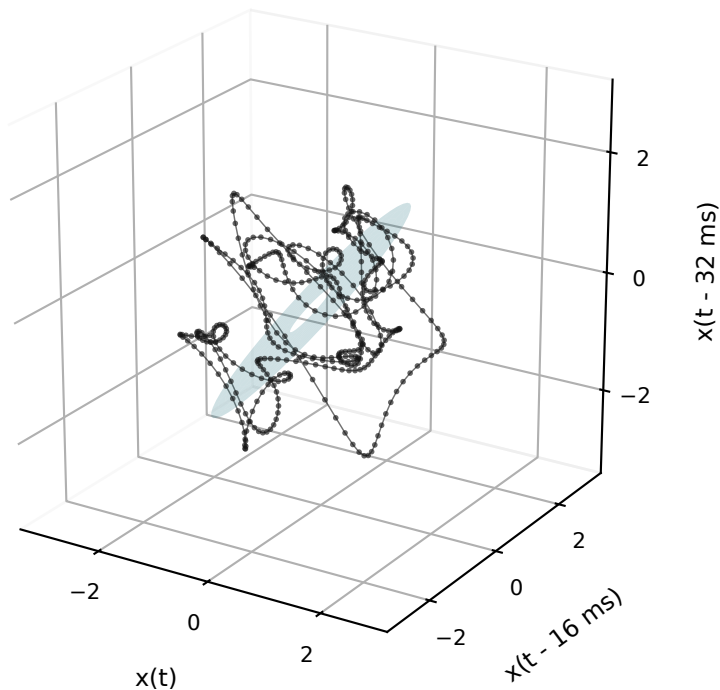
Normalized 3D set error: 0.0012  
Native coverage: 0.908 | nfev: 59  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 2 | original trial 579

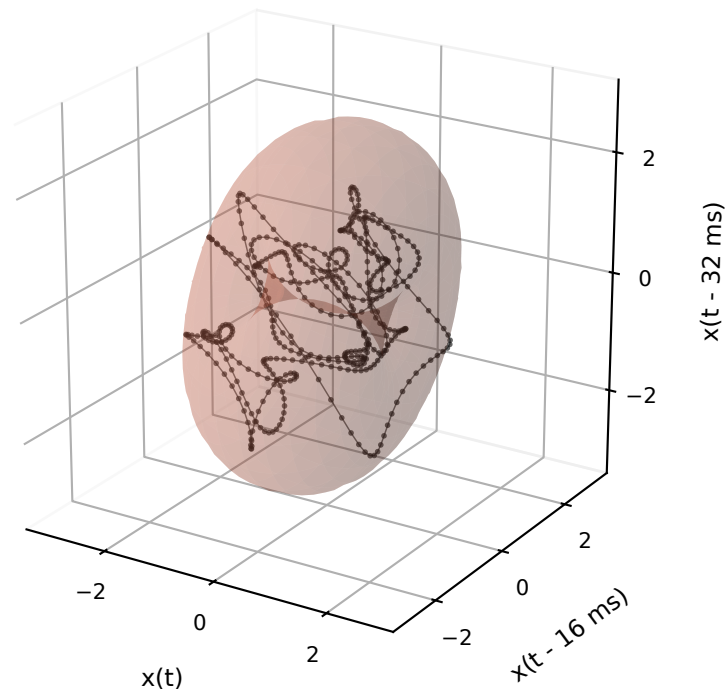
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2954  
Native coverage: 0.900 | nfev: 102  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

2-torus: donut volume



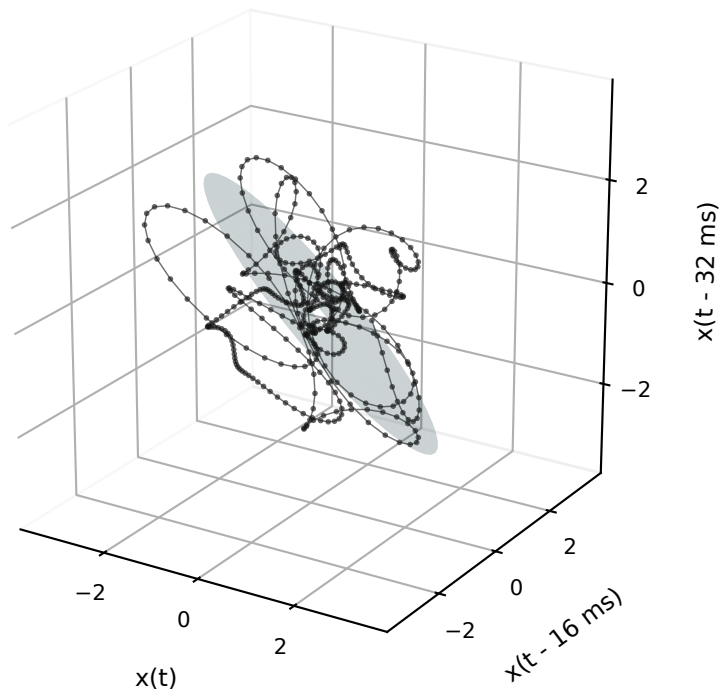
Normalized 3D set error: 0.0014  
Native coverage: 0.857 | nfev: 5404  
Flags: collapsed hole; active optimizer bound; near  
optimizer bound; nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 2 | original trial 419

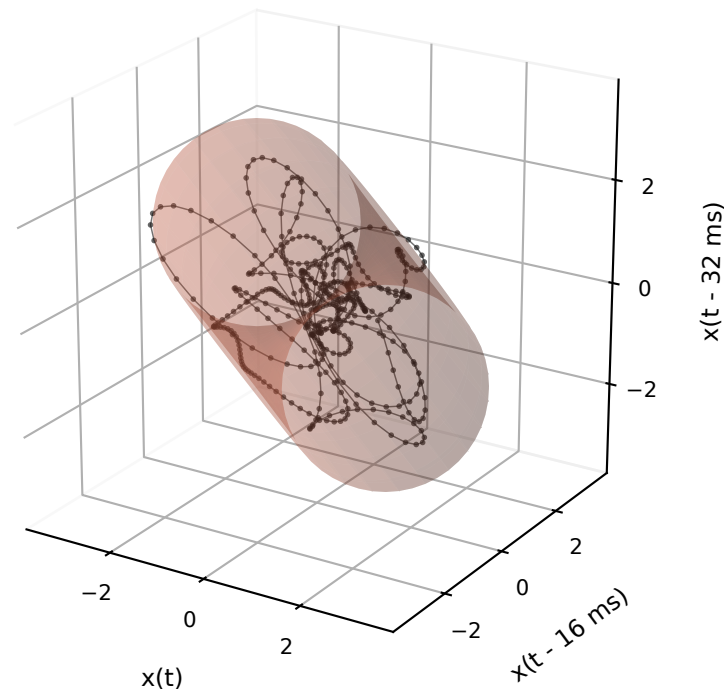
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1969  
Native coverage: 0.902 | nfev: 29  
Flags: none

2-torus: donut volume



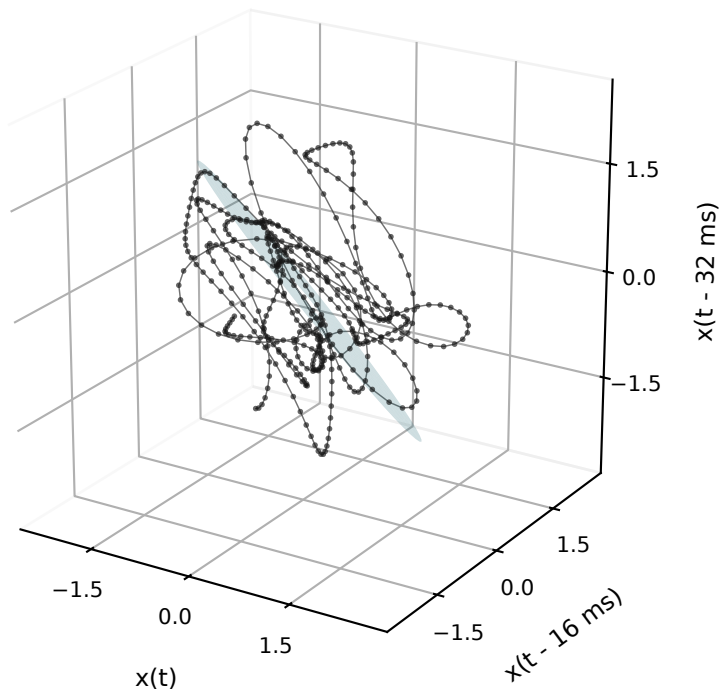
Normalized 3D set error: 0.0010  
Native coverage: 0.882 | nfev: 28  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 2 | original trial 580

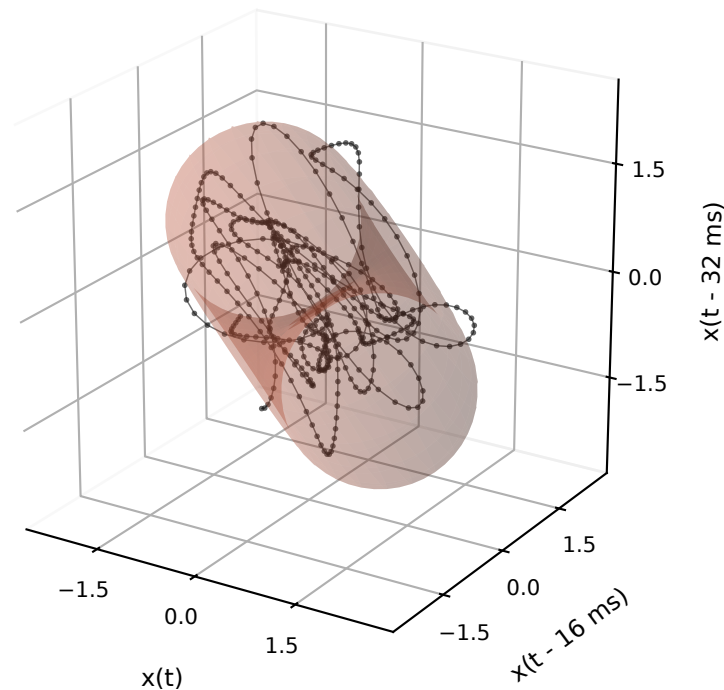
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1608  
Native coverage: 0.923 | nfev: 18  
Flags: active optimizer bound; near optimizer bound

2-torus: donut volume



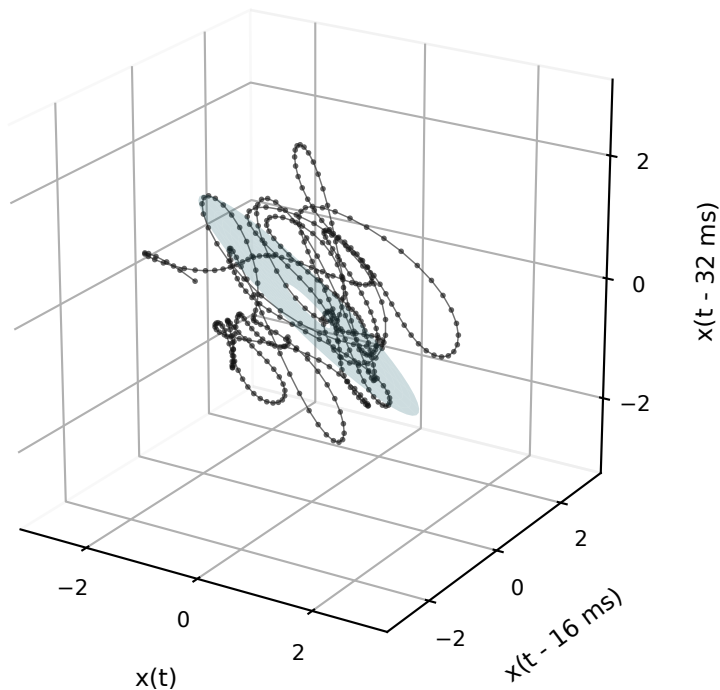
Normalized 3D set error: 0.0013  
Native coverage: 0.882 | nfev: 24  
Flags: near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 2 | original trial 428

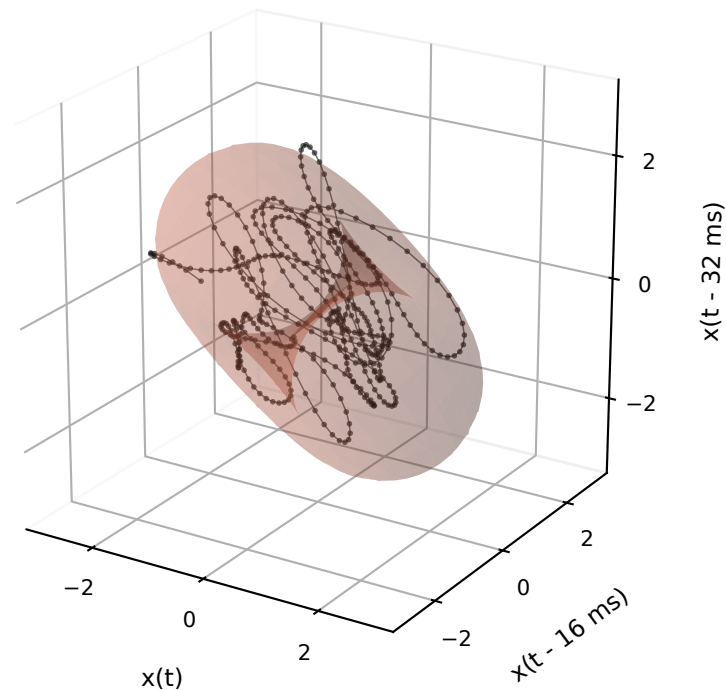
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1841  
Native coverage: 0.927 | nfev: 81  
Flags: none

2-torus: donut volume



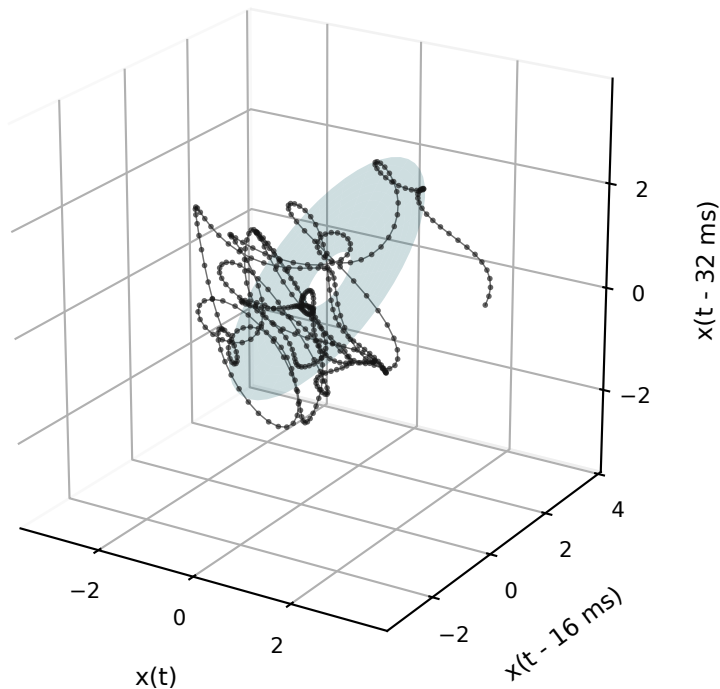
Normalized 3D set error: 0.0010  
Native coverage: 0.865 | nfev: 40  
Flags: none

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 2 | original trial 543

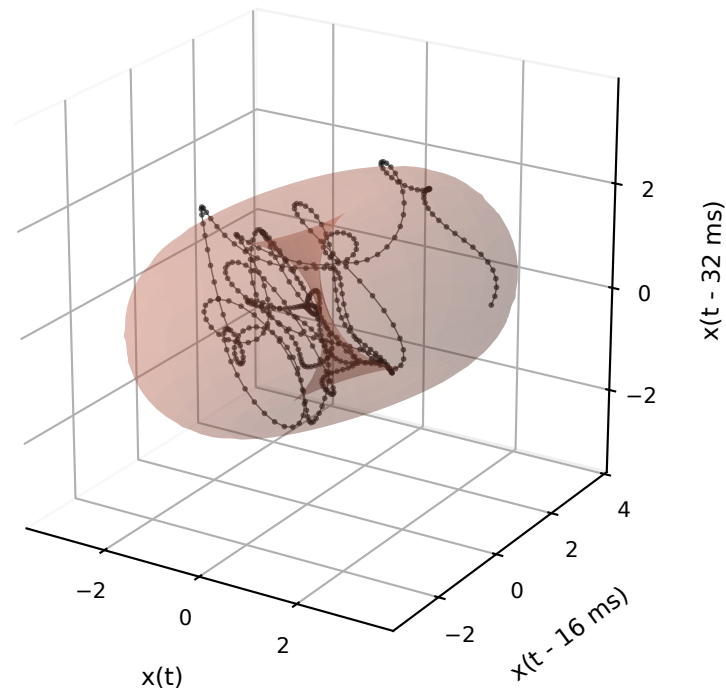
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.3154  
Native coverage: 0.885 | nfev: 64  
Flags: none

2-torus: donut volume



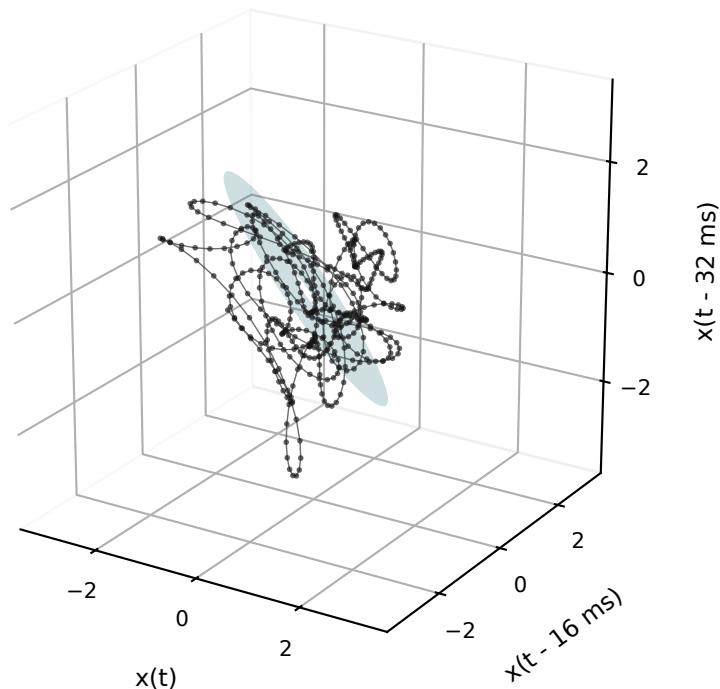
Normalized 3D set error: 0.0011  
Native coverage: 0.865 | nfev: 28  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 2 | original trial 387

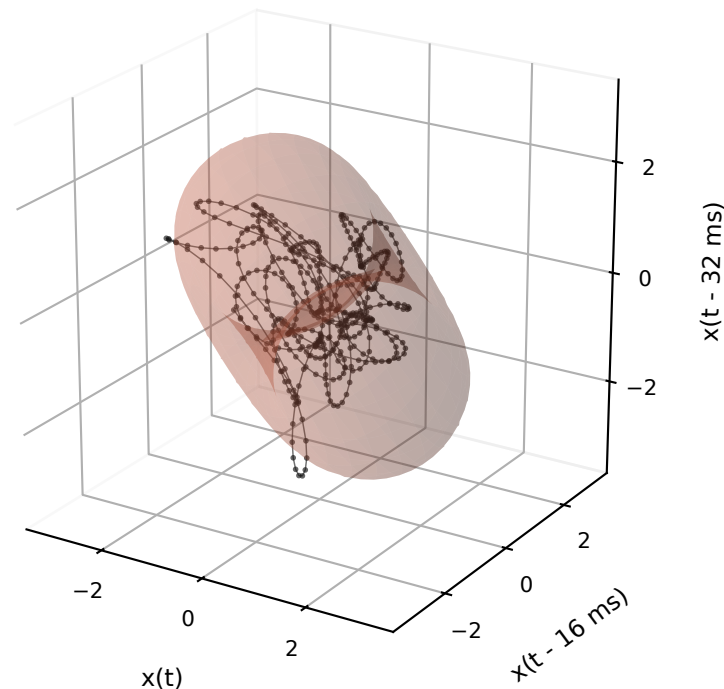
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2296  
Native coverage: 0.900 | nfev: 39  
Flags: none

2-torus: donut volume



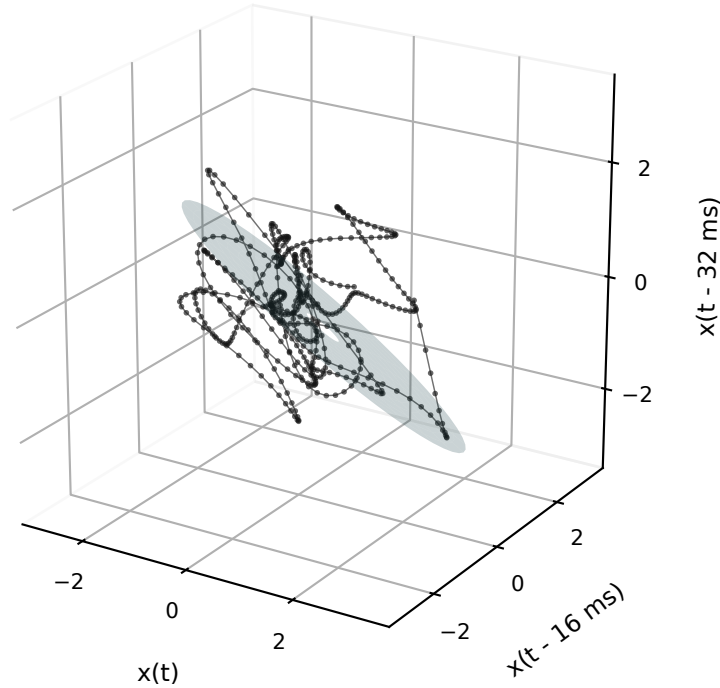
Normalized 3D set error: 0.0019  
Native coverage: 0.887 | nfev: 41  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 2 | original trial 158

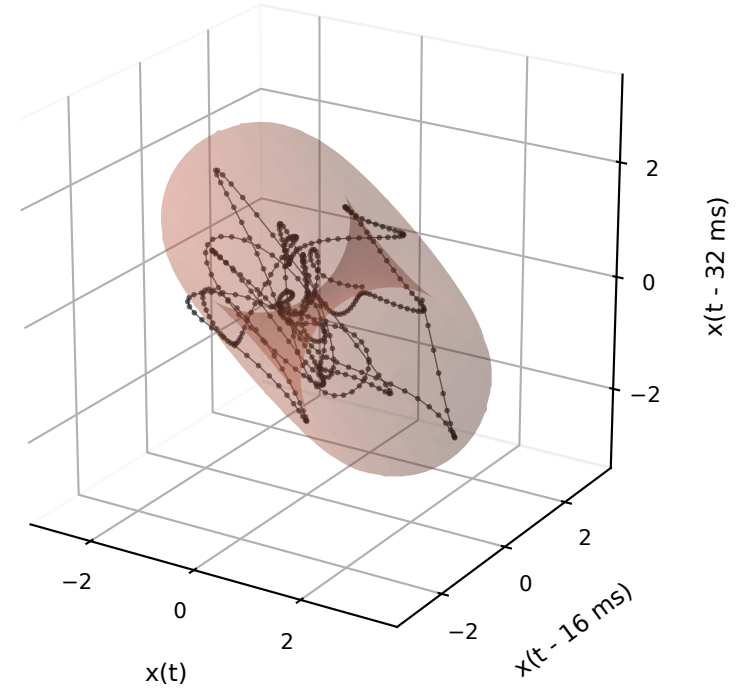
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1706  
Native coverage: 0.936 | nfev: 28  
Flags: near optimizer bound; nearly circular orientation

2-torus: donut volume



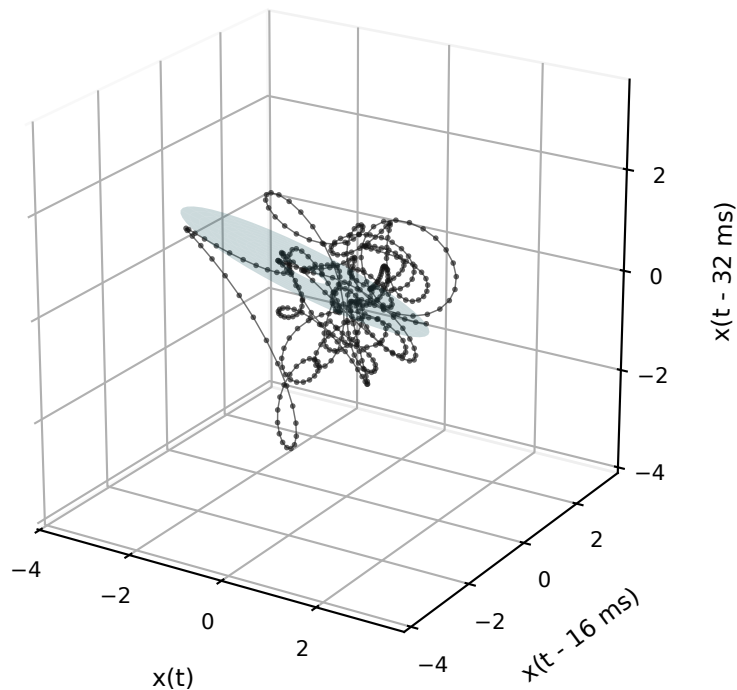
Normalized 3D set error: 0.0007  
Native coverage: 0.870 | nfev: 18  
Flags: active optimizer bound; near optimizer bound; nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 2 | original trial 275

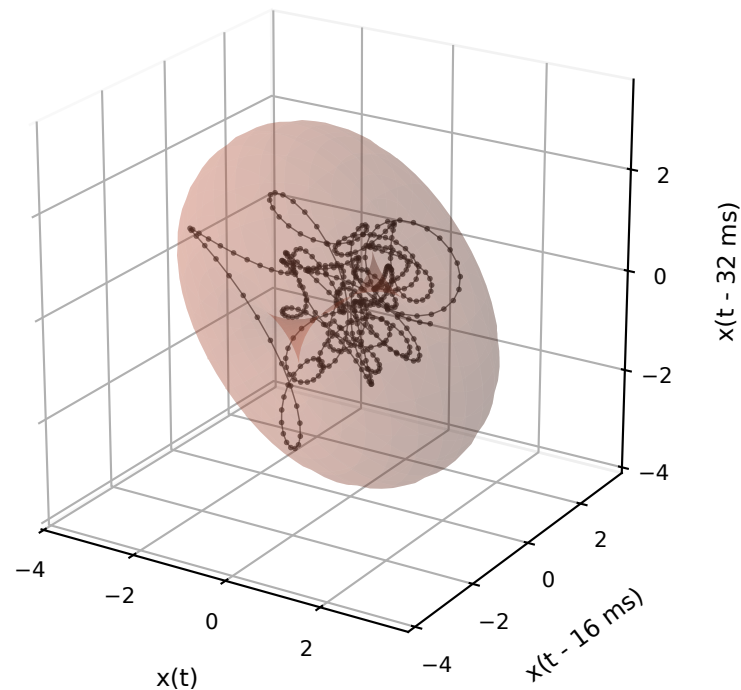
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.3154  
Native coverage: 0.906 | nfev: 75  
Flags: nearly circular orientation

2-torus: donut volume



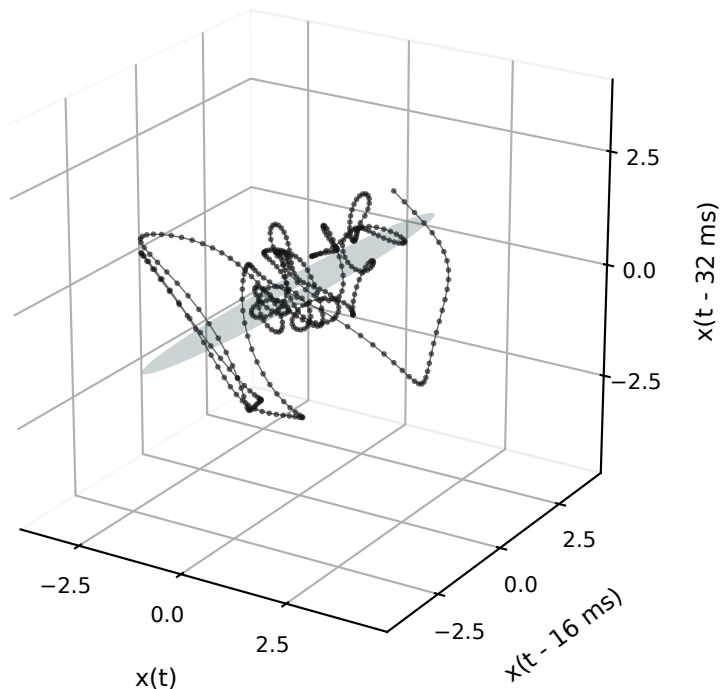
Normalized 3D set error: 0.0013  
Native coverage: 0.917 | nfev: 23  
Flags: collapsed hole; active optimizer bound; near optimizer bound; nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 2 | original trial 254

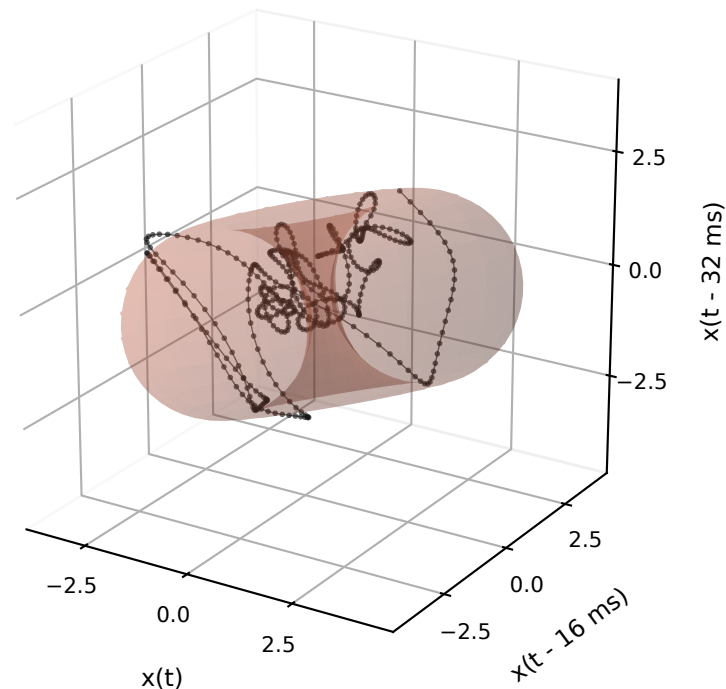
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2656  
Native coverage: 0.906 | nfev: 42  
Flags: none

2-torus: donut volume



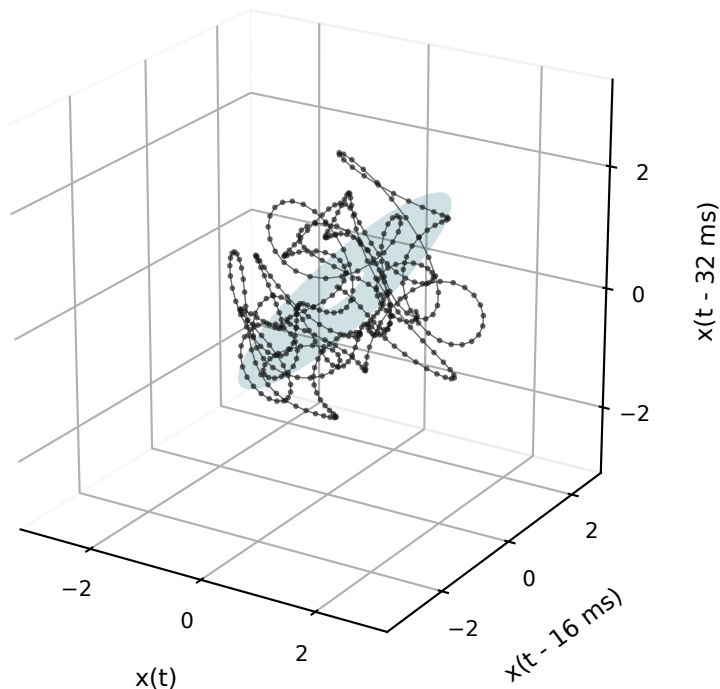
Normalized 3D set error: 0.0013  
Native coverage: 0.912 | nfev: 42  
Flags: collapsed hole

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 2 | original trial 220

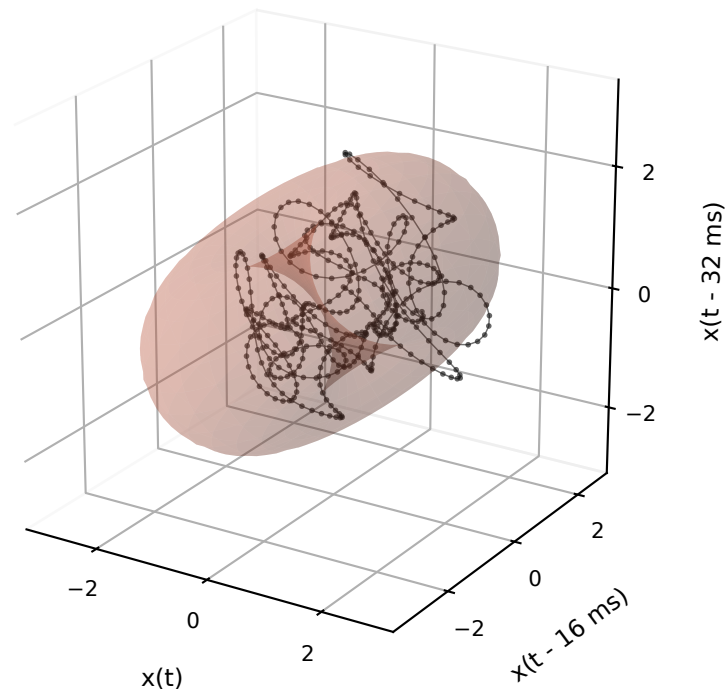
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2841  
Native coverage: 0.893 | nfev: 53  
Flags: none

2-torus: donut volume



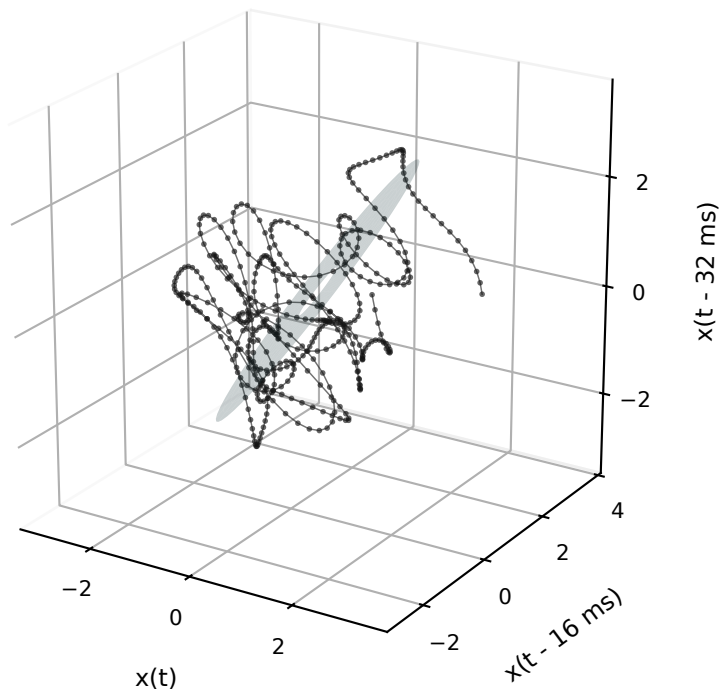
Normalized 3D set error: 0.0023  
Native coverage: 0.882 | nfev: 21  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 2 | original trial 589

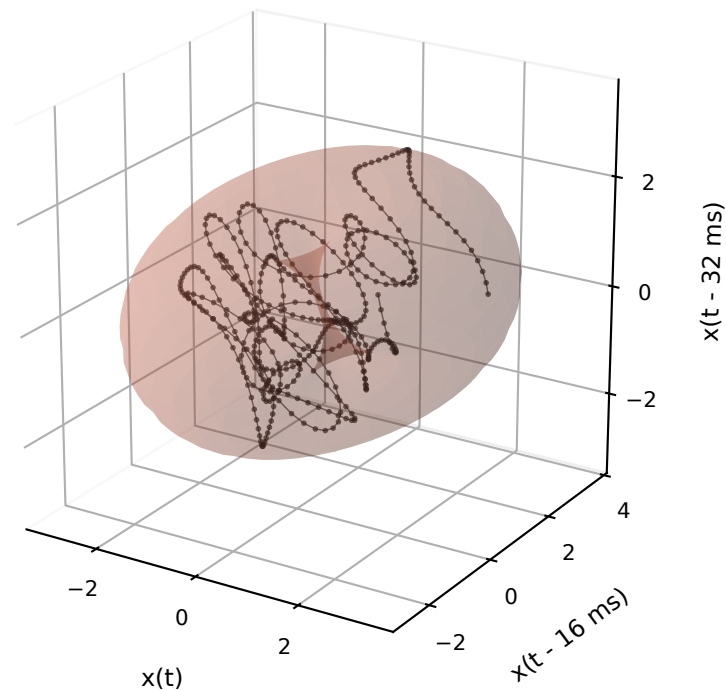
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2542  
Native coverage: 0.904 | nfev: 57  
Flags: none

2-torus: donut volume



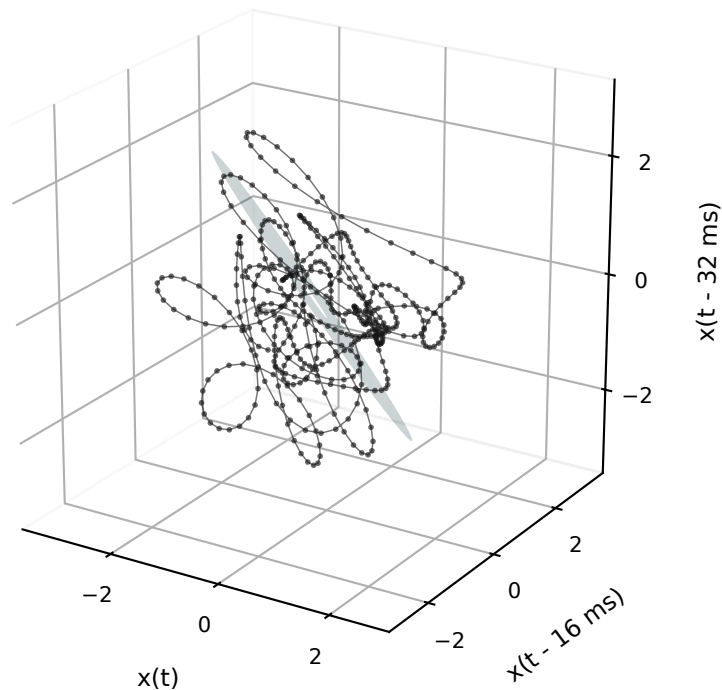
Normalized 3D set error: 0.0008  
Native coverage: 0.880 | nfev: 38  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 2 | original trial 530

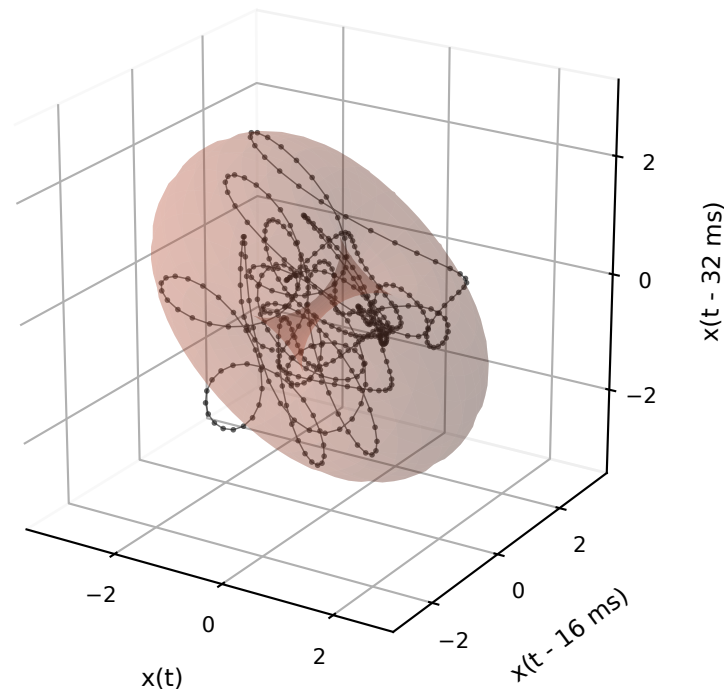
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2742  
Native coverage: 0.925 | nfev: 39  
Flags: none

2-torus: donut volume



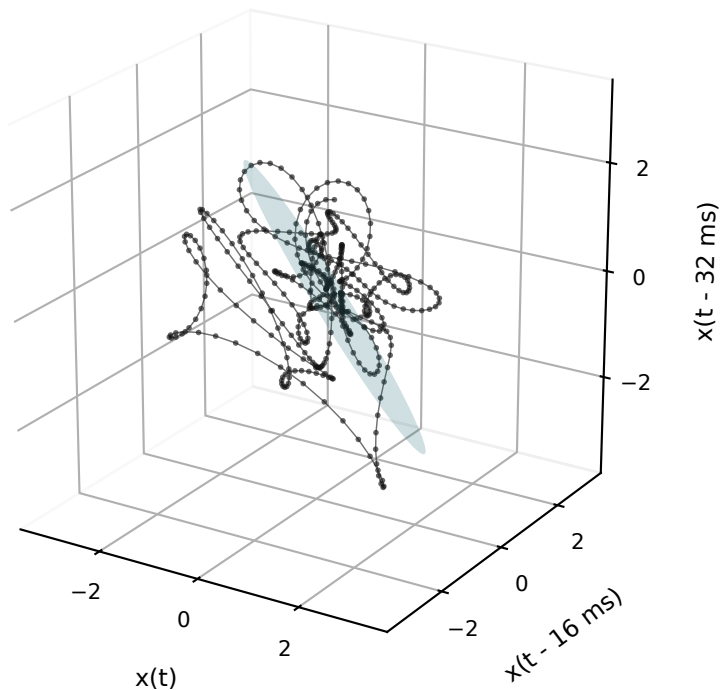
Normalized 3D set error: 0.0013  
Native coverage: 0.857 | nfev: 33  
Flags: near optimizer bound; nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 2 | original trial 531

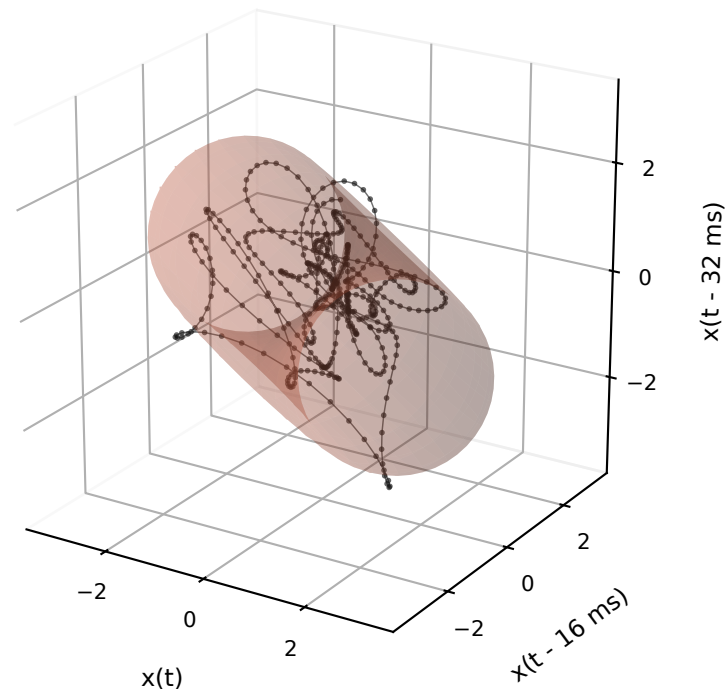
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2207  
Native coverage: 0.917 | nfev: 60  
Flags: none

2-torus: donut volume



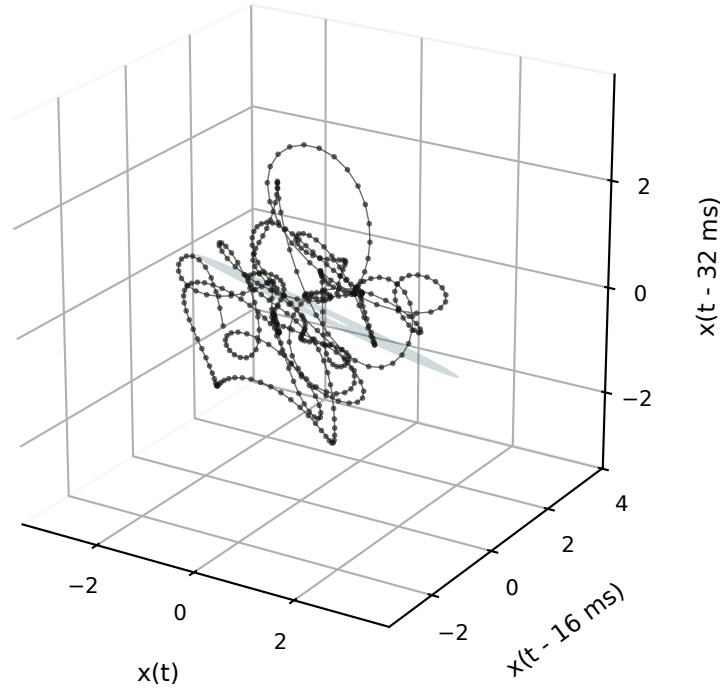
Normalized 3D set error: 0.0010  
Native coverage: 0.861 | nfev: 37  
Flags: active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 2 | original trial 91

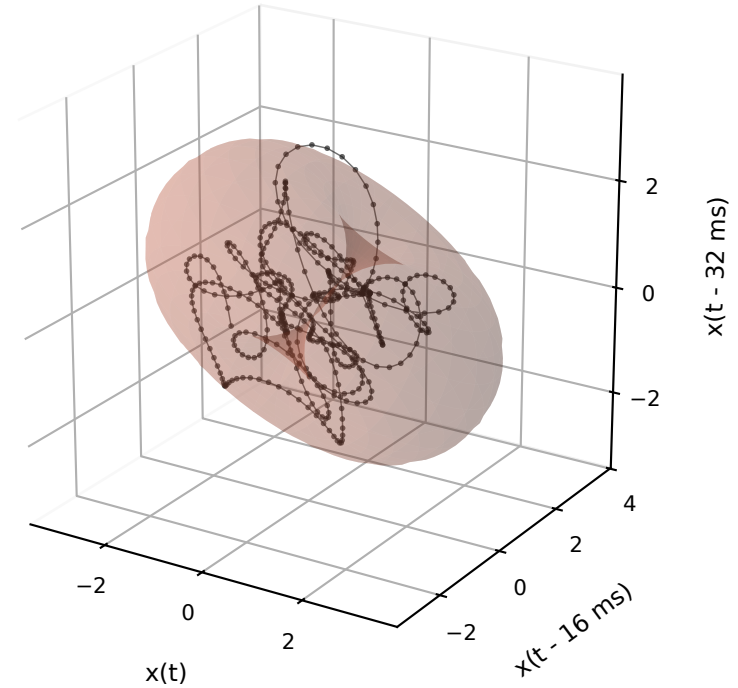
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2936  
Native coverage: 0.906 | nfev: 75  
Flags: near optimizer bound; nearly circular orientation

2-torus: donut volume



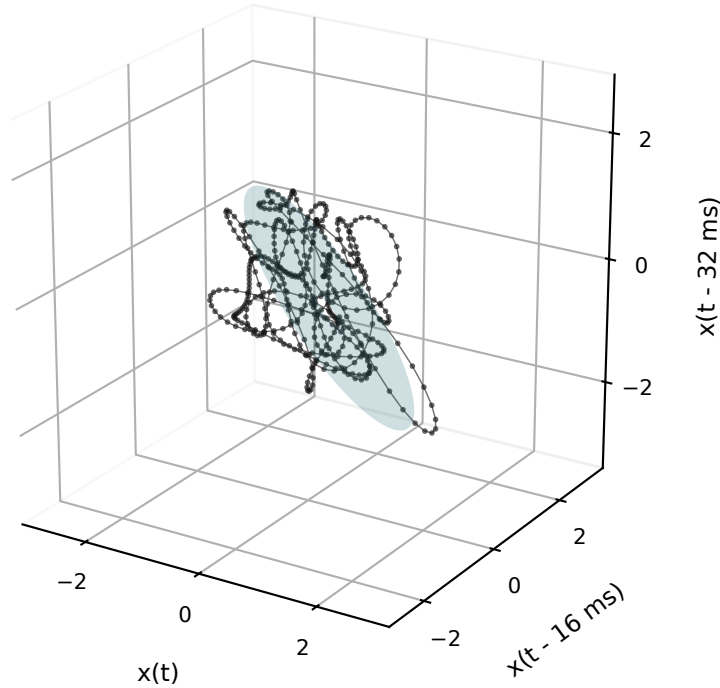
Normalized 3D set error: 0.0020  
Native coverage: 0.932 | nfev: 38  
Flags: collapsed hole; active optimizer bound; near optimizer bound; nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 2 | original trial 96

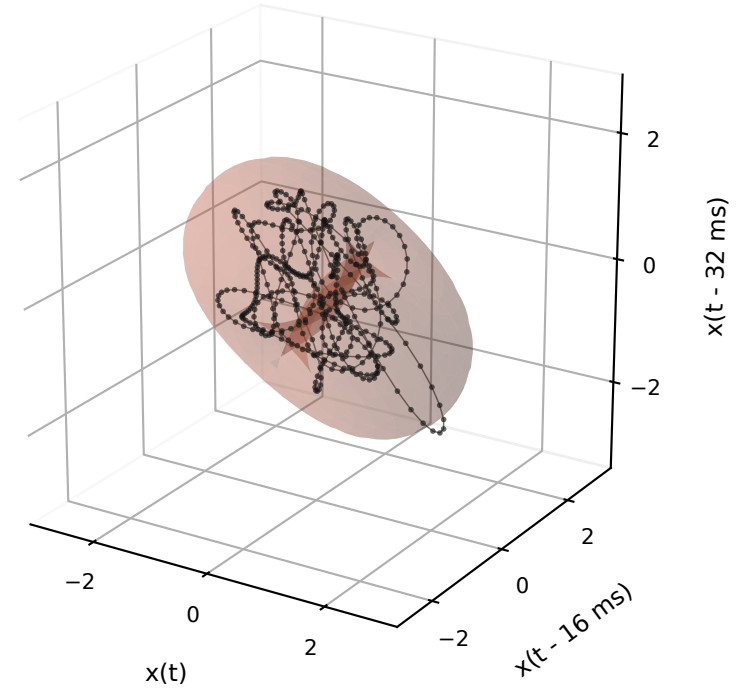
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1301  
Native coverage: 0.910 | nfev: 28  
Flags: none

2-torus: donut volume



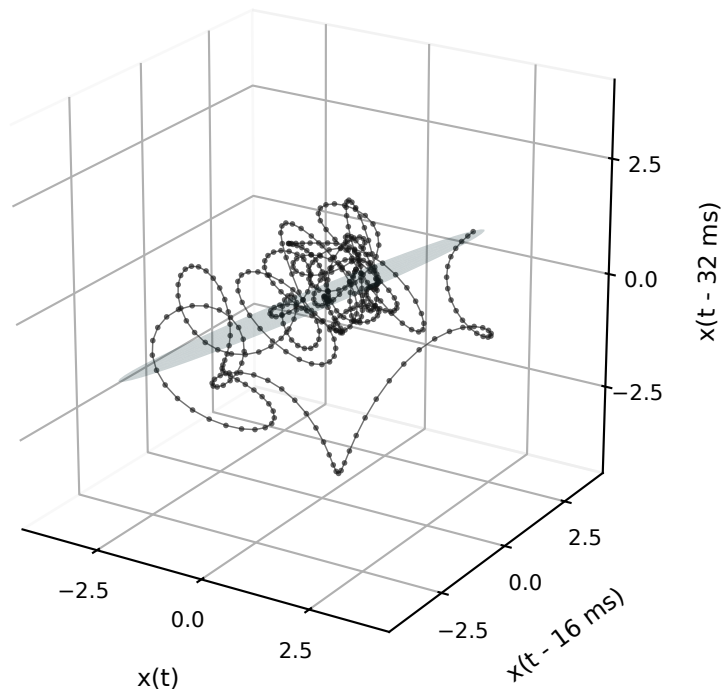
Normalized 3D set error: 0.0063  
Native coverage: 0.827 | nfev: 39  
Flags: near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 2 | original trial 197

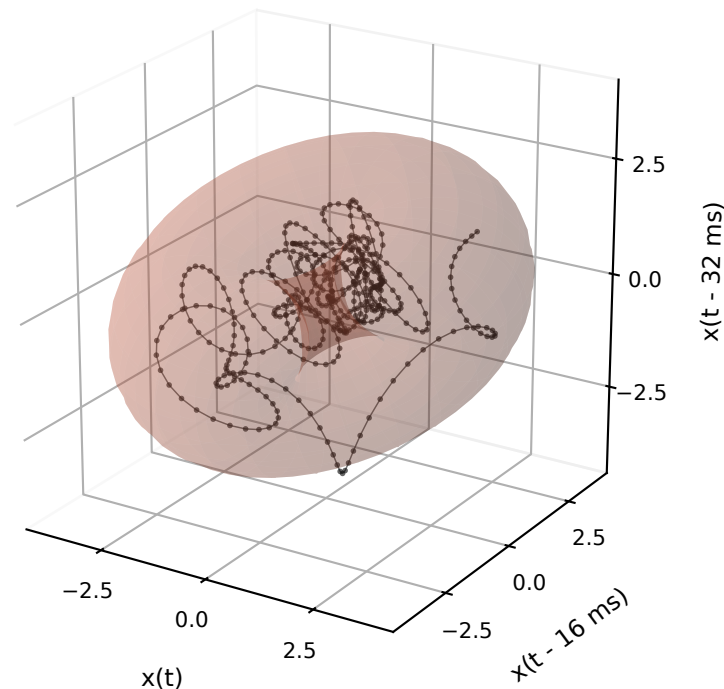
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

**1-torus: planar band**



Normalized 3D set error: 0.2392  
Native coverage: 0.932 | nfev: 72  
Flags: active optimizer bound; near optimizer bound

**2-torus: donut volume**



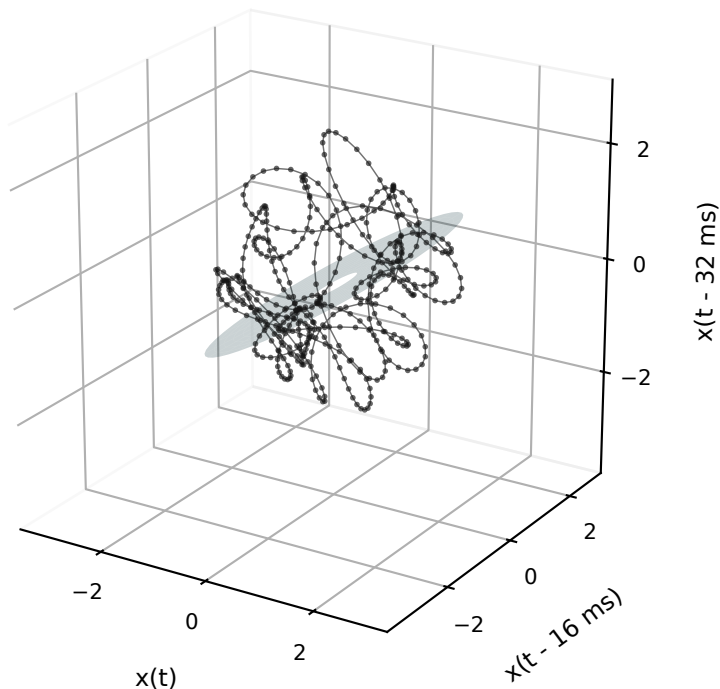
Normalized 3D set error: 0.0009  
Native coverage: 0.874 | nfev: 43  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 2 | original trial 281

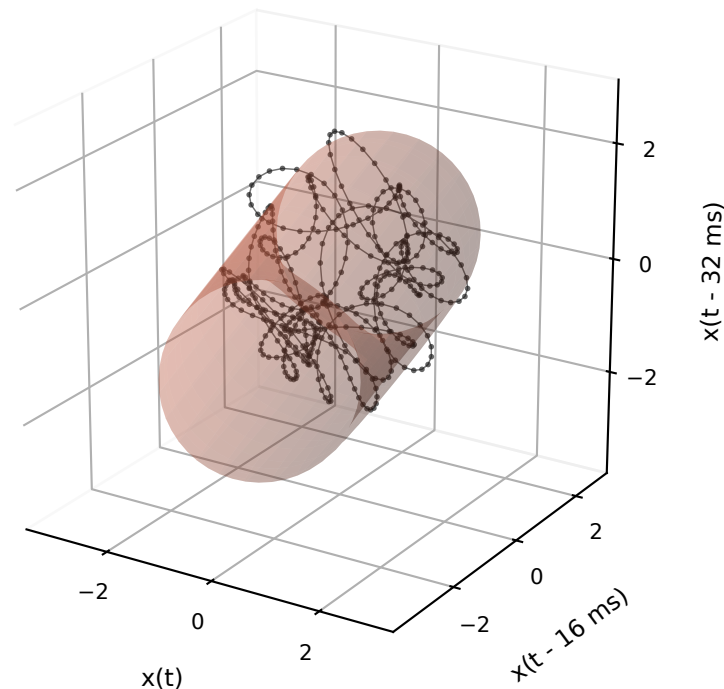
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2824  
Native coverage: 0.919 | nfev: 53  
Flags: none

2-torus: donut volume



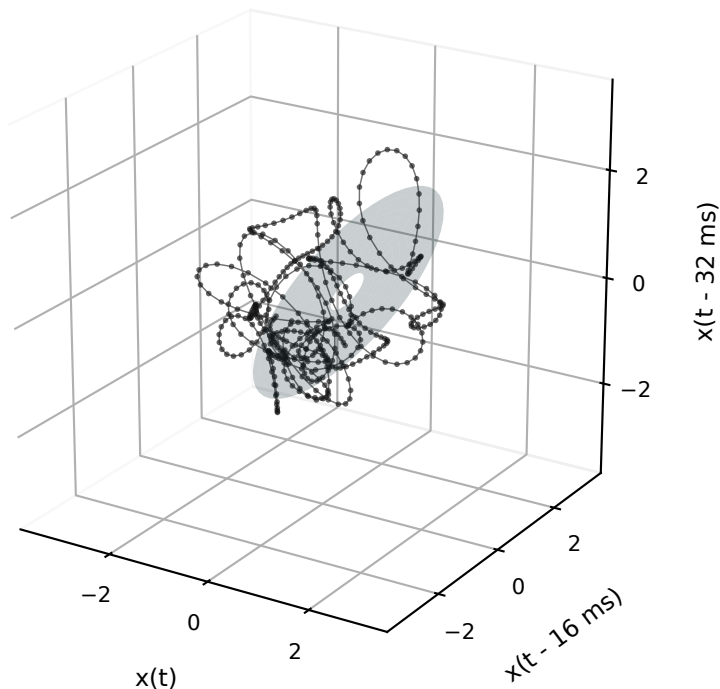
Normalized 3D set error: 0.0019  
Native coverage: 0.889 | nfev: 45  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 2 | original trial 401

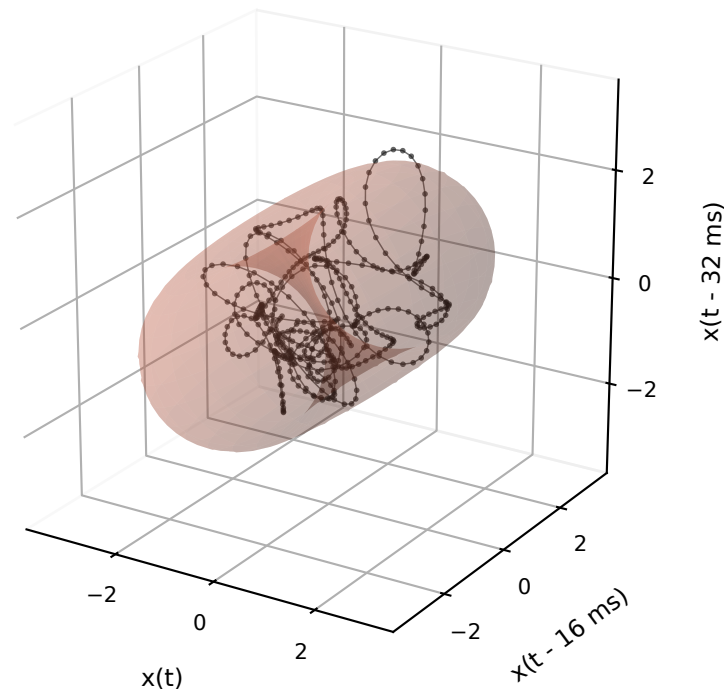
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2925  
Native coverage: 0.880 | nfev: 111  
Flags: nearly circular orientation

2-torus: donut volume



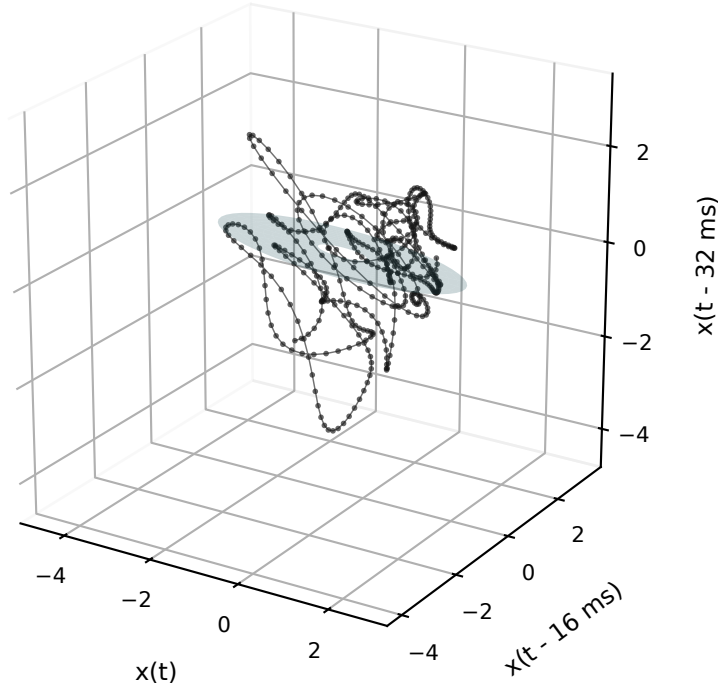
Normalized 3D set error: 0.0014  
Native coverage: 0.889 | nfev: 23  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 2 | original trial 34

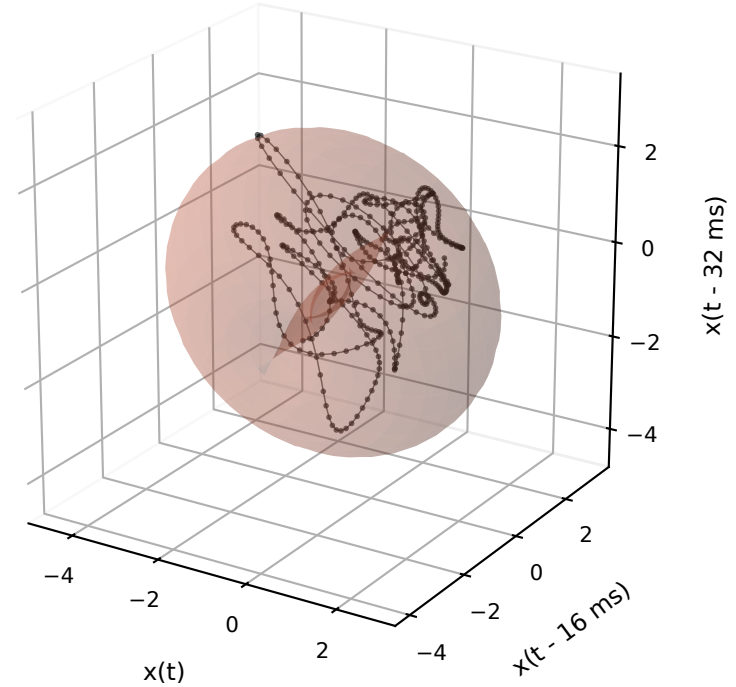
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.3059  
Native coverage: 0.919 | nfev: 57  
Flags: active optimizer bound; near optimizer bound

2-torus: donut volume



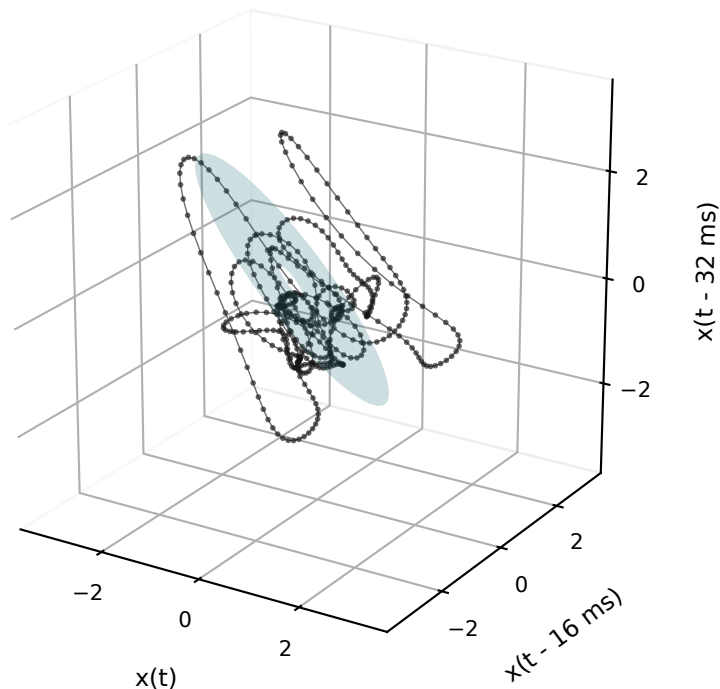
Normalized 3D set error: 0.0016  
Native coverage: 0.831 | nfev: 49  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 2 | original trial 479

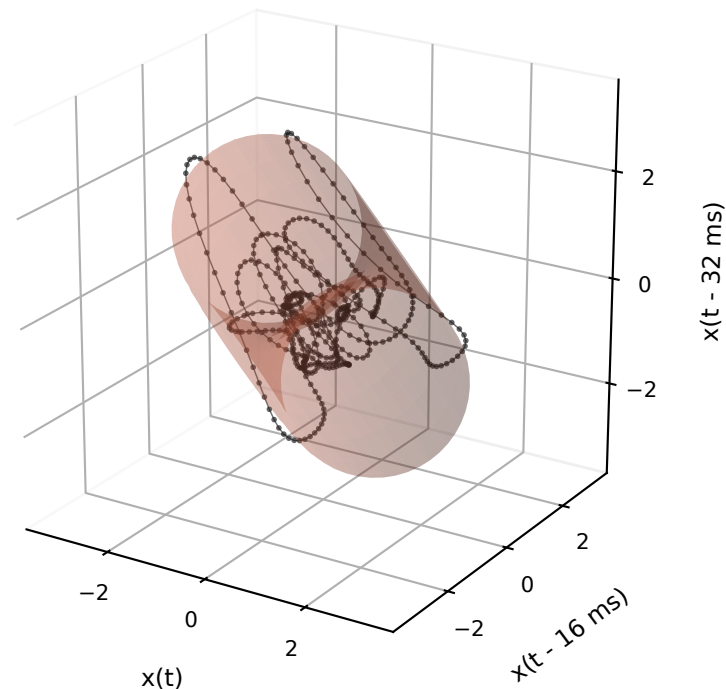
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1878  
Native coverage: 0.902 | nfev: 38  
Flags: none

2-torus: donut volume



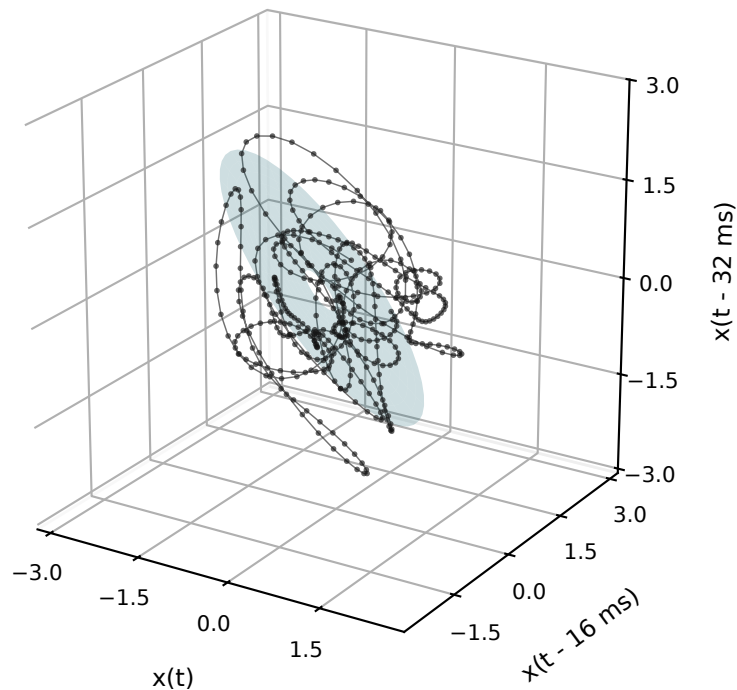
Normalized 3D set error: 0.0010  
Native coverage: 0.878 | nfev: 40  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 2 | original trial 117

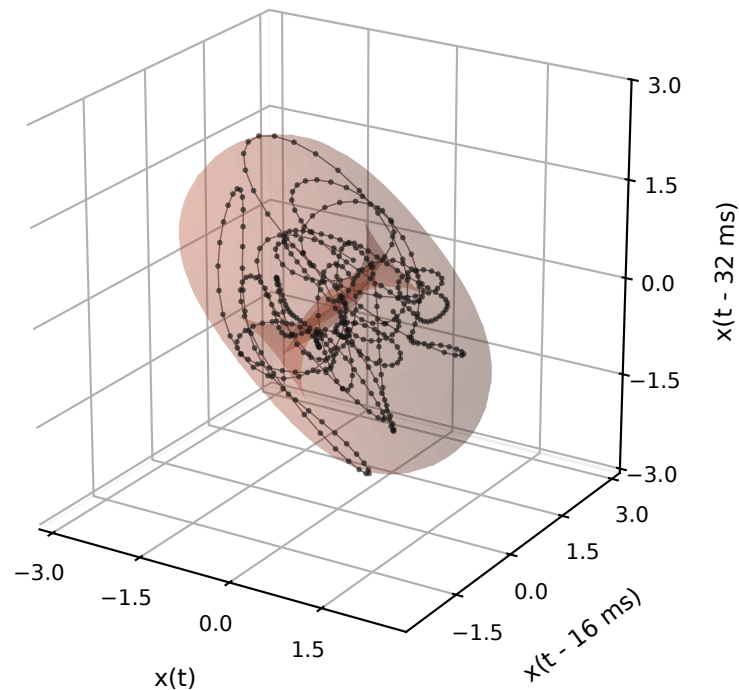
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1713  
Native coverage: 0.923 | nfev: 45  
Flags: none

2-torus: donut volume



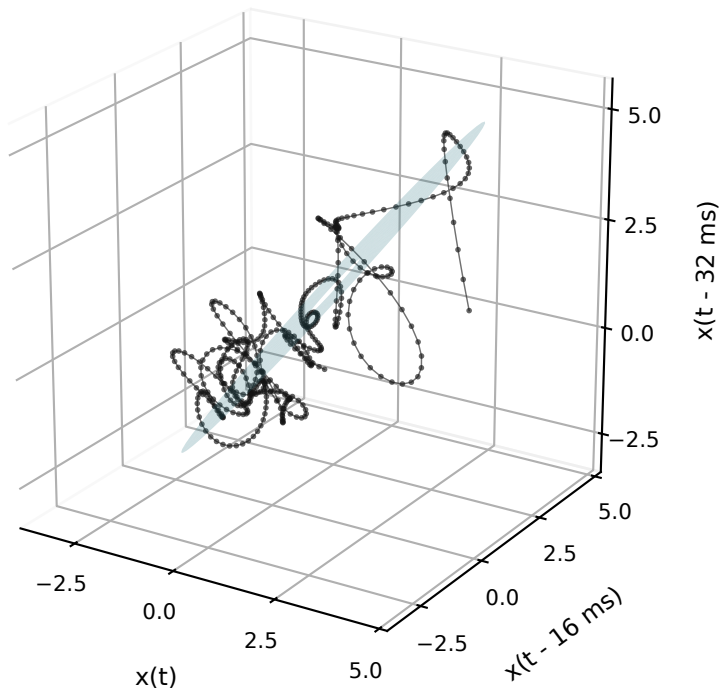
Normalized 3D set error: 0.0018  
Native coverage: 0.859 | nfev: 36  
Flags: near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 3 | original trial 663

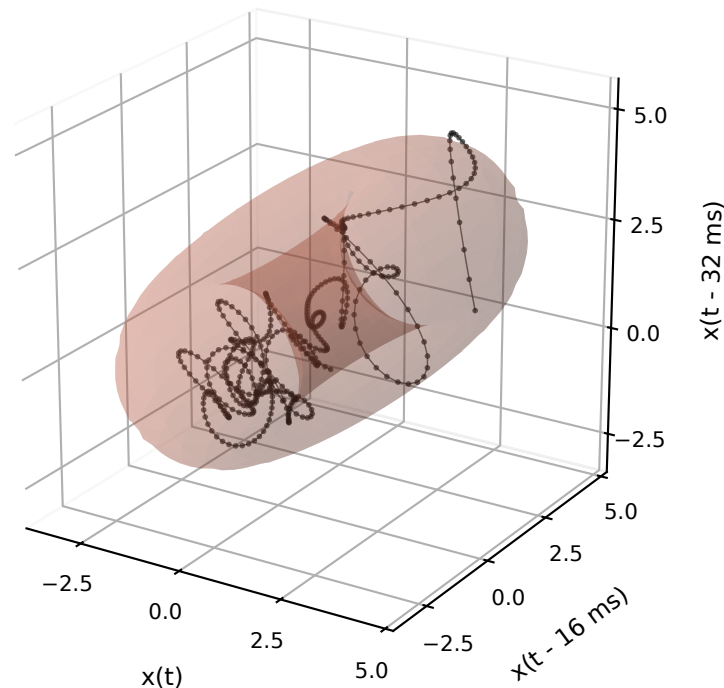
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1042  
Native coverage: 0.938 | nfev: 37  
Flags: none

2-torus: donut volume



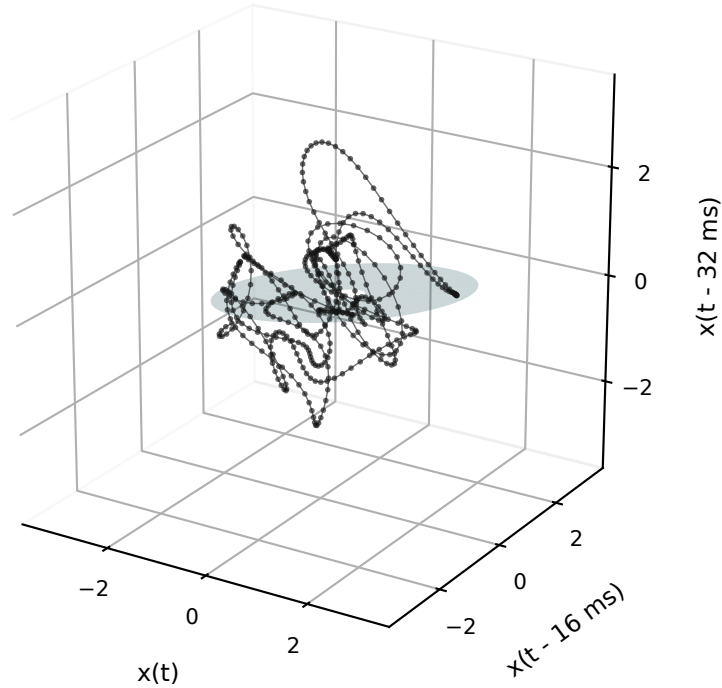
Normalized 3D set error: 0.0011  
Native coverage: 0.891 | nfev: 30  
Flags: collapsed hole; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 3 | original trial 14

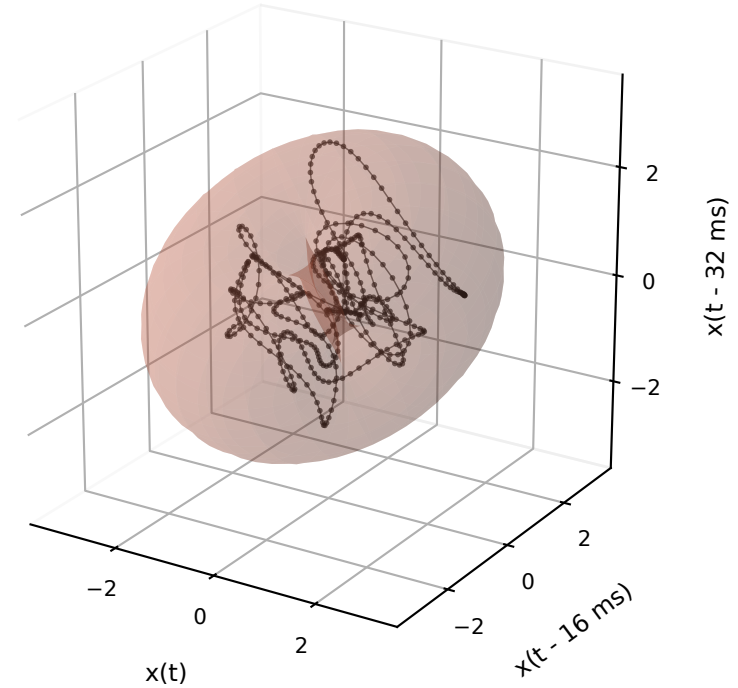
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2656  
Native coverage: 0.910 | nfev: 34  
Flags: none

2-torus: donut volume



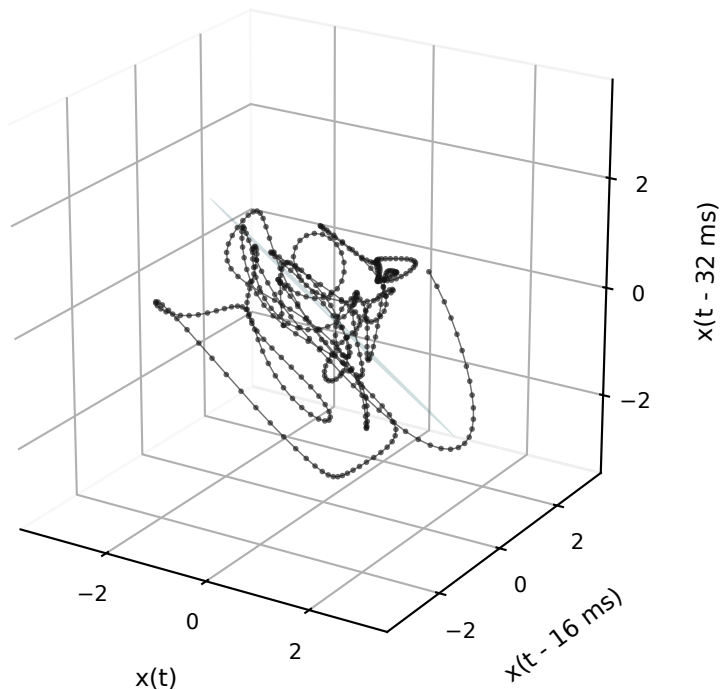
Normalized 3D set error: 0.0011  
Native coverage: 0.897 | nfev: 25  
Flags: collapsed hole; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 3 | original trial 540

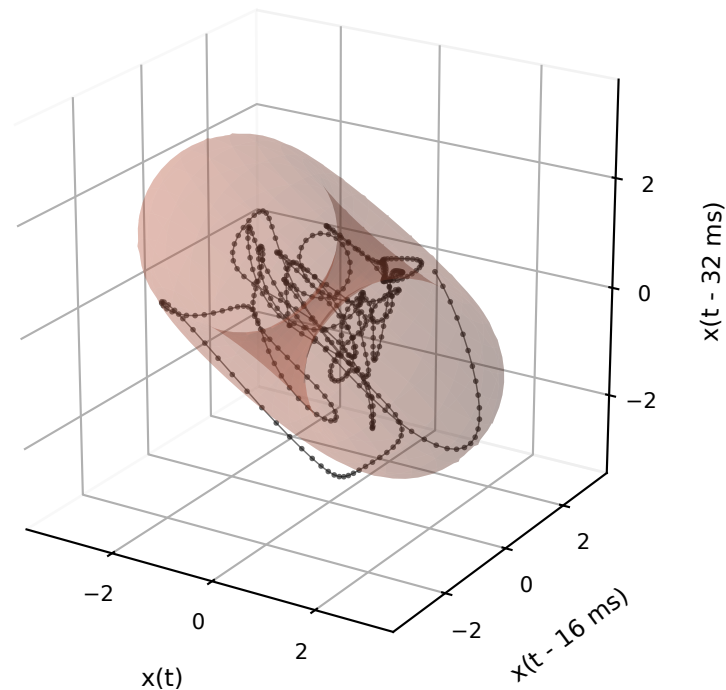
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2454  
Native coverage: 0.900 | nfev: 36  
Flags: none

2-torus: donut volume



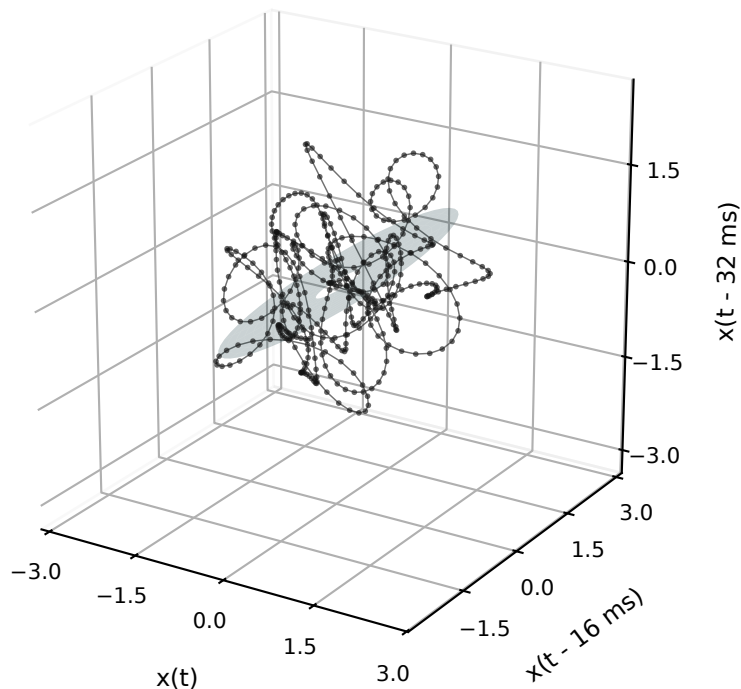
Normalized 3D set error: 0.0011  
Native coverage: 0.885 | nfev: 18  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 3 | original trial 128

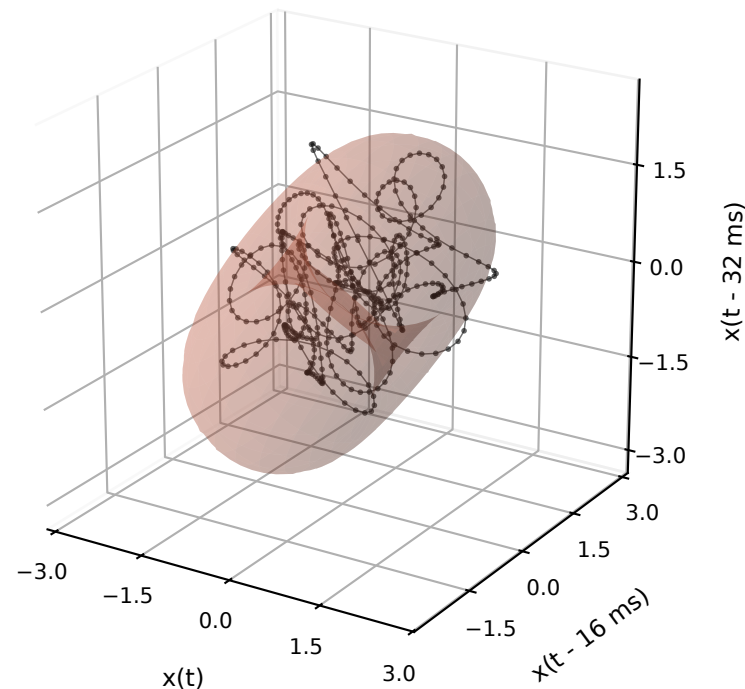
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2237  
Native coverage: 0.895 | nfev: 32  
Flags: none

2-torus: donut volume



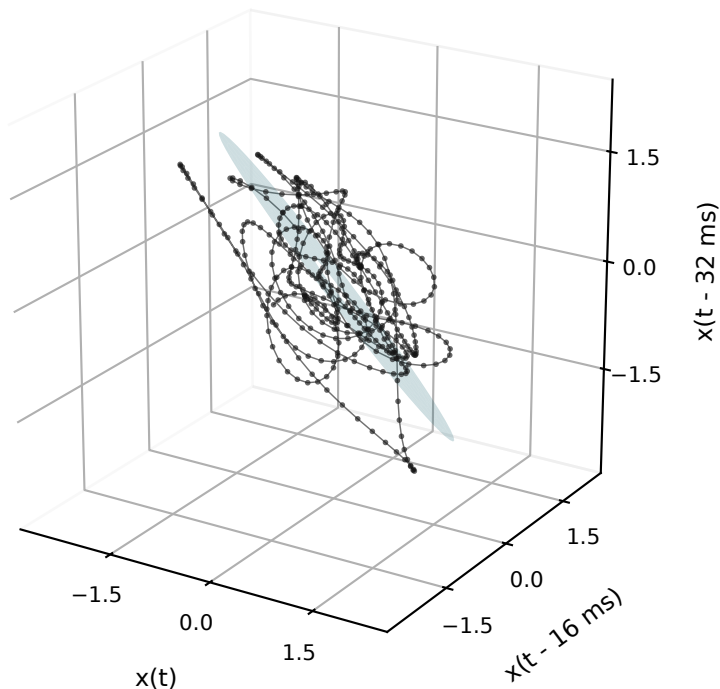
Normalized 3D set error: 0.0017  
Native coverage: 0.872 | nfev: 21  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 3 | original trial 420

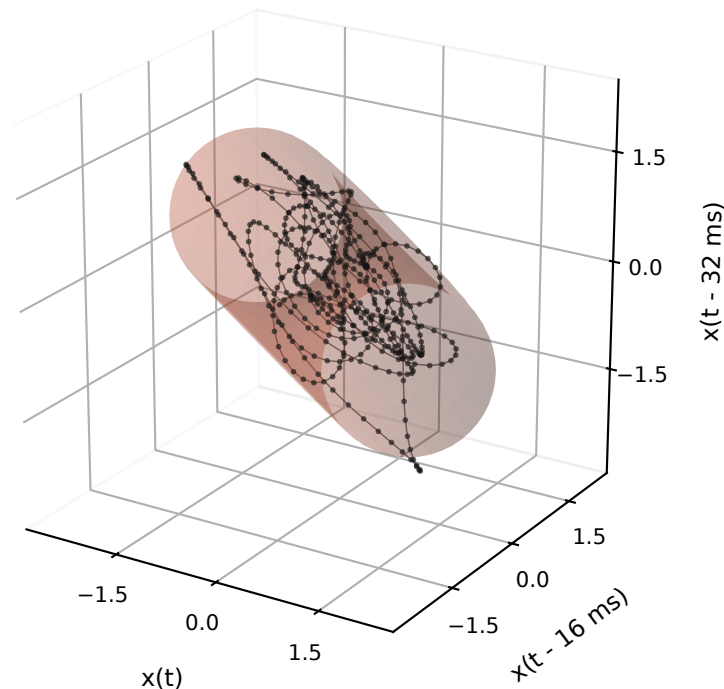
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.0916  
Native coverage: 0.915 | nfev: 73  
Flags: nearly circular orientation

2-torus: donut volume



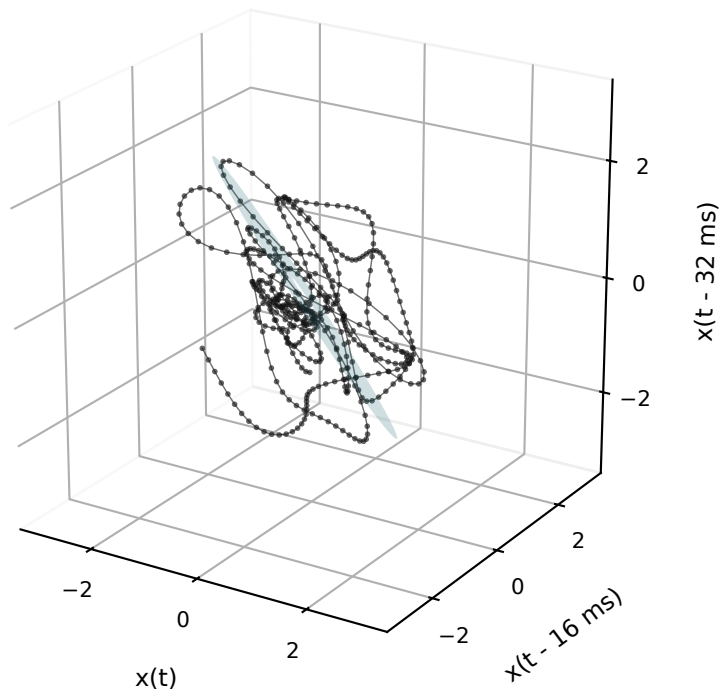
Normalized 3D set error: 0.0019  
Native coverage: 0.872 | nfev: 19  
Flags: active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 3 | original trial 503

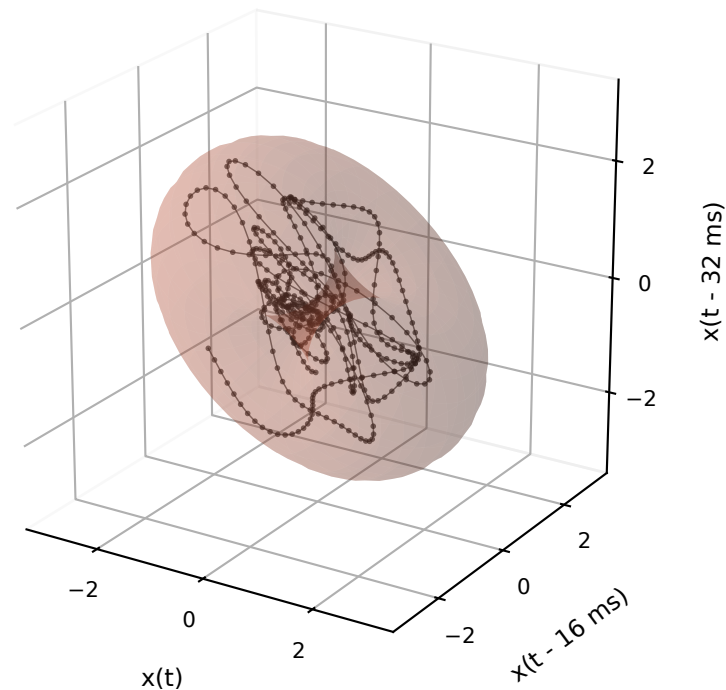
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1624  
Native coverage: 0.902 | nfev: 16  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

2-torus: donut volume



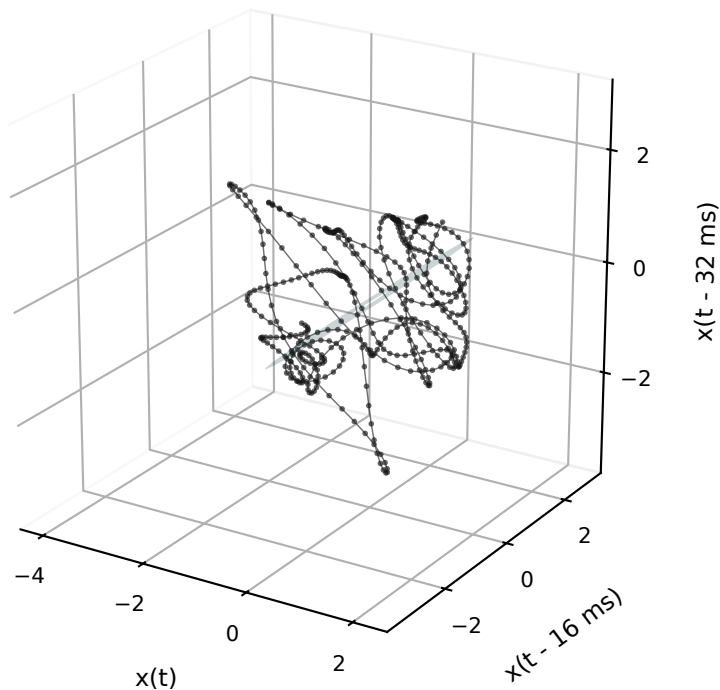
Normalized 3D set error: 0.0007  
Native coverage: 0.859 | nfev: 13  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 3 | original trial 636

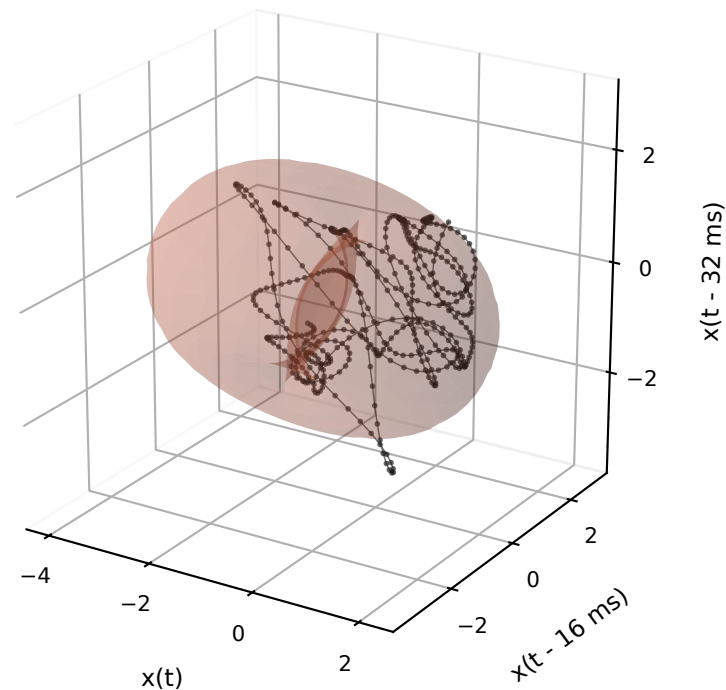
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.3027  
Native coverage: 0.921 | nfev: 64  
Flags: none

2-torus: donut volume



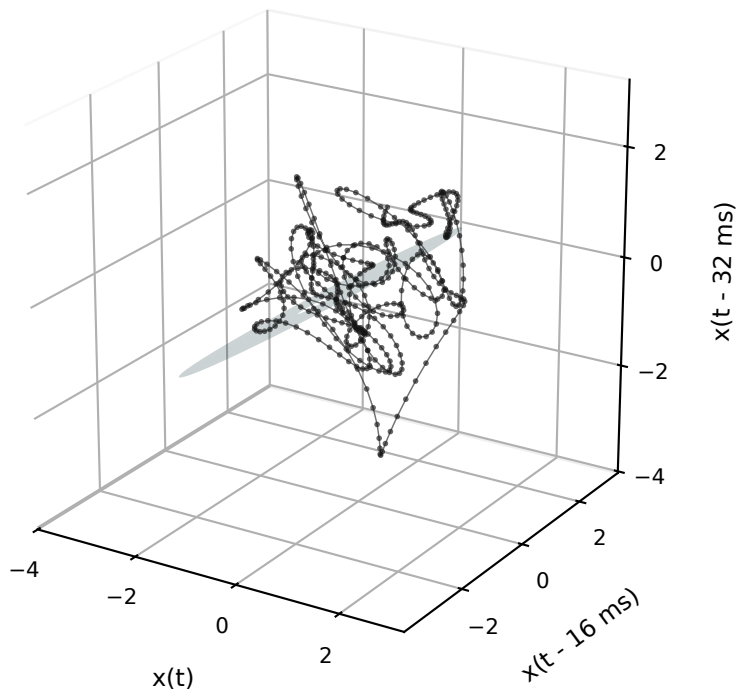
Normalized 3D set error: 0.0021  
Native coverage: 0.835 | nfev: 41  
Flags: collapsed hole

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 3 | original trial 307

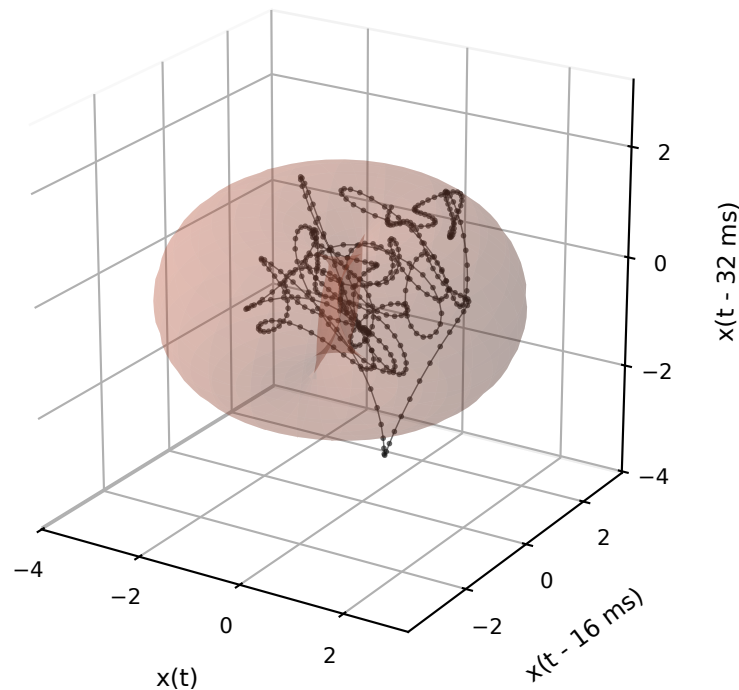
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2732  
Native coverage: 0.887 | nfev: 40  
Flags: none

2-torus: donut volume



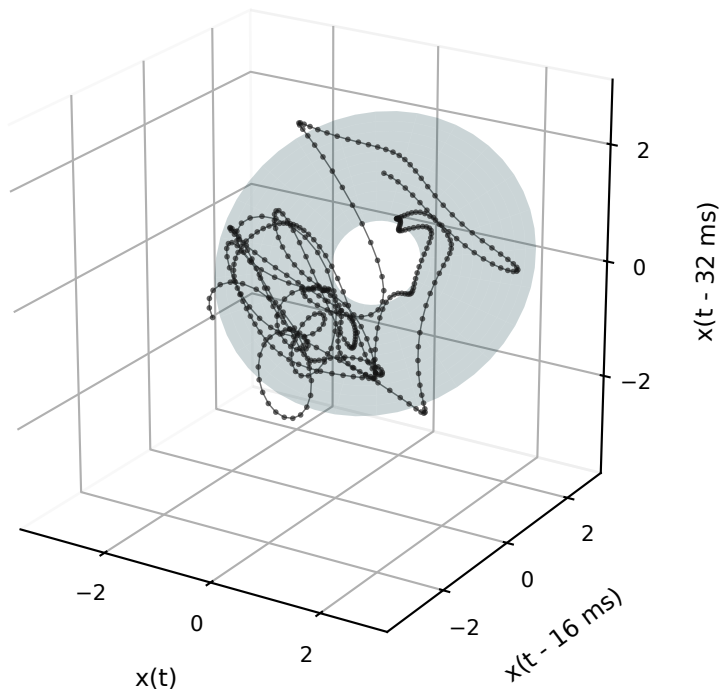
Normalized 3D set error: 0.0021  
Native coverage: 0.844 | nfev: 39  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 3 | original trial 612

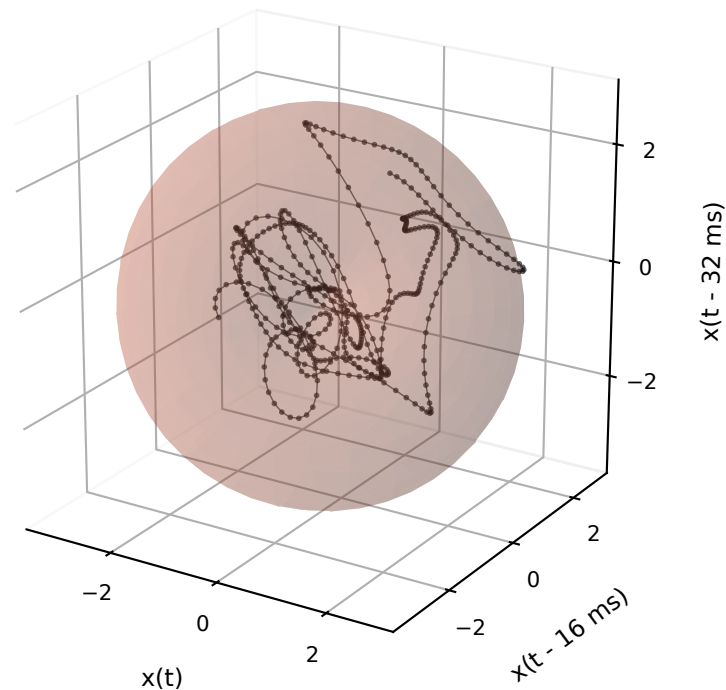
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2368  
Native coverage: 0.878 | nfev: 71  
Flags: none

2-torus: donut volume



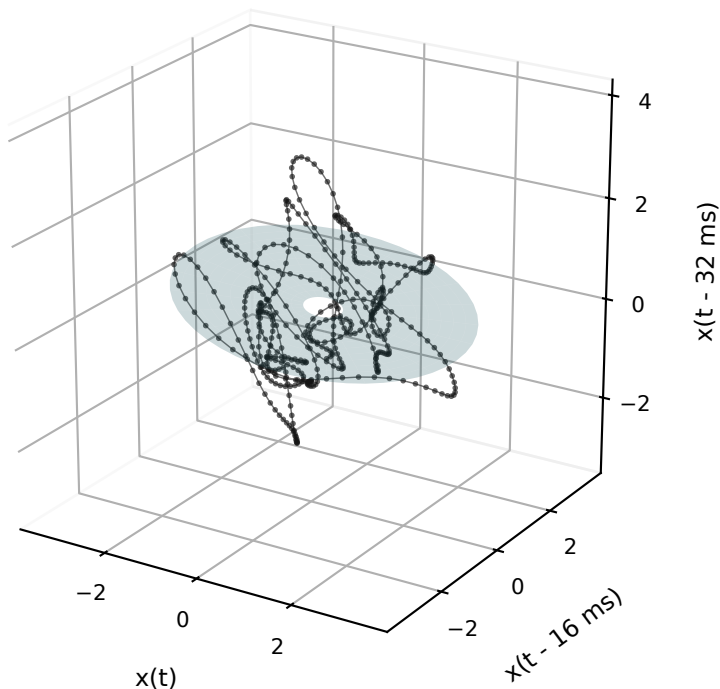
Normalized 3D set error: 0.0011  
Native coverage: 0.865 | nfev: 17  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 3 | original trial 609

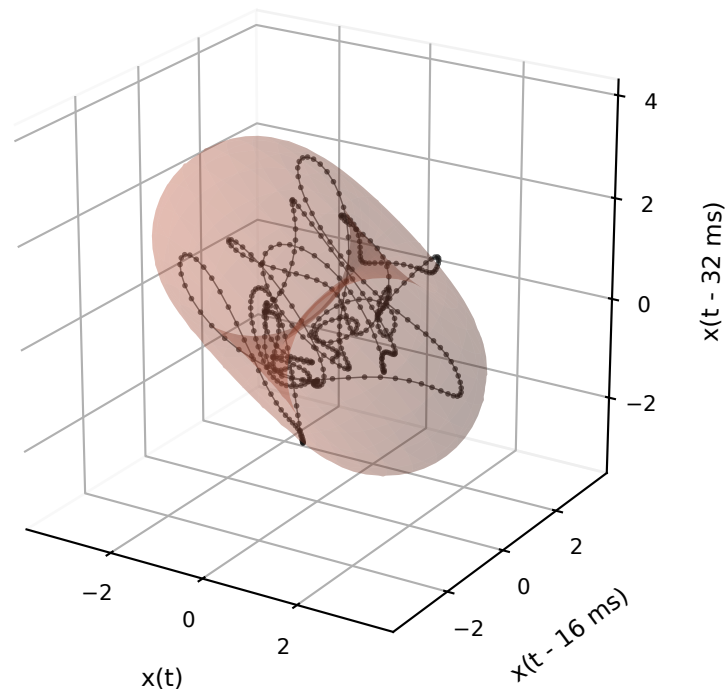
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2886  
Native coverage: 0.912 | nfev: 92  
Flags: none

2-torus: donut volume



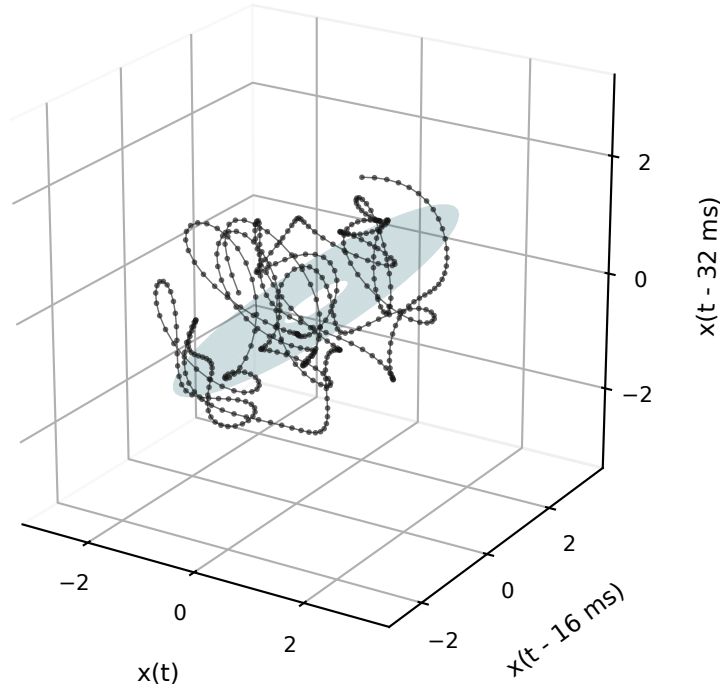
Normalized 3D set error: 0.0008  
Native coverage: 0.848 | nfev: 19  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 3 | original trial 46

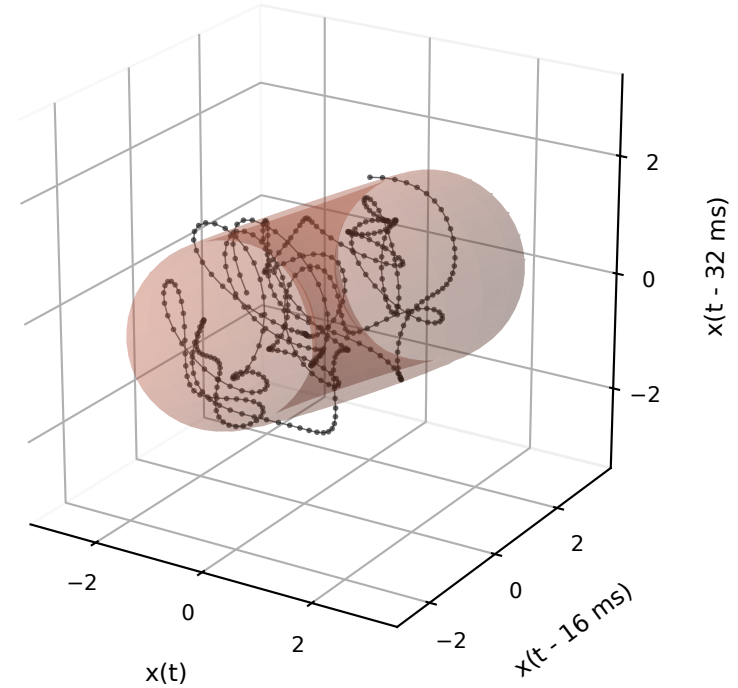
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2270  
Native coverage: 0.891 | nfev: 103  
Flags: none

2-torus: donut volume



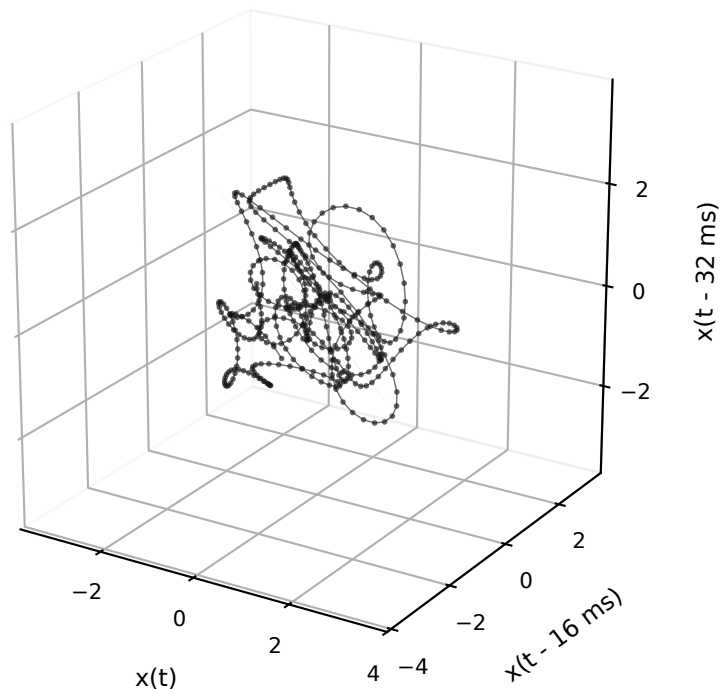
Normalized 3D set error: 0.0012  
Native coverage: 0.885 | nfev: 30  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 3 | original trial 615

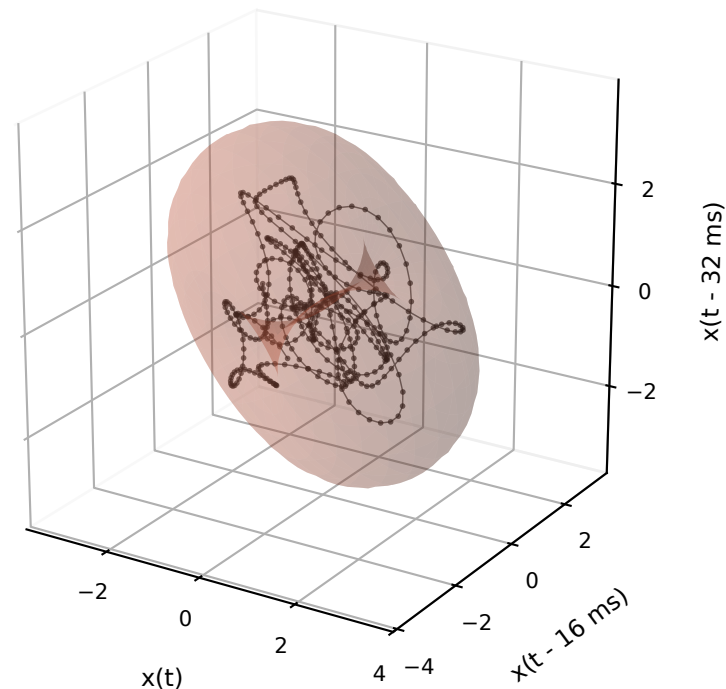
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2457  
Native coverage: 0.921 | nfev: 31  
Flags: none

2-torus: donut volume



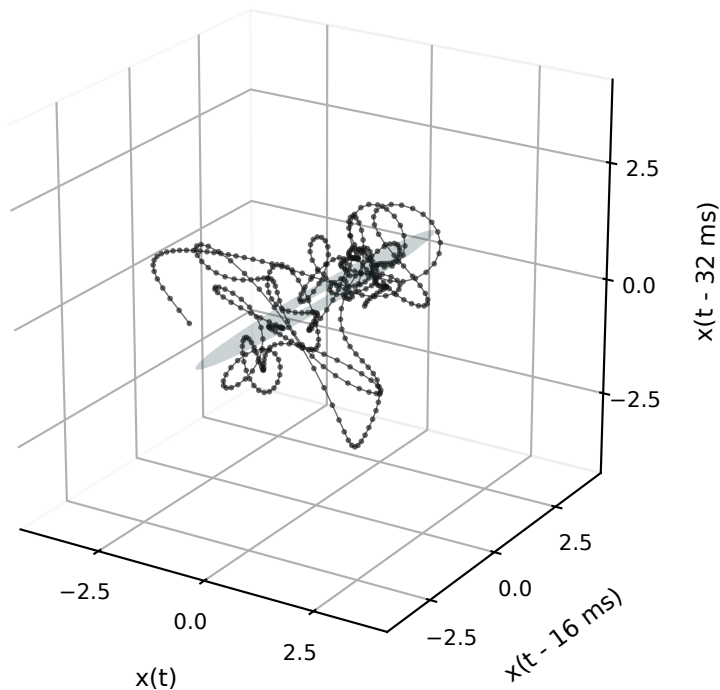
Normalized 3D set error: 0.0008  
Native coverage: 0.870 | nfev: 32  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 3 | original trial 510

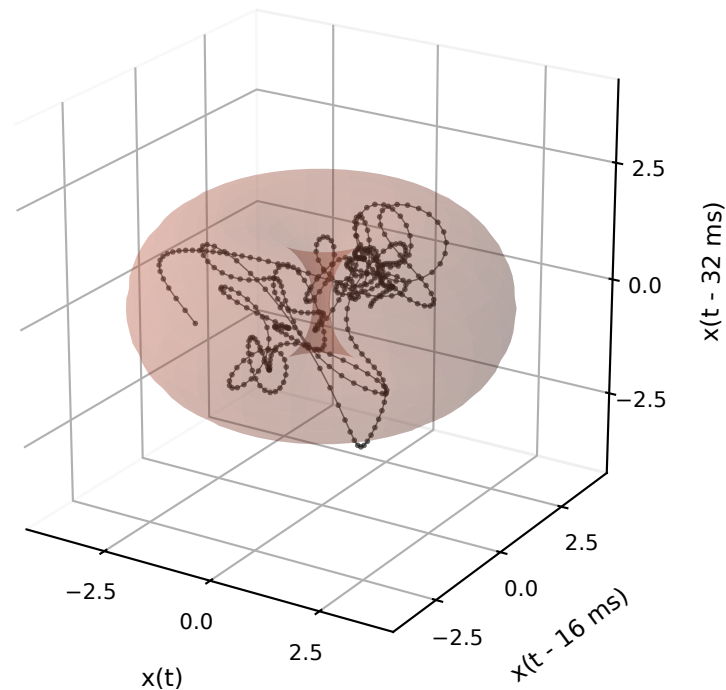
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2554  
Native coverage: 0.923 | nfev: 107  
Flags: active optimizer bound; near optimizer bound

2-torus: donut volume



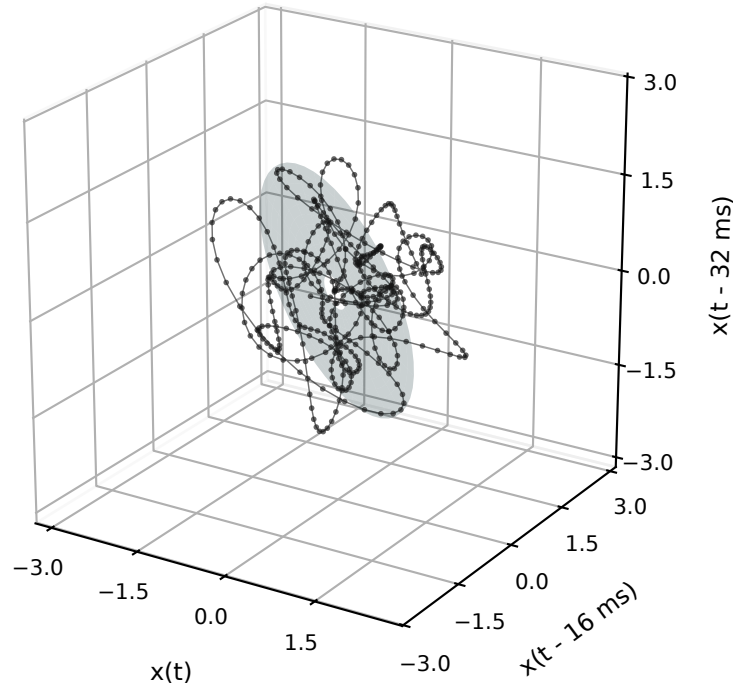
Normalized 3D set error: 0.0011  
Native coverage: 0.919 | nfev: 35  
Flags: collapsed hole

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 3 | original trial 43

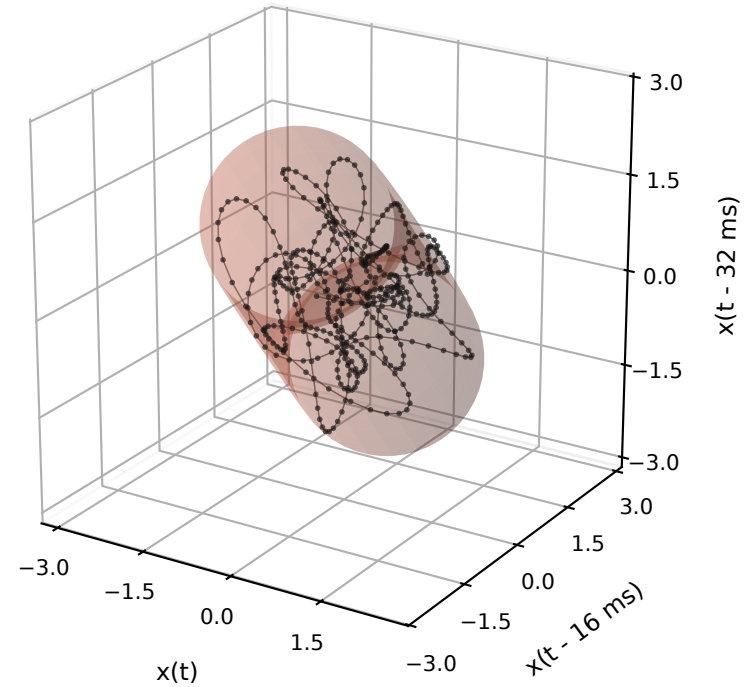
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2253  
Native coverage: 0.895 | nfev: 42  
Flags: none

2-torus: donut volume



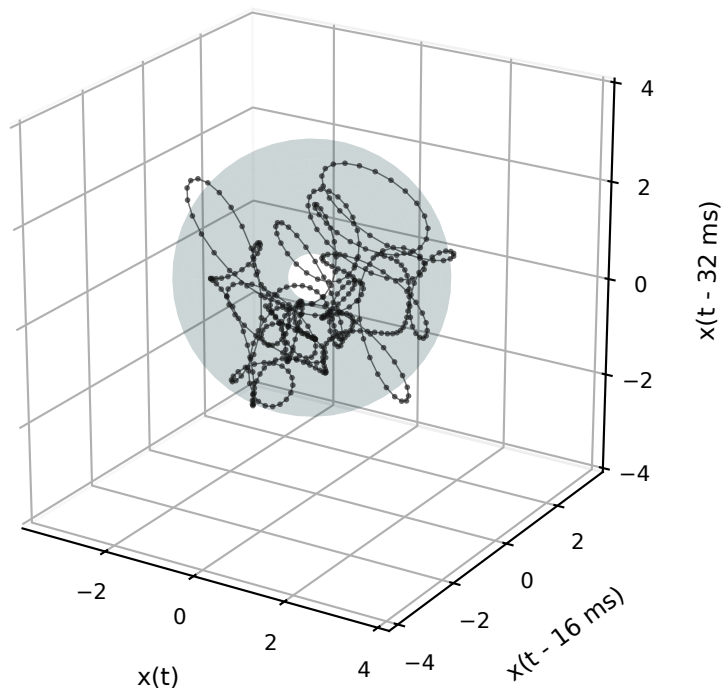
Normalized 3D set error: 0.0031  
Native coverage: 0.859 | nfev: 39  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 3 | original trial 147

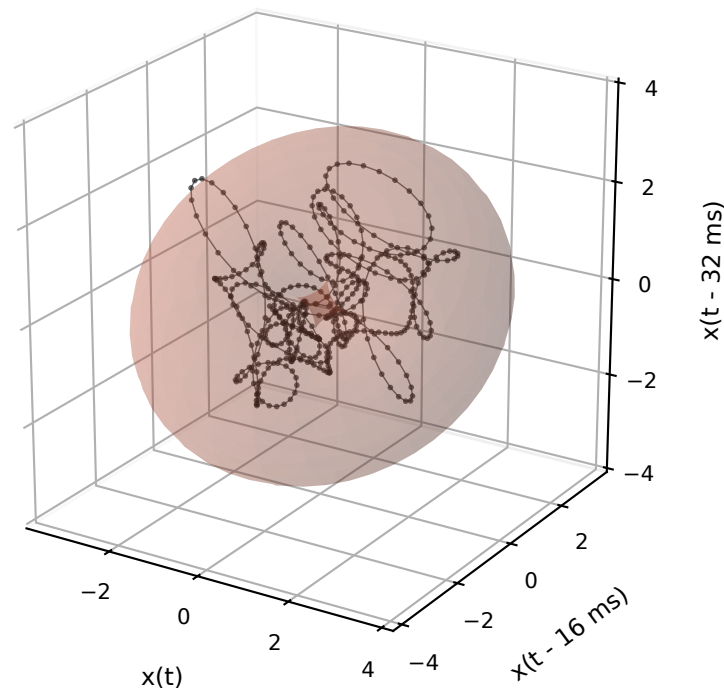
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2311  
Native coverage: 0.932 | nfev: 69  
Flags: nearly circular orientation

2-torus: donut volume



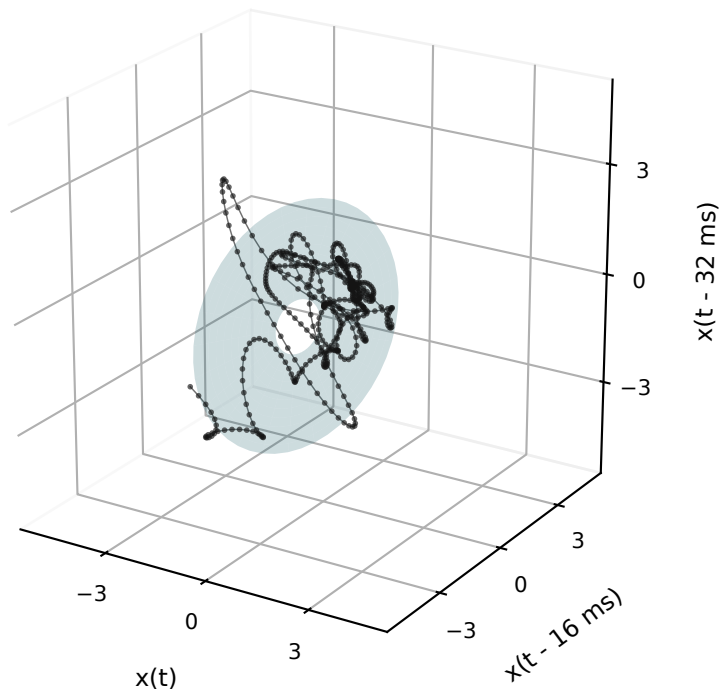
Normalized 3D set error: 0.0012  
Native coverage: 0.891 | nfev: 23  
Flags: none

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 3 | original trial 406

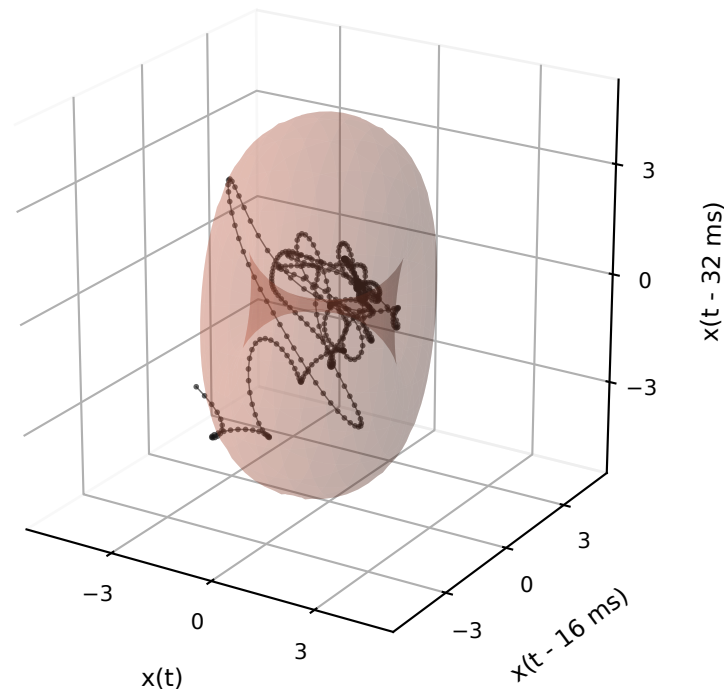
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2622  
Native coverage: 0.932 | nfev: 47  
Flags: near optimizer bound; nearly circular orientation

2-torus: donut volume



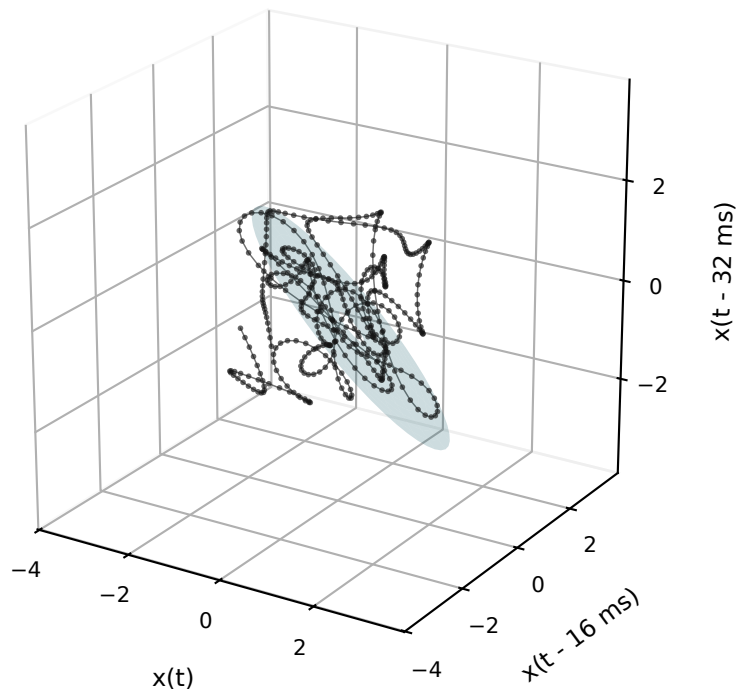
Normalized 3D set error: 0.0005  
Native coverage: 0.921 | nfev: 30  
Flags: collapsed hole

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 3 | original trial 200

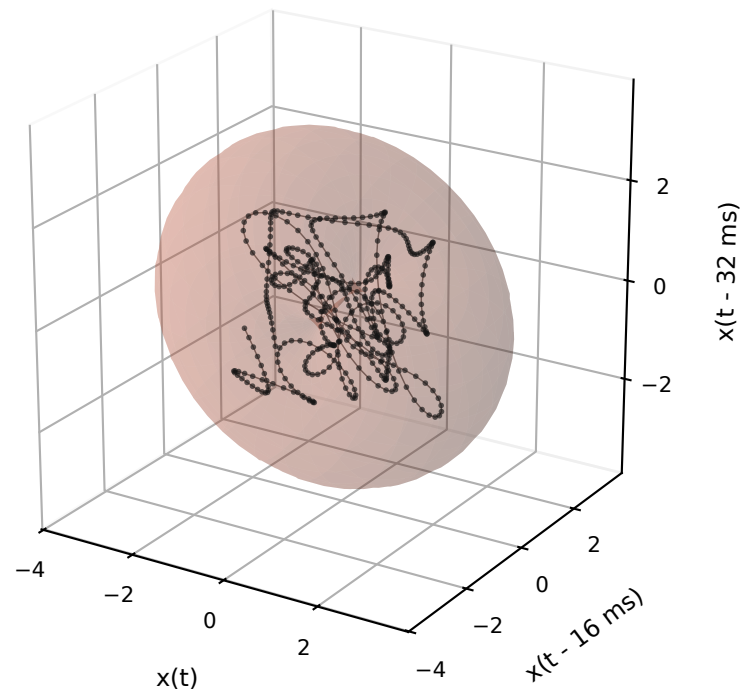
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2938  
Native coverage: 0.910 | nfev: 52  
Flags: none

2-torus: donut volume



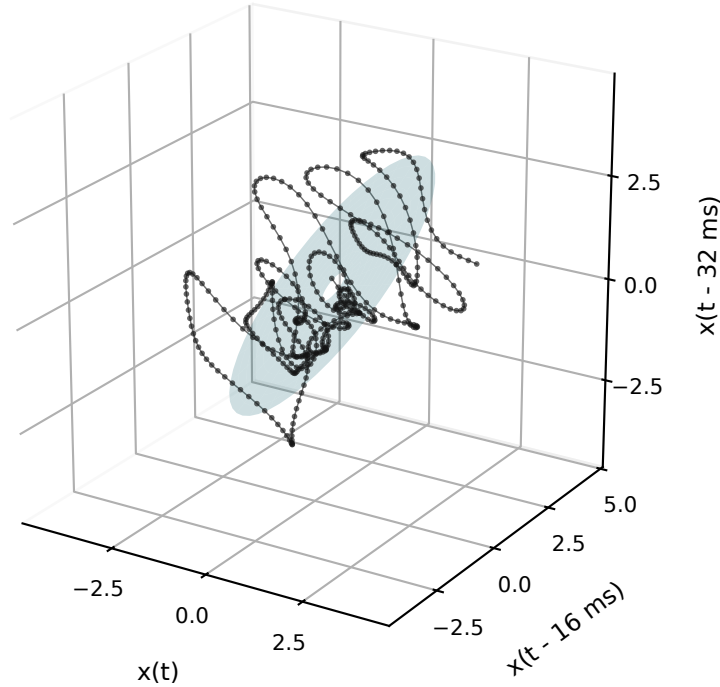
Normalized 3D set error: 0.0011  
Native coverage: 0.919 | nfev: 36  
Flags: collapsed hole; active optimizer bound; near optimizer bound; nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 3 | original trial 48

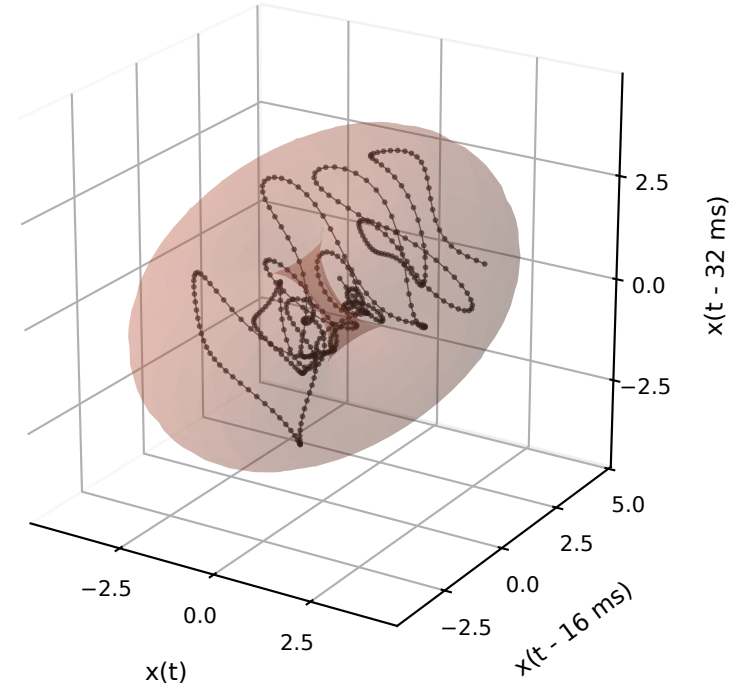
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2388  
Native coverage: 0.902 | nfev: 64  
Flags: near optimizer bound

2-torus: donut volume



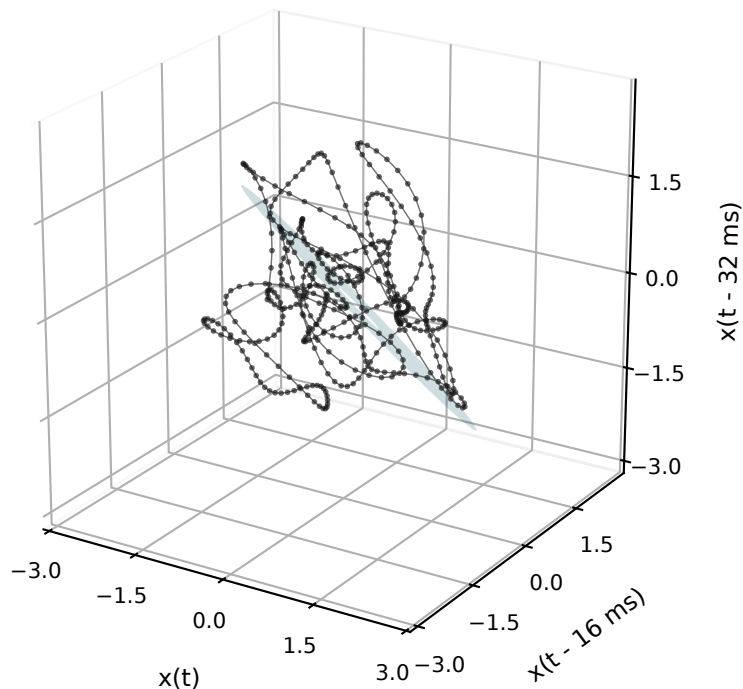
Normalized 3D set error: 0.0004  
Native coverage: 0.919 | nfev: 35  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 3 | original trial 177

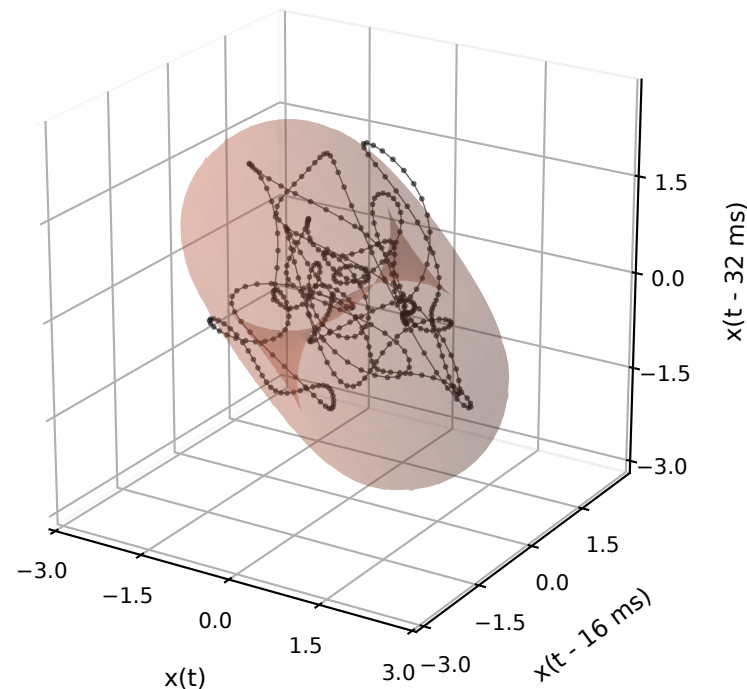
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2472  
Native coverage: 0.917 | nfev: 31  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

2-torus: donut volume



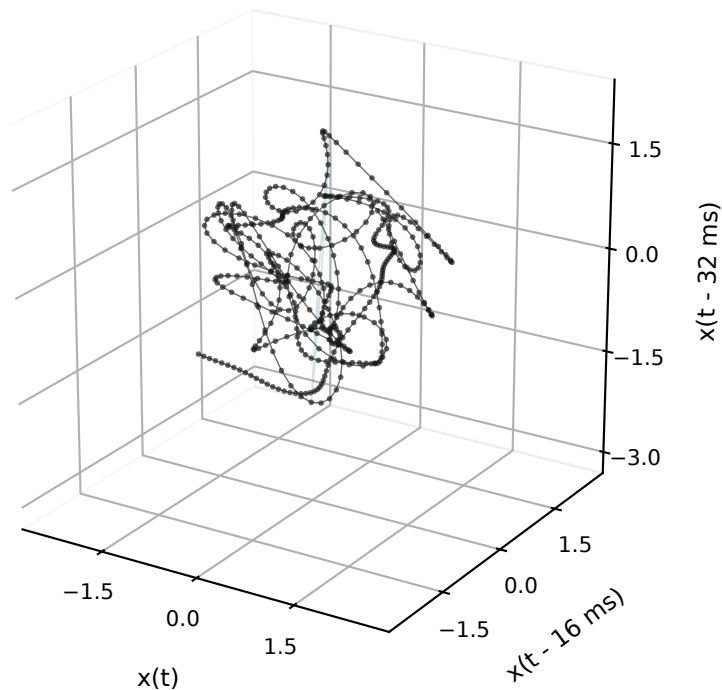
Normalized 3D set error: 0.0016  
Native coverage: 0.885 | nfev: 302  
Flags: collapsed hole; active optimizer bound; near  
optimizer bound; nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 3 | original trial 319

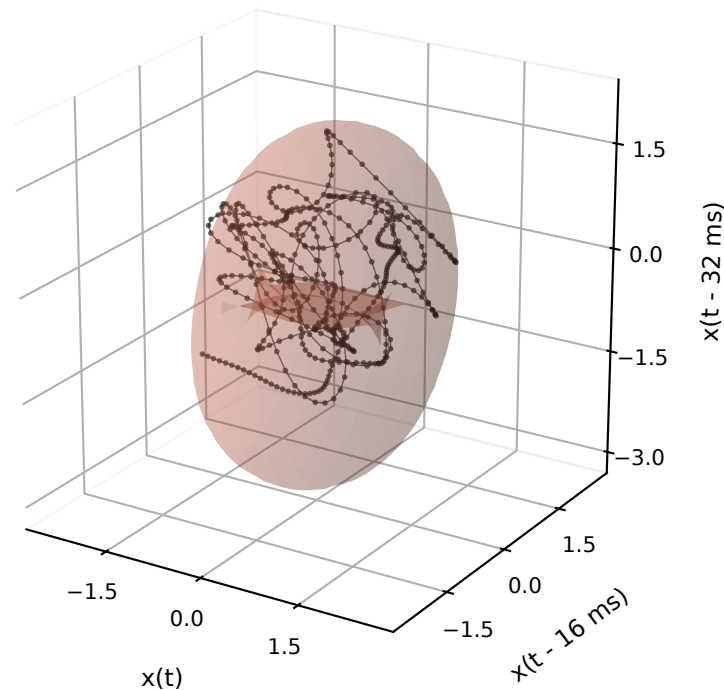
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.3024  
Native coverage: 0.889 | nfev: 99  
Flags: none

2-torus: donut volume



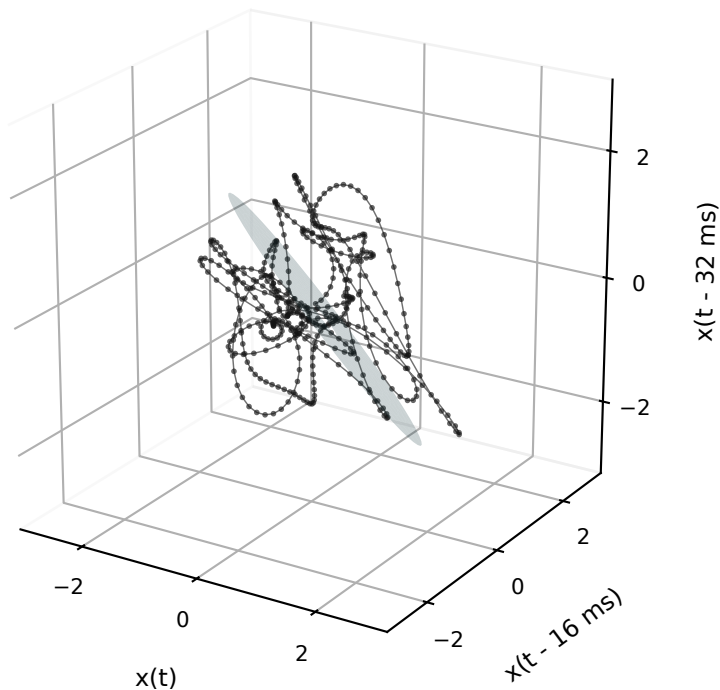
Normalized 3D set error: 0.0035  
Native coverage: 0.855 | nfev: 30  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 3 | original trial 137

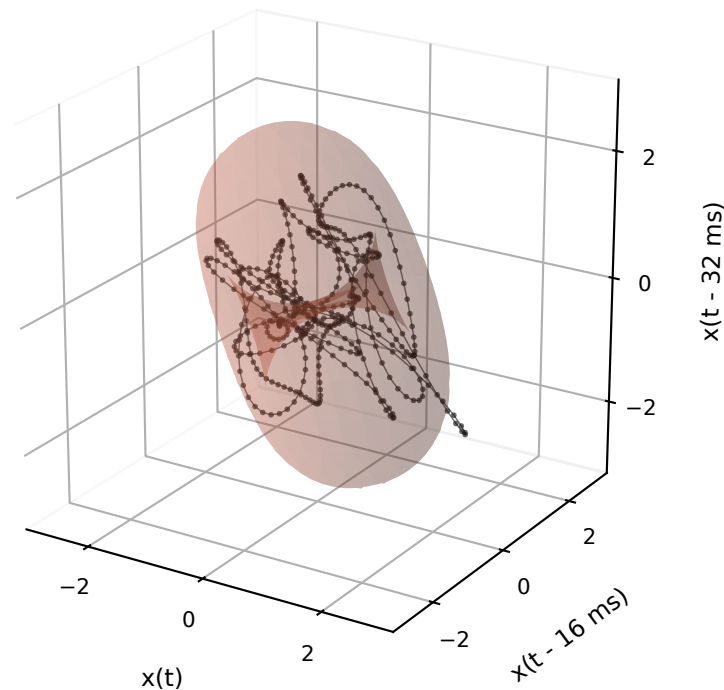
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1991  
Native coverage: 0.912 | nfev: 41  
Flags: near optimizer bound

2-torus: donut volume



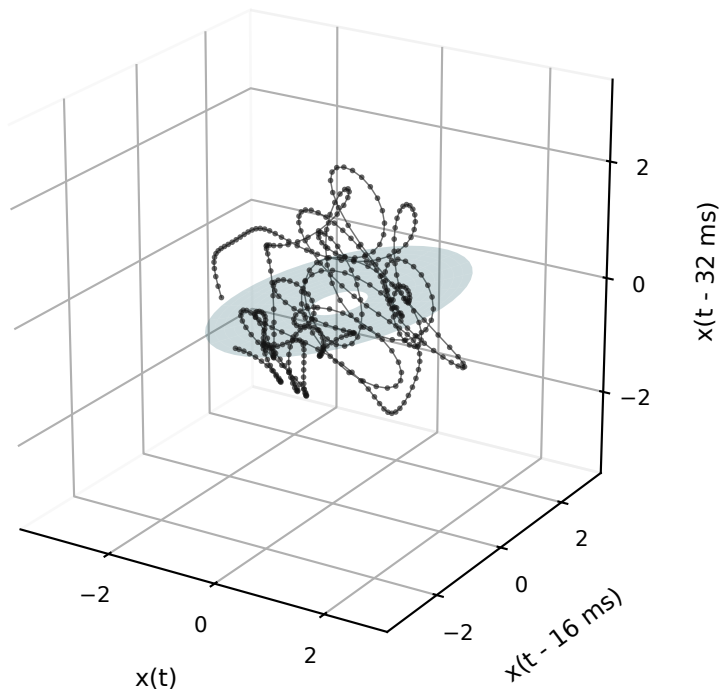
Normalized 3D set error: 0.0015  
Native coverage: 0.872 | nfev: 30  
Flags: near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 3 | original trial 309

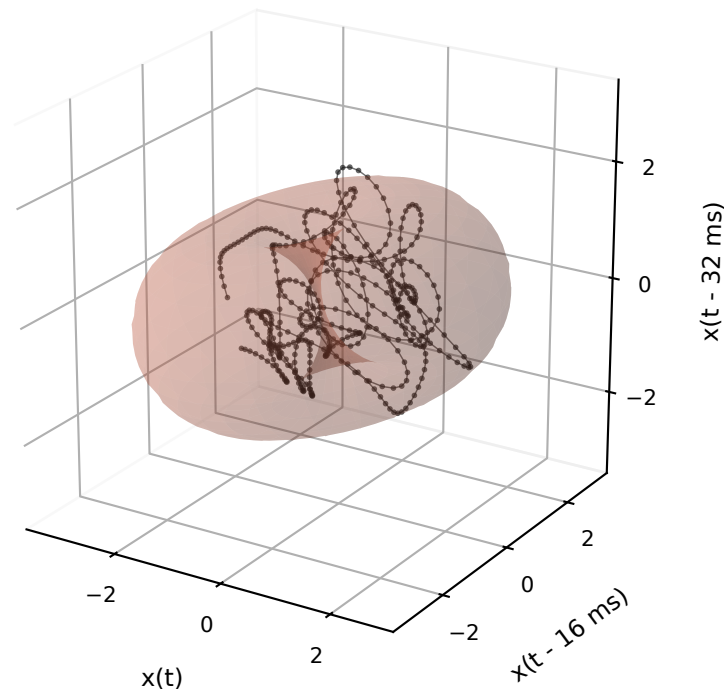
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.3168  
Native coverage: 0.882 | nfev: 79  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

2-torus: donut volume



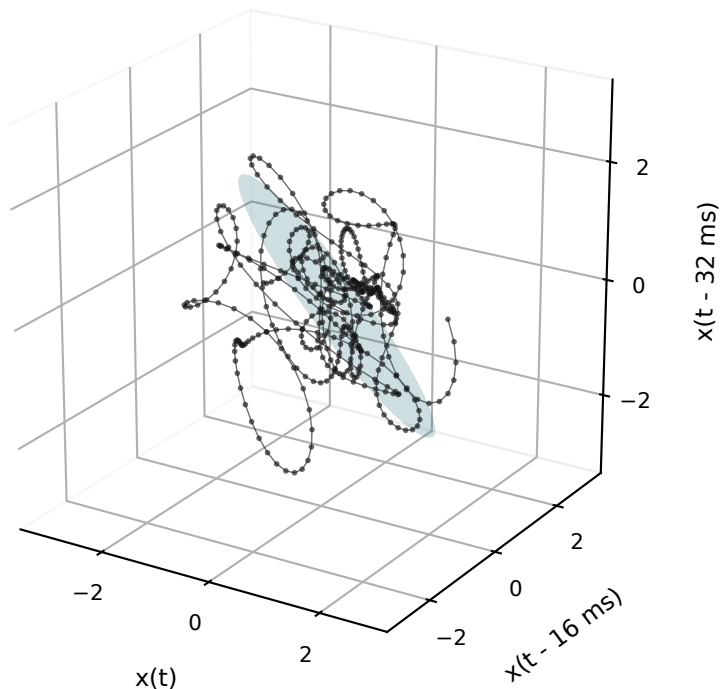
Normalized 3D set error: 0.0013  
Native coverage: 0.868 | nfev: 20  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 3 | original trial 416

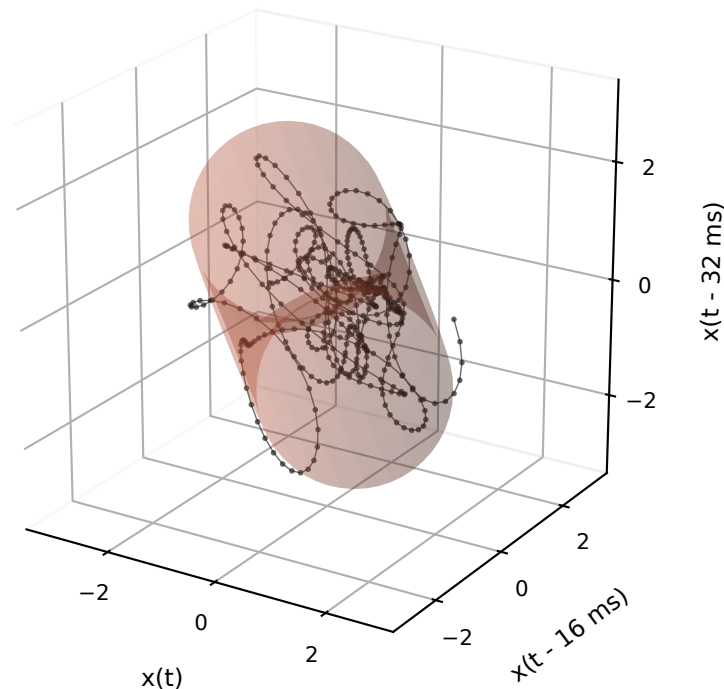
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1999  
Native coverage: 0.917 | nfev: 30  
Flags: active optimizer bound; near optimizer bound

2-torus: donut volume



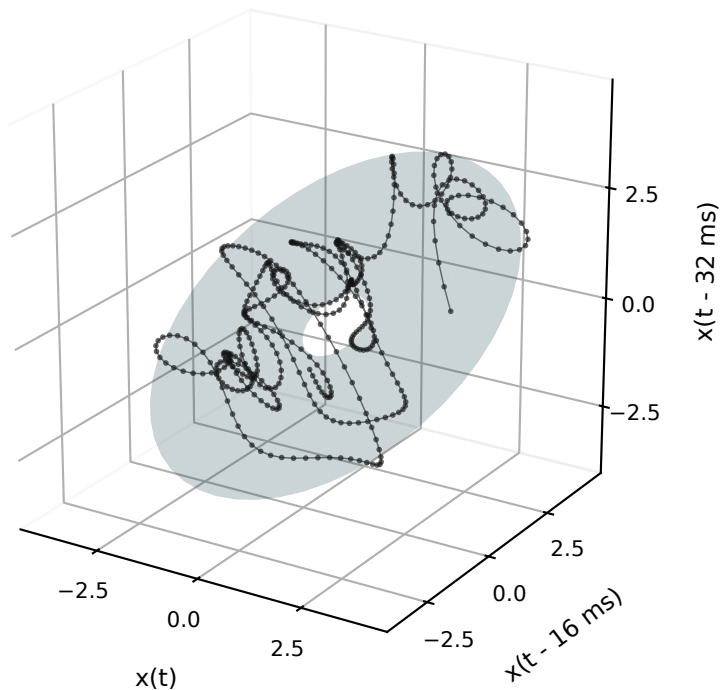
Normalized 3D set error: 0.0011  
Native coverage: 0.870 | nfev: 32  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 3 | original trial 669

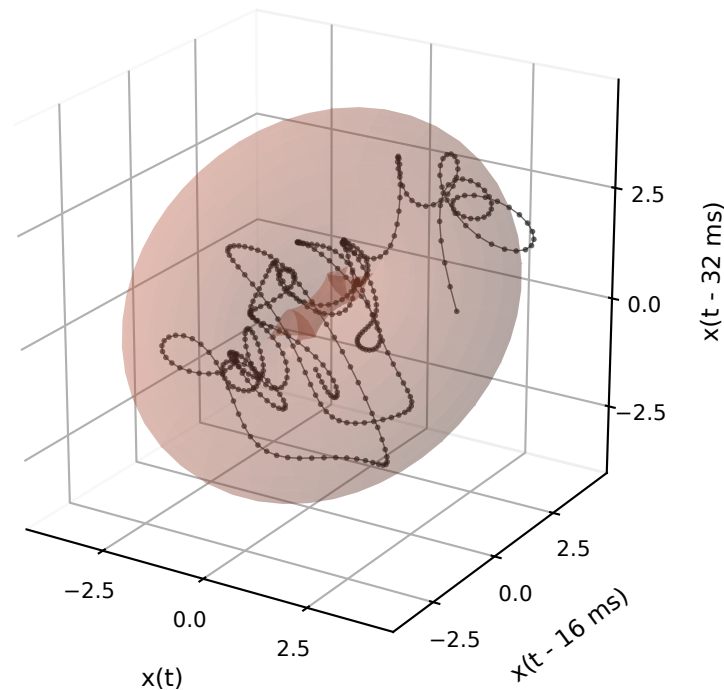
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1418  
Native coverage: 0.917 | nfev: 66  
Flags: none

2-torus: donut volume



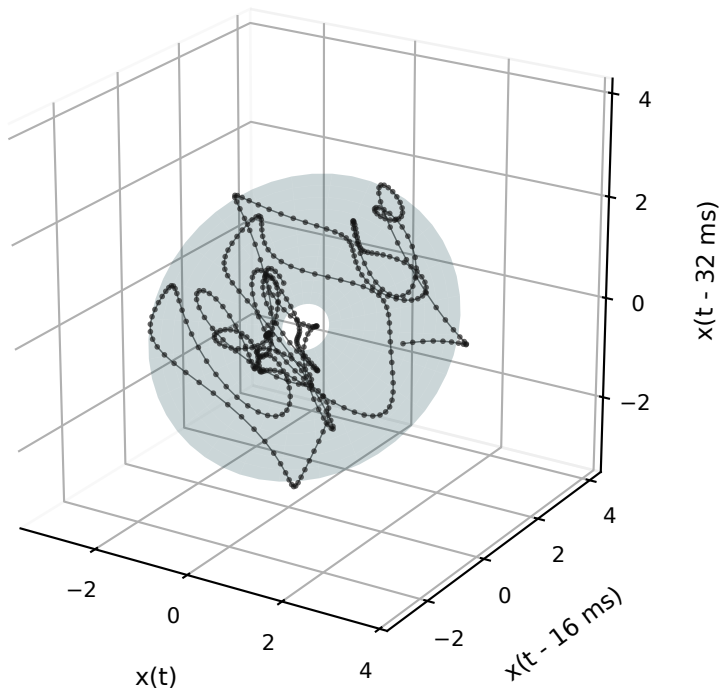
Normalized 3D set error: 0.0013  
Native coverage: 0.897 | nfev: 39  
Flags: collapsed hole

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 3 | original trial 572

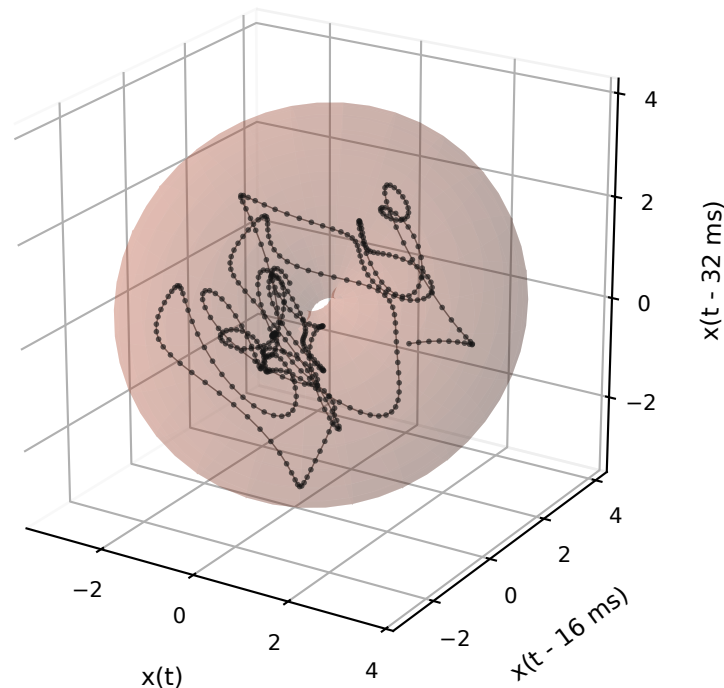
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2310  
Native coverage: 0.889 | nfev: 33  
Flags: none

2-torus: donut volume



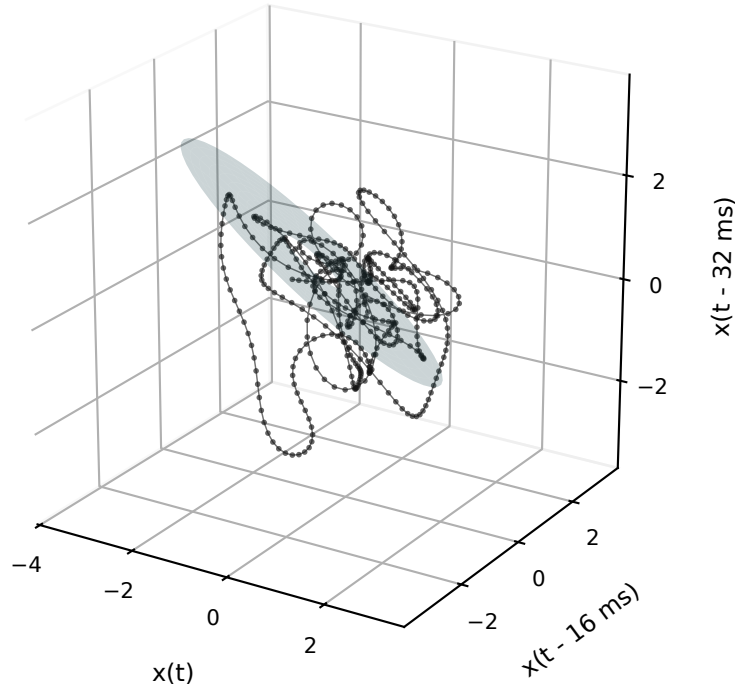
Normalized 3D set error: 0.0011  
Native coverage: 0.897 | nfev: 21  
Flags: active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 3 | original trial 477

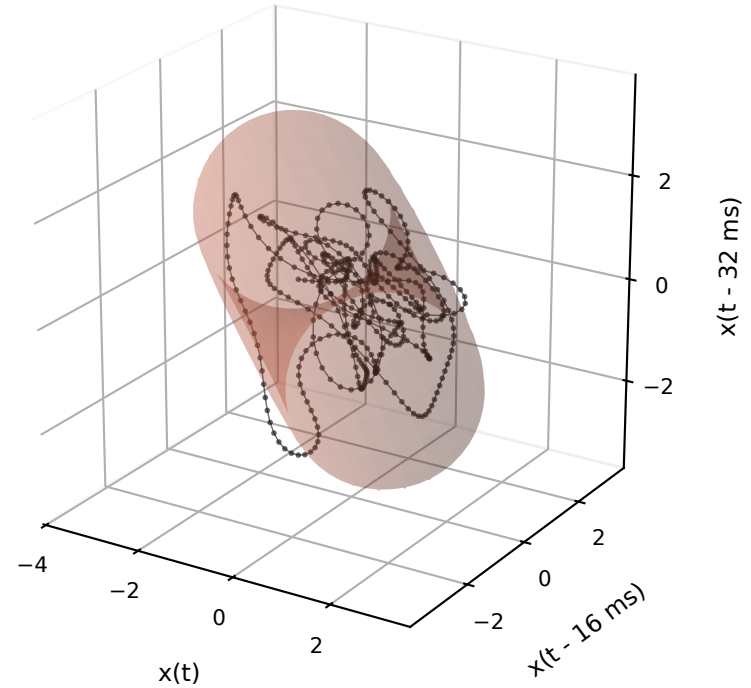
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2506  
Native coverage: 0.897 | nfev: 91  
Flags: none

2-torus: donut volume



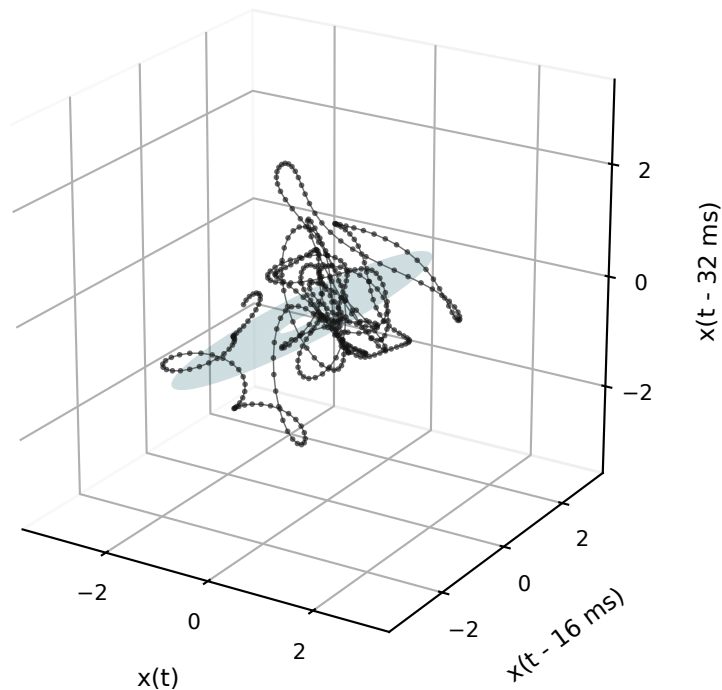
Normalized 3D set error: 0.0013  
Native coverage: 0.893 | nfev: 6000  
Flags: nonconvergence; collapsed hole; active optimizer  
bound; near optimizer bound; nearly circular  
orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 3 | original trial 323

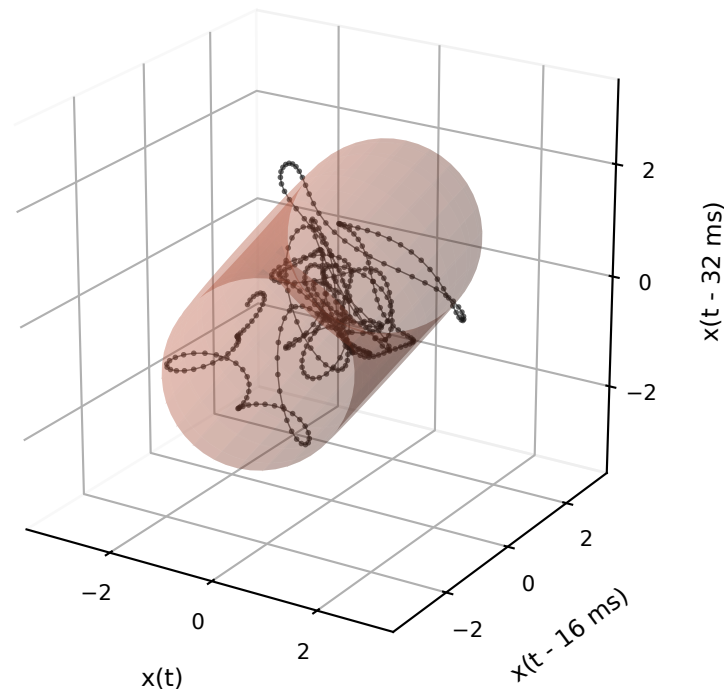
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2354  
Native coverage: 0.895 | nfev: 69  
Flags: none

2-torus: donut volume



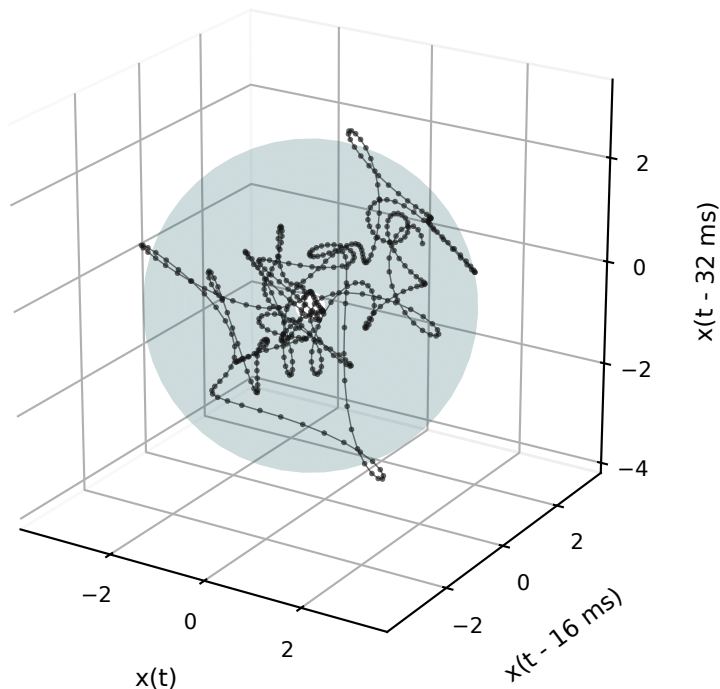
Normalized 3D set error: 0.0014  
Native coverage: 0.876 | nfev: 36  
Flags: collapsed hole

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 3 | original trial 196

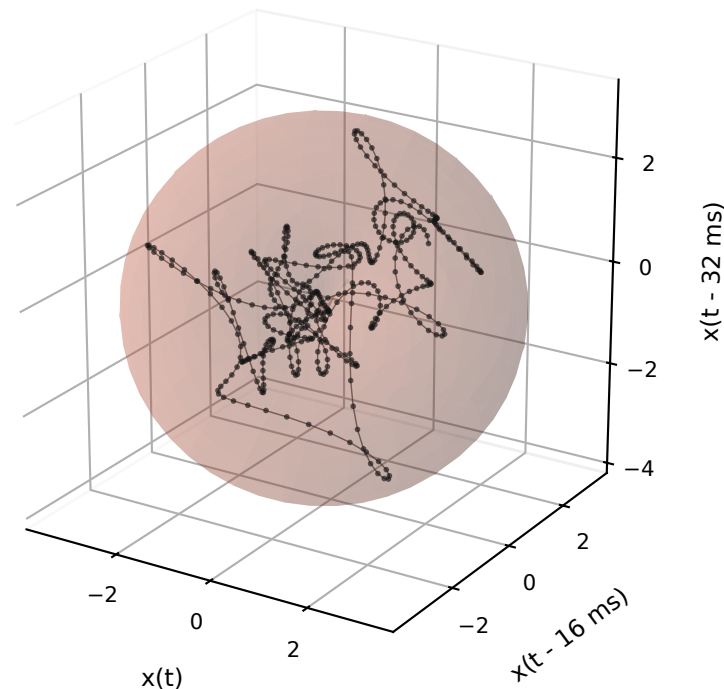
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2283  
Native coverage: 0.887 | nfev: 51  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

2-torus: donut volume



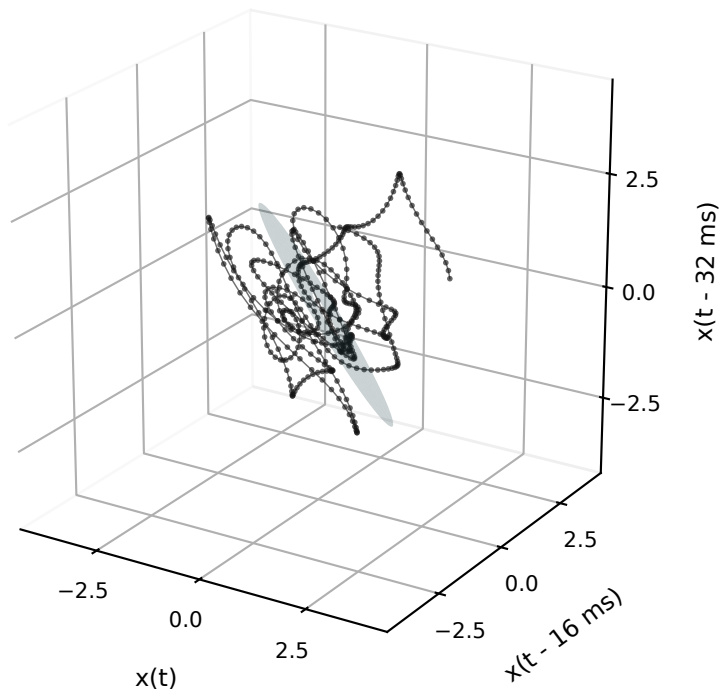
Normalized 3D set error: 0.0011  
Native coverage: 0.870 | nfev: 31  
Flags: none

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 3 | original trial 494

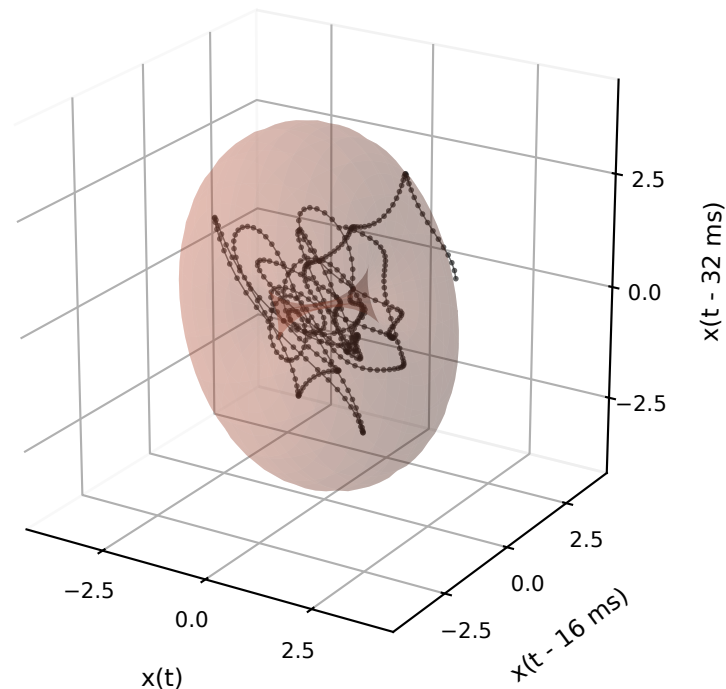
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2593  
Native coverage: 0.915 | nfev: 87  
Flags: none

2-torus: donut volume



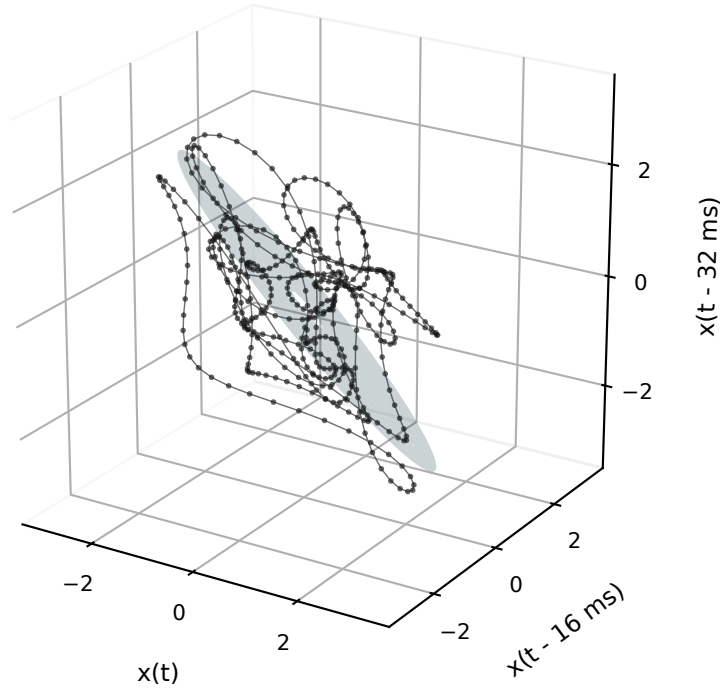
Normalized 3D set error: 0.0008  
Native coverage: 0.893 | nfev: 23  
Flags: active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 3 | original trial 51

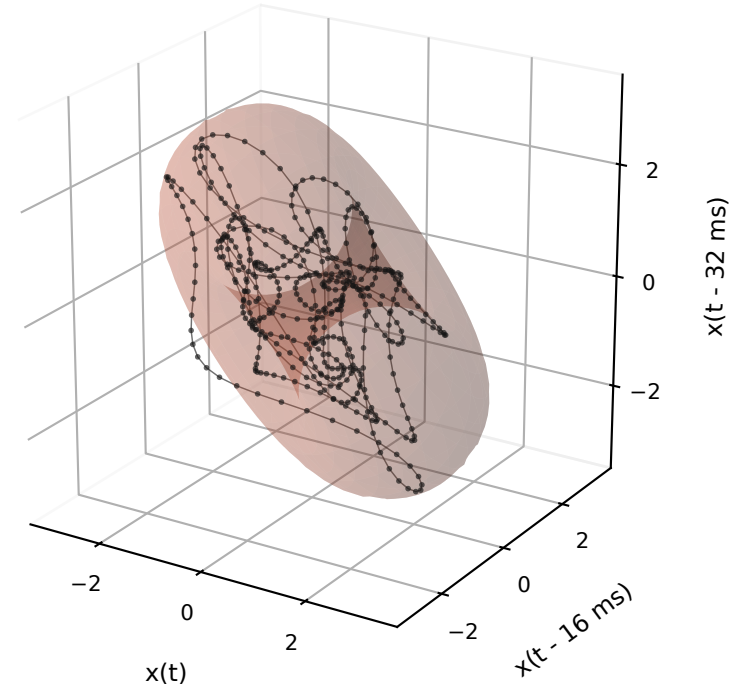
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1468  
Native coverage: 0.927 | nfev: 85  
Flags: none

2-torus: donut volume



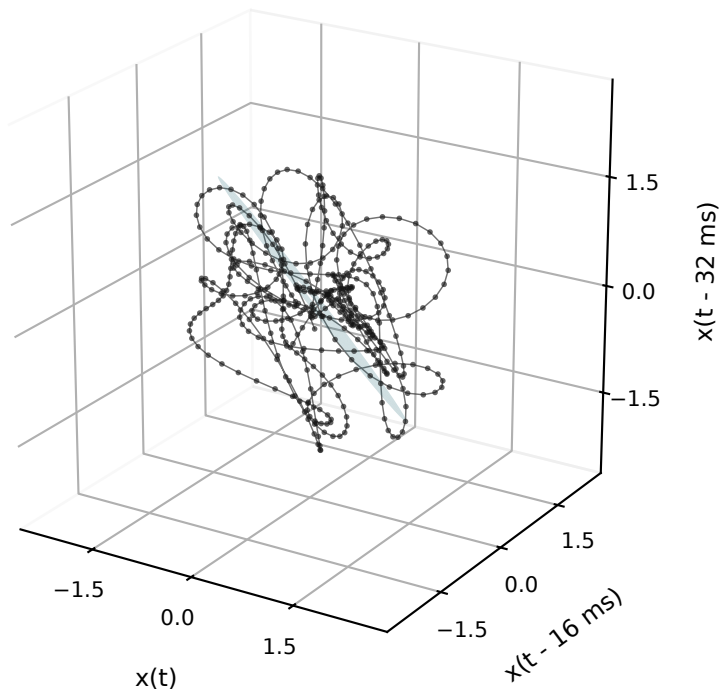
Normalized 3D set error: 0.0005  
Native coverage: 0.878 | nfev: 31  
Flags: near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 3 | original trial 432

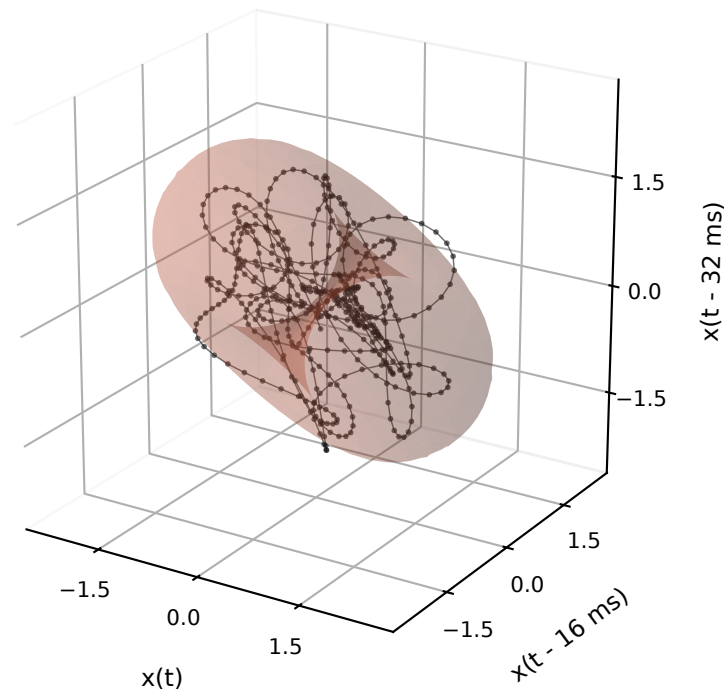
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1895  
Native coverage: 0.889 | nfev: 23  
Flags: none

2-torus: donut volume



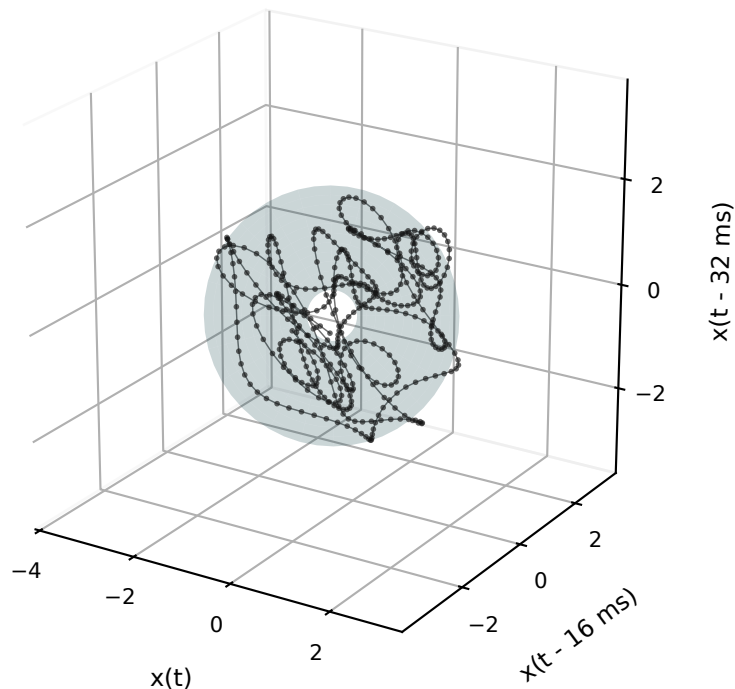
Normalized 3D set error: 0.0016  
Native coverage: 0.872 | nfev: 42  
Flags: active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 3 | original trial 144

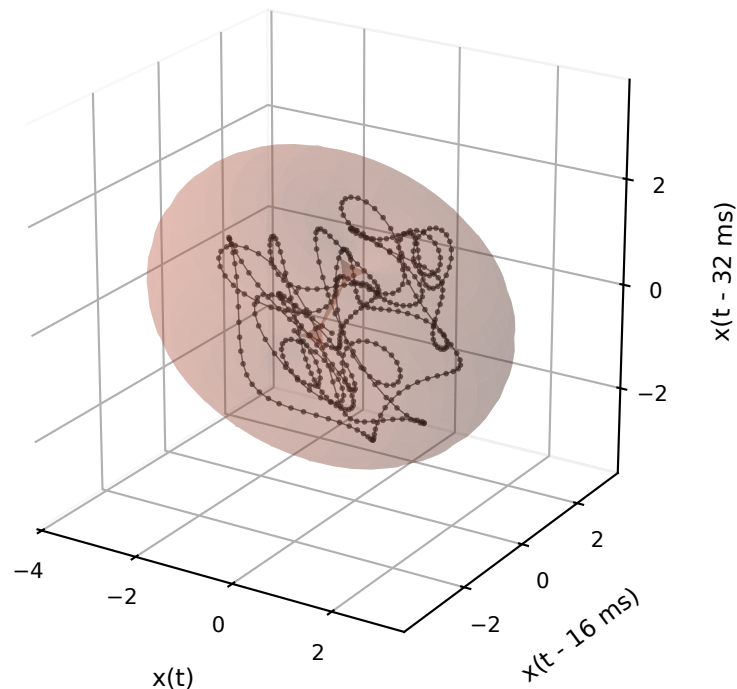
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.3176  
Native coverage: 0.887 | nfev: 76  
Flags: nearly circular orientation

2-torus: donut volume



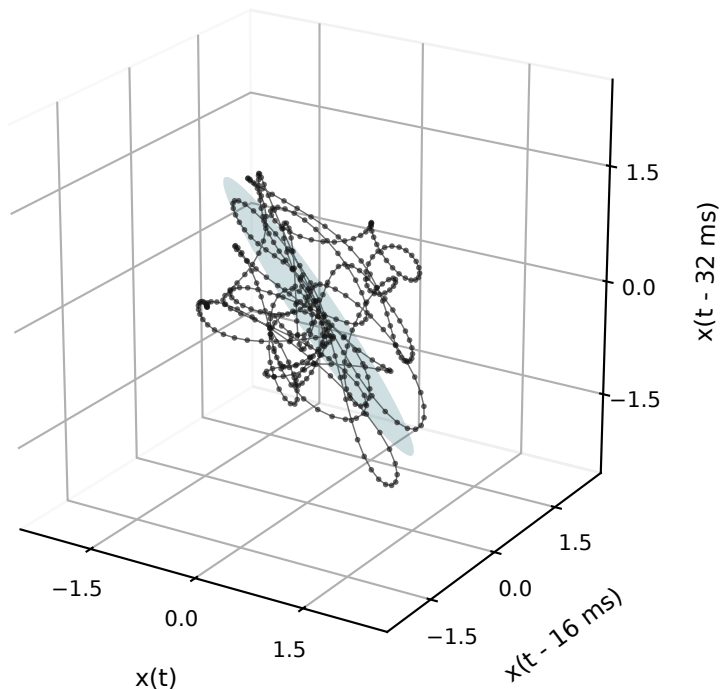
Normalized 3D set error: 0.0010  
Native coverage: 0.876 | nfev: 29  
Flags: collapsed hole; active optimizer bound; near optimizer bound; nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 3 | original trial 216

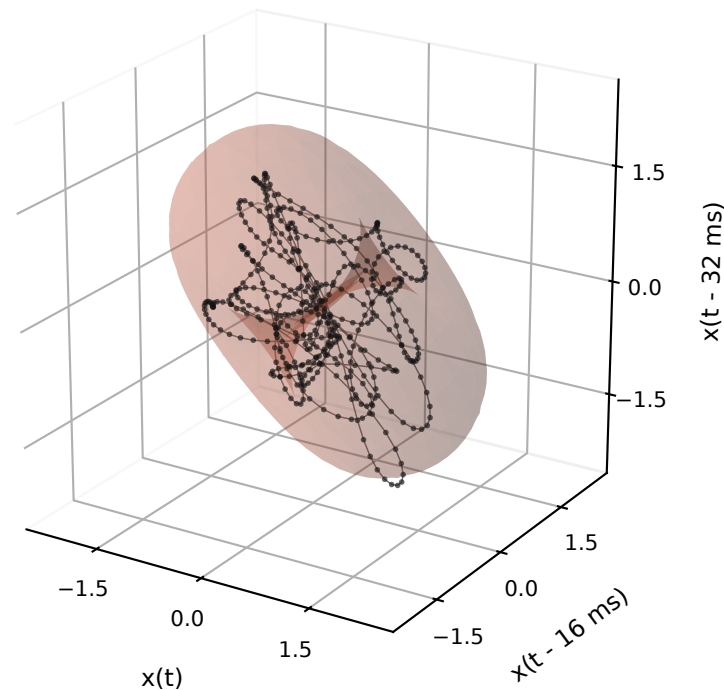
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1592  
Native coverage: 0.887 | nfev: 148  
Flags: none

2-torus: donut volume



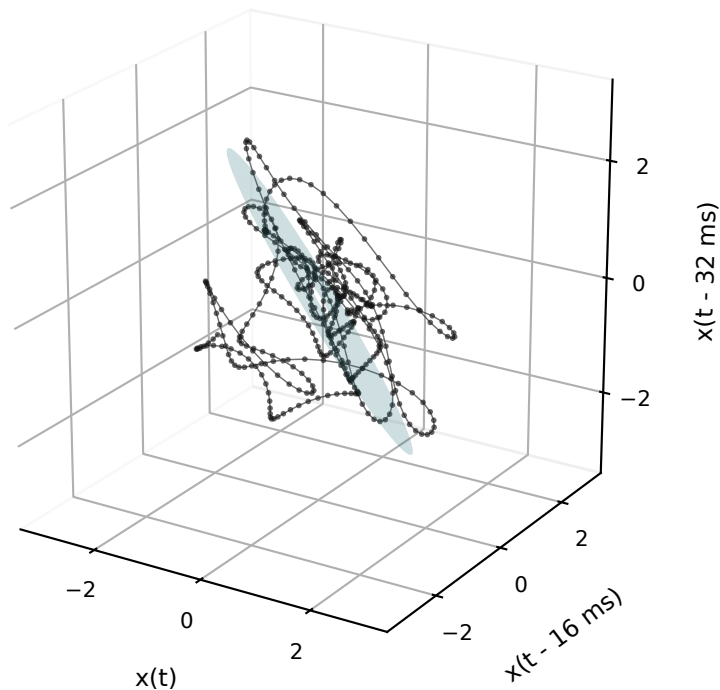
Normalized 3D set error: 0.0021  
Native coverage: 0.840 | nfev: 39  
Flags: none

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 4 | original trial 312

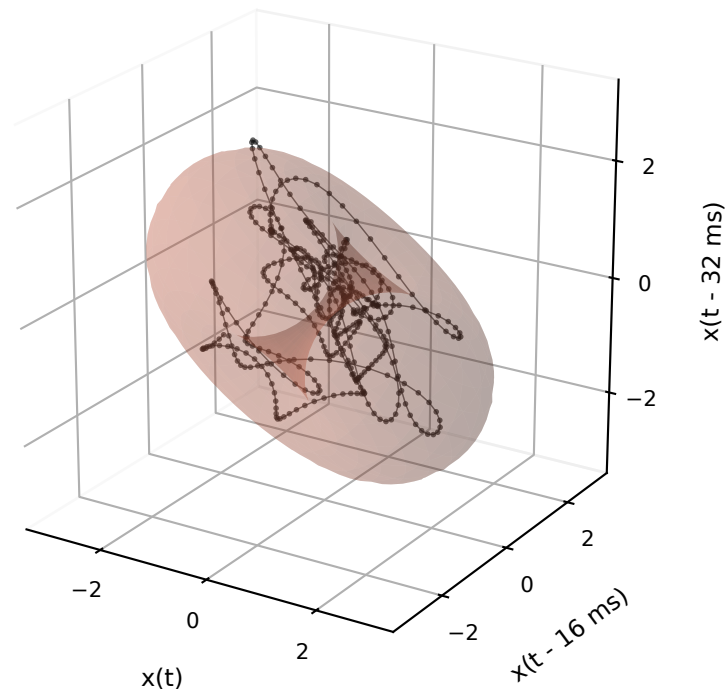
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2048  
Native coverage: 0.904 | nfev: 53  
Flags: nearly circular orientation

2-torus: donut volume



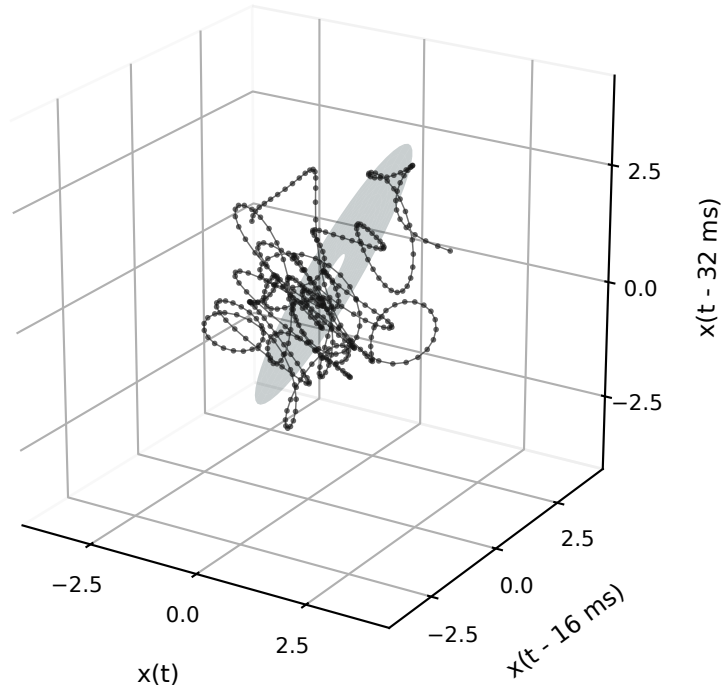
Normalized 3D set error: 0.0009  
Native coverage: 0.887 | nfev: 19  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 4 | original trial 400

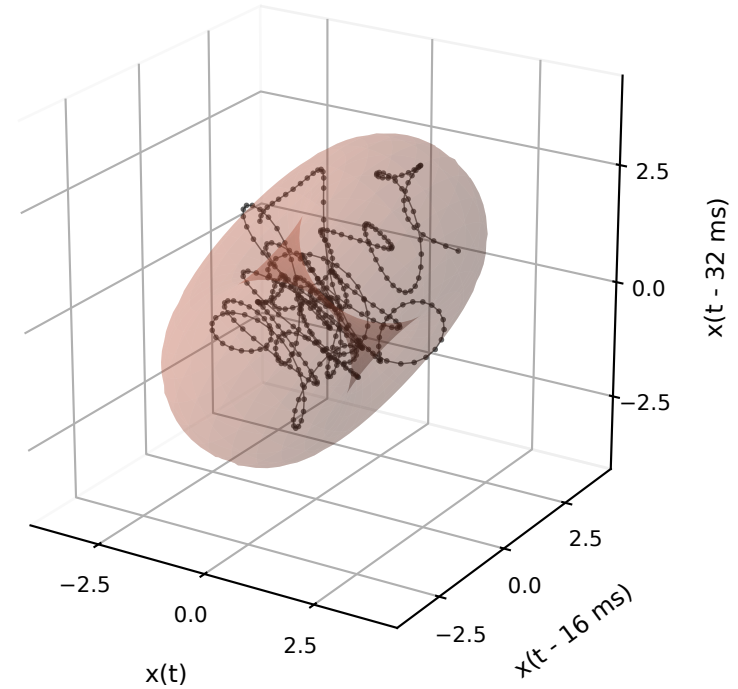
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2664  
Native coverage: 0.915 | nfev: 63  
Flags: near optimizer bound; nearly circular orientation

2-torus: donut volume



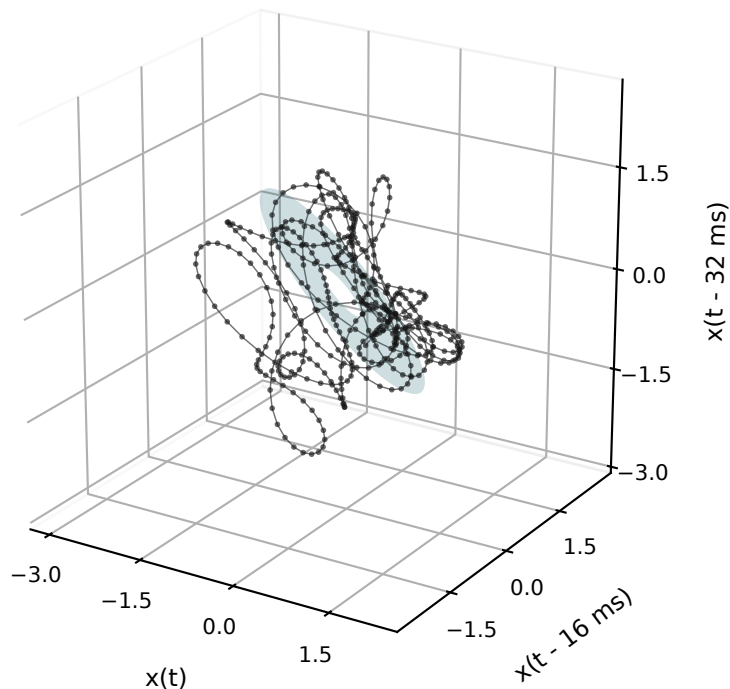
Normalized 3D set error: 0.0008  
Native coverage: 0.904 | nfev: 20  
Flags: active optimizer bound; near optimizer bound; nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 4 | original trial 333

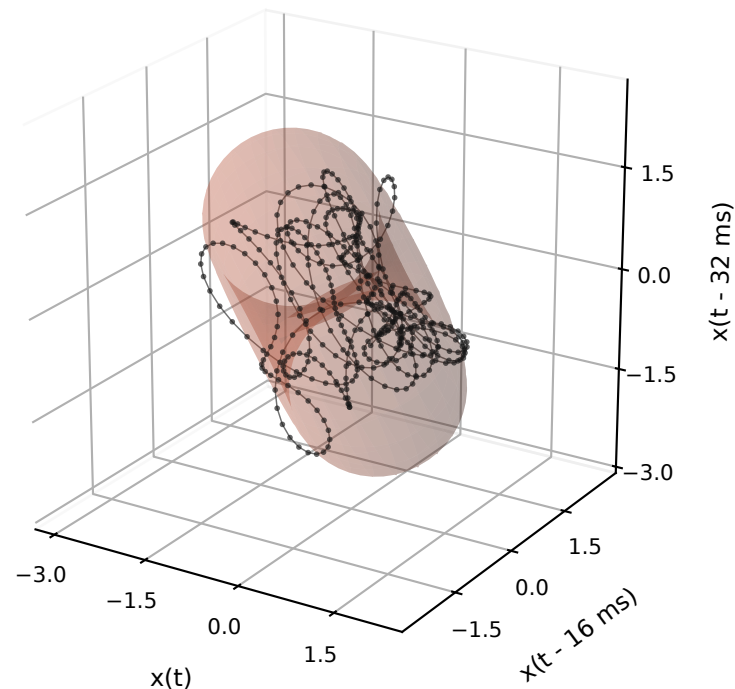
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1786  
Native coverage: 0.919 | nfev: 26  
Flags: none

2-torus: donut volume



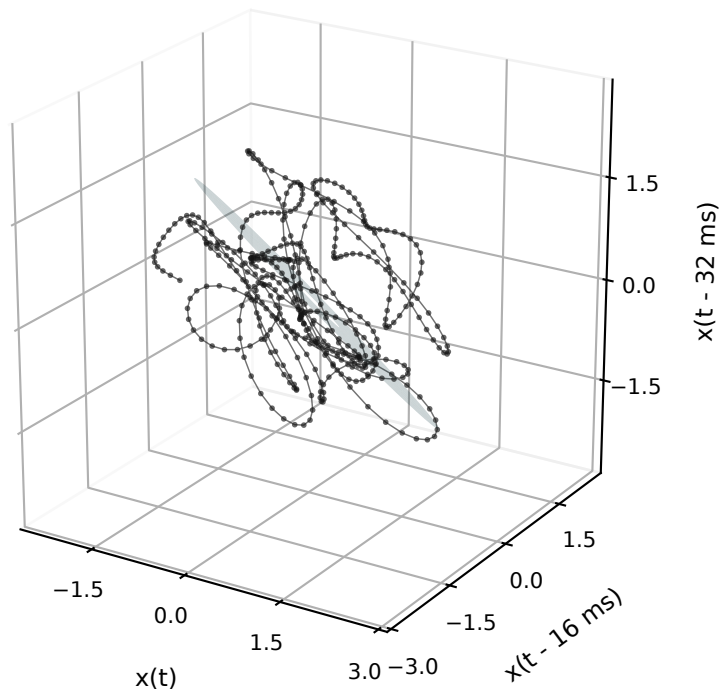
Normalized 3D set error: 0.0014  
Native coverage: 0.855 | nfev: 39  
Flags: active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 4 | original trial 459

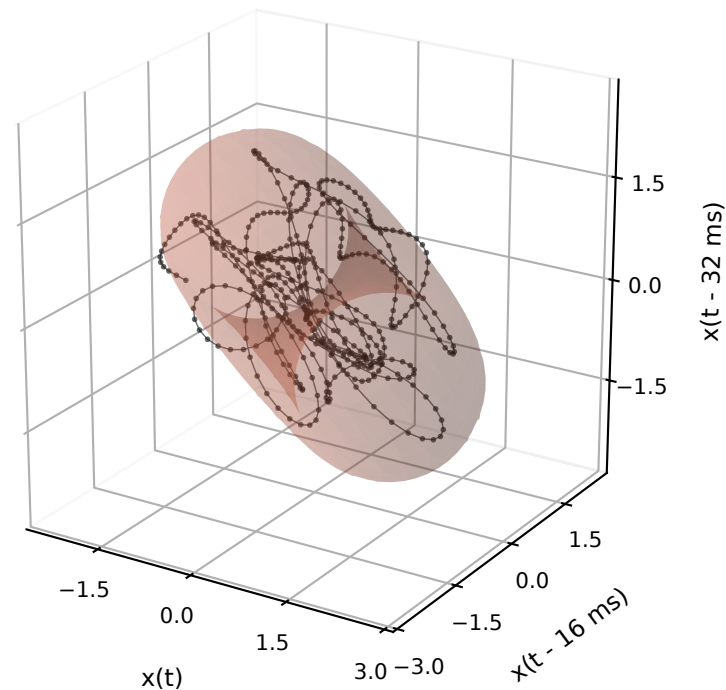
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1811  
Native coverage: 0.908 | nfev: 45  
Flags: none

2-torus: donut volume



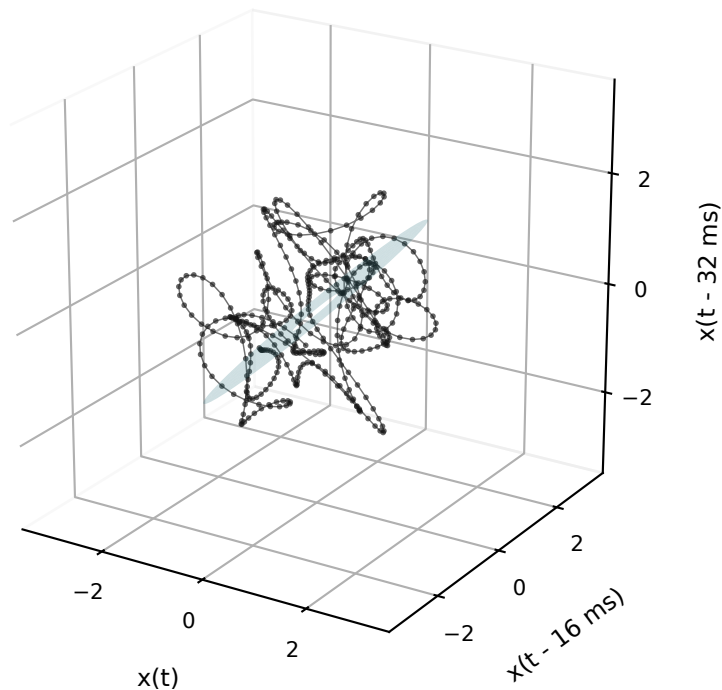
Normalized 3D set error: 0.0011  
Native coverage: 0.861 | nfev: 31  
Flags: near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 4 | original trial 606

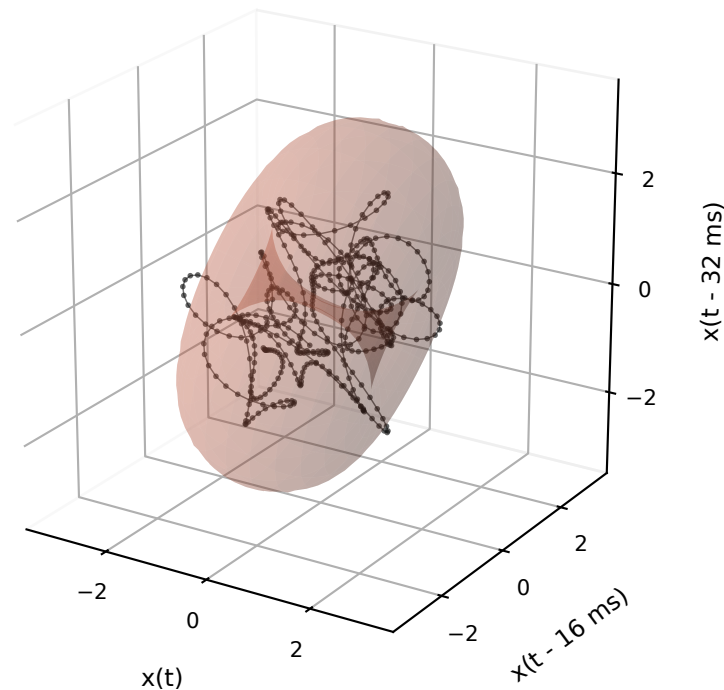
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2764  
Native coverage: 0.908 | nfev: 102  
Flags: none

2-torus: donut volume



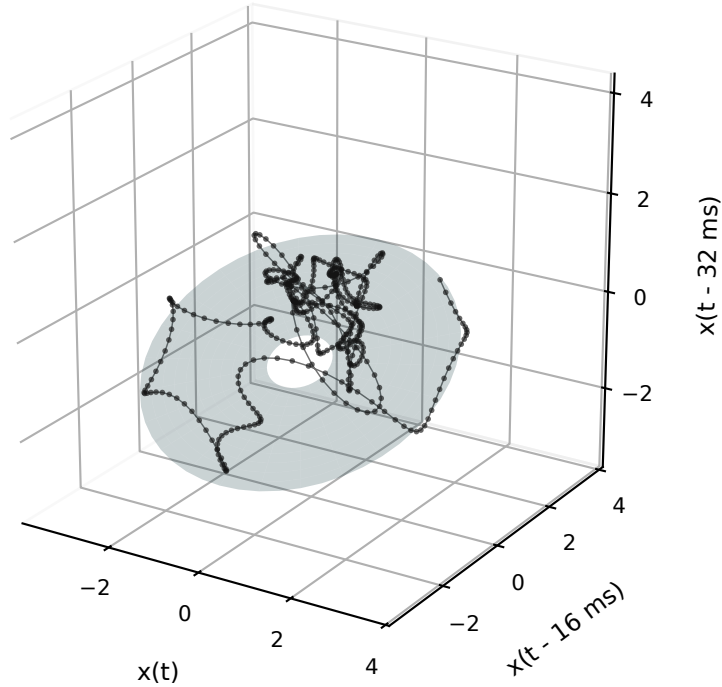
Normalized 3D set error: 0.0015  
Native coverage: 0.891 | nfev: 40  
Flags: collapsed hole; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 4 | original trial 86

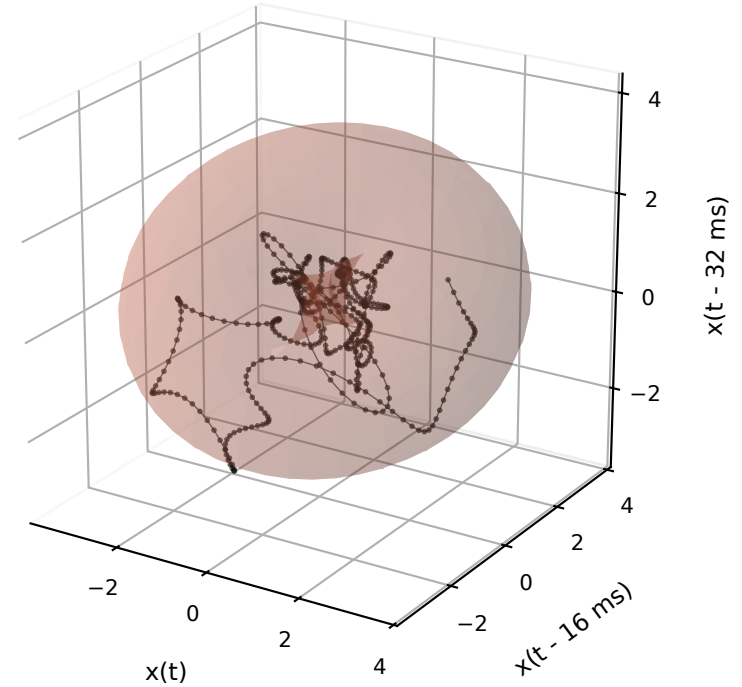
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2622  
Native coverage: 0.906 | nfev: 40  
Flags: none

2-torus: donut volume



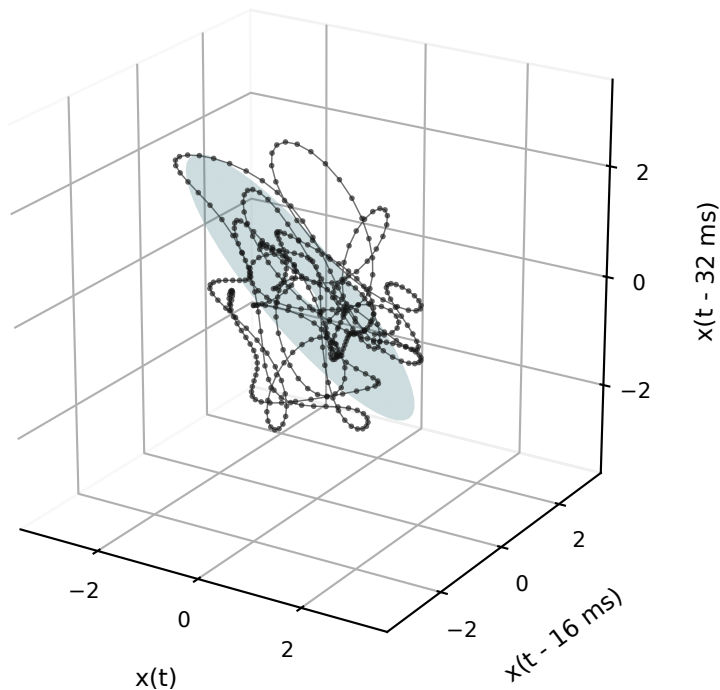
Normalized 3D set error: 0.0020  
Native coverage: 0.889 | nfev: 54  
Flags: collapsed hole

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 4 | original trial 242

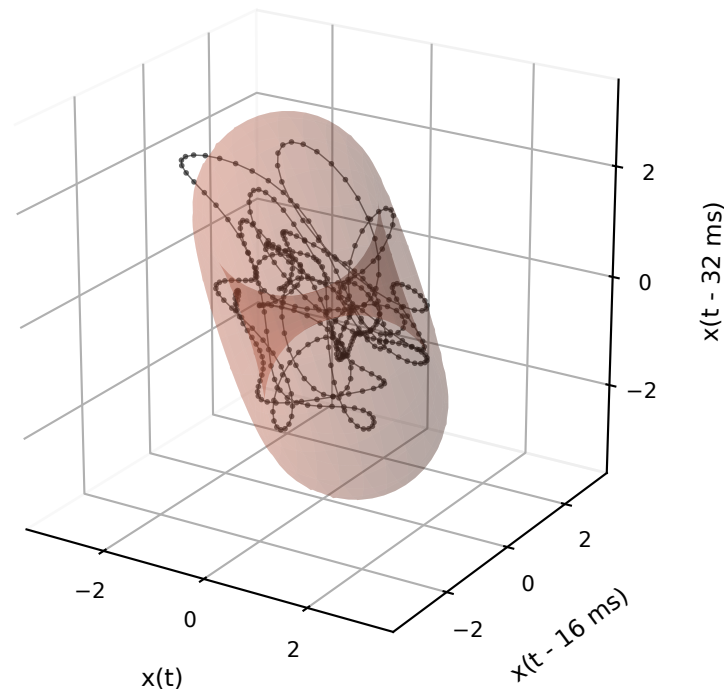
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1952  
Native coverage: 0.897 | nfev: 70  
Flags: near optimizer bound

2-torus: donut volume



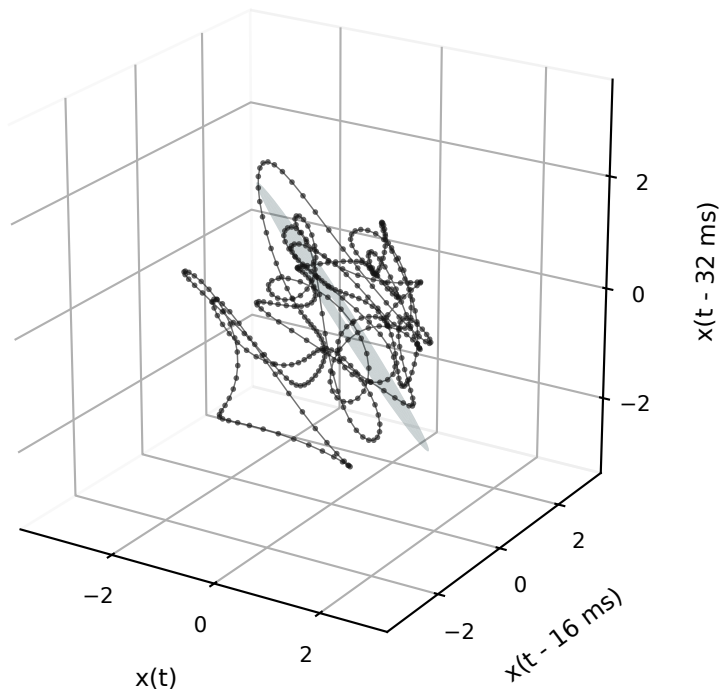
Normalized 3D set error: 0.0010  
Native coverage: 0.904 | nfev: 15  
Flags: active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 4 | original trial 248

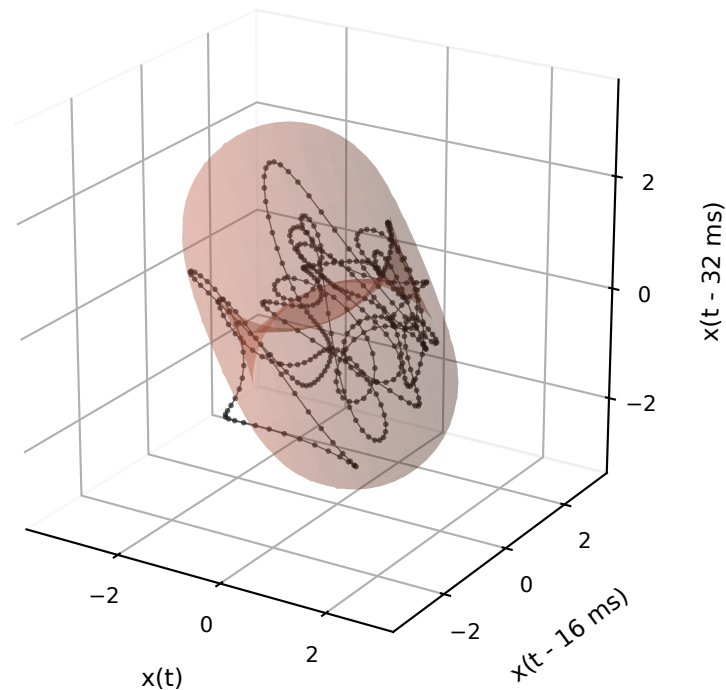
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2833  
Native coverage: 0.923 | nfev: 75  
Flags: near optimizer bound

2-torus: donut volume



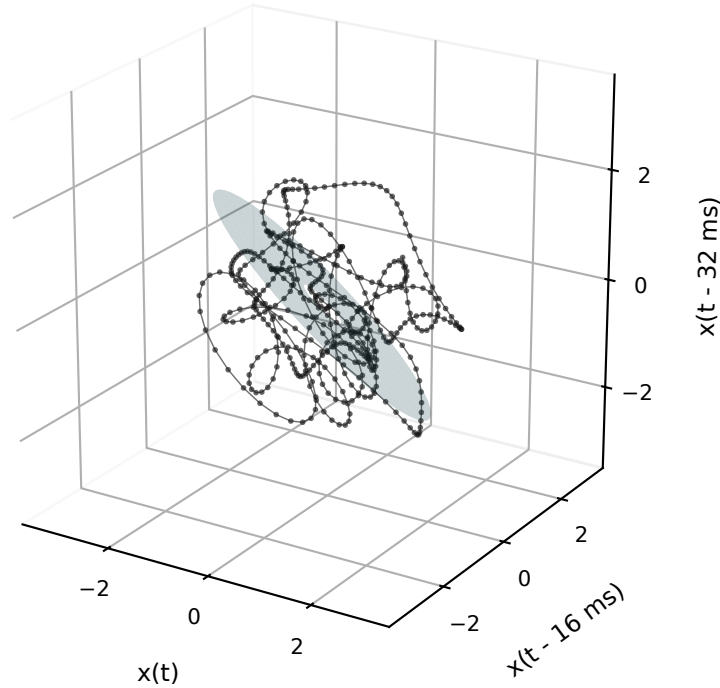
Normalized 3D set error: 0.0015  
Native coverage: 0.868 | nfev: 33  
Flags: collapsed hole

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 4 | original trial 89

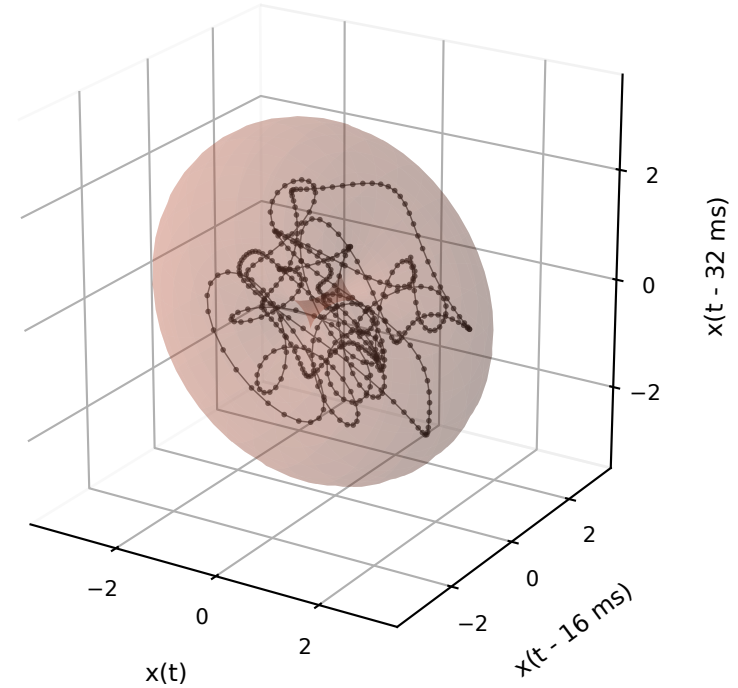
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2478  
Native coverage: 0.906 | nfev: 88  
Flags: nearly circular orientation

2-torus: donut volume



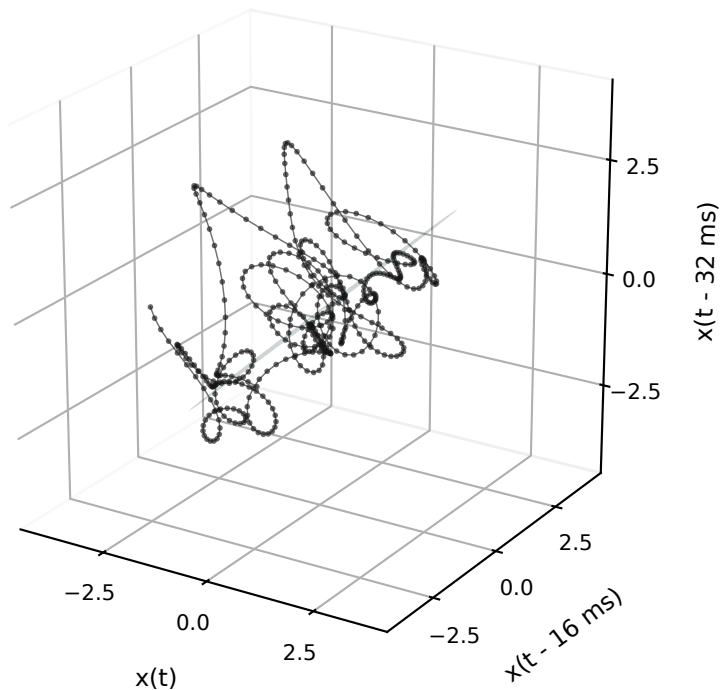
Normalized 3D set error: 0.0011  
Native coverage: 0.891 | nfev: 32  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 4 | original trial 691

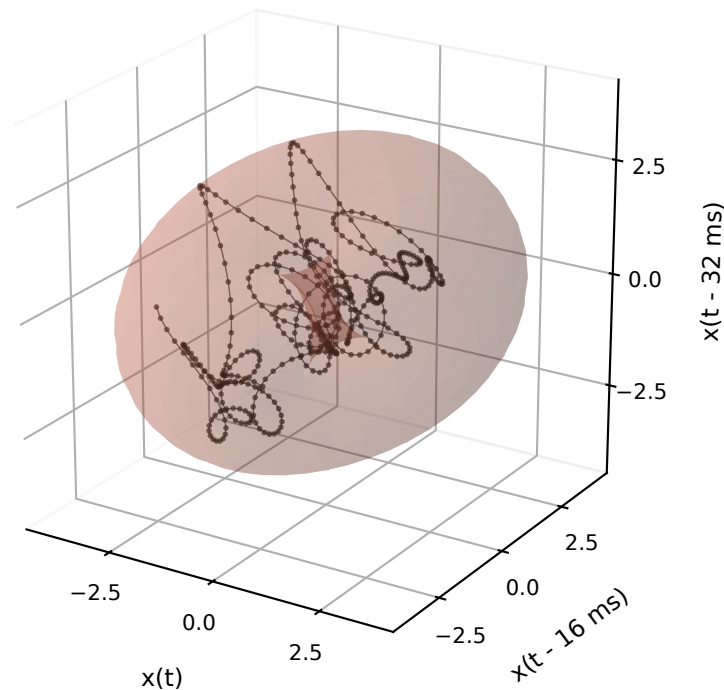
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2255  
Native coverage: 0.906 | nfev: 39  
Flags: none

2-torus: donut volume



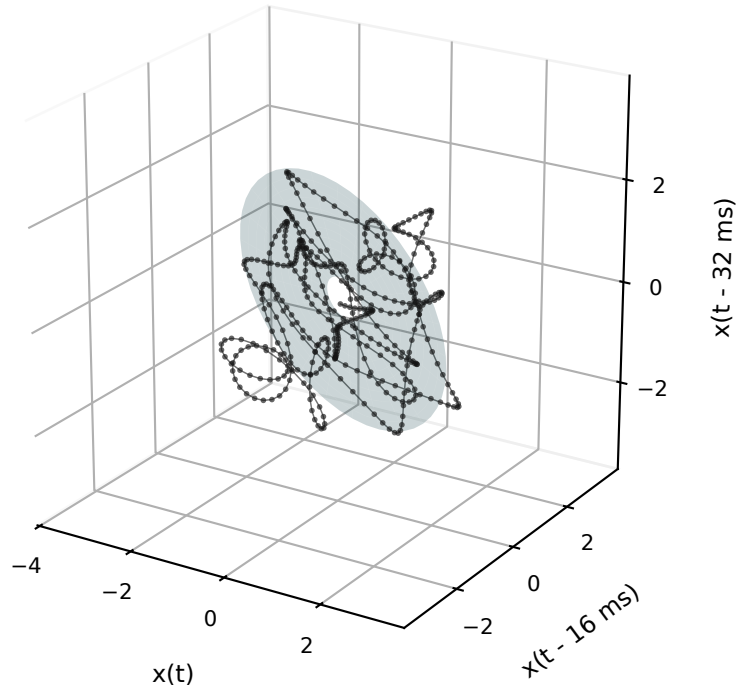
Normalized 3D set error: 0.0005  
Native coverage: 0.906 | nfev: 30  
Flags: collapsed hole; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 4 | original trial 94

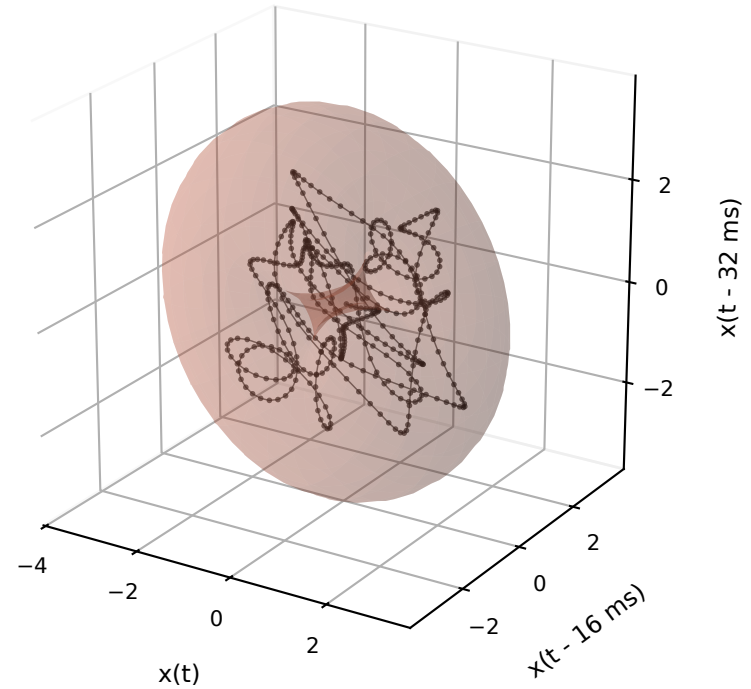
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2861  
Native coverage: 0.906 | nfev: 42  
Flags: none

2-torus: donut volume



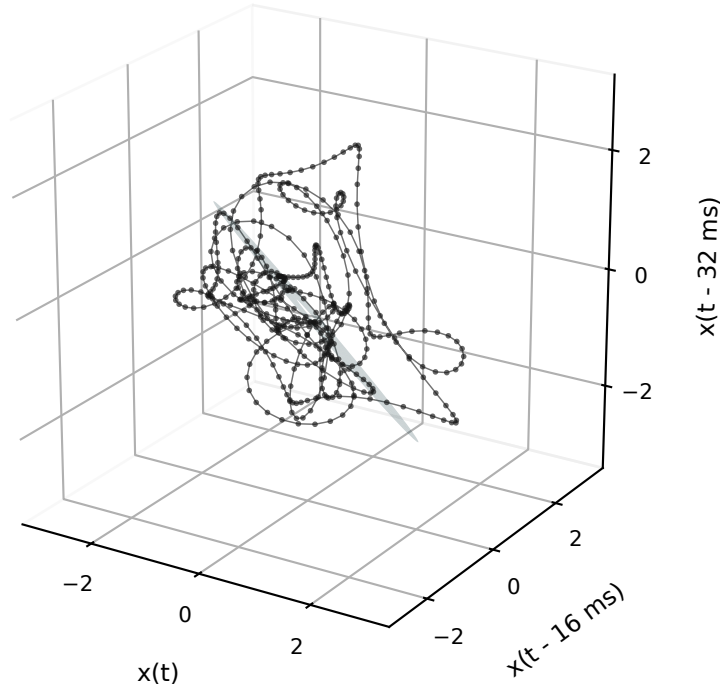
Normalized 3D set error: 0.0011  
Native coverage: 0.891 | nfev: 51  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 4 | original trial 17

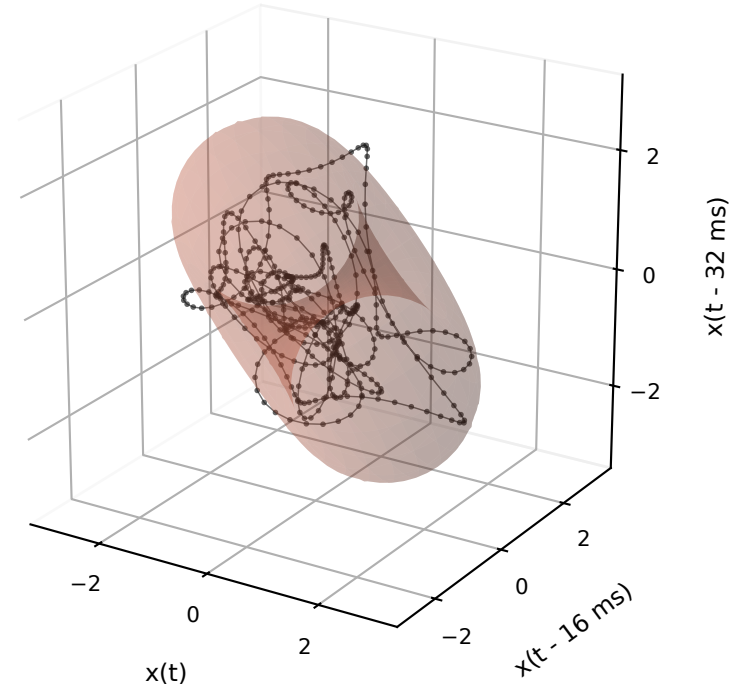
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1967  
Native coverage: 0.900 | nfev: 38  
Flags: none

2-torus: donut volume



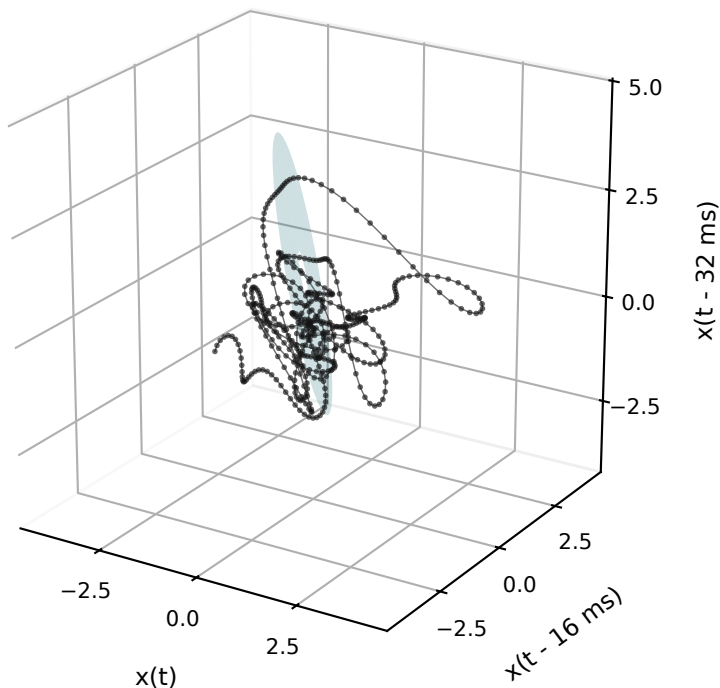
Normalized 3D set error: 0.0013  
Native coverage: 0.893 | nfev: 24  
Flags: near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 4 | original trial 109

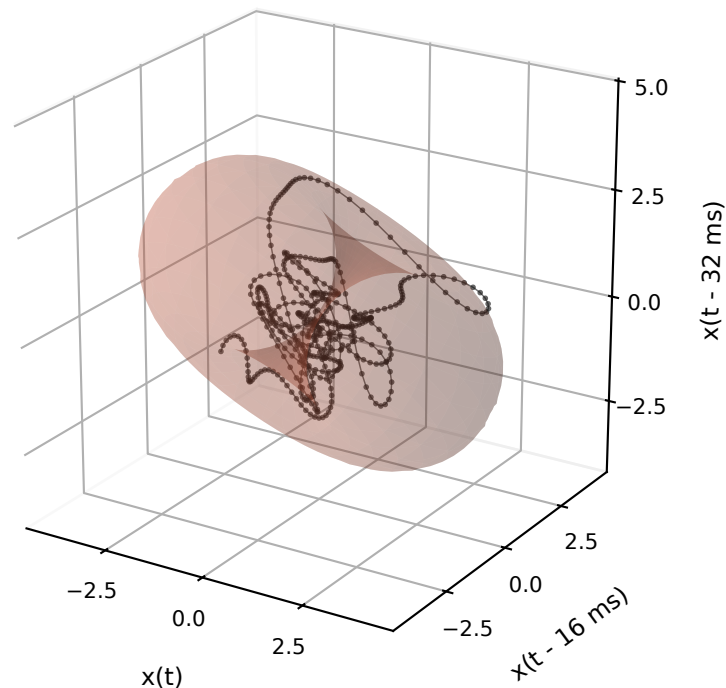
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.3152  
Native coverage: 0.938 | nfev: 46  
Flags: nearly circular orientation

2-torus: donut volume



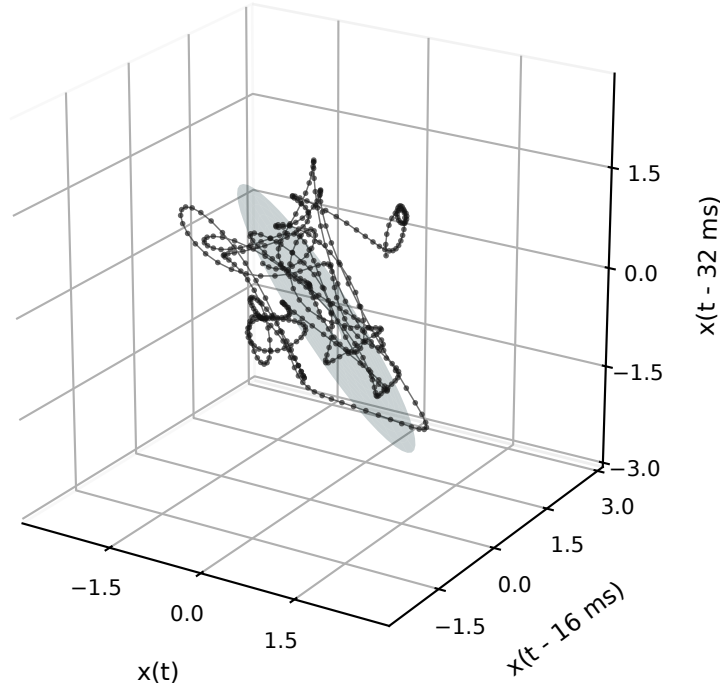
Normalized 3D set error: 0.0005  
Native coverage: 0.893 | nfev: 23  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 4 | original trial 695

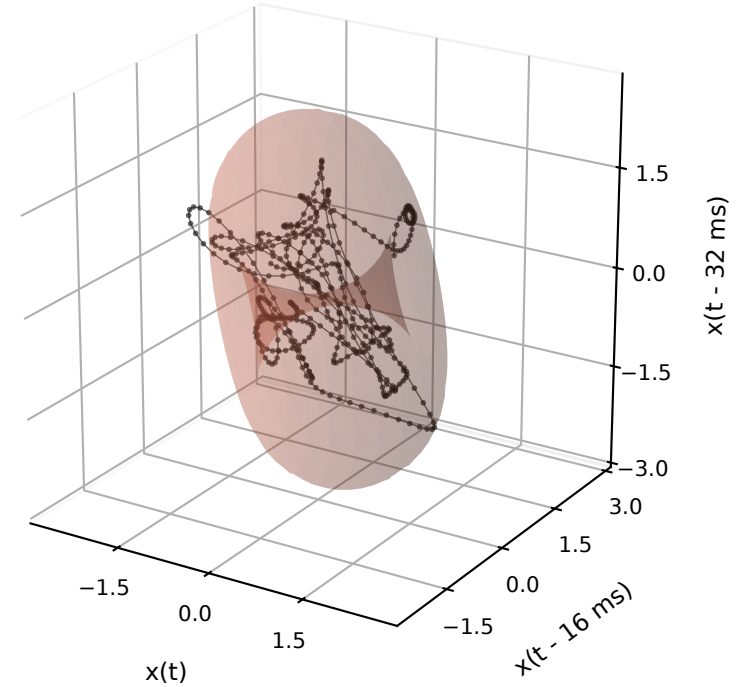
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2297  
Native coverage: 0.887 | nfev: 68  
Flags: nearly circular orientation

2-torus: donut volume



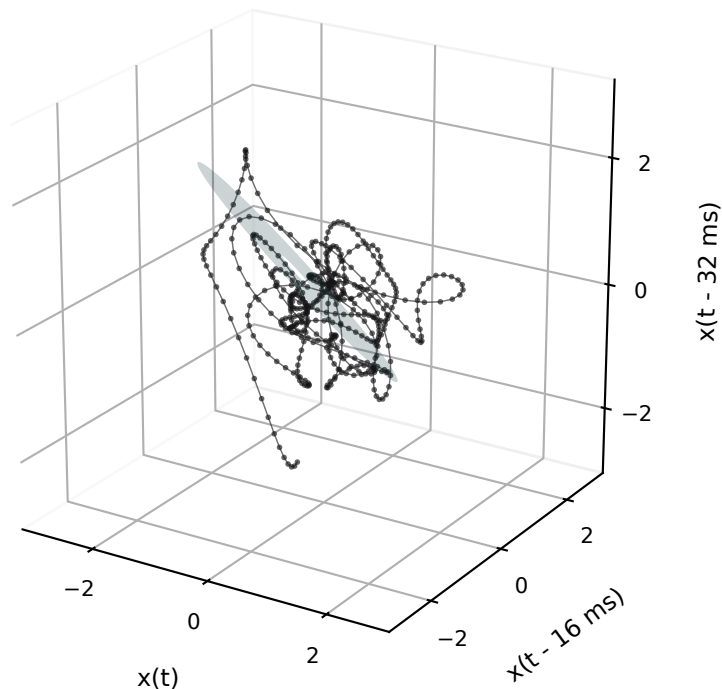
Normalized 3D set error: 0.0014  
Native coverage: 0.842 | nfev: 35  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 4 | original trial 116

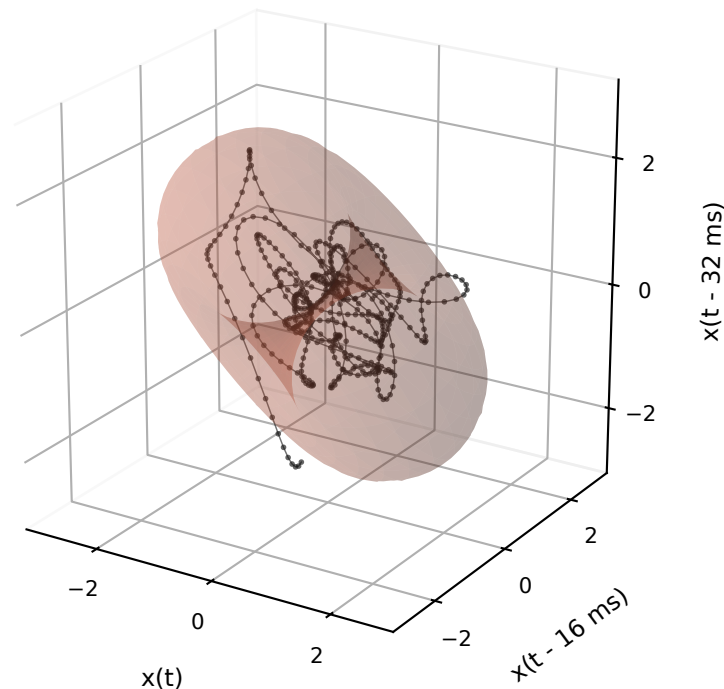
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1981  
Native coverage: 0.889 | nfev: 48  
Flags: none

2-torus: donut volume



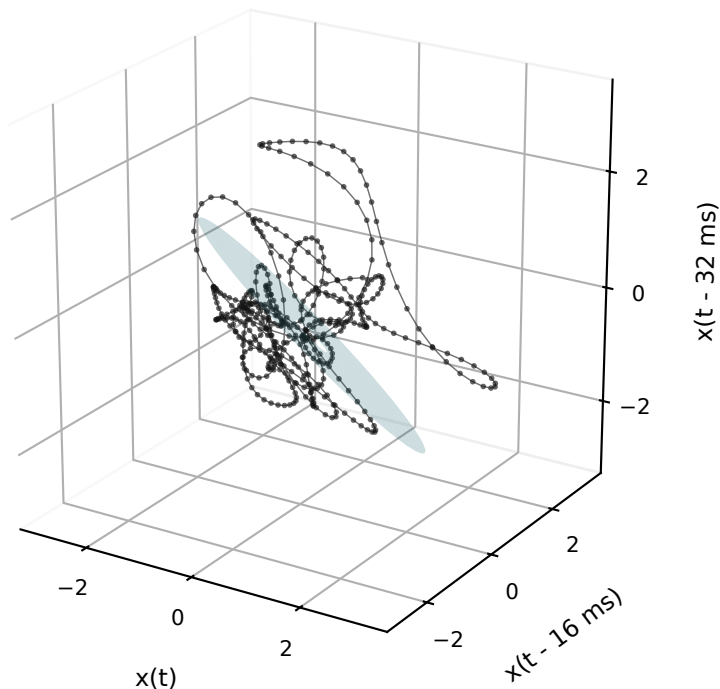
Normalized 3D set error: 0.0024  
Native coverage: 0.880 | nfev: 59  
Flags: nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 4 | original trial 361

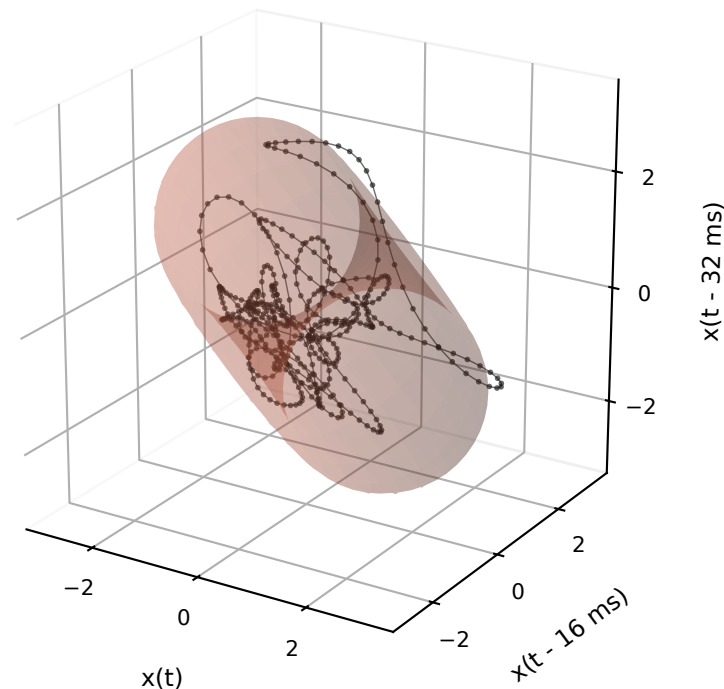
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2481  
Native coverage: 0.906 | nfev: 74  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

2-torus: donut volume



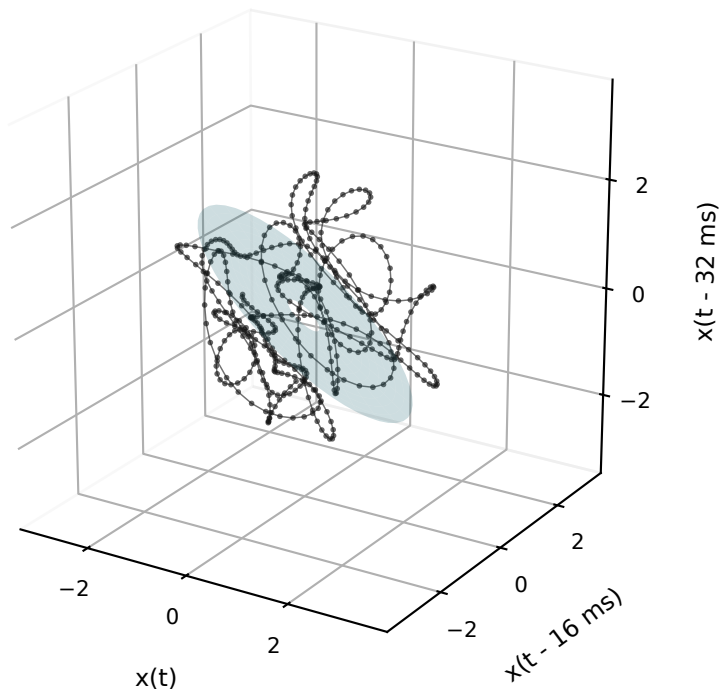
Normalized 3D set error: 0.0009  
Native coverage: 0.878 | nfev: 28  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 4 | original trial 283

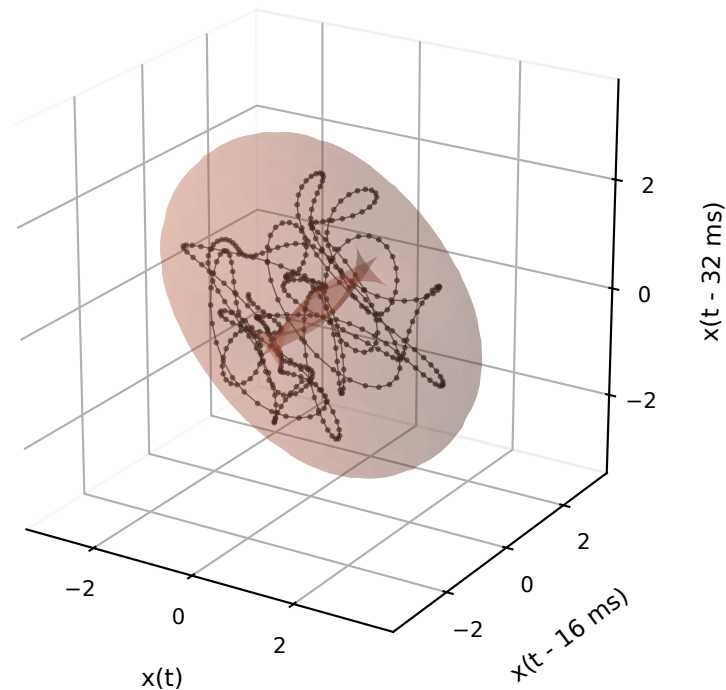
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2523  
Native coverage: 0.917 | nfev: 137  
Flags: none

2-torus: donut volume



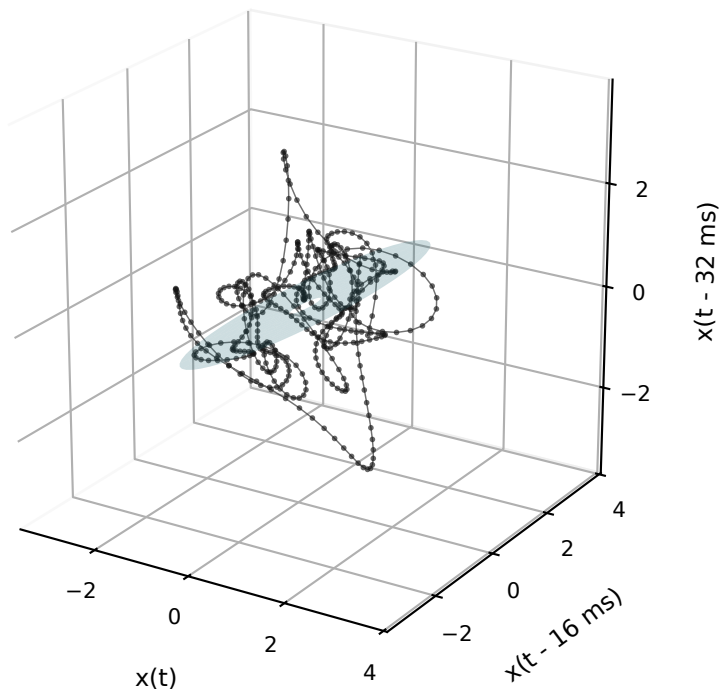
Normalized 3D set error: 0.0010  
Native coverage: 0.876 | nfev: 22  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 4 | original trial 511

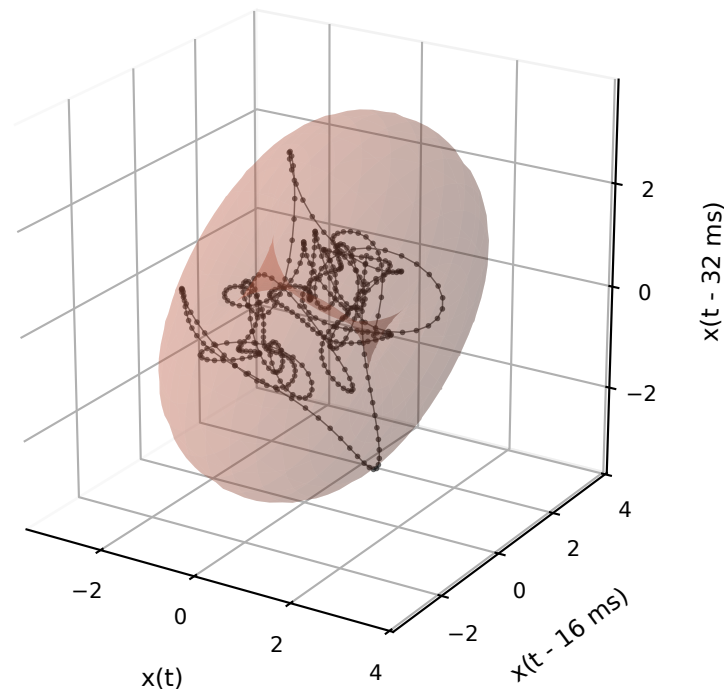
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2679  
Native coverage: 0.917 | nfev: 42  
Flags: none

2-torus: donut volume



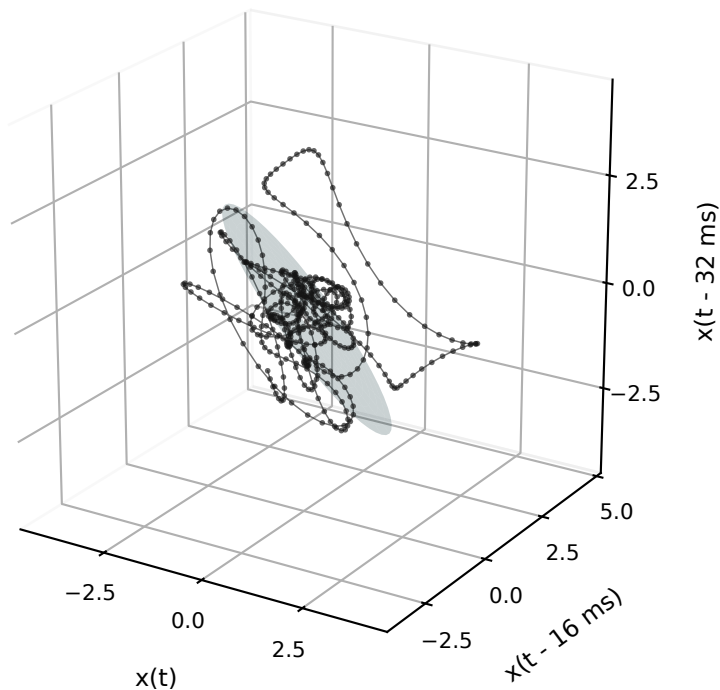
Normalized 3D set error: 0.0011  
Native coverage: 0.917 | nfev: 35  
Flags: collapsed hole; nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 4 | original trial 206

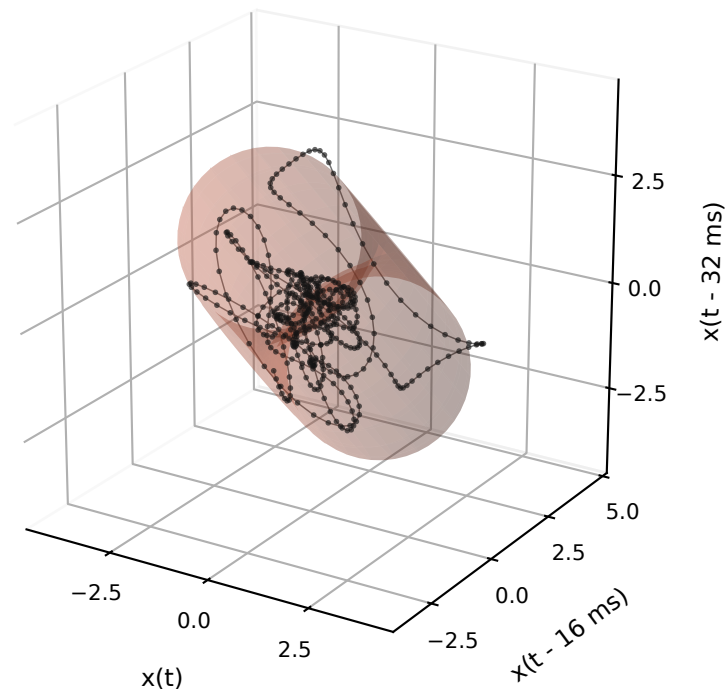
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1796  
Native coverage: 0.923 | nfev: 30  
Flags: none

2-torus: donut volume



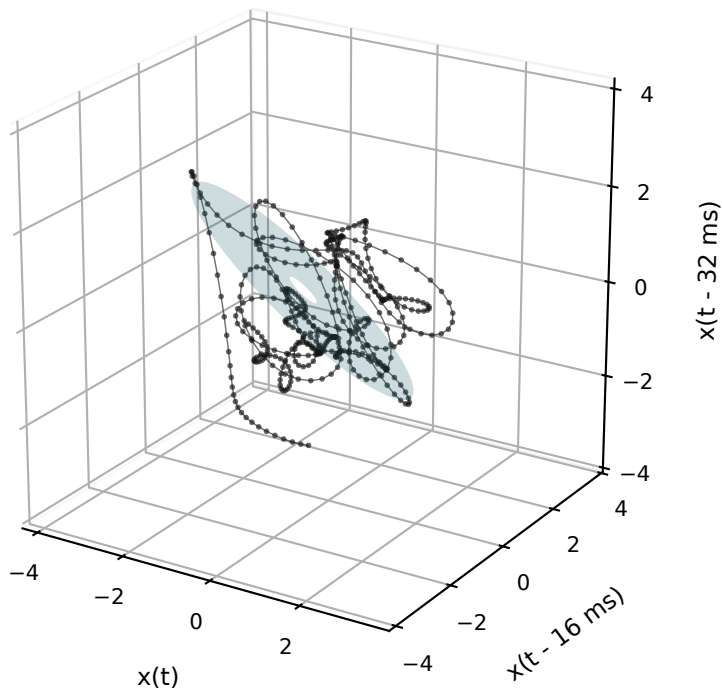
Normalized 3D set error: 0.0018  
Native coverage: 0.865 | nfev: 66  
Flags: collapsed hole

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 4 | original trial 520

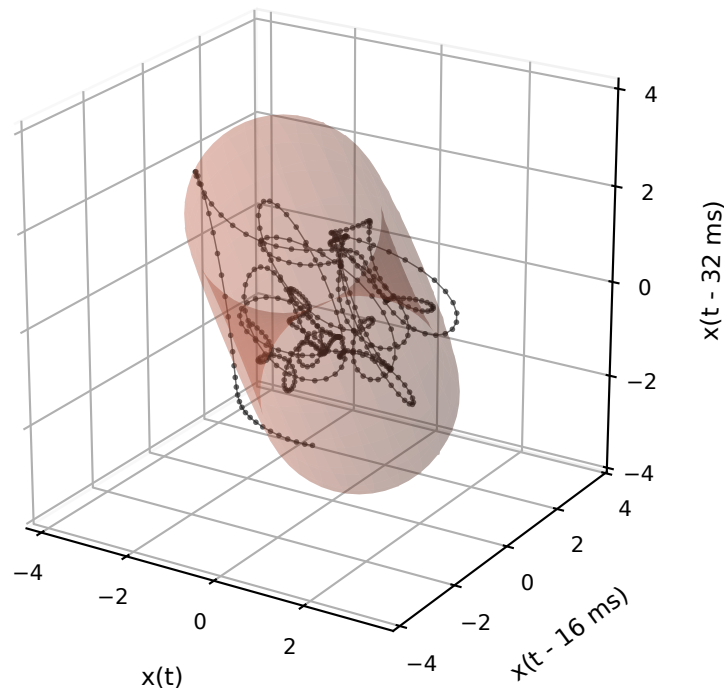
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2853  
Native coverage: 0.917 | nfev: 29  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

2-torus: donut volume



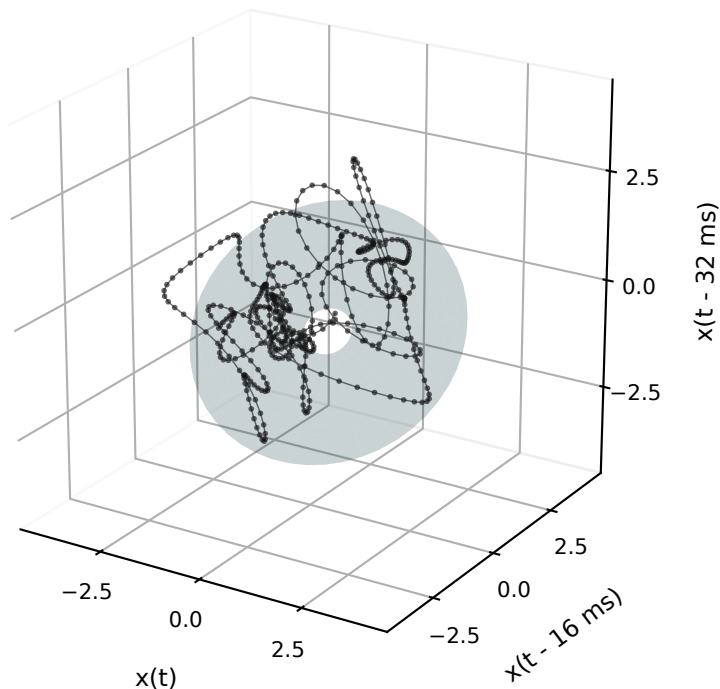
Normalized 3D set error: 0.0010  
Native coverage: 0.917 | nfev: 20  
Flags: collapsed hole; active optimizer bound; near  
optimizer bound; nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 4 | original trial 119

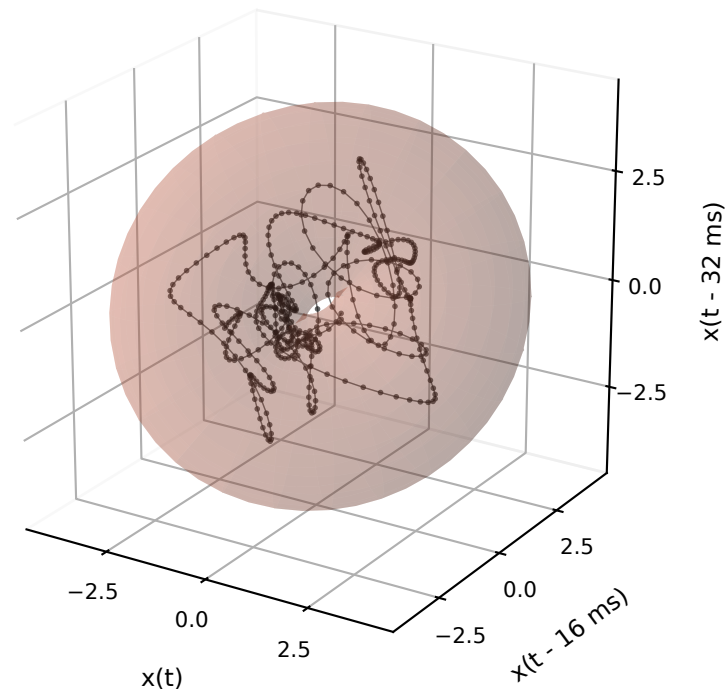
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2387  
Native coverage: 0.917 | nfev: 52  
Flags: none

2-torus: donut volume



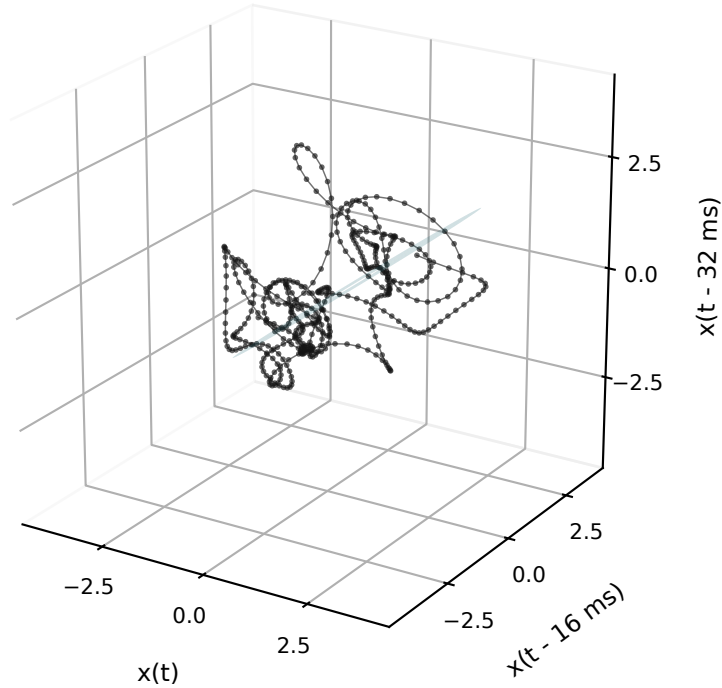
Normalized 3D set error: 0.0009  
Native coverage: 0.910 | nfev: 33  
Flags: none

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 4 | original trial 85

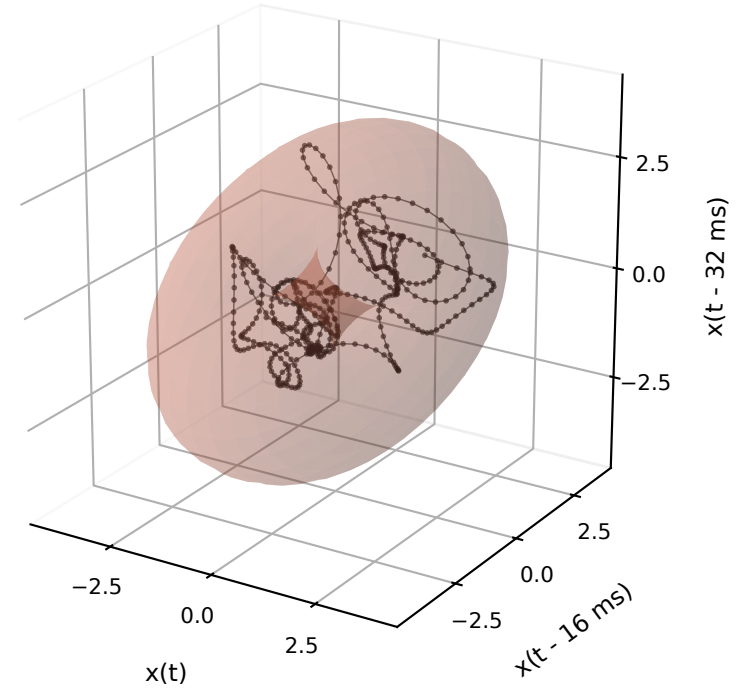
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2271  
Native coverage: 0.919 | nfev: 61  
Flags: none

2-torus: donut volume



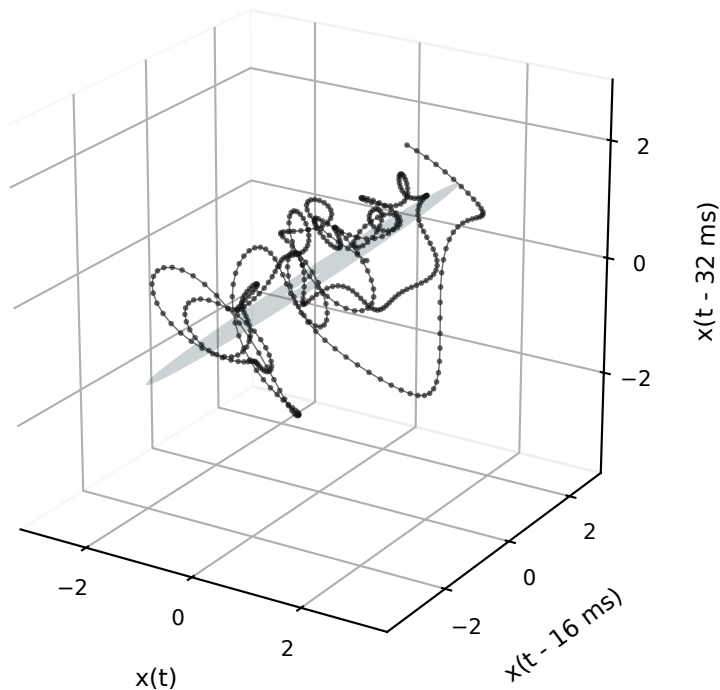
Normalized 3D set error: 0.0009  
Native coverage: 0.912 | nfev: 53  
Flags: none

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 4 | original trial 199

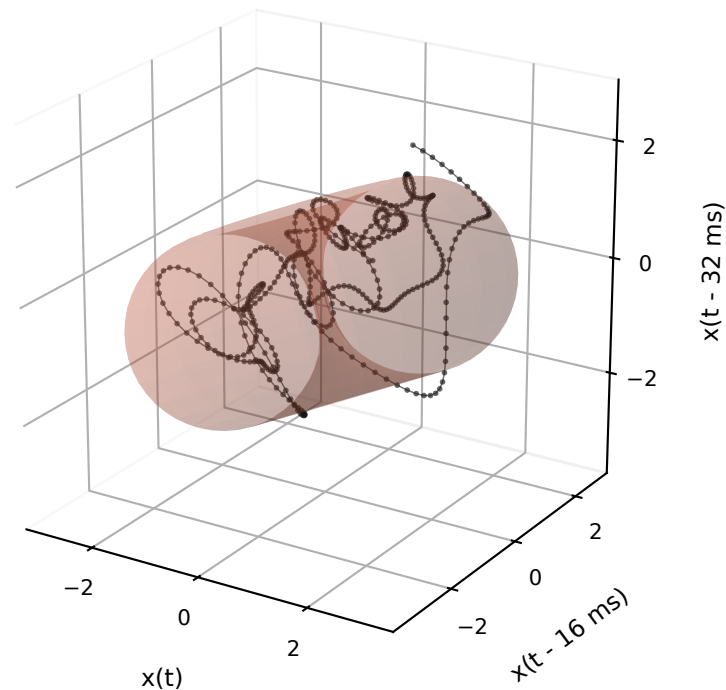
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1841  
Native coverage: 0.906 | nfev: 36  
Flags: none

2-torus: donut volume



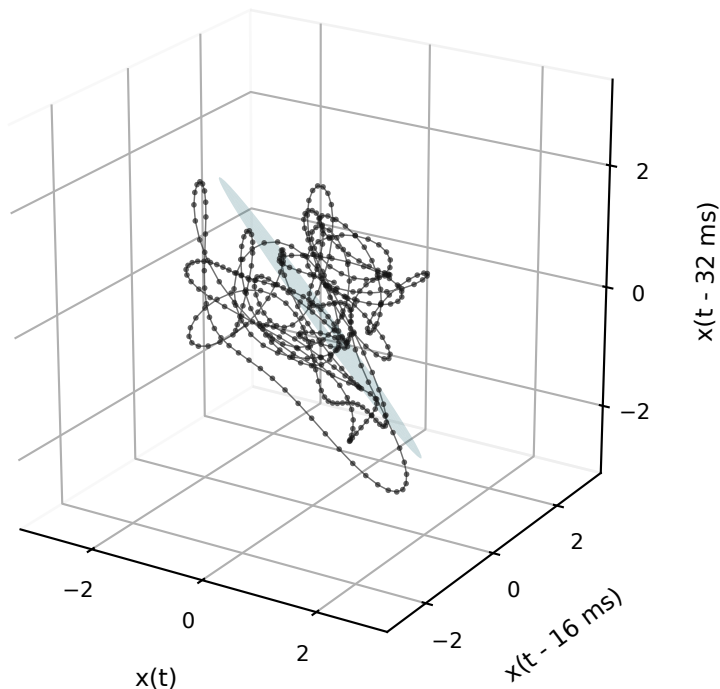
Normalized 3D set error: 0.0016  
Native coverage: 0.868 | nfev: 36  
Flags: collapsed hole; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 4 | original trial 536

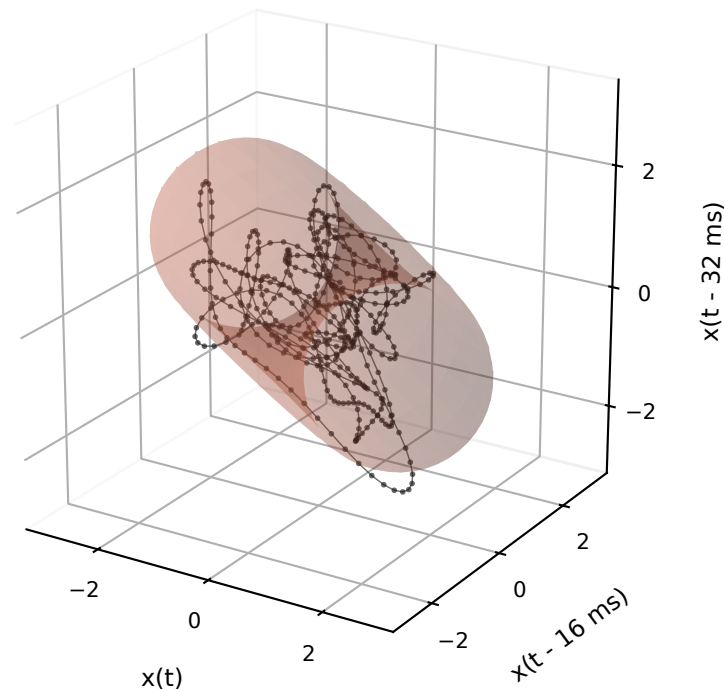
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1779  
Native coverage: 0.915 | nfev: 81  
Flags: near optimizer bound; nearly circular orientation

2-torus: donut volume



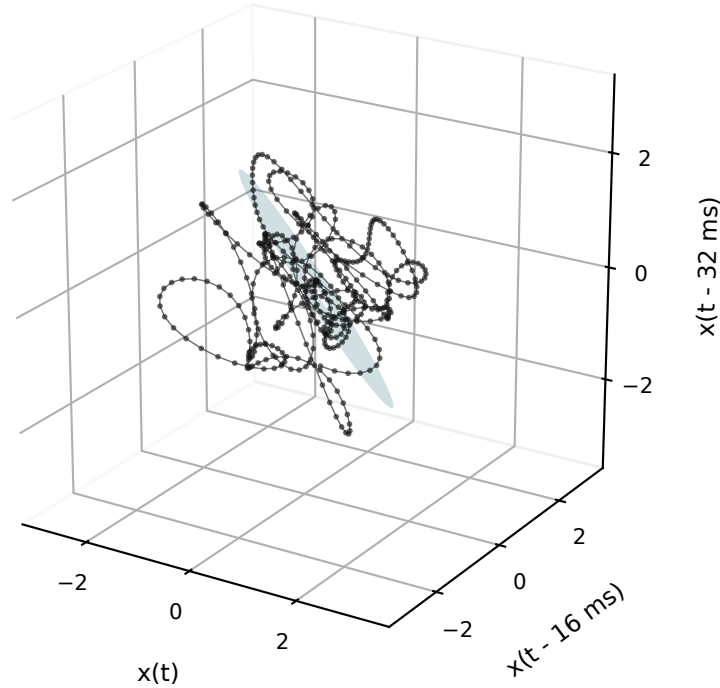
Normalized 3D set error: 0.0011  
Native coverage: 0.885 | nfev: 16  
Flags: active optimizer bound; near optimizer bound; nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 4 | original trial 339

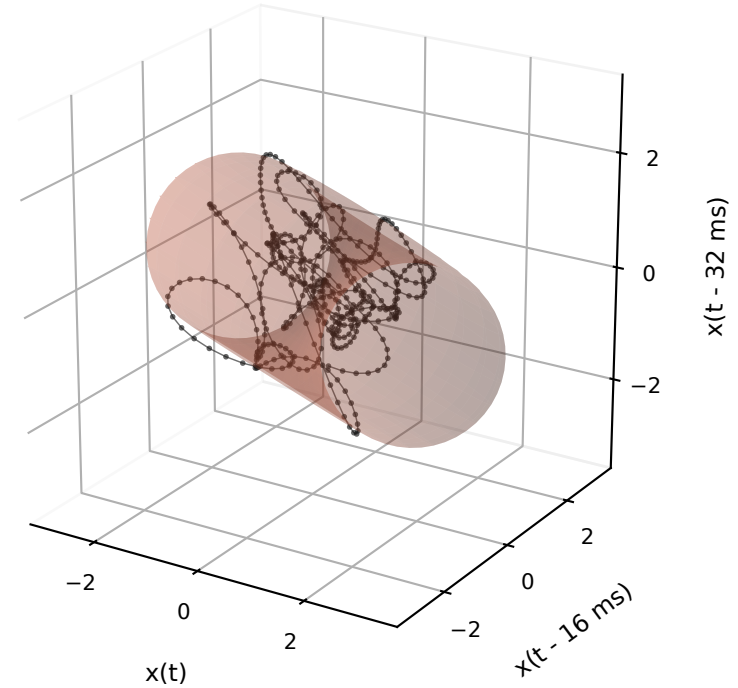
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2625  
Native coverage: 0.897 | nfev: 35  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

2-torus: donut volume



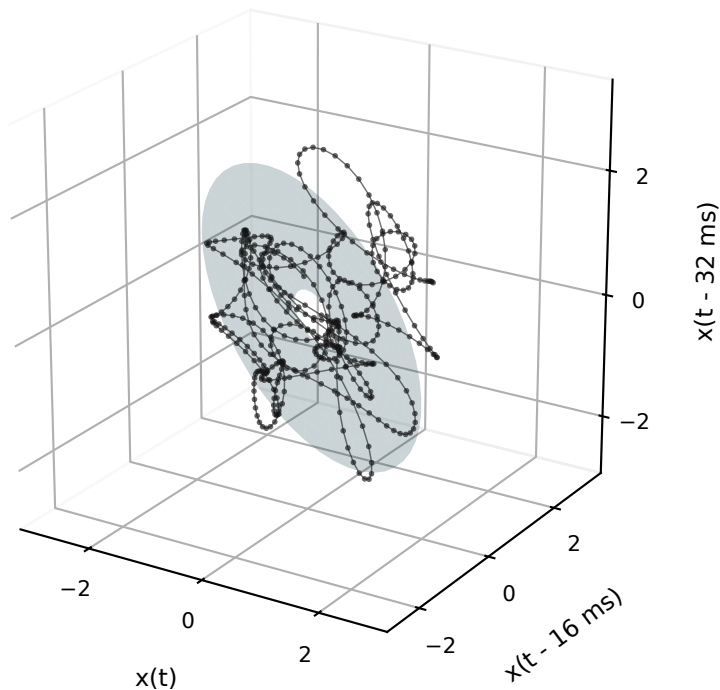
Normalized 3D set error: 0.0010  
Native coverage: 0.865 | nfev: 30  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 4 | original trial 701

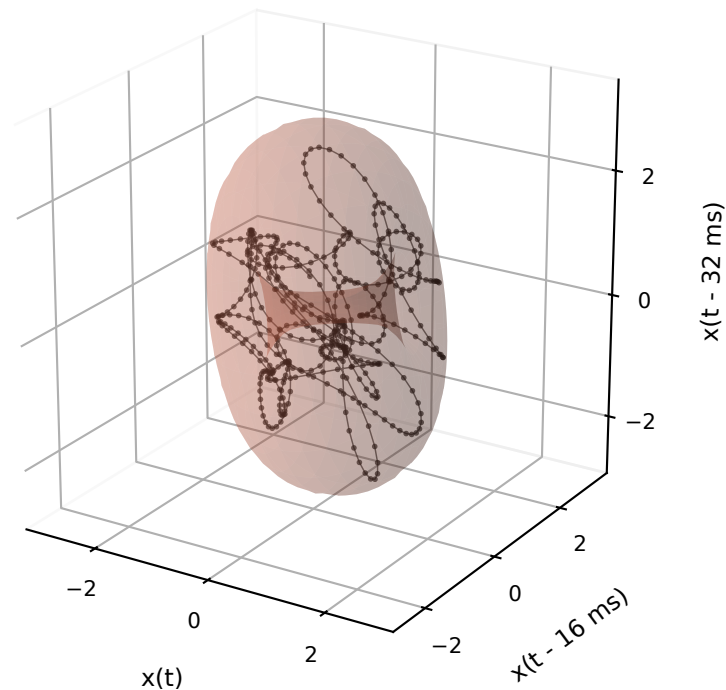
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2812  
Native coverage: 0.910 | nfev: 44  
Flags: active optimizer bound; near optimizer bound

2-torus: donut volume



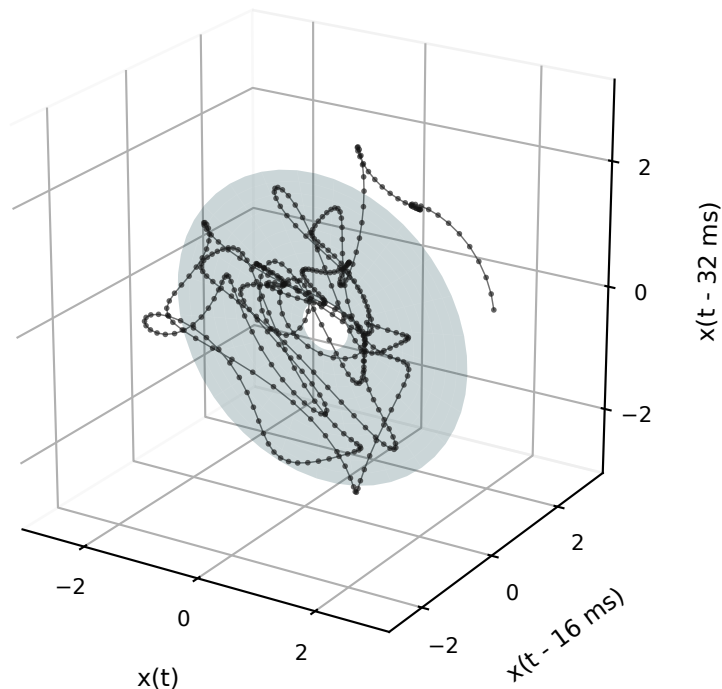
Normalized 3D set error: 0.0016  
Native coverage: 0.855 | nfev: 43  
Flags: collapsed hole; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 4 | original trial 704

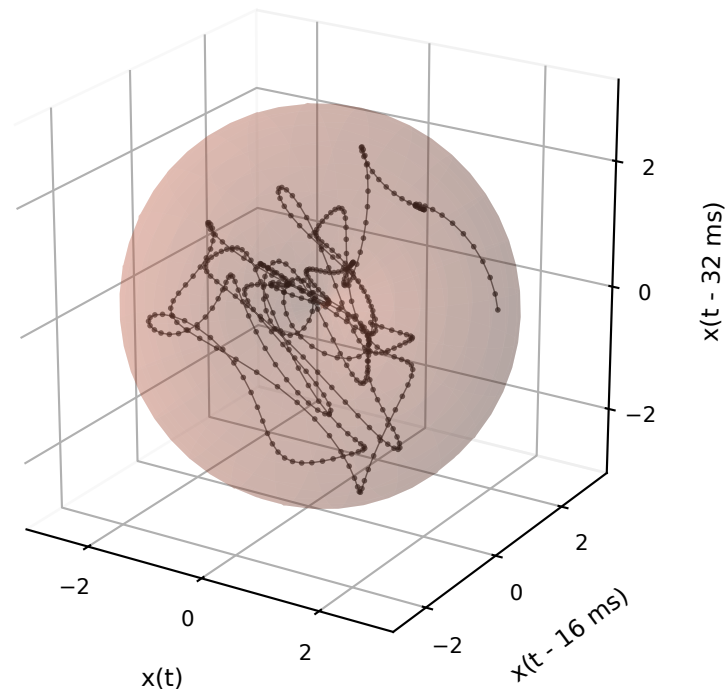
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2503  
Native coverage: 0.917 | nfev: 29  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

2-torus: donut volume



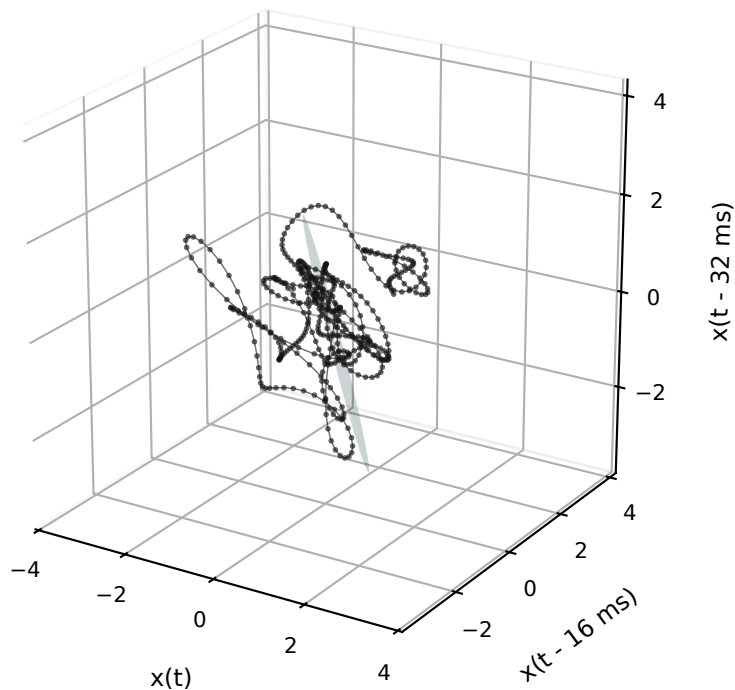
Normalized 3D set error: 0.0011  
Native coverage: 0.863 | nfev: 22  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 4 | original trial 592

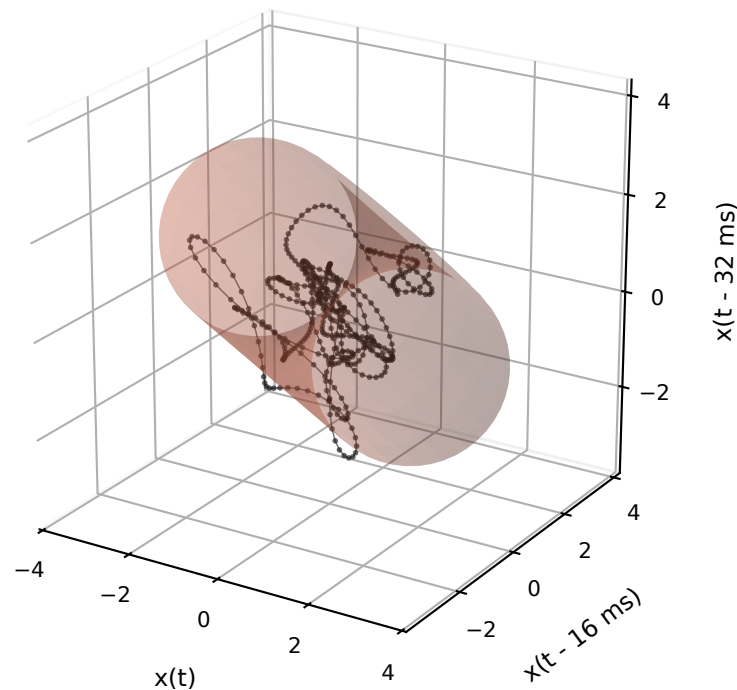
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.3109  
Native coverage: 0.893 | nfev: 64  
Flags: near optimizer bound; nearly circular orientation

2-torus: donut volume



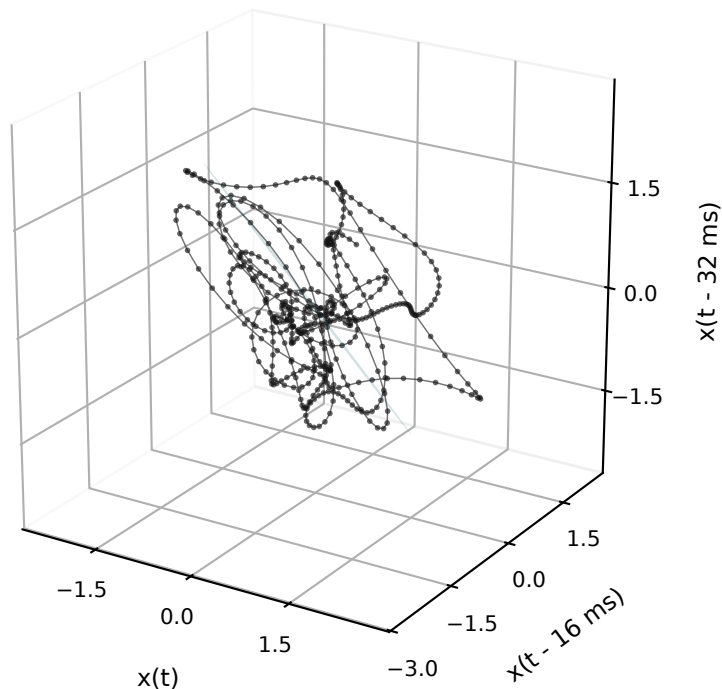
Normalized 3D set error: 0.0014  
Native coverage: 0.908 | nfev: 39  
Flags: active optimizer bound; near optimizer bound; nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 4 | original trial 440

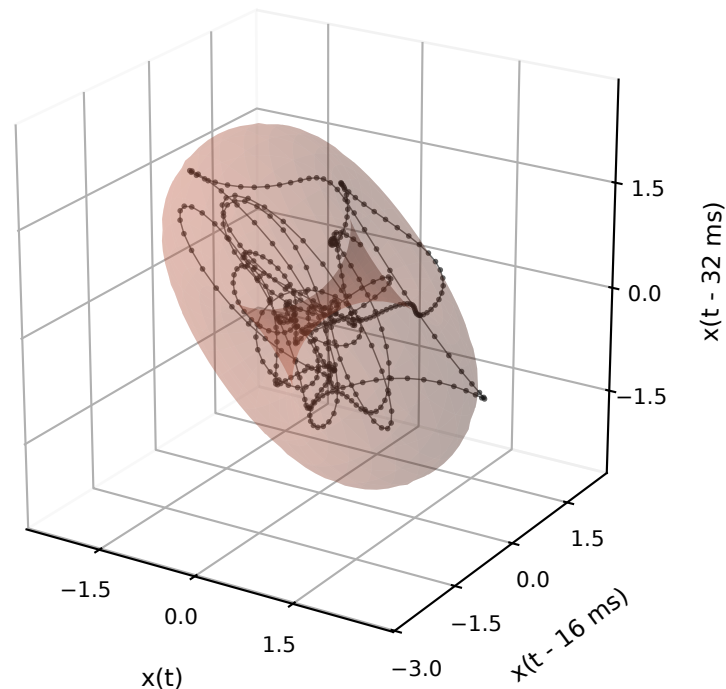
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1880  
Native coverage: 0.910 | nfev: 18  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

2-torus: donut volume



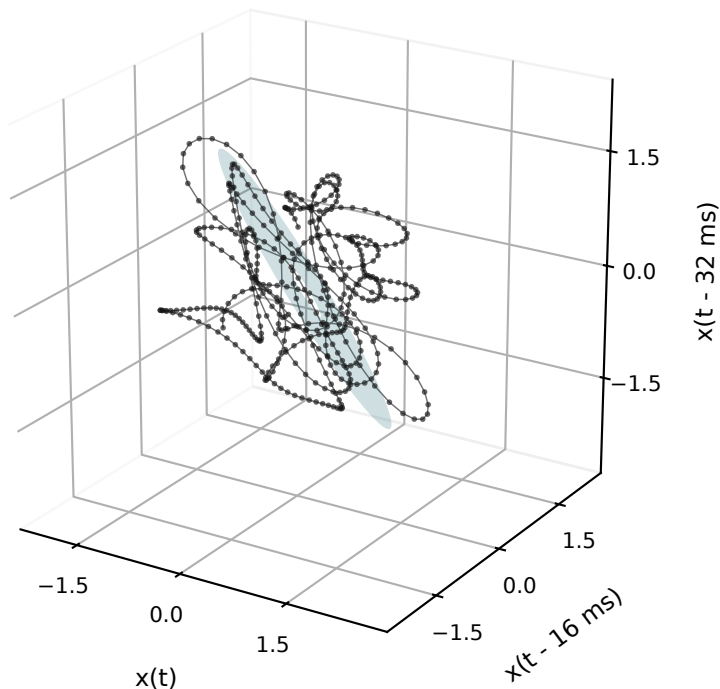
Normalized 3D set error: 0.0009  
Native coverage: 0.855 | nfev: 16  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 4 | original trial 388

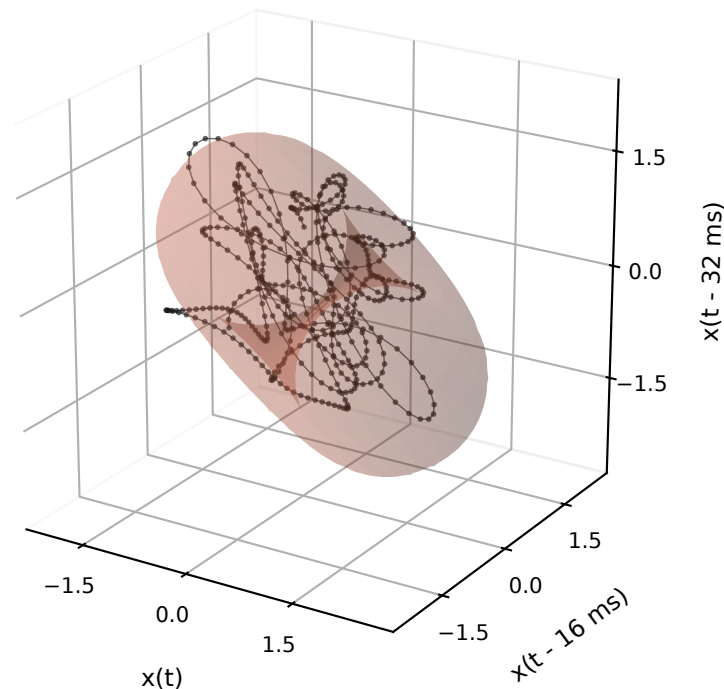
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1932  
Native coverage: 0.910 | nfev: 21  
Flags: none

2-torus: donut volume



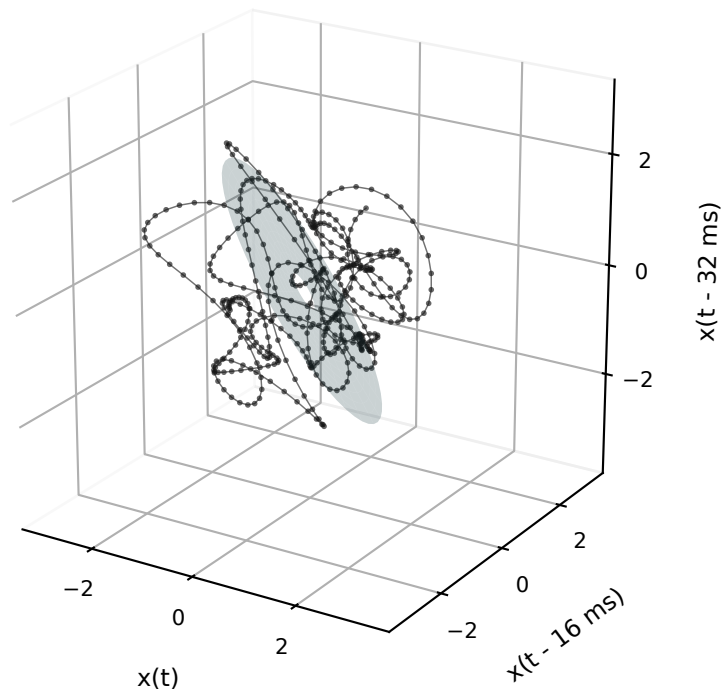
Normalized 3D set error: 0.0011  
Native coverage: 0.835 | nfev: 19  
Flags: active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 4 | original trial 121

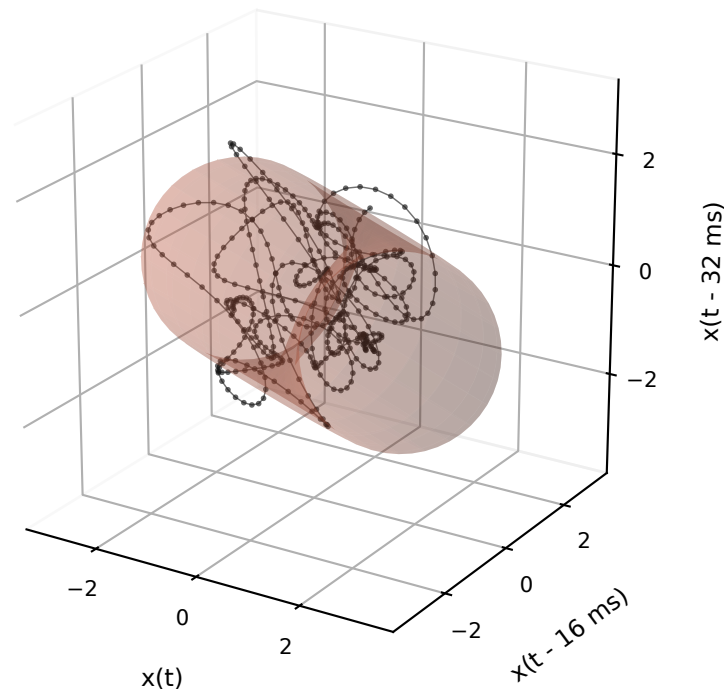
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.3023  
Native coverage: 0.912 | nfev: 58  
Flags: none

2-torus: donut volume



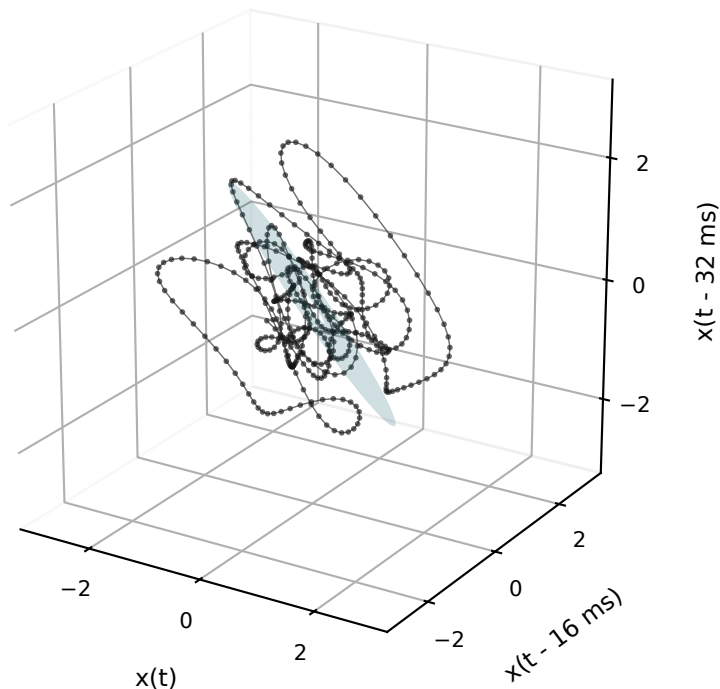
Normalized 3D set error: 0.0016  
Native coverage: 0.895 | nfev: 46  
Flags: collapsed hole; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 4 | original trial 291

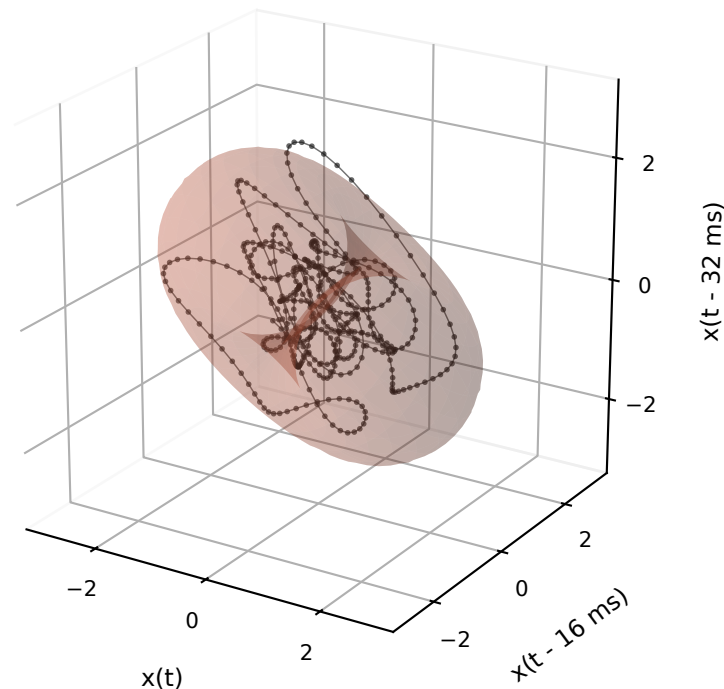
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1861  
Native coverage: 0.906 | nfev: 28  
Flags: active optimizer bound; near optimizer bound

2-torus: donut volume



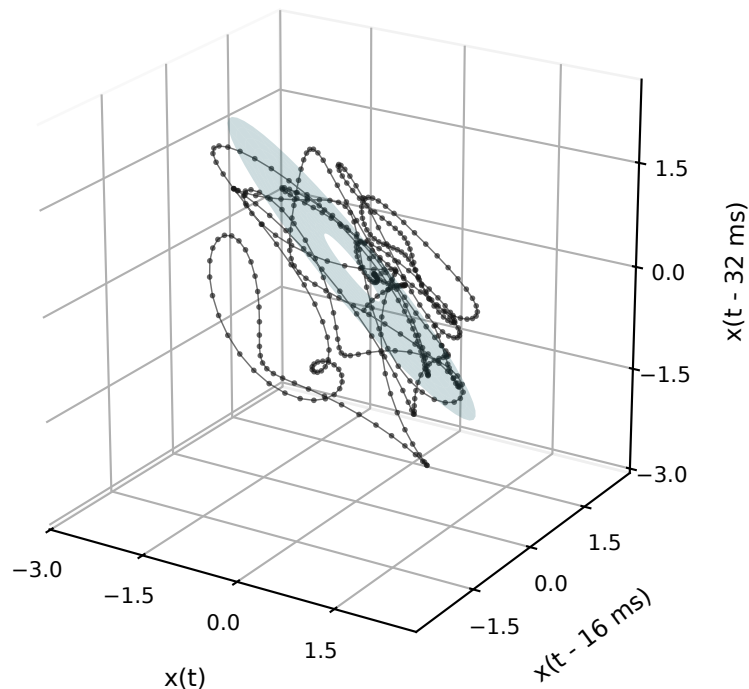
Normalized 3D set error: 0.0018  
Native coverage: 0.902 | nfev: 36  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 4 | original trial 573

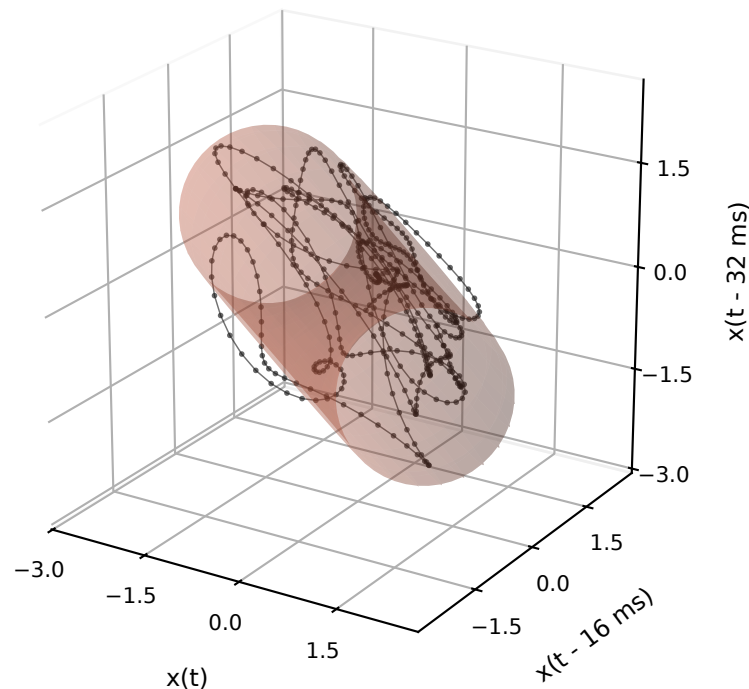
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1519  
Native coverage: 0.917 | nfev: 27  
Flags: active optimizer bound; near optimizer bound

2-torus: donut volume



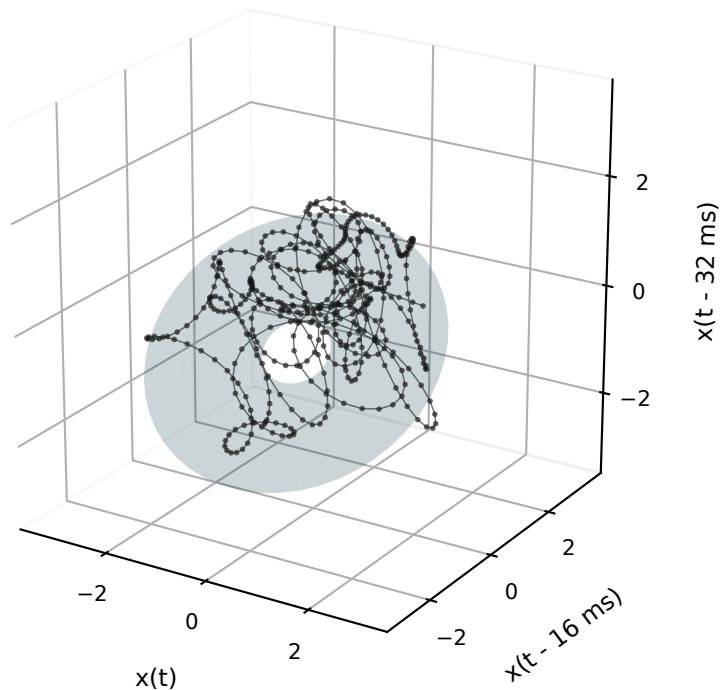
Normalized 3D set error: 0.0012  
Native coverage: 0.865 | nfev: 40  
Flags: active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 5 | original trial 705

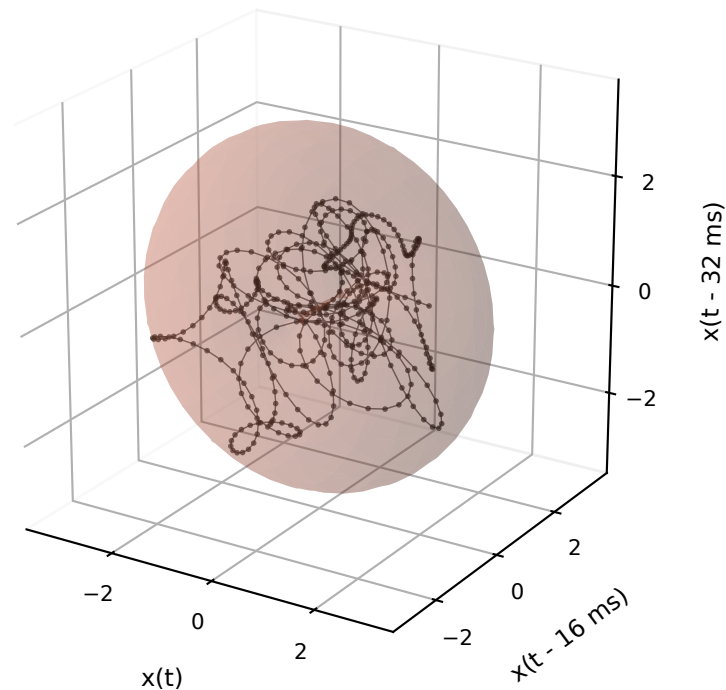
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2509  
Native coverage: 0.902 | nfev: 47  
Flags: none

2-torus: donut volume



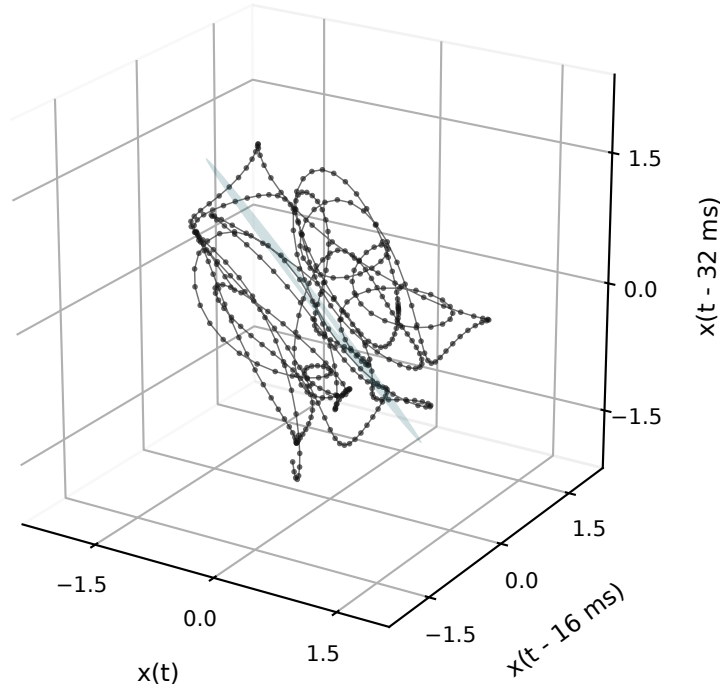
Normalized 3D set error: 0.0008  
Native coverage: 0.891 | nfev: 39  
Flags: collapsed hole; active optimizer bound; near optimizer bound; nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 5 | original trial 45

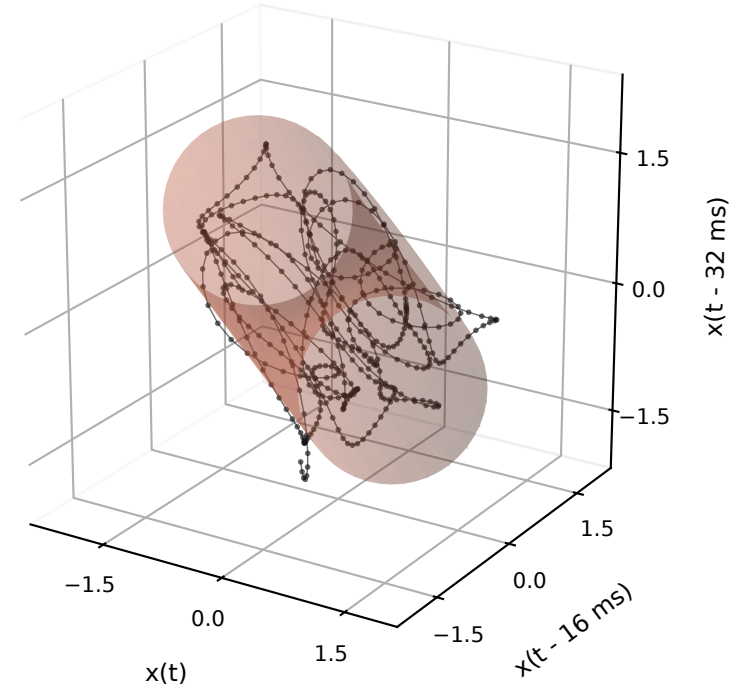
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1752  
Native coverage: 0.895 | nfev: 20  
Flags: none

2-torus: donut volume



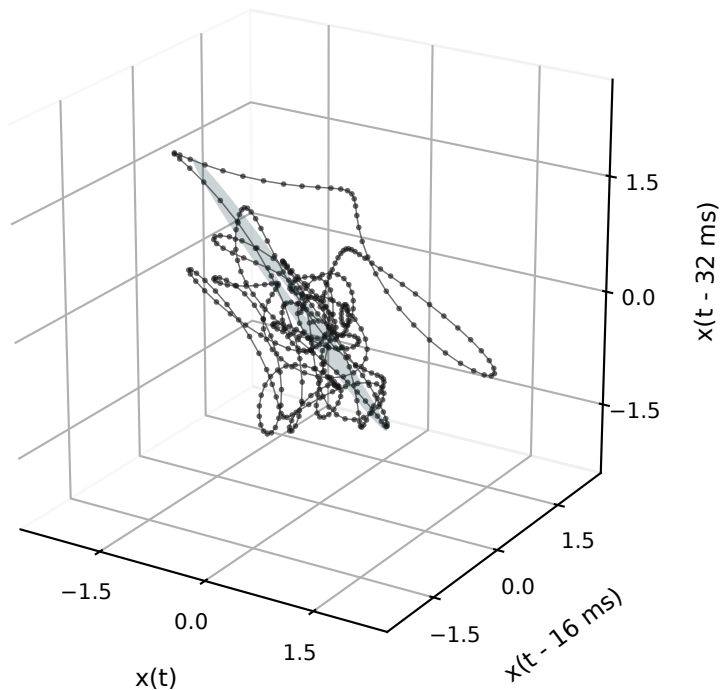
Normalized 3D set error: 0.0013  
Native coverage: 0.865 | nfev: 17  
Flags: active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 5 | original trial 410

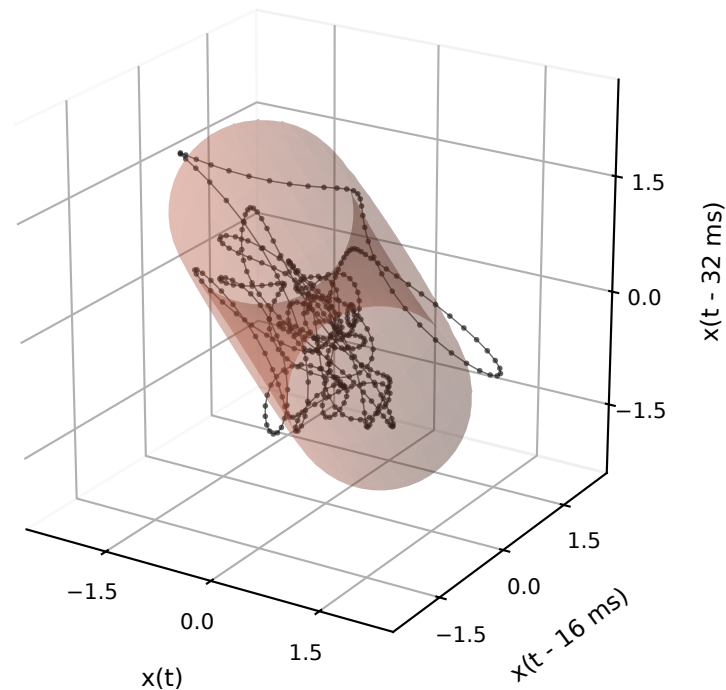
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1652  
Native coverage: 0.850 | nfev: 21  
Flags: active optimizer bound; near optimizer bound

2-torus: donut volume



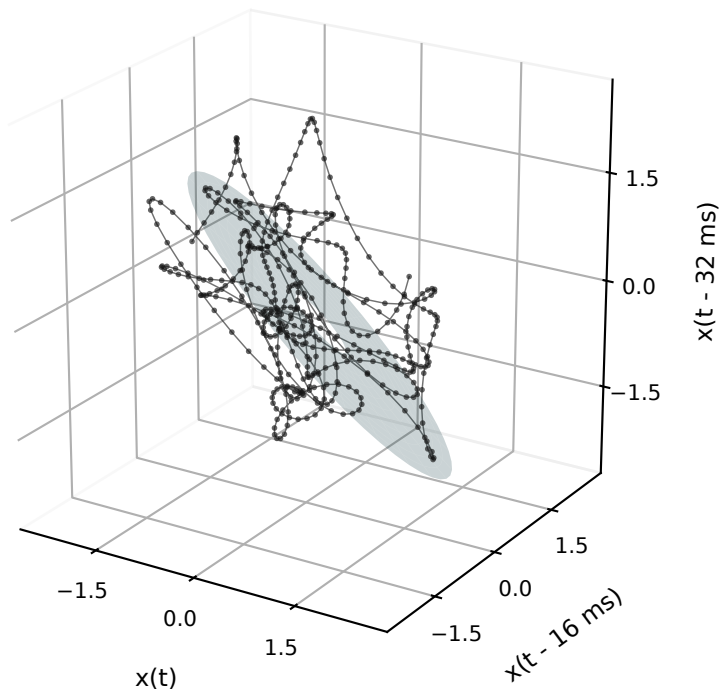
Normalized 3D set error: 0.0024  
Native coverage: 0.844 | nfev: 30  
Flags: active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 5 | original trial 142

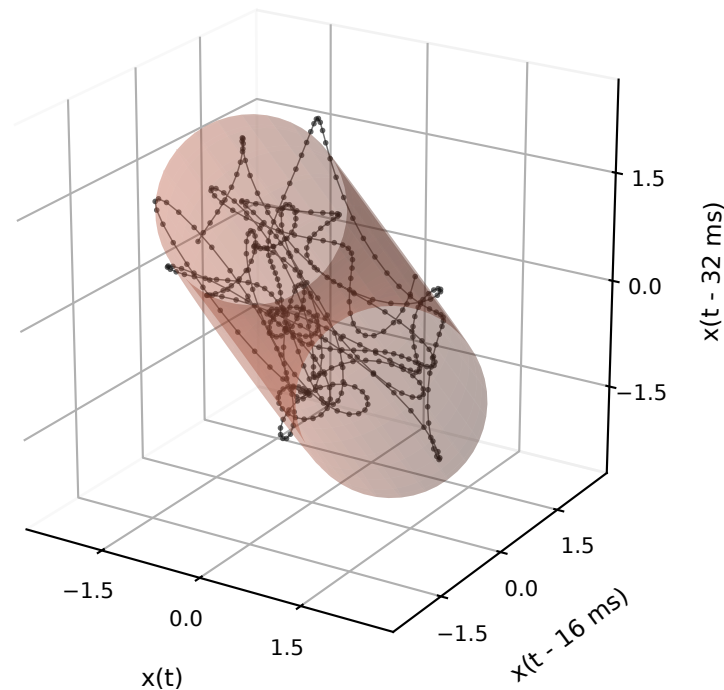
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

**1-torus: planar band**



Normalized 3D set error: 0.1606  
Native coverage: 0.878 | nfev: 44  
Flags: none

**2-torus: donut volume**



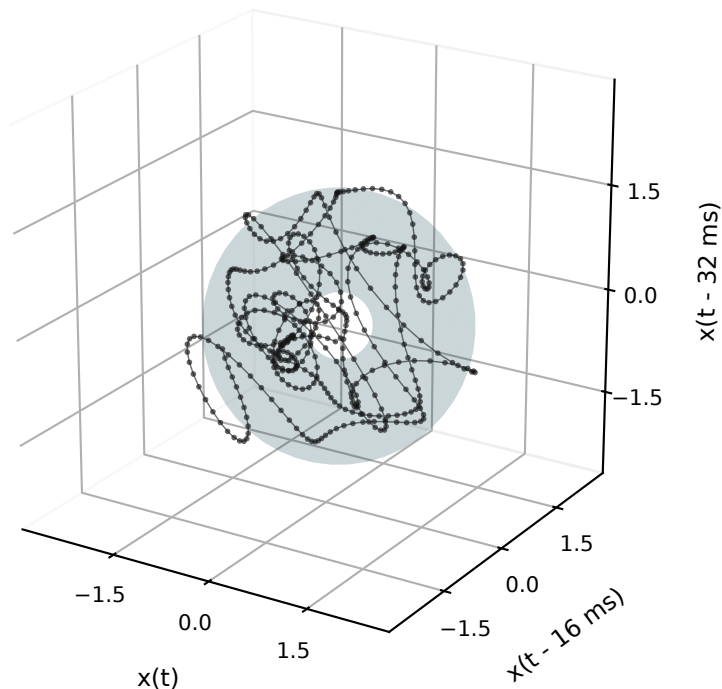
Normalized 3D set error: 0.0010  
Native coverage: 0.842 | nfev: 27  
Flags: none

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 5 | original trial 706

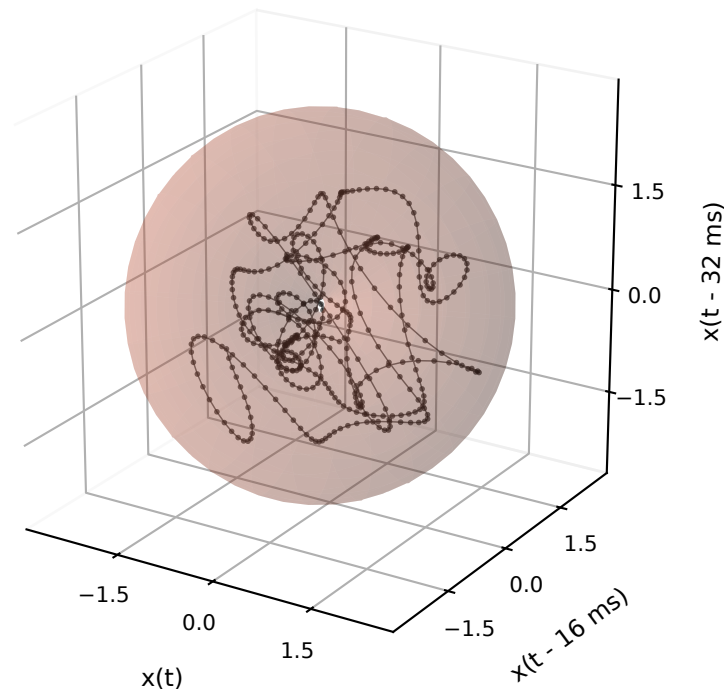
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2302  
Native coverage: 0.863 | nfev: 104  
Flags: nearly circular orientation

2-torus: donut volume



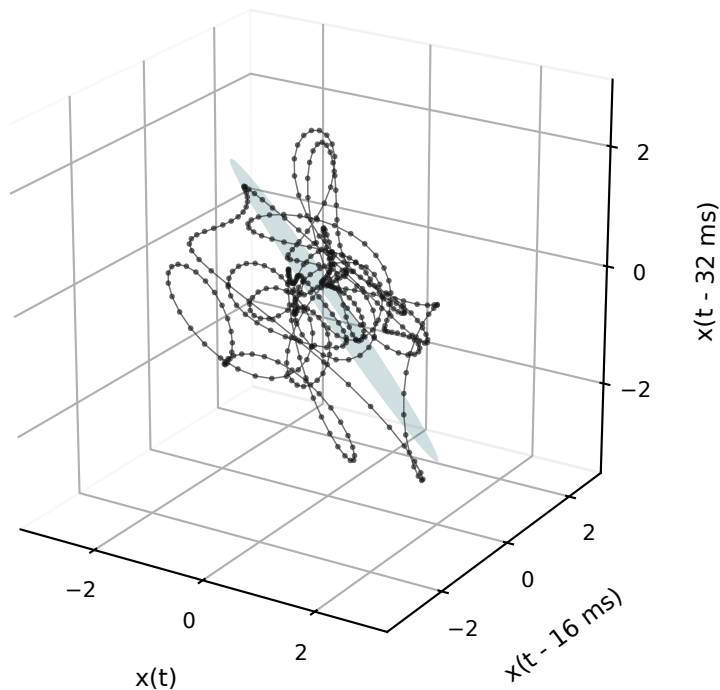
Normalized 3D set error: 0.0020  
Native coverage: 0.882 | nfev: 21  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 5 | original trial 328

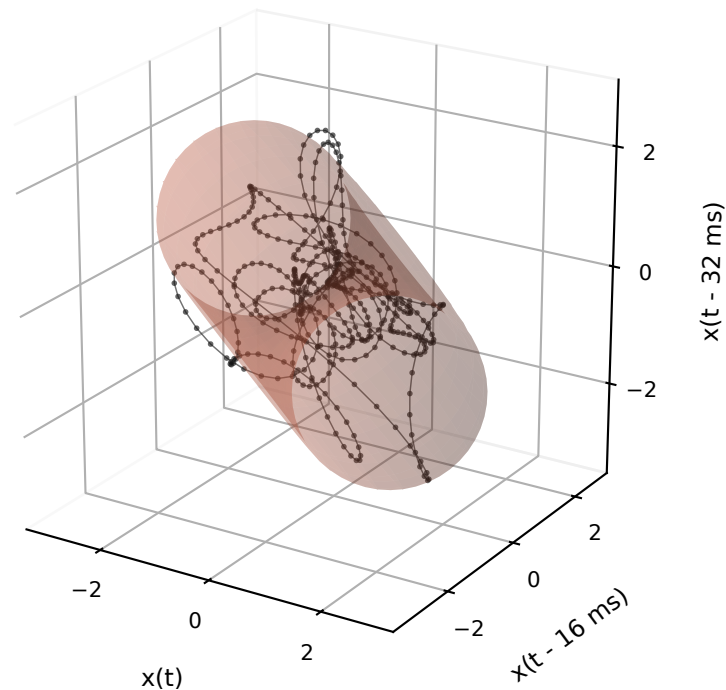
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1976  
Native coverage: 0.927 | nfev: 37  
Flags: near optimizer bound; nearly circular orientation

2-torus: donut volume



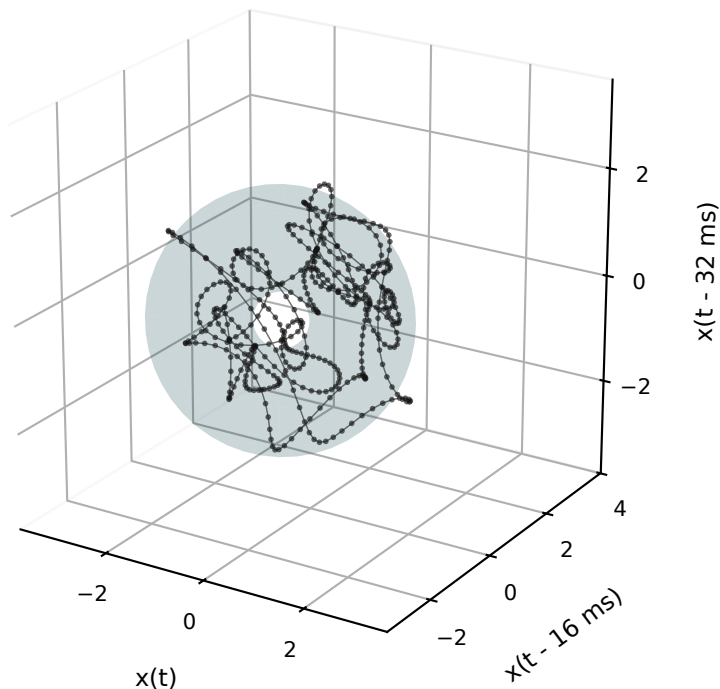
Normalized 3D set error: 0.0007  
Native coverage: 0.863 | nfev: 11  
Flags: active optimizer bound; near optimizer bound; nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 5 | original trial 143

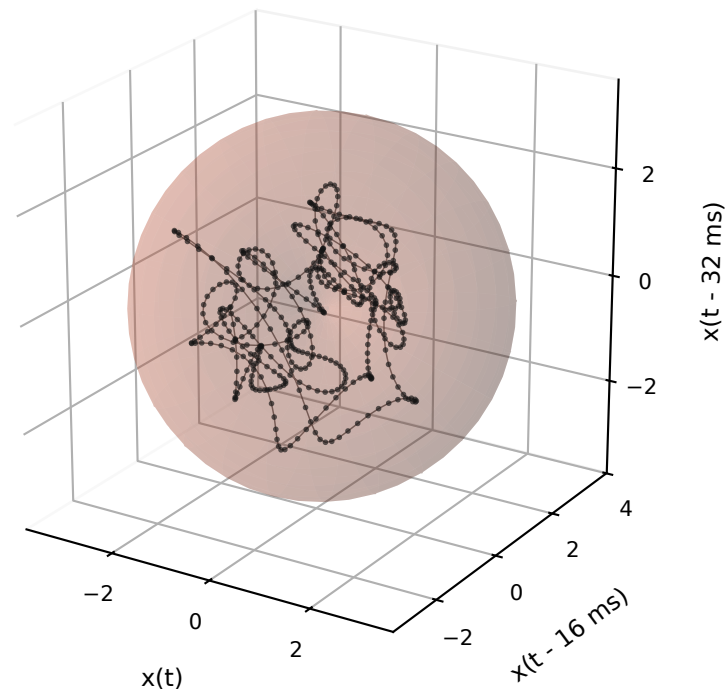
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2646  
Native coverage: 0.912 | nfev: 38  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

2-torus: donut volume



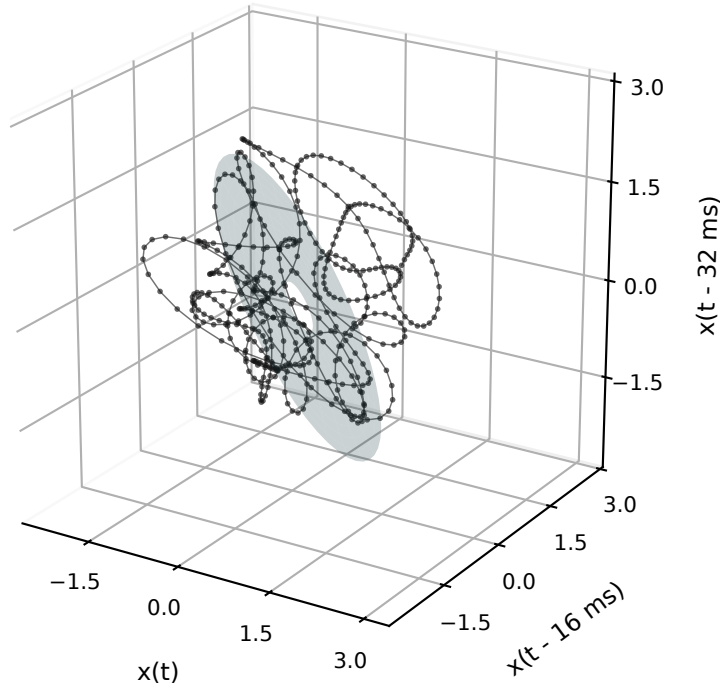
Normalized 3D set error: 0.0013  
Native coverage: 0.891 | nfev: 23  
Flags: collapsed hole; active optimizer bound; near  
optimizer bound; nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 5 | original trial 44

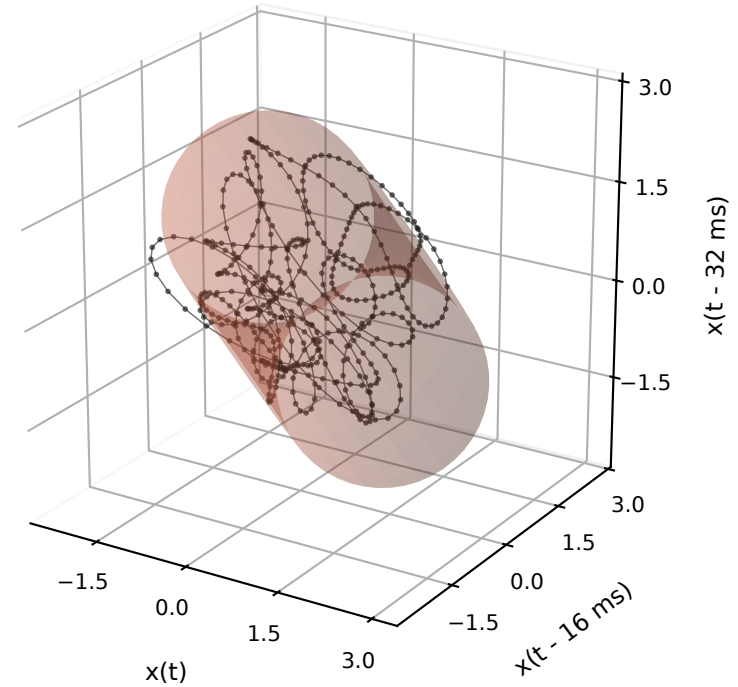
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2808  
Native coverage: 0.900 | nfev: 62  
Flags: near optimizer bound; nearly circular orientation

2-torus: donut volume



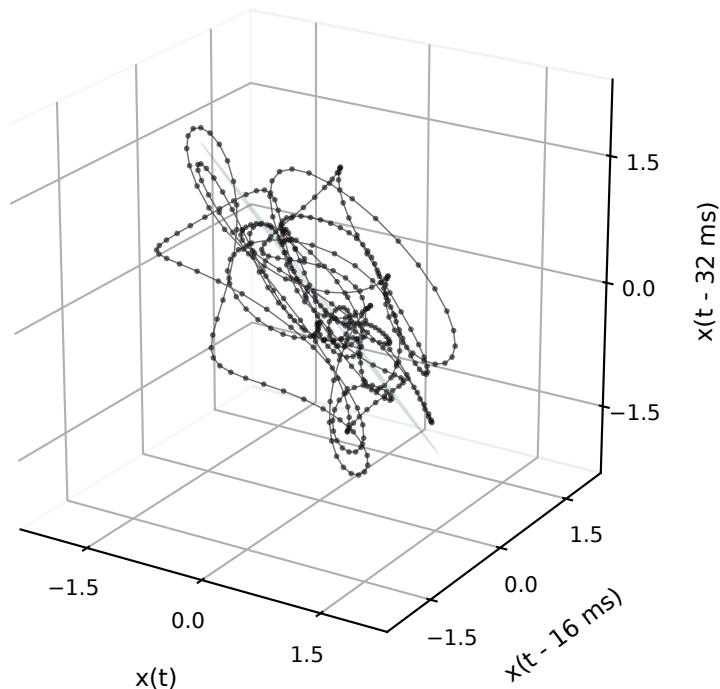
Normalized 3D set error: 0.0016  
Native coverage: 0.885 | nfev: 23  
Flags: collapsed hole; active optimizer bound; near optimizer bound; nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 5 | original trial 164

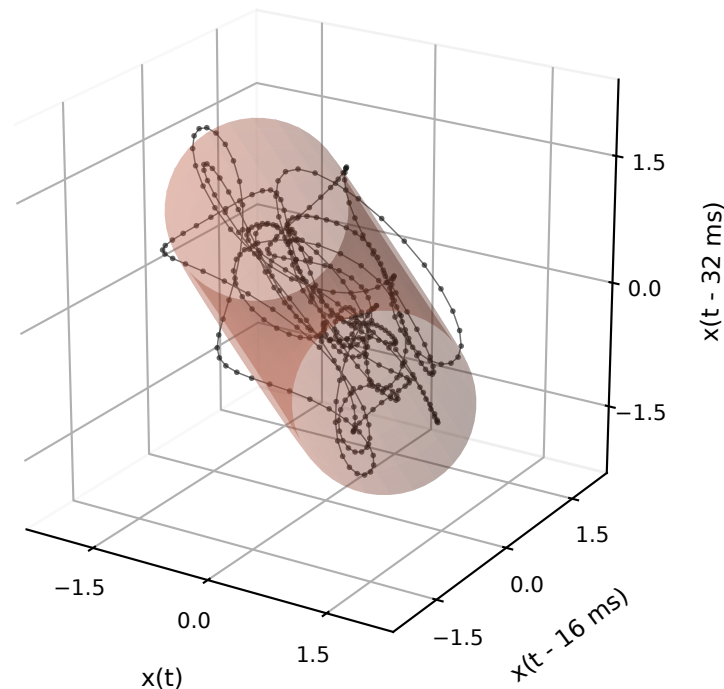
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1037  
Native coverage: 0.912 | nfev: 25  
Flags: nearly circular orientation

2-torus: donut volume



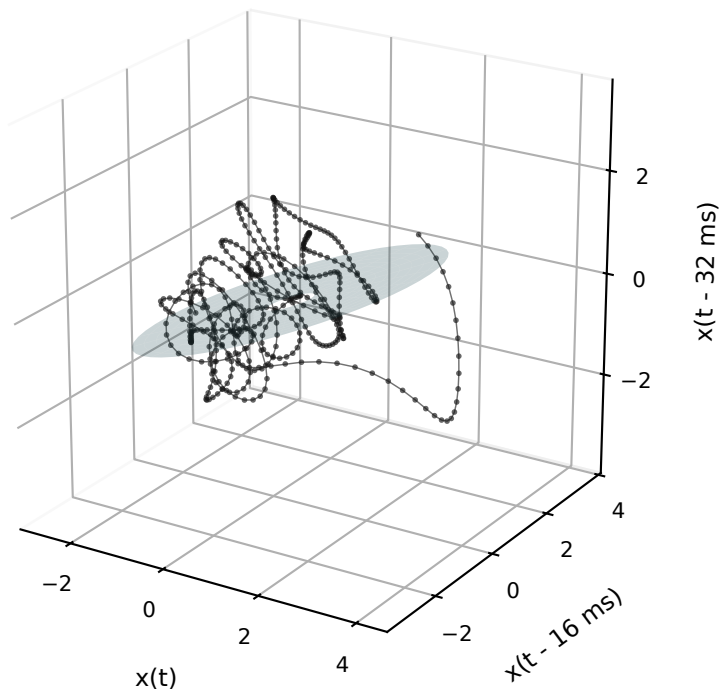
Normalized 3D set error: 0.0006  
Native coverage: 0.891 | nfev: 15  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 5 | original trial 560

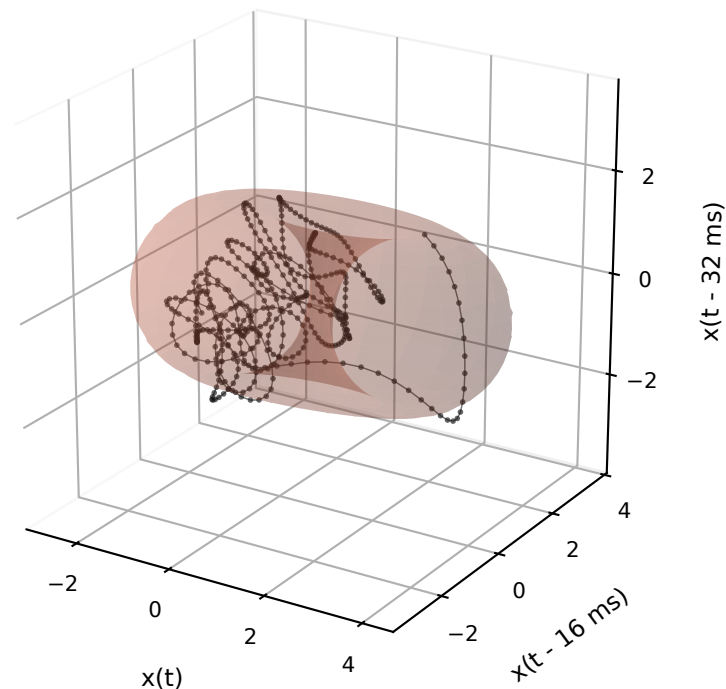
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2646  
Native coverage: 0.912 | nfev: 79  
Flags: none

2-torus: donut volume



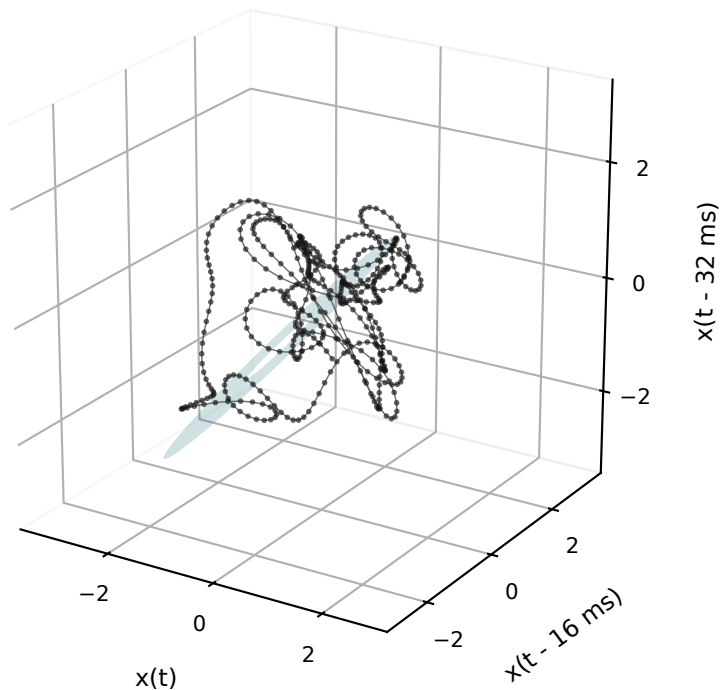
Normalized 3D set error: 0.0015  
Native coverage: 0.865 | nfev: 33  
Flags: collapsed hole; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 5 | original trial 194

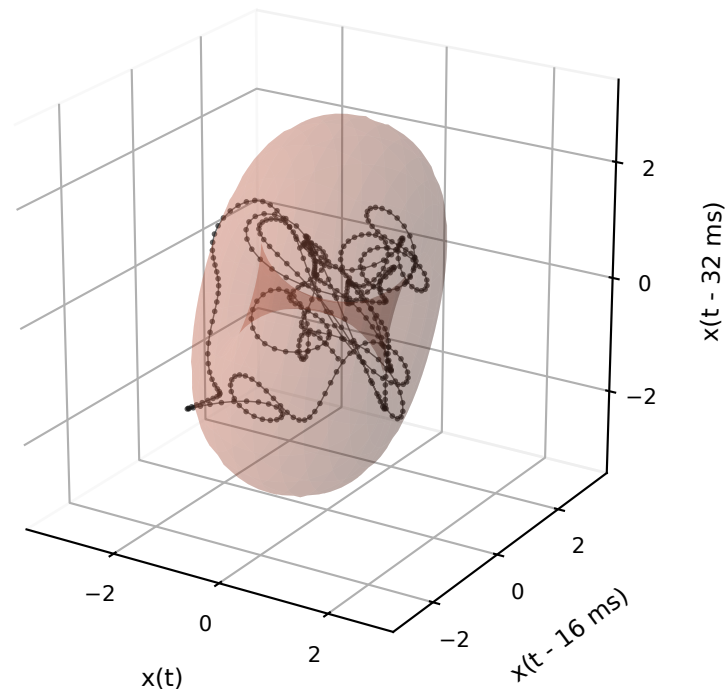
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2991  
Native coverage: 0.855 | nfev: 54  
Flags: none

2-torus: donut volume



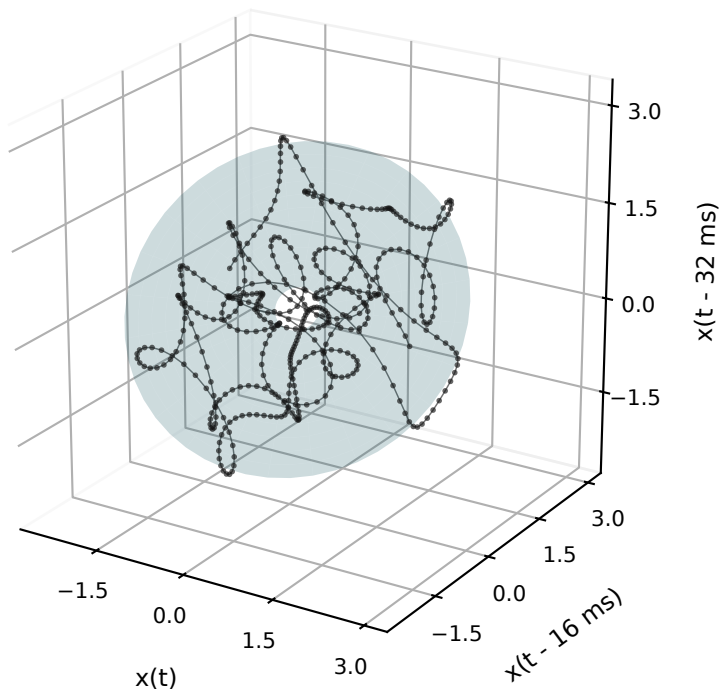
Normalized 3D set error: 0.0015  
Native coverage: 0.882 | nfev: 21  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 5 | original trial 243

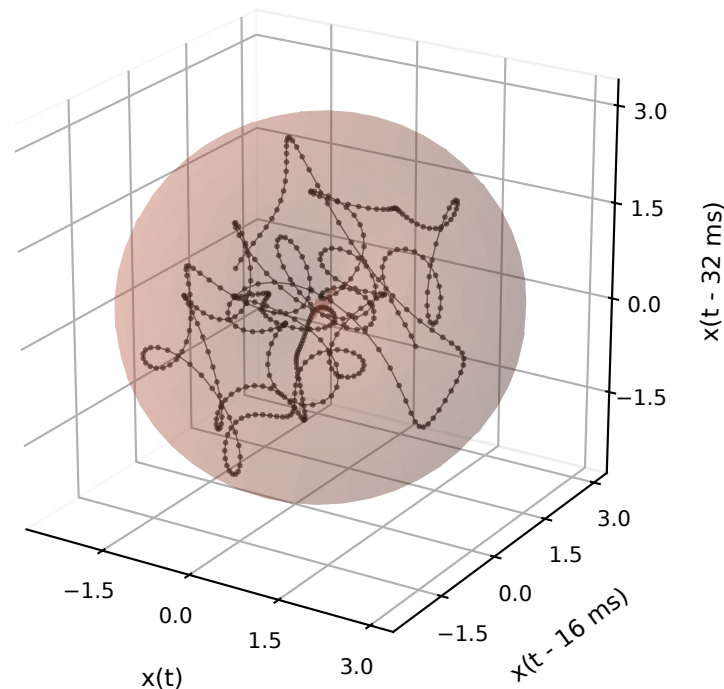
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2095  
Native coverage: 0.882 | nfev: 41  
Flags: none

2-torus: donut volume



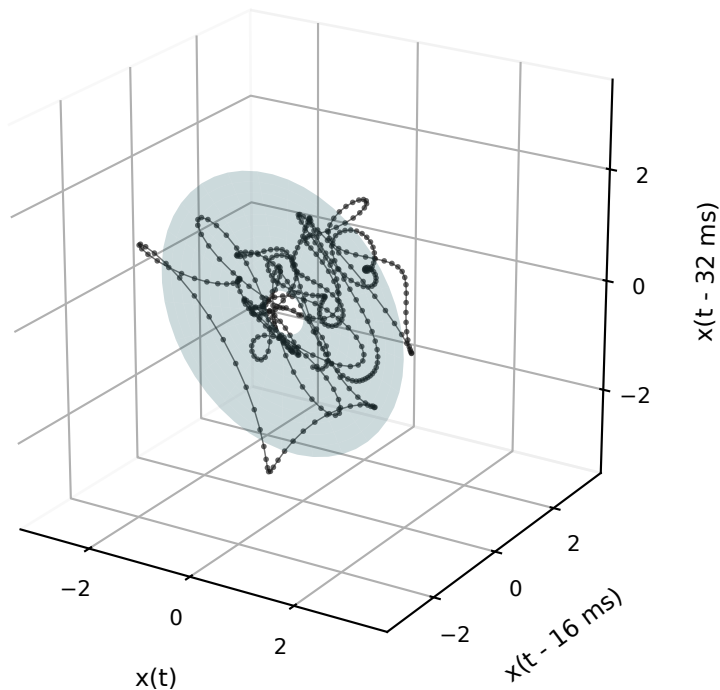
Normalized 3D set error: 0.0011  
Native coverage: 0.878 | nfev: 23  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 5 | original trial 403

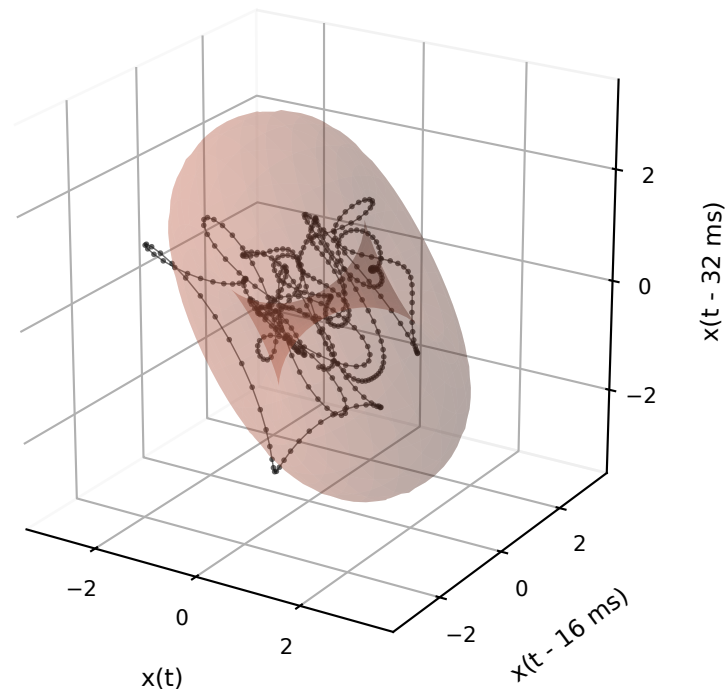
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2808  
Native coverage: 0.906 | nfev: 41  
Flags: none

2-torus: donut volume



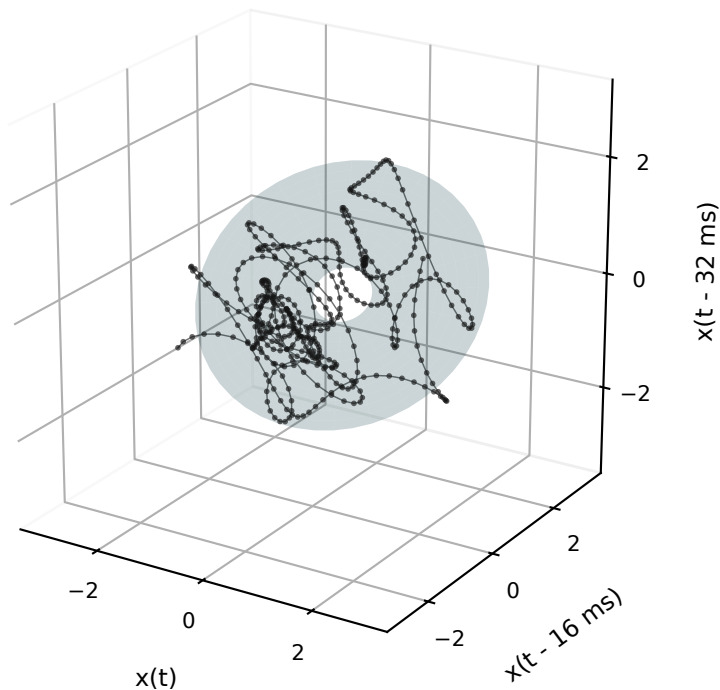
Normalized 3D set error: 0.0013  
Native coverage: 0.874 | nfev: 31  
Flags: active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 5 | original trial 568

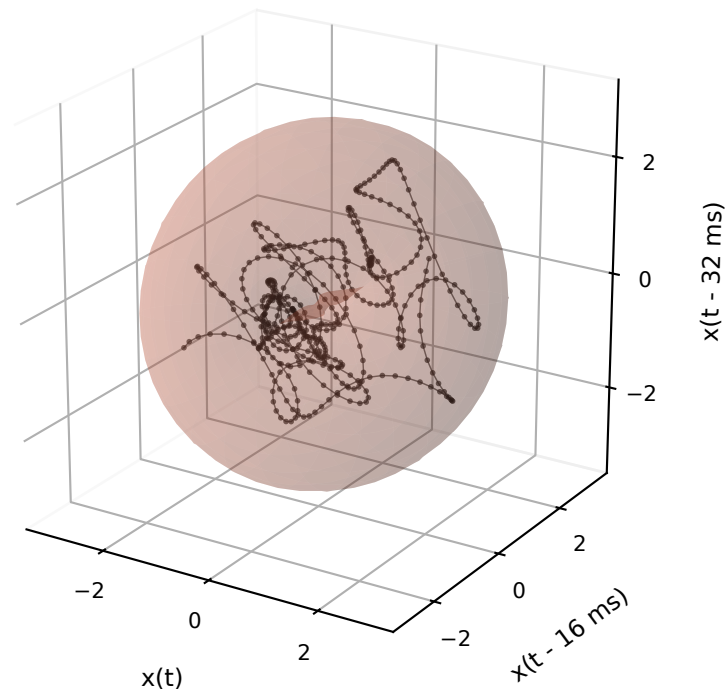
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2714  
Native coverage: 0.908 | nfev: 215  
Flags: active optimizer bound; near optimizer bound

2-torus: donut volume



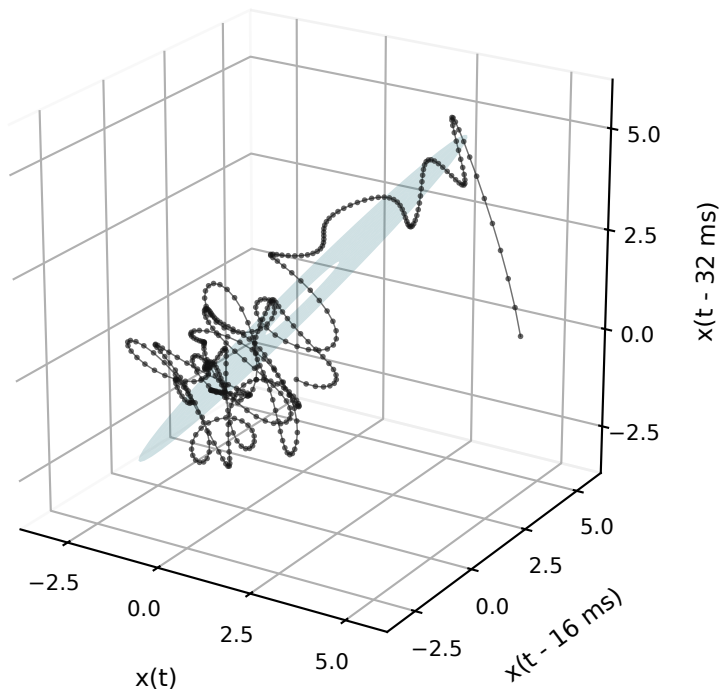
Normalized 3D set error: 0.0012  
Native coverage: 0.870 | nfev: 34  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 5 | original trial 581

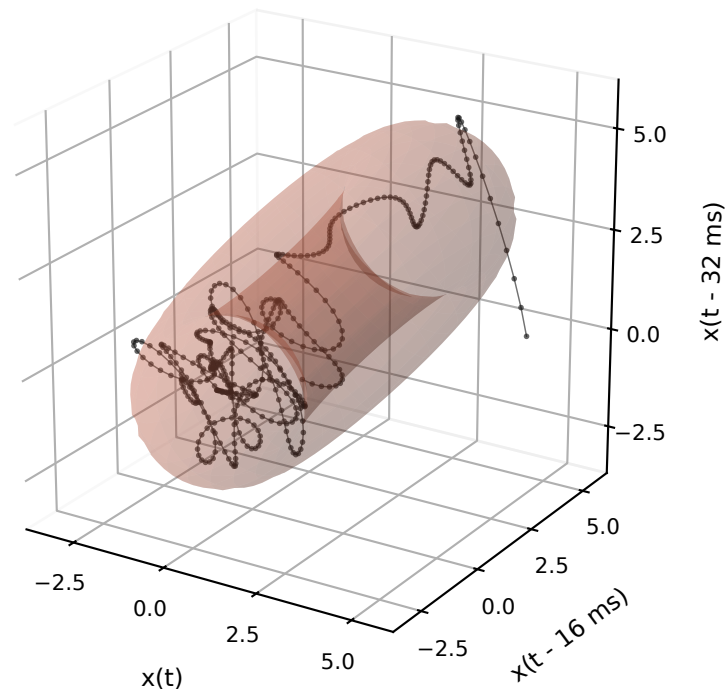
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1225  
Native coverage: 0.927 | nfev: 55  
Flags: none

2-torus: donut volume



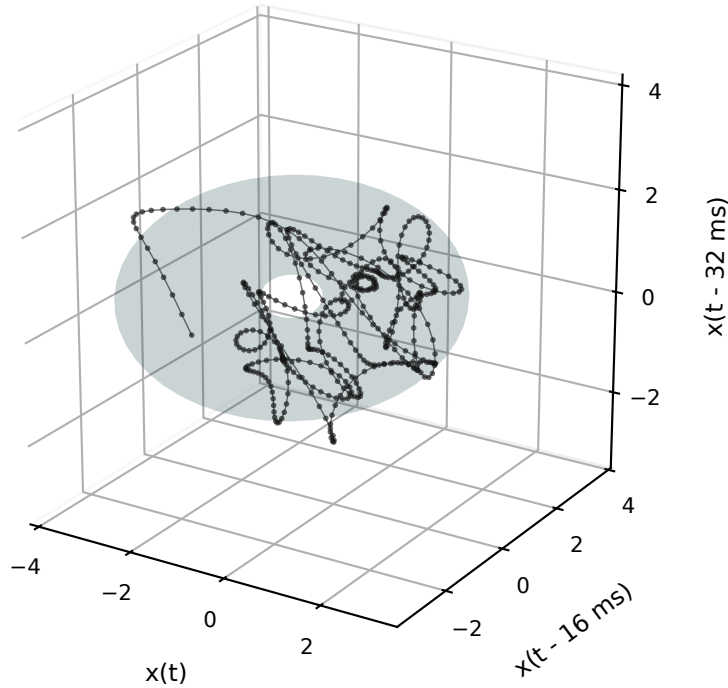
Normalized 3D set error: 0.0017  
Native coverage: 0.906 | nfev: 46  
Flags: collapsed hole

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 5 | original trial 63

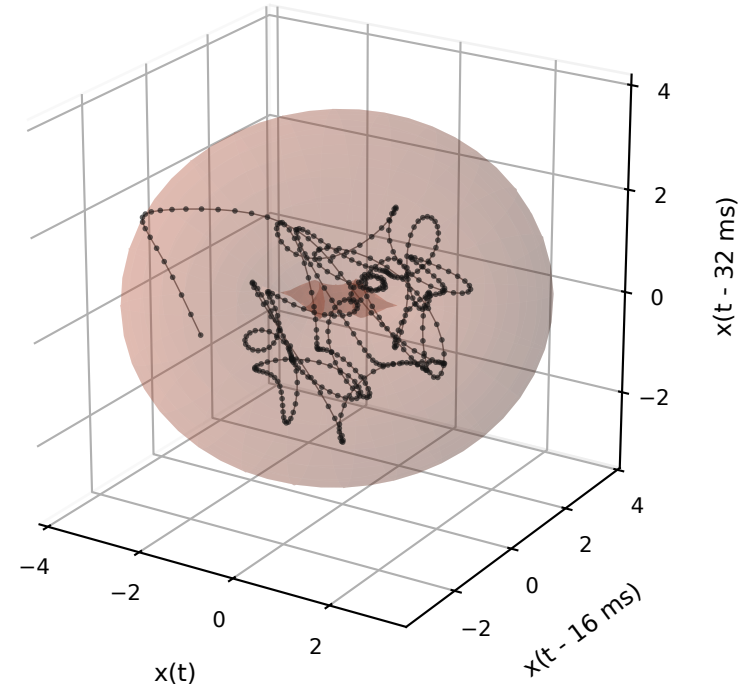
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2567  
Native coverage: 0.906 | nfev: 26  
Flags: active optimizer bound; near optimizer bound

2-torus: donut volume



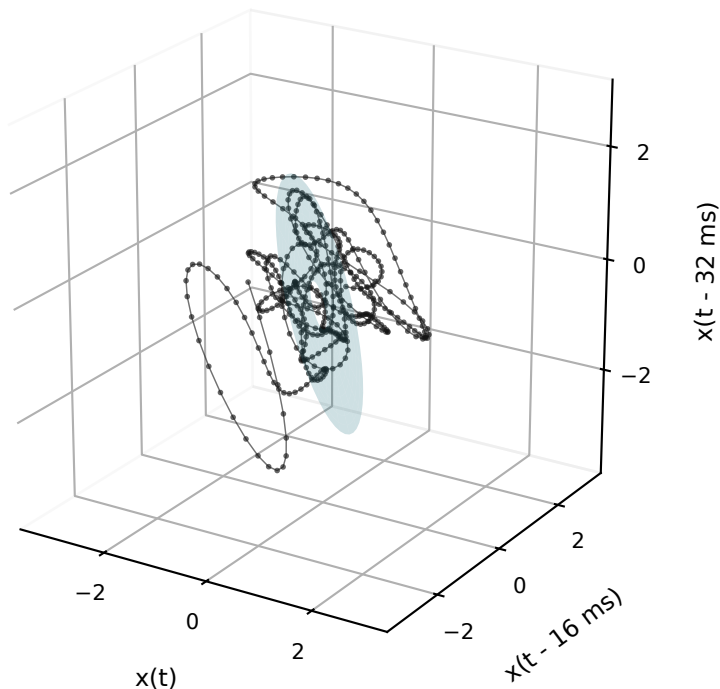
Normalized 3D set error: 0.0015  
Native coverage: 0.865 | nfev: 40  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 5 | original trial 294

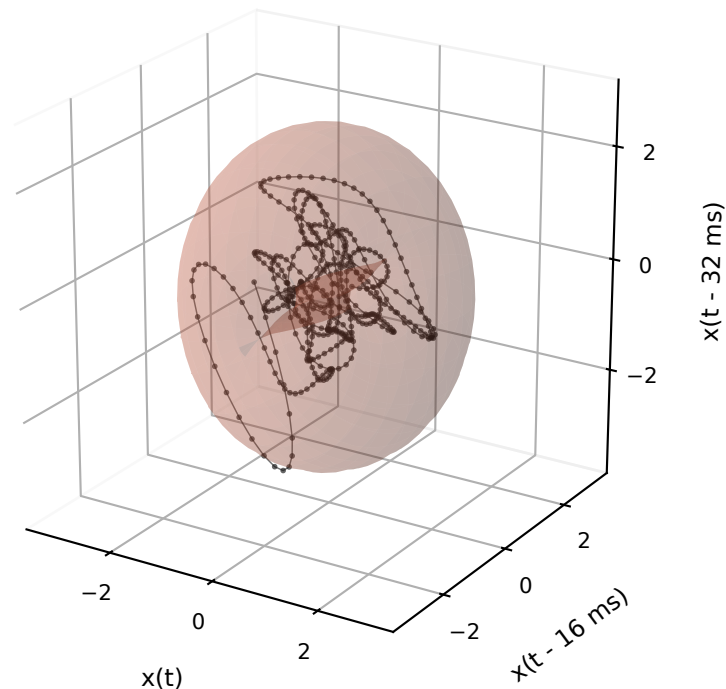
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

**1-torus: planar band**



Normalized 3D set error: 0.2300  
Native coverage: 0.906 | nfev: 41  
Flags: near optimizer bound

**2-torus: donut volume**



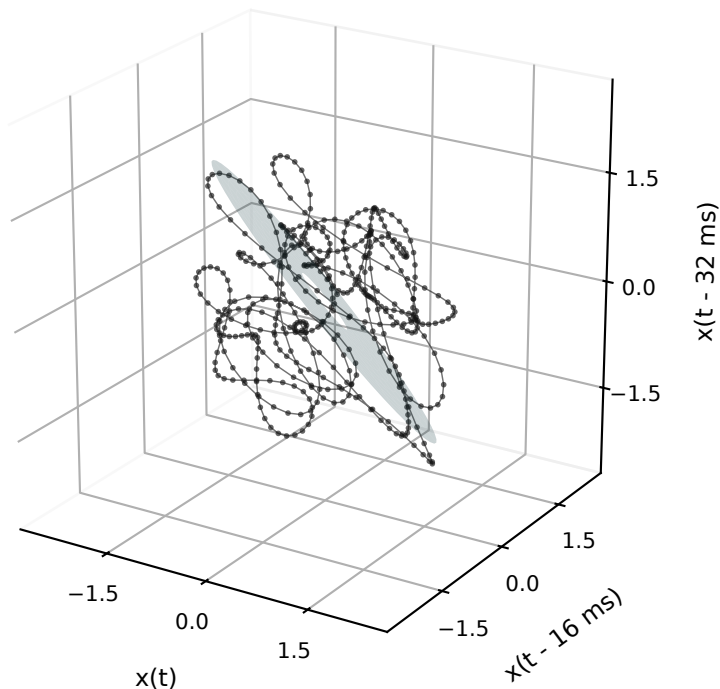
Normalized 3D set error: 0.0019  
Native coverage: 0.878 | nfev: 43  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 5 | original trial 367

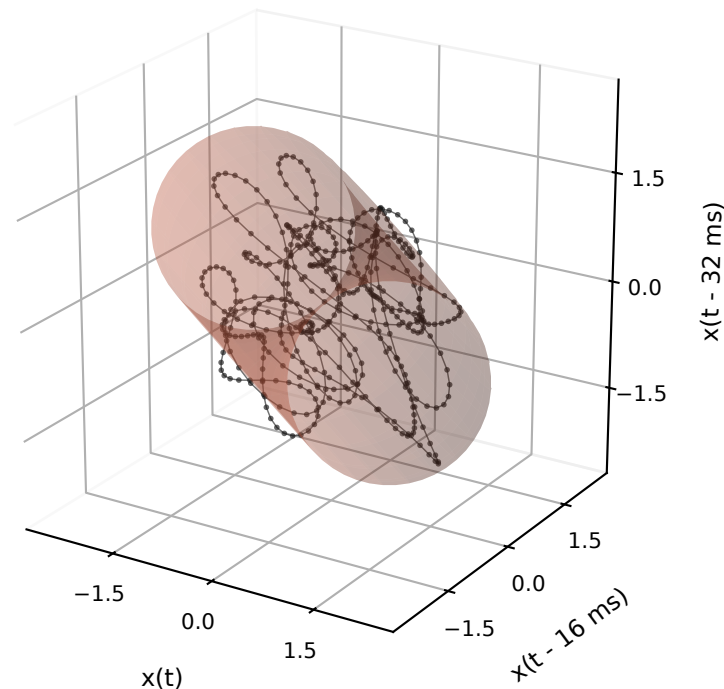
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2630  
Native coverage: 0.929 | nfev: 29  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

2-torus: donut volume



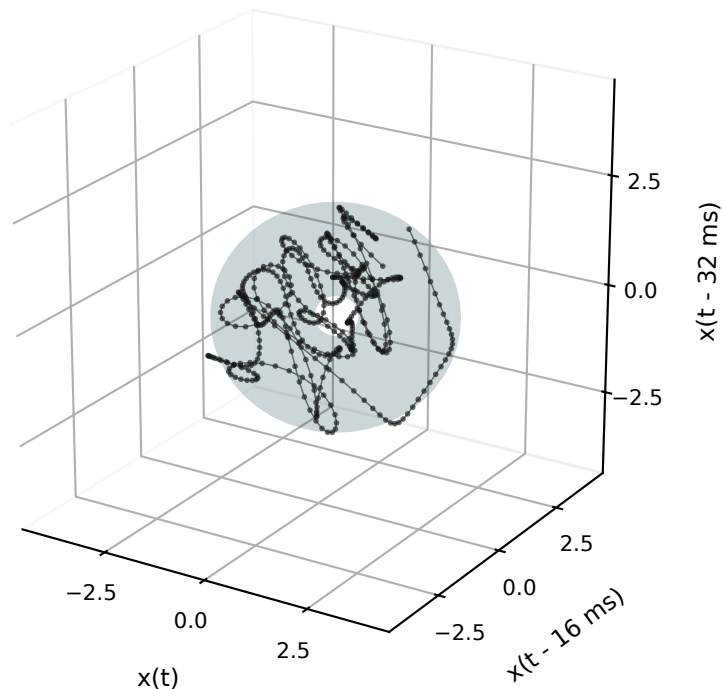
Normalized 3D set error: 0.0015  
Native coverage: 0.868 | nfev: 29  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 5 | original trial 436

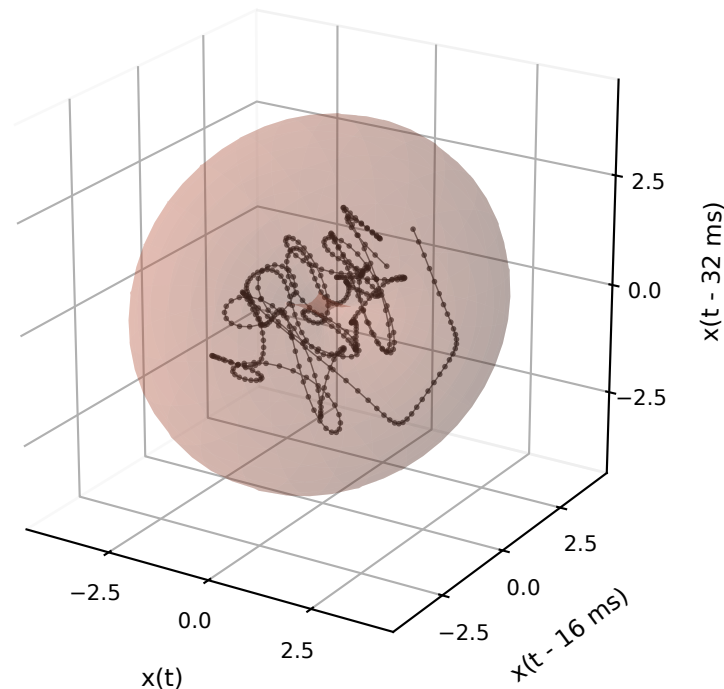
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2712  
Native coverage: 0.921 | nfev: 50  
Flags: none

2-torus: donut volume



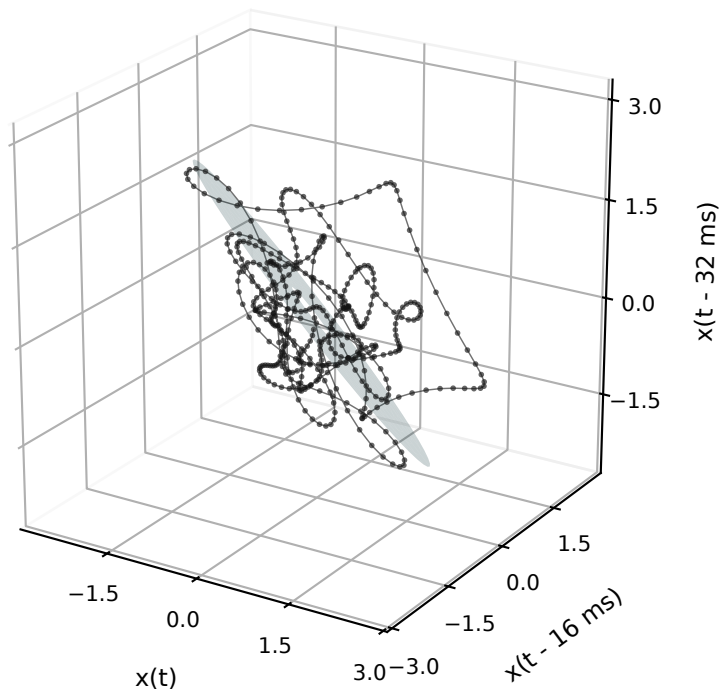
Normalized 3D set error: 0.0009  
Native coverage: 0.927 | nfev: 33  
Flags: collapsed hole; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 5 | original trial 241

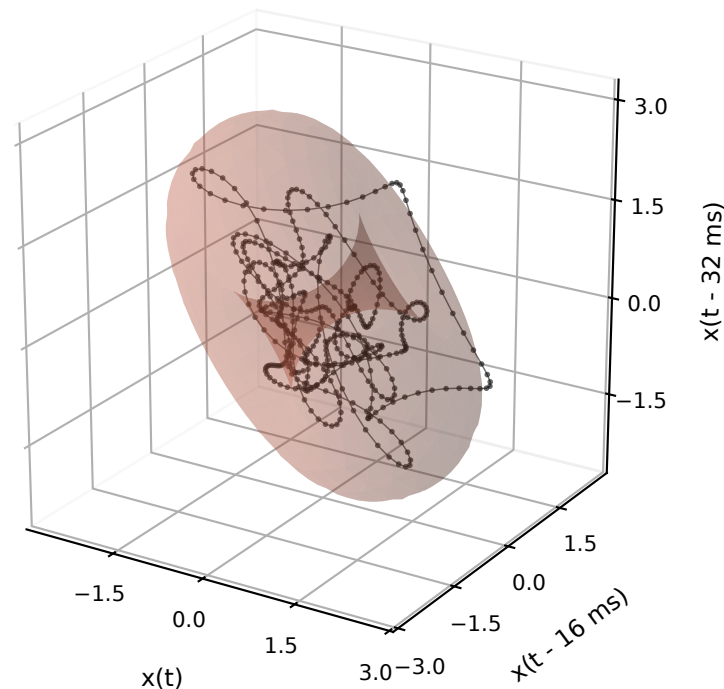
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

**1-torus: planar band**



Normalized 3D set error: 0.1871  
Native coverage: 0.897 | nfev: 21  
Flags: none

**2-torus: donut volume**



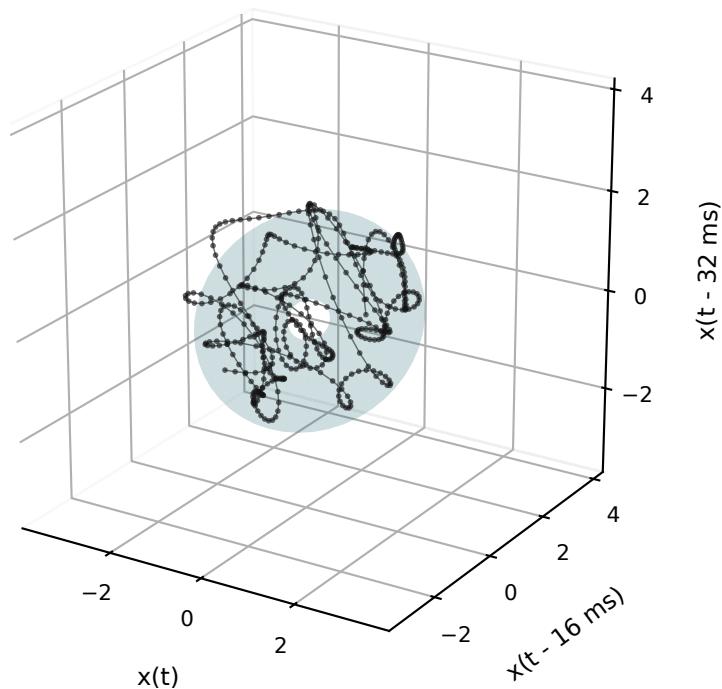
Normalized 3D set error: 0.0015  
Native coverage: 0.876 | nfev: 23  
Flags: active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 5 | original trial 457

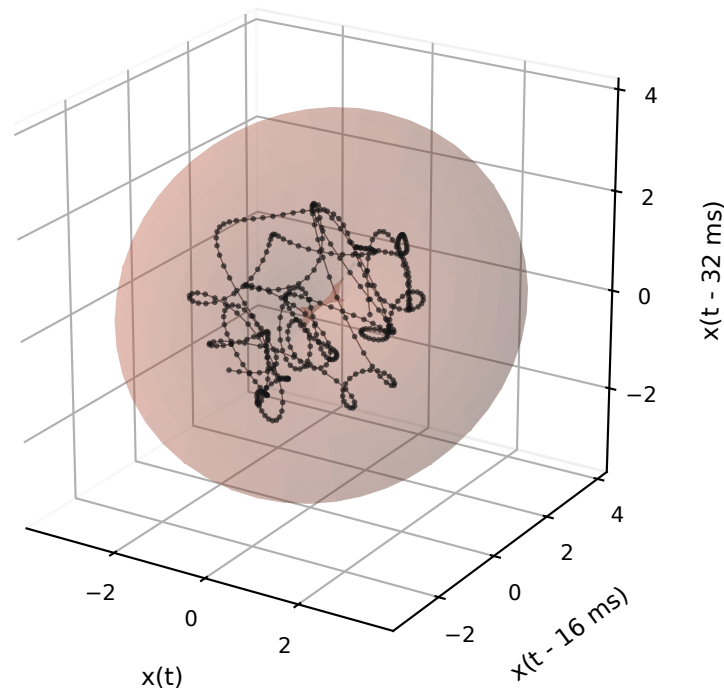
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2589  
Native coverage: 0.906 | nfev: 106  
Flags: nearly circular orientation

2-torus: donut volume



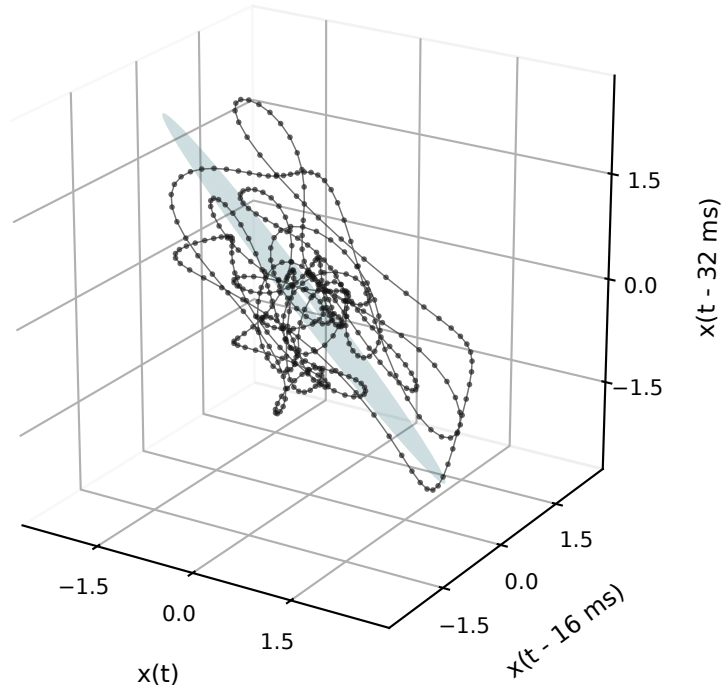
Normalized 3D set error: 0.0011  
Native coverage: 0.908 | nfev: 41  
Flags: collapsed hole

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 5 | original trial 36

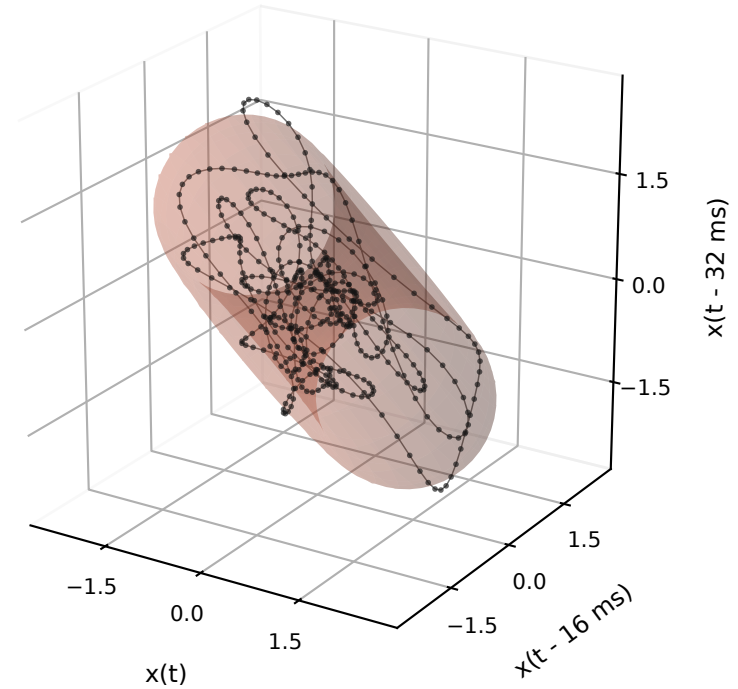
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

**1-torus: planar band**



Normalized 3D set error: 0.1202  
Native coverage: 0.900 | nfev: 38  
Flags: none

**2-torus: donut volume**



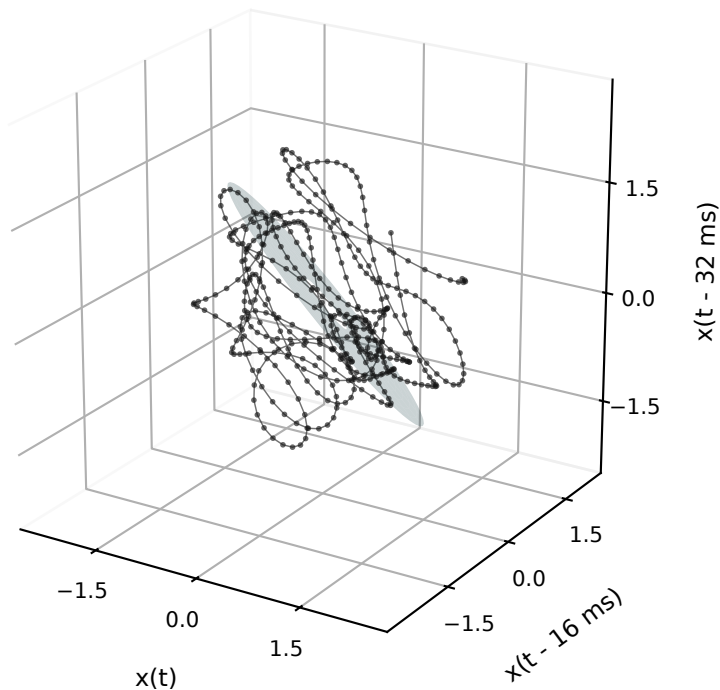
Normalized 3D set error: 0.0027  
Native coverage: 0.831 | nfev: 30  
Flags: active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 5 | original trial 485

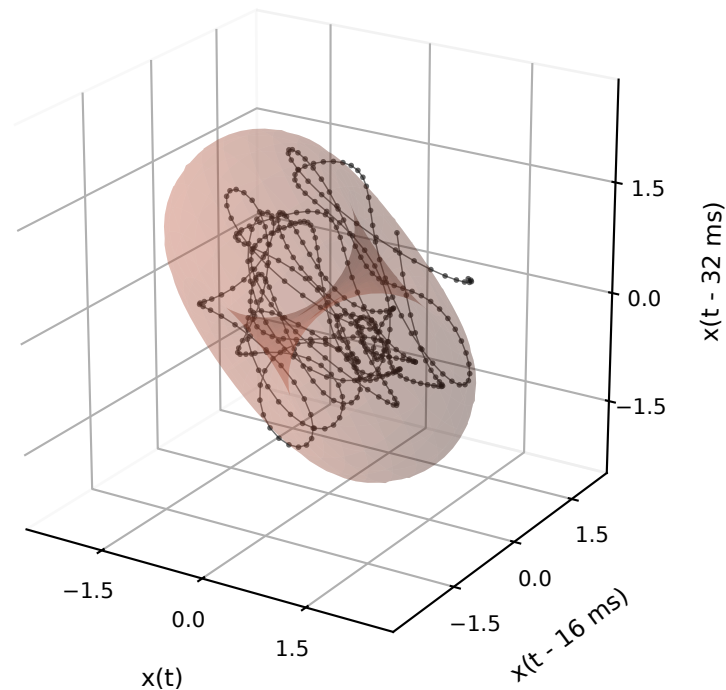
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

**1-torus: planar band**



Normalized 3D set error: 0.1922  
Native coverage: 0.910 | nfev: 22  
Flags: none

**2-torus: donut volume**



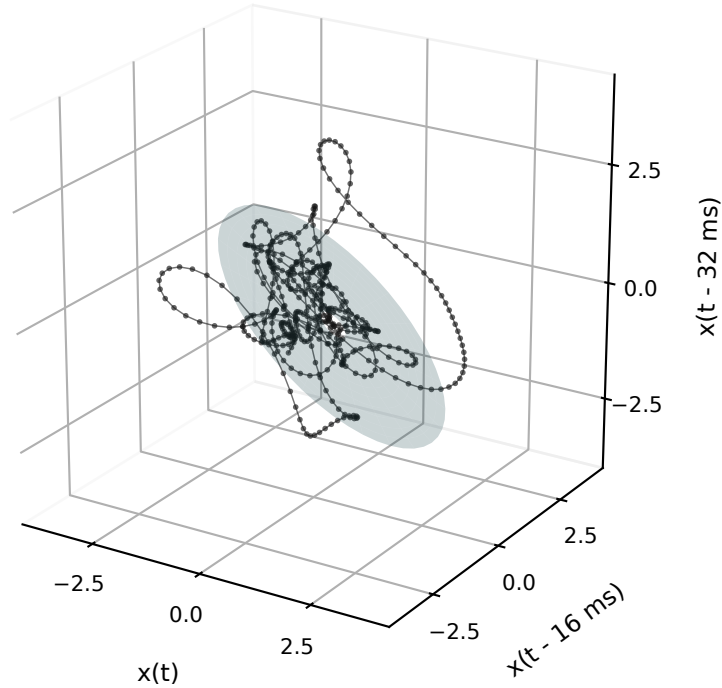
Normalized 3D set error: 0.0020  
Native coverage: 0.853 | nfev: 16  
Flags: active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 5 | original trial 700

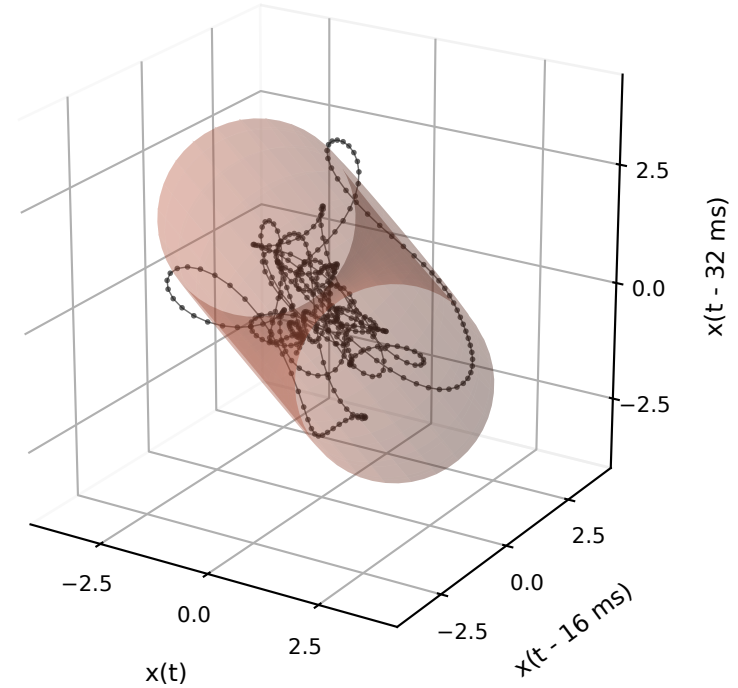
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

**1-torus: planar band**



Normalized 3D set error: 0.2372  
Native coverage: 0.904 | nfev: 41  
Flags: near optimizer bound; nearly circular orientation

**2-torus: donut volume**



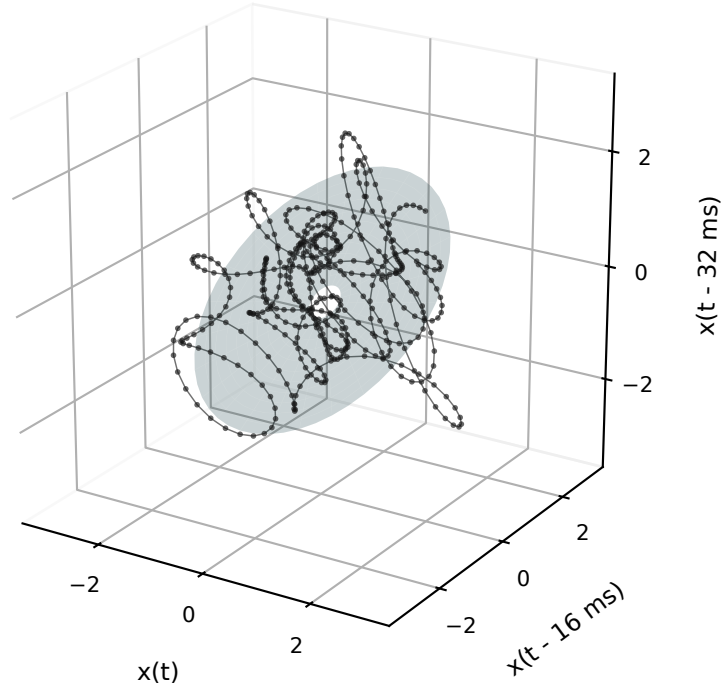
Normalized 3D set error: 0.0009  
Native coverage: 0.893 | nfev: 13  
Flags: nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 5 | original trial 6

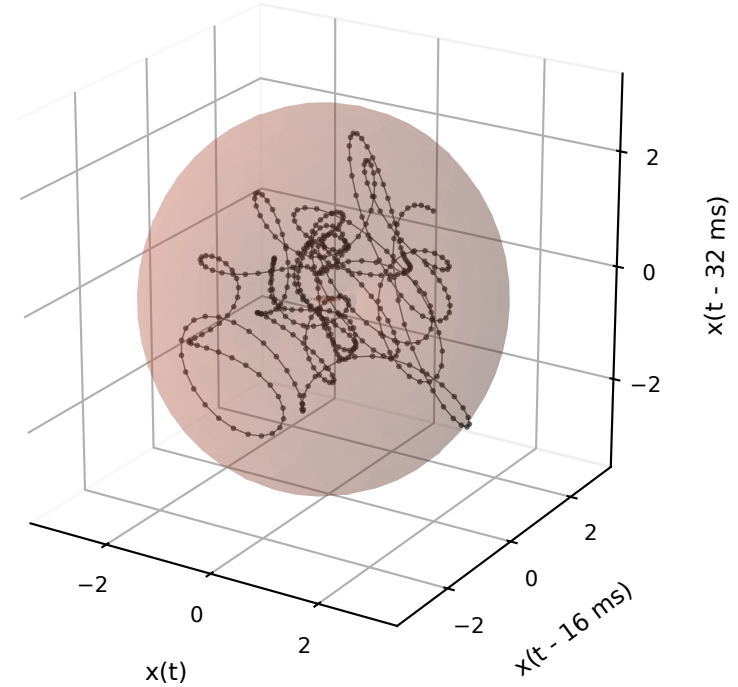
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.3013  
Native coverage: 0.897 | nfev: 71  
Flags: none

2-torus: donut volume



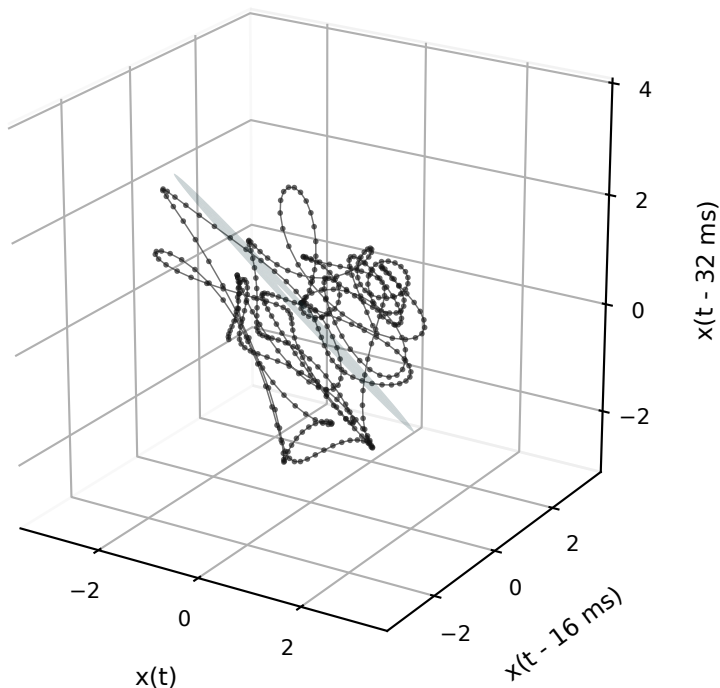
Normalized 3D set error: 0.0007  
Native coverage: 0.865 | nfev: 25  
Flags: collapsed hole; active optimizer bound; near optimizer bound; nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 5 | original trial 157

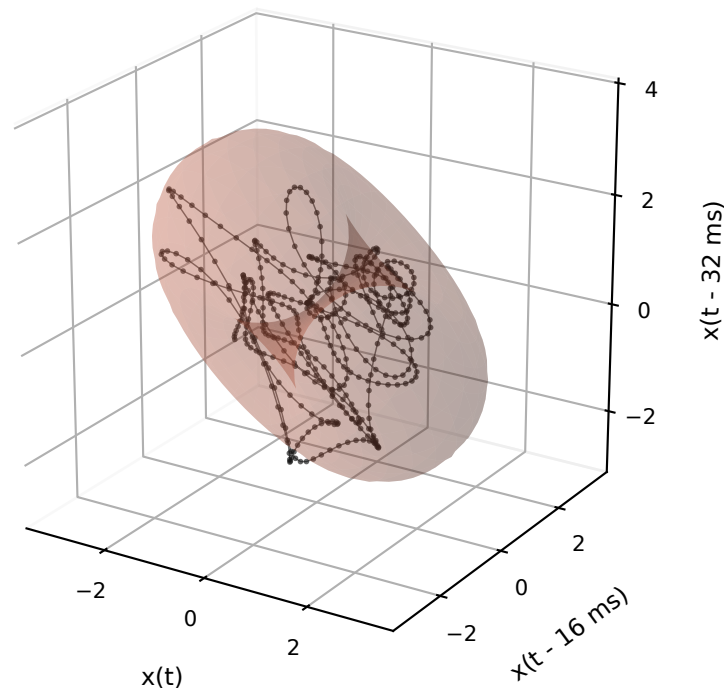
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2388  
Native coverage: 0.910 | nfev: 104  
Flags: nearly circular orientation

2-torus: donut volume



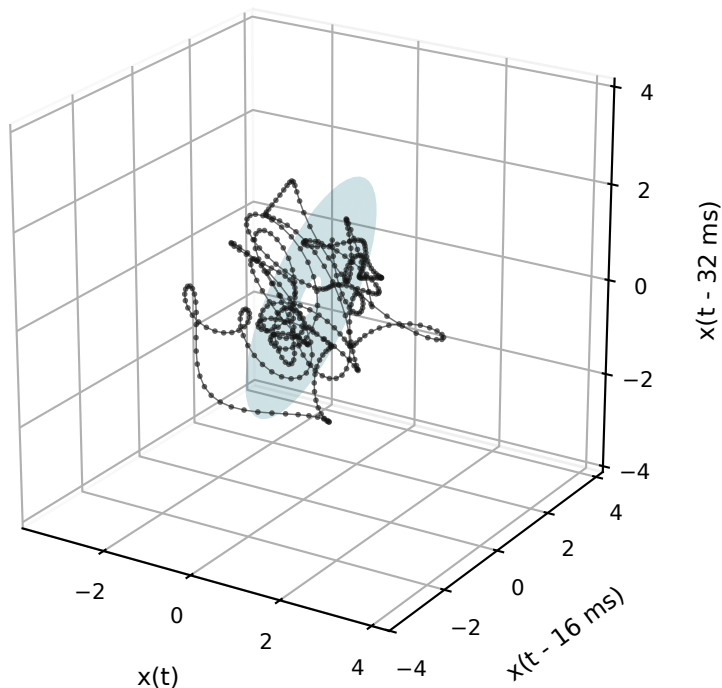
Normalized 3D set error: 0.0007  
Native coverage: 0.887 | nfev: 25  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 5 | original trial 476

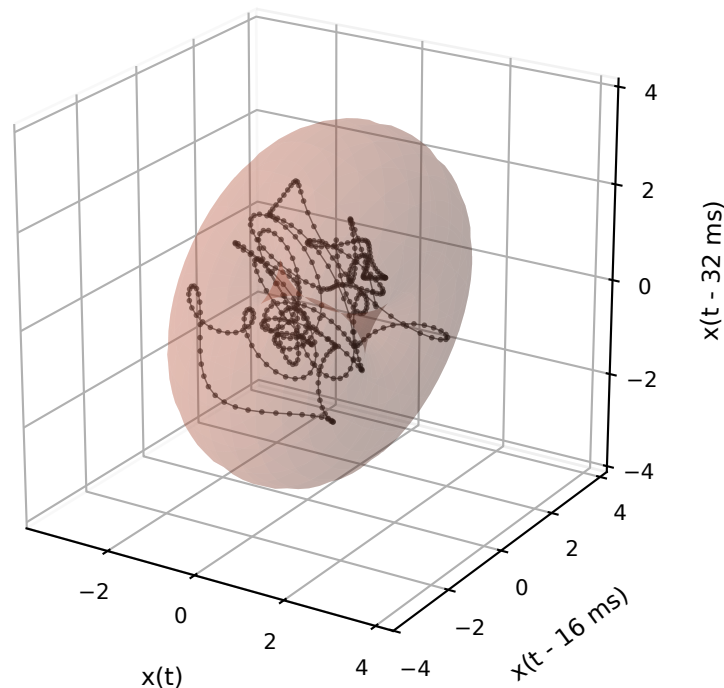
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.3212  
Native coverage: 0.912 | nfev: 99  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

2-torus: donut volume



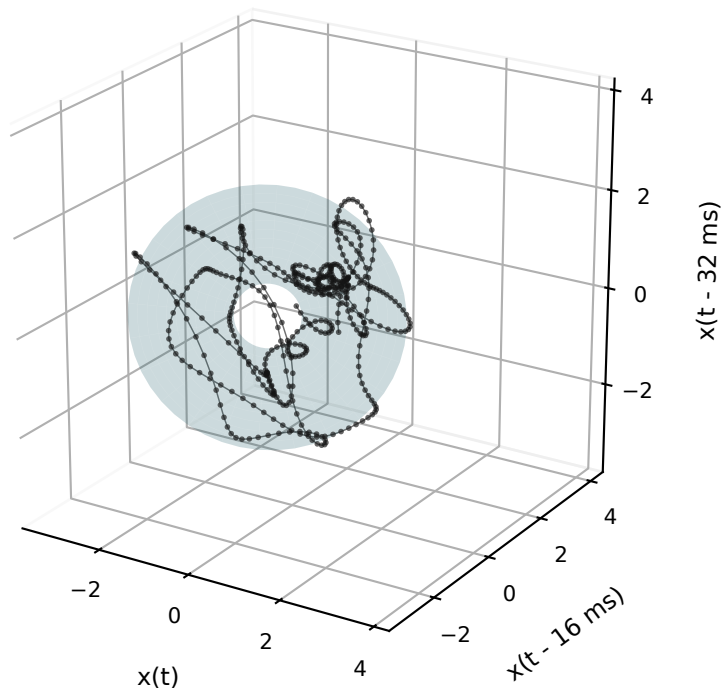
Normalized 3D set error: 0.0016  
Native coverage: 0.936 | nfev: 33  
Flags: collapsed hole; active optimizer bound; near  
optimizer bound; nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 5 | original trial 271

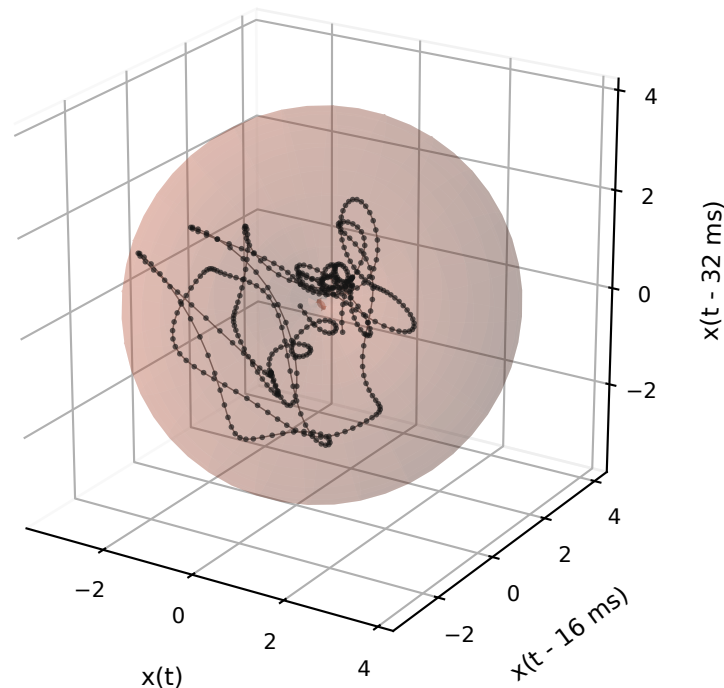
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2828  
Native coverage: 0.900 | nfev: 49  
Flags: near optimizer bound; nearly circular orientation

2-torus: donut volume



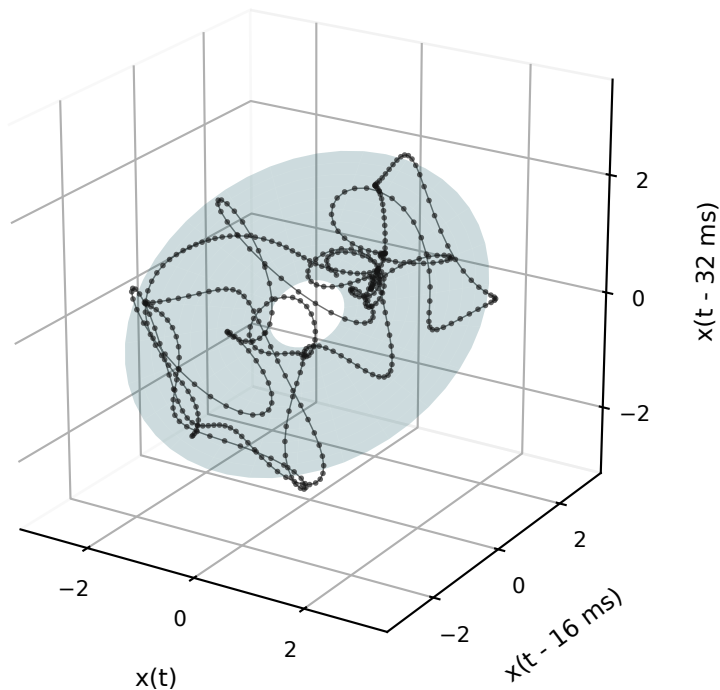
Normalized 3D set error: 0.0011  
Native coverage: 0.910 | nfev: 18  
Flags: collapsed hole; nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 5 | original trial 399

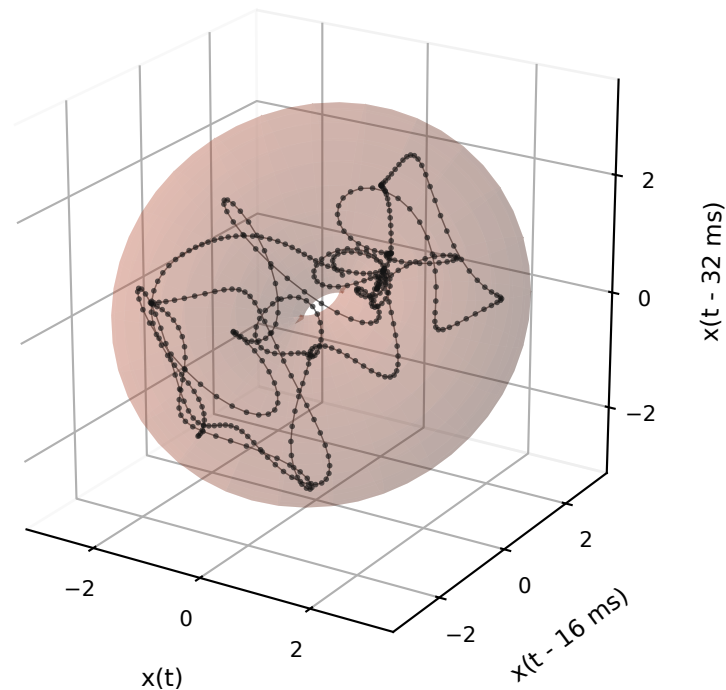
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1690  
Native coverage: 0.908 | nfev: 25  
Flags: none

2-torus: donut volume



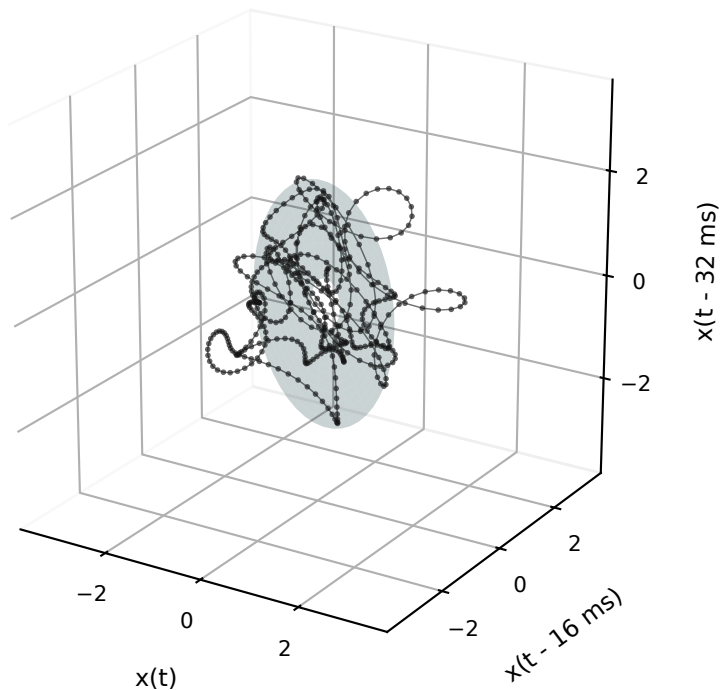
Normalized 3D set error: 0.0007  
Native coverage: 0.876 | nfev: 37  
Flags: none

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 5 | original trial 300

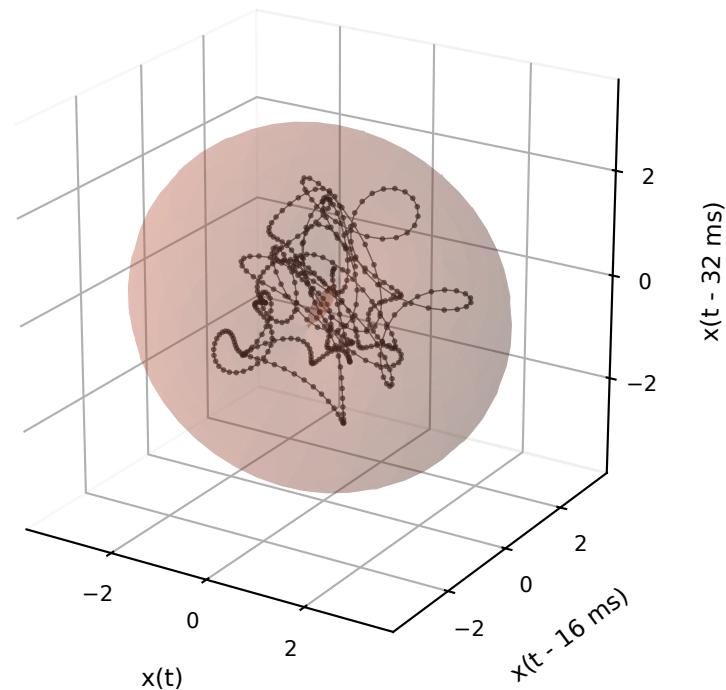
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.3182  
Native coverage: 0.895 | nfev: 77  
Flags: nearly circular orientation

2-torus: donut volume



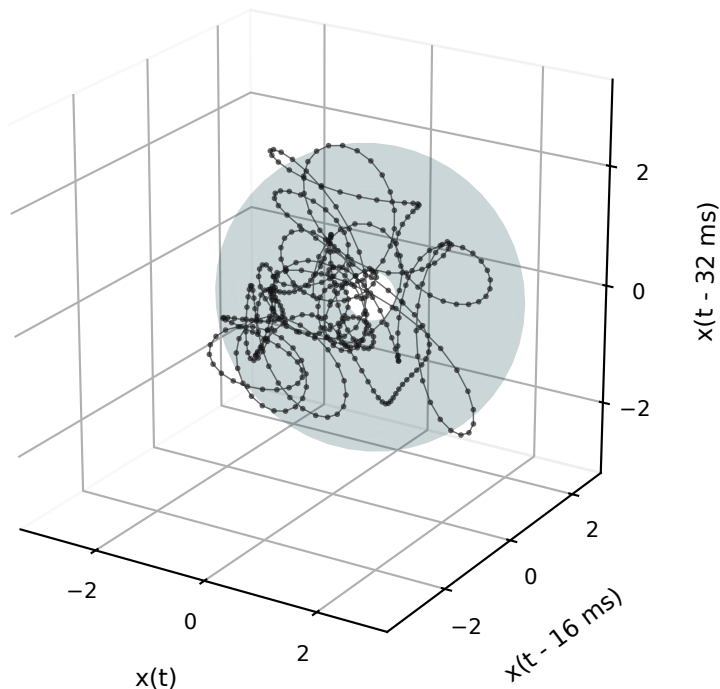
Normalized 3D set error: 0.0016  
Native coverage: 0.921 | nfev: 37  
Flags: collapsed hole; active optimizer bound; near optimizer bound; nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 5 | original trial 446

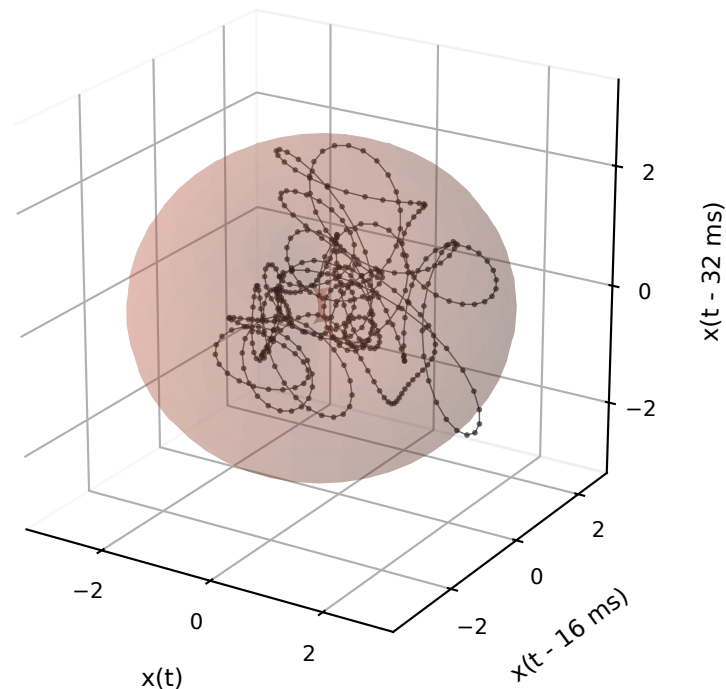
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2682  
Native coverage: 0.906 | nfev: 65  
Flags: near optimizer bound; nearly circular orientation

2-torus: donut volume



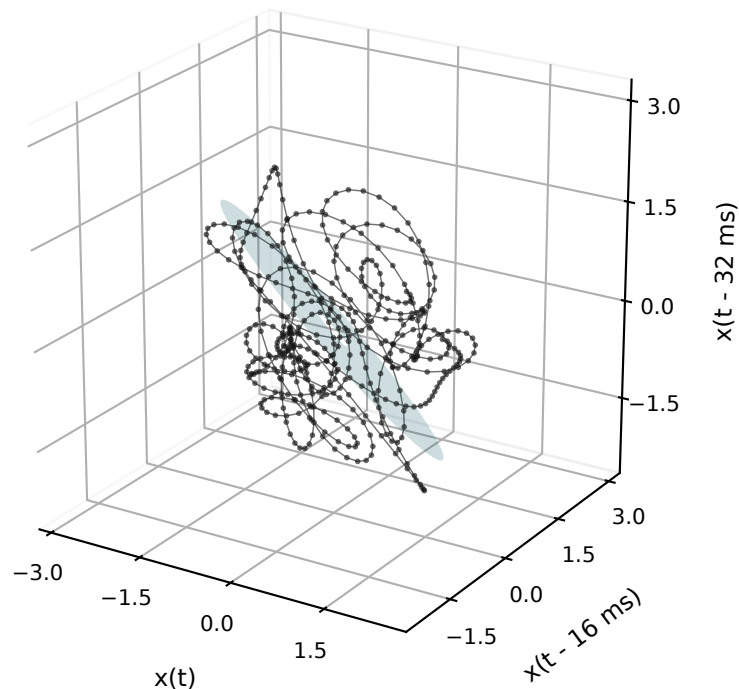
Normalized 3D set error: 0.0010  
Native coverage: 0.846 | nfev: 24  
Flags: active optimizer bound; near optimizer bound; nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 5 | original trial 542

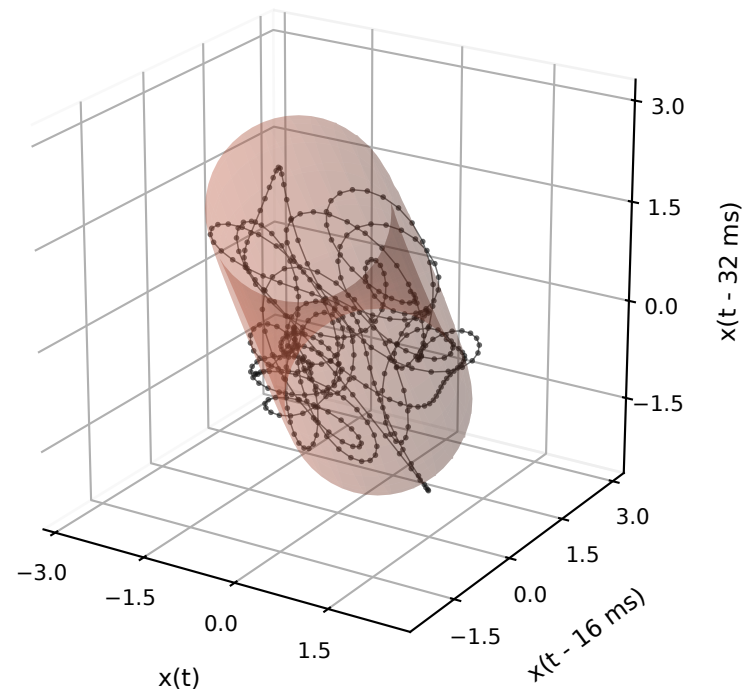
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2395  
Native coverage: 0.891 | nfev: 131  
Flags: nearly circular orientation

2-torus: donut volume



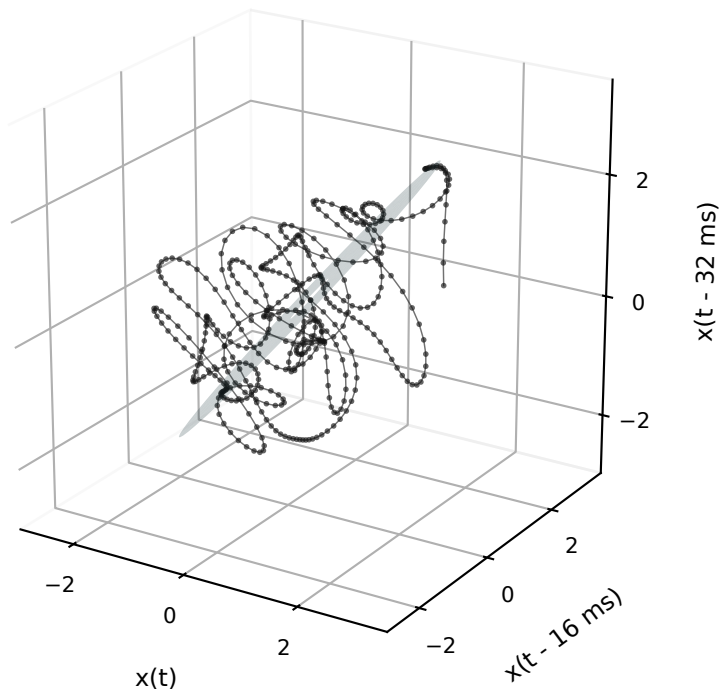
Normalized 3D set error: 0.0012  
Native coverage: 0.840 | nfev: 19  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 5 | original trial 693

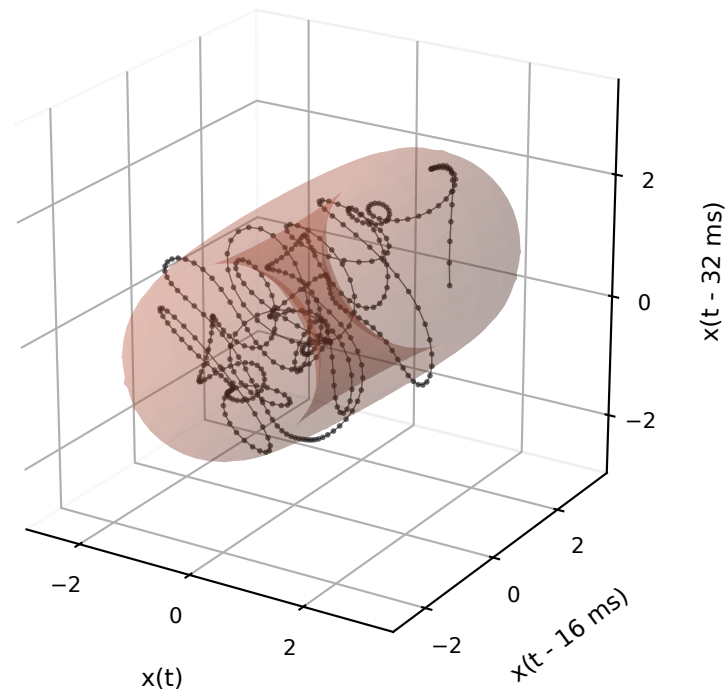
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2187  
Native coverage: 0.902 | nfev: 49  
Flags: none

2-torus: donut volume



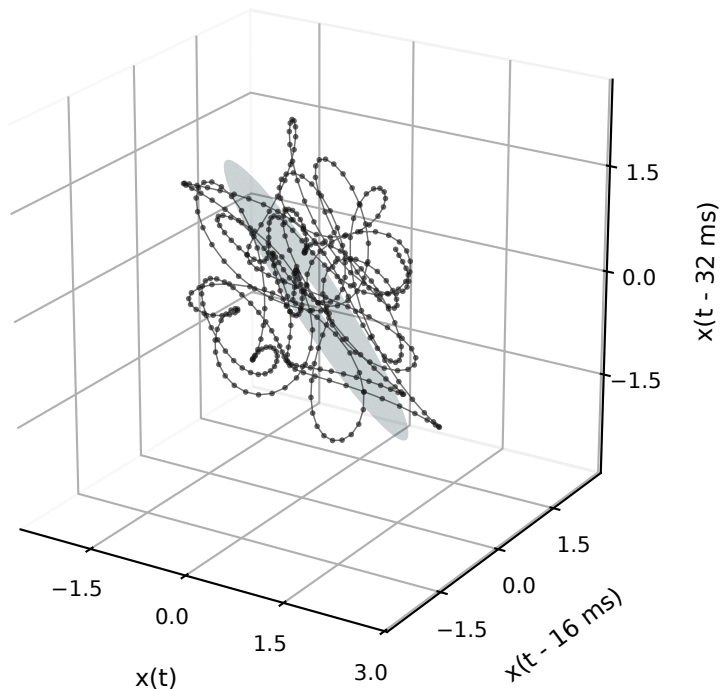
Normalized 3D set error: 0.0012  
Native coverage: 0.857 | nfev: 28  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 6 | original trial 258

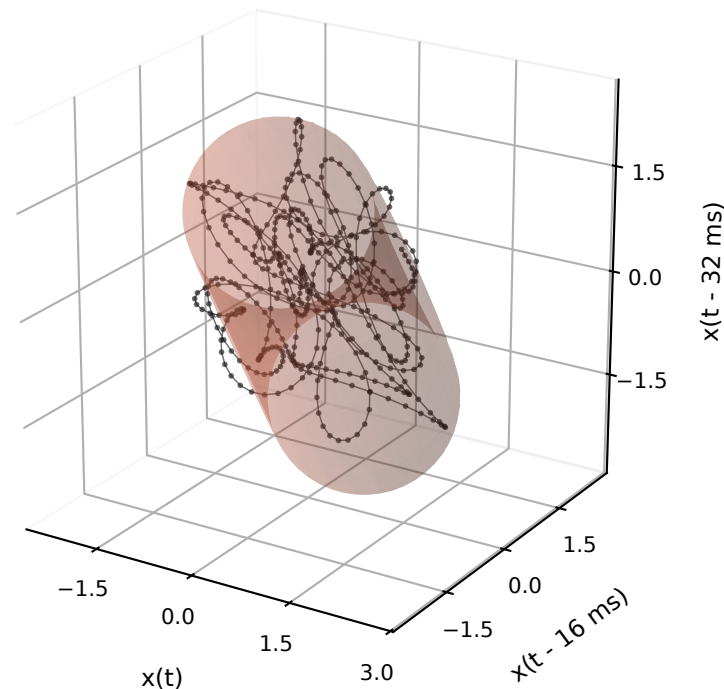
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2110  
Native coverage: 0.906 | nfev: 35  
Flags: near optimizer bound; nearly circular orientation

2-torus: donut volume



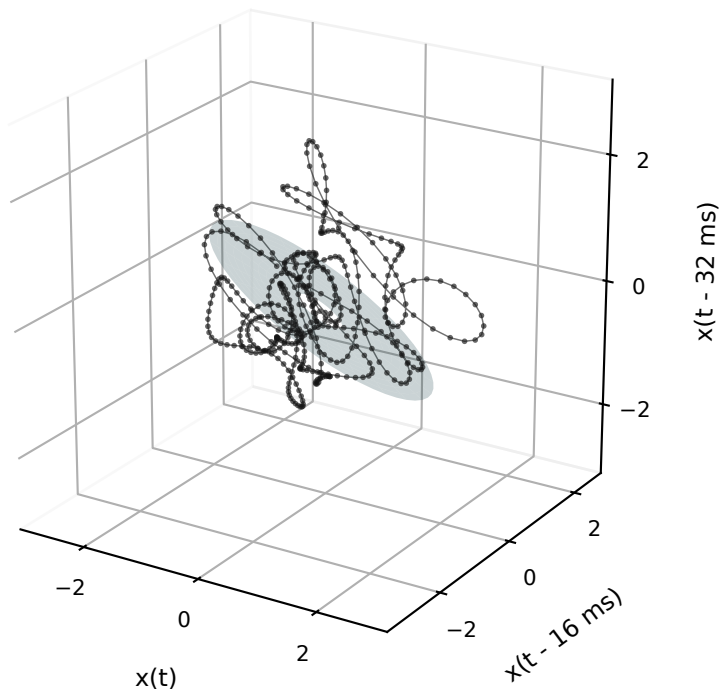
Normalized 3D set error: 0.0015  
Native coverage: 0.872 | nfev: 16  
Flags: active optimizer bound; near optimizer bound; nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 6 | original trial 550

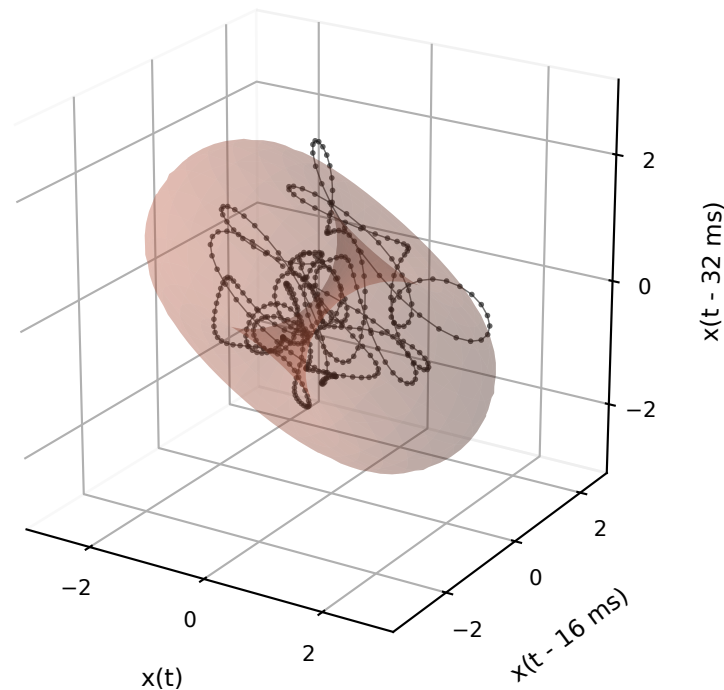
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2750  
Native coverage: 0.885 | nfev: 63  
Flags: nearly circular orientation

2-torus: donut volume



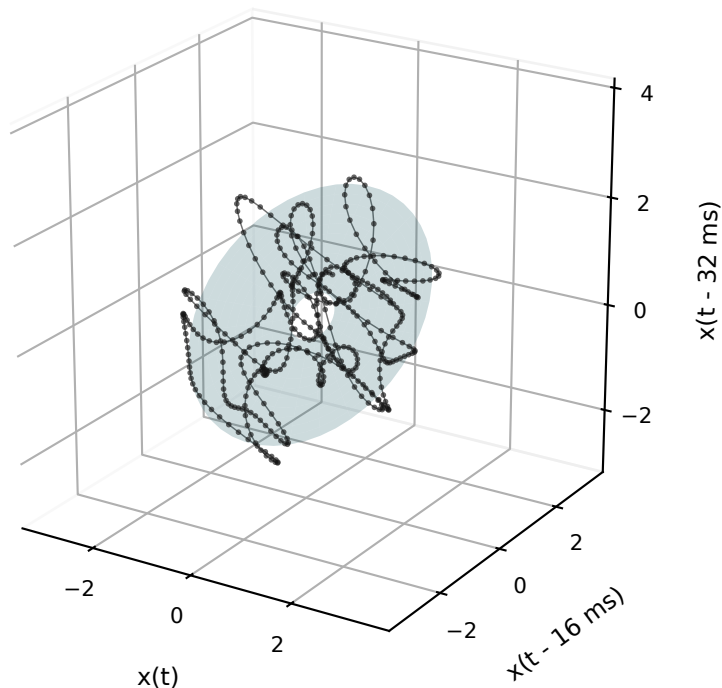
Normalized 3D set error: 0.0014  
Native coverage: 0.863 | nfev: 21  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 6 | original trial 100

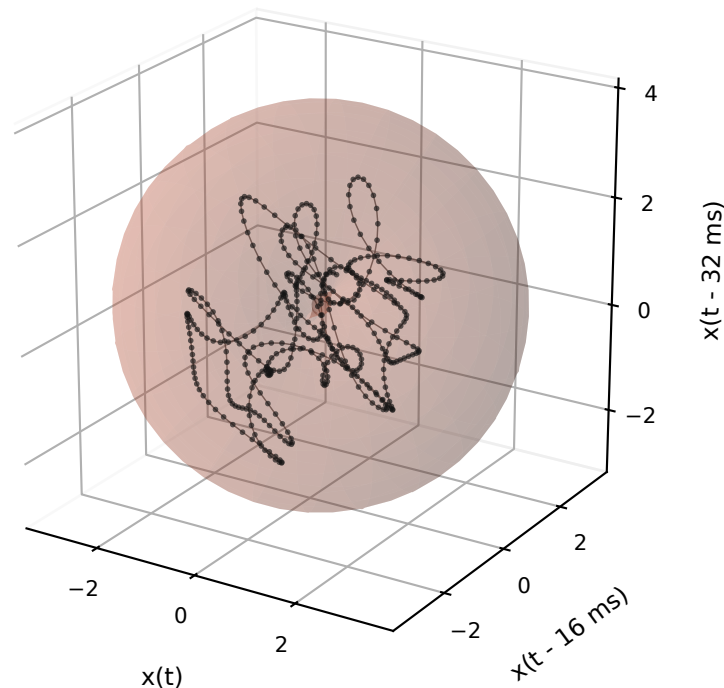
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2891  
Native coverage: 0.868 | nfev: 258  
Flags: nearly circular orientation

2-torus: donut volume



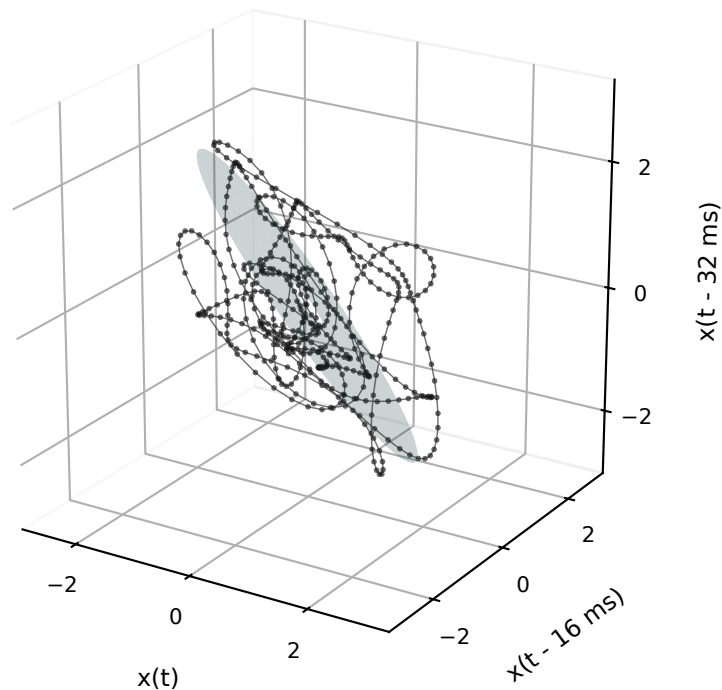
Normalized 3D set error: 0.0012  
Native coverage: 0.919 | nfev: 47  
Flags: collapsed hole; active optimizer bound; near optimizer bound; nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 6 | original trial 129

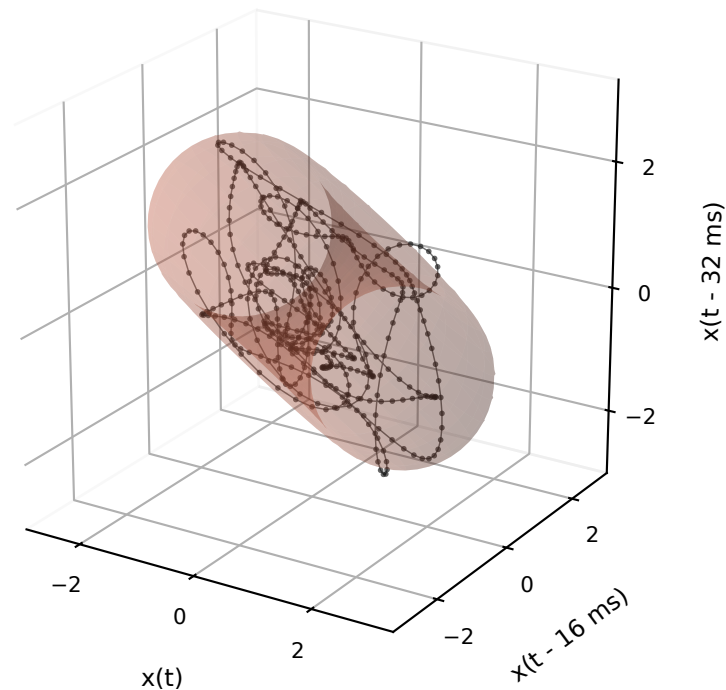
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1526  
Native coverage: 0.897 | nfev: 76  
Flags: none

2-torus: donut volume



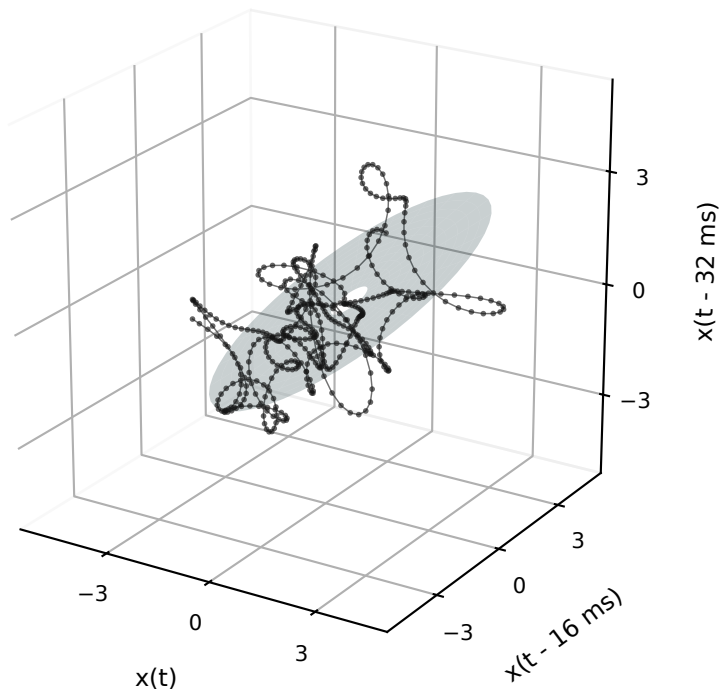
Normalized 3D set error: 0.0007  
Native coverage: 0.850 | nfev: 36  
Flags: none

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 6 | original trial 186

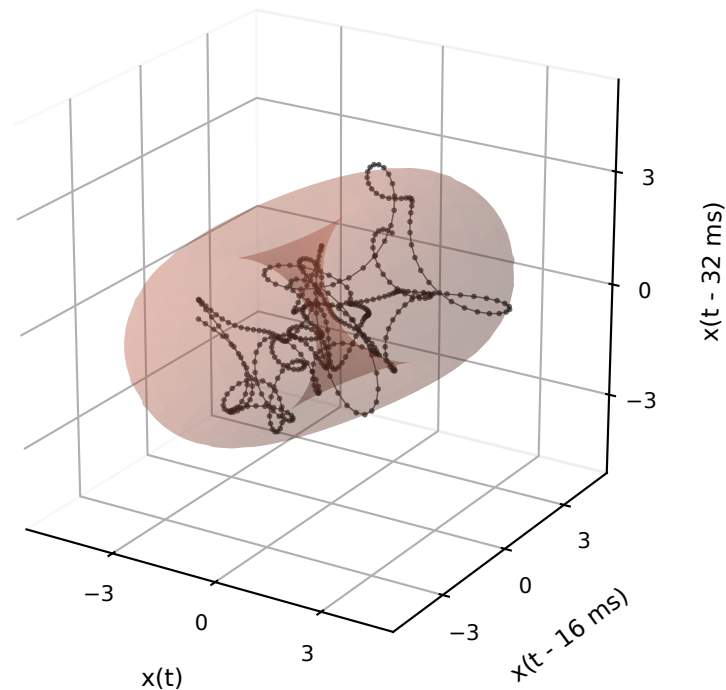
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2131  
Native coverage: 0.938 | nfev: 75  
Flags: near optimizer bound

2-torus: donut volume



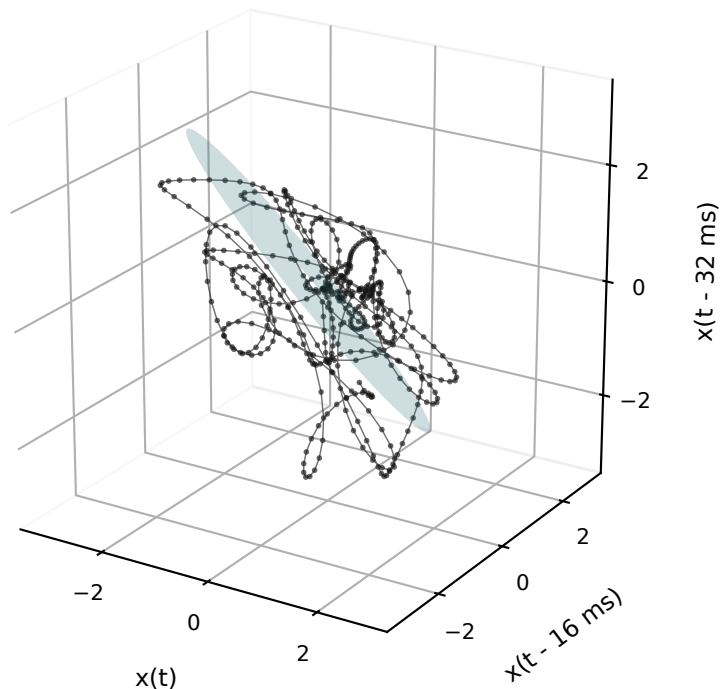
Normalized 3D set error: 0.0004  
Native coverage: 0.904 | nfev: 46  
Flags: none

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 6 | original trial 650

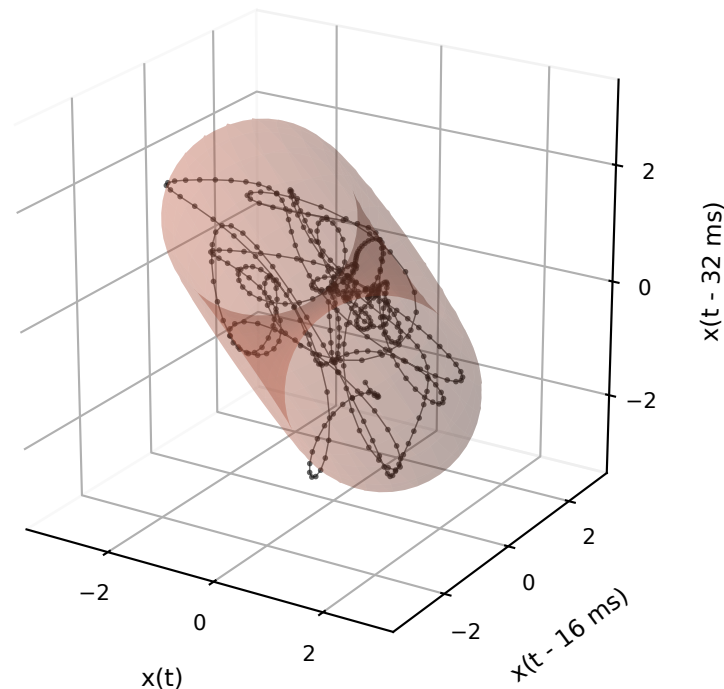
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

**1-torus: planar band**



Normalized 3D set error: 0.1528  
Native coverage: 0.897 | nfev: 27  
Flags: nearly circular orientation

**2-torus: donut volume**



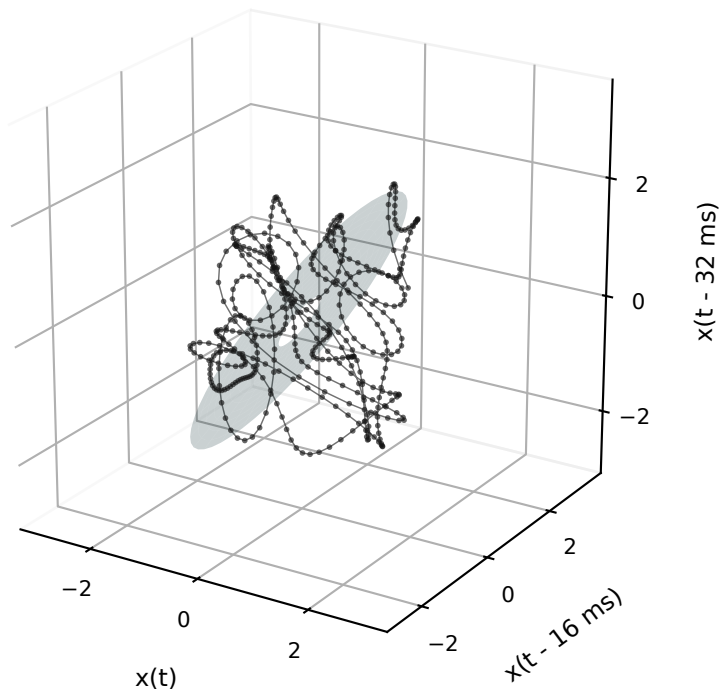
Normalized 3D set error: 0.0008  
Native coverage: 0.878 | nfev: 18  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 6 | original trial 438

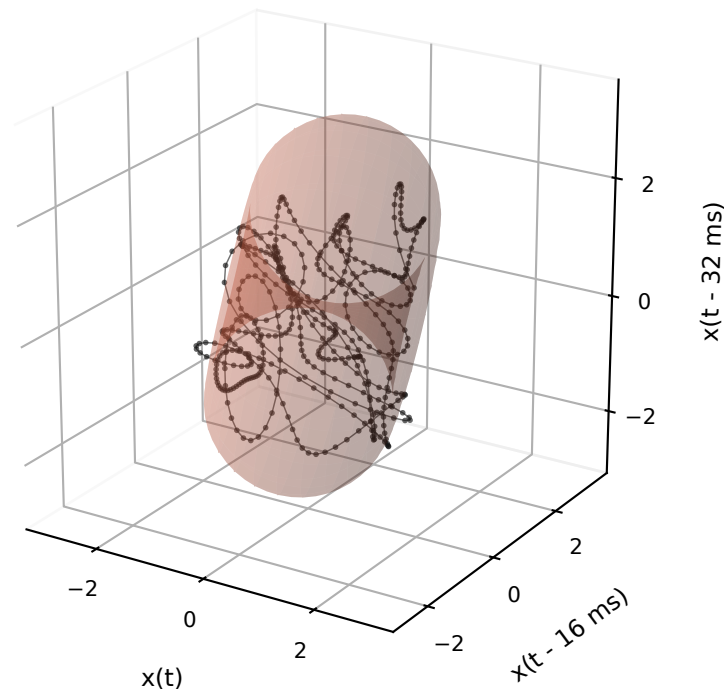
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.3093  
Native coverage: 0.902 | nfev: 73  
Flags: none

2-torus: donut volume



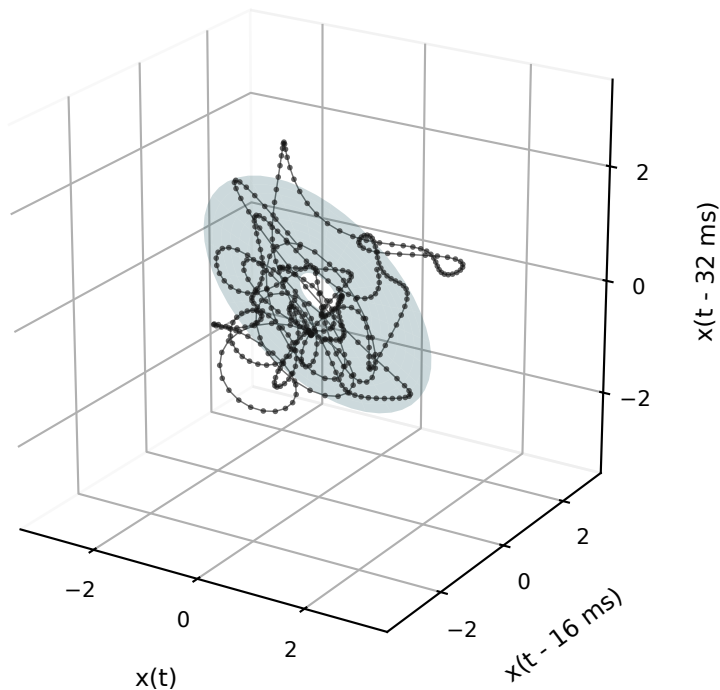
Normalized 3D set error: 0.0009  
Native coverage: 0.878 | nfev: 14  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 6 | original trial 290

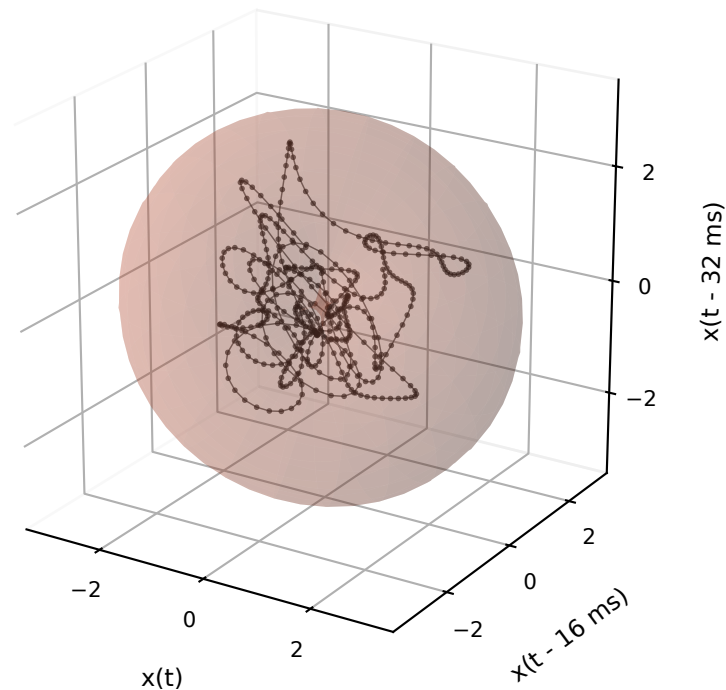
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.3051  
Native coverage: 0.895 | nfev: 69  
Flags: none

2-torus: donut volume



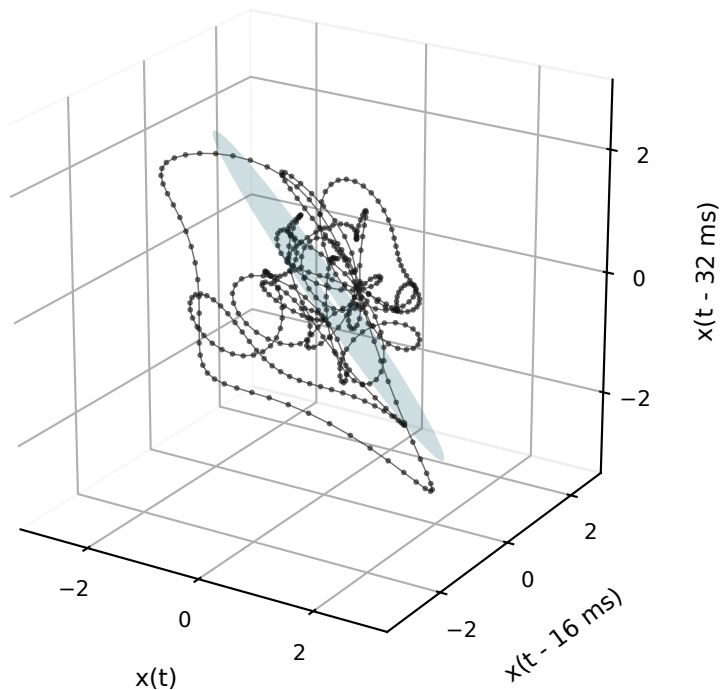
Normalized 3D set error: 0.0011  
Native coverage: 0.870 | nfev: 18  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 6 | original trial 289

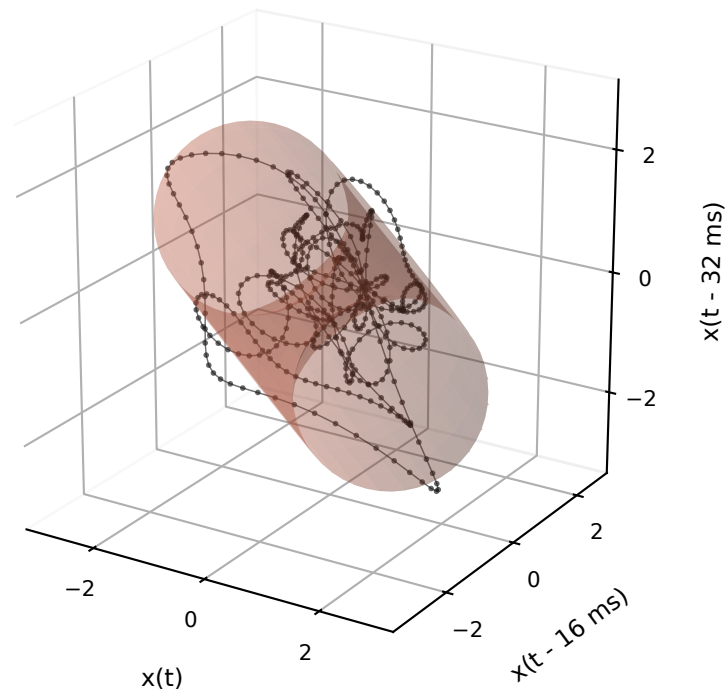
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2156  
Native coverage: 0.910 | nfev: 101  
Flags: none

2-torus: donut volume



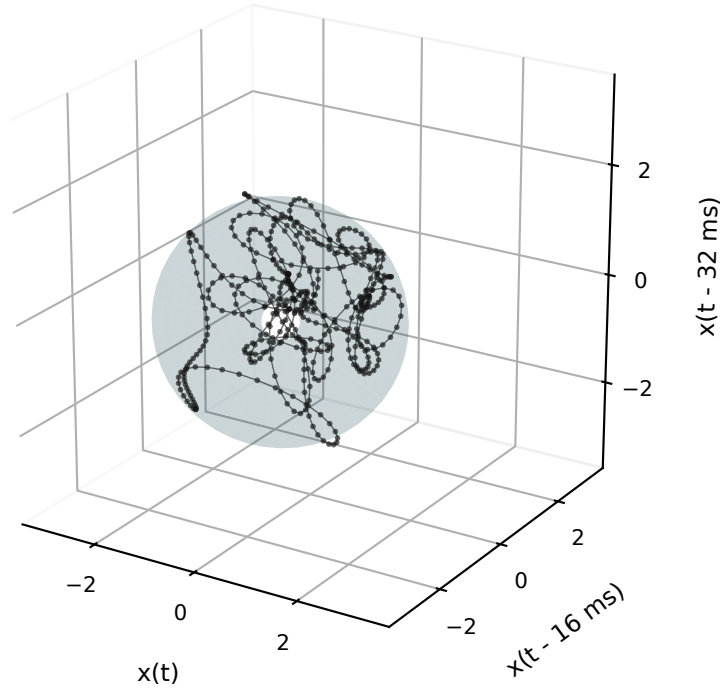
Normalized 3D set error: 0.0017  
Native coverage: 0.863 | nfev: 36  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 6 | original trial 67

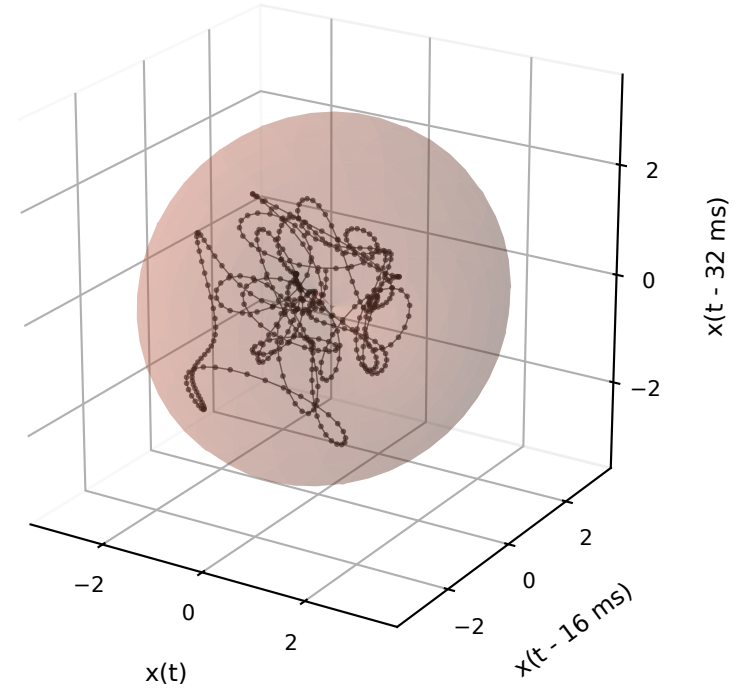
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2949  
Native coverage: 0.893 | nfev: 76  
Flags: nearly circular orientation

2-torus: donut volume



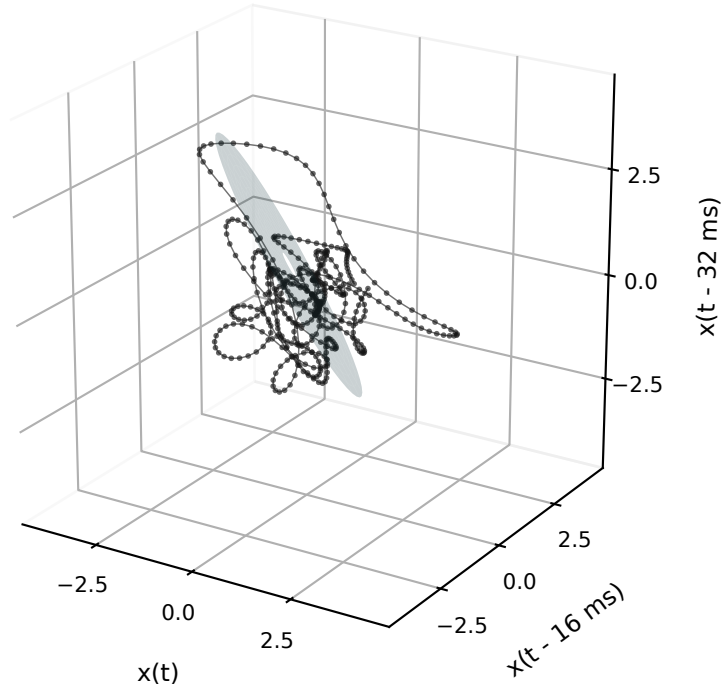
Normalized 3D set error: 0.0012  
Native coverage: 0.882 | nfev: 22  
Flags: collapsed hole; active optimizer bound; near optimizer bound; nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 6 | original trial 458

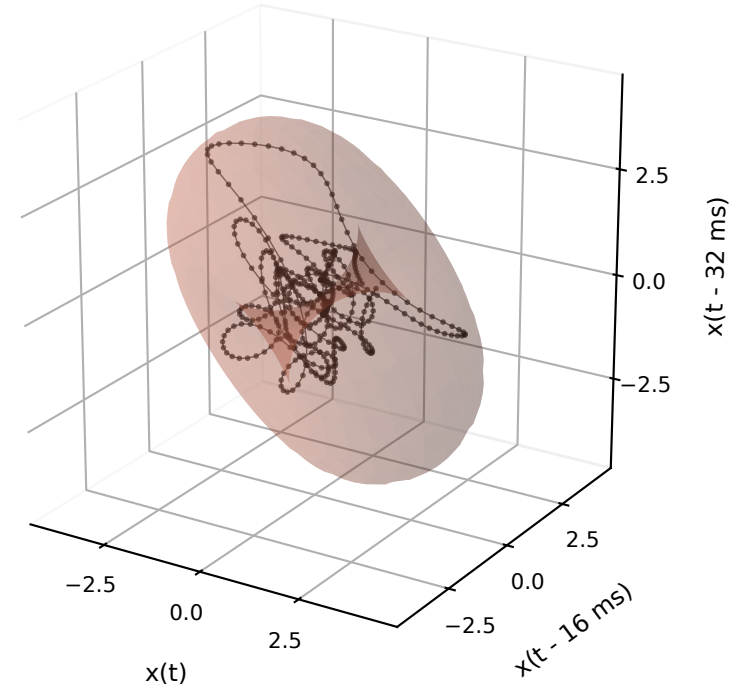
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

**1-torus: planar band**



Normalized 3D set error: 0.2318  
Native coverage: 0.904 | nfev: 197  
Flags: nearly circular orientation

**2-torus: donut volume**



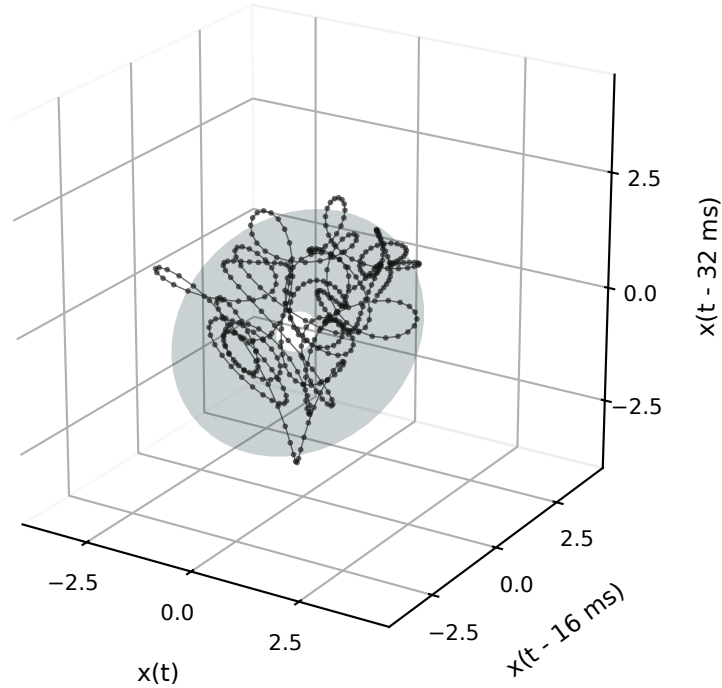
Normalized 3D set error: 0.0010  
Native coverage: 0.880 | nfev: 15  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 6 | original trial 90

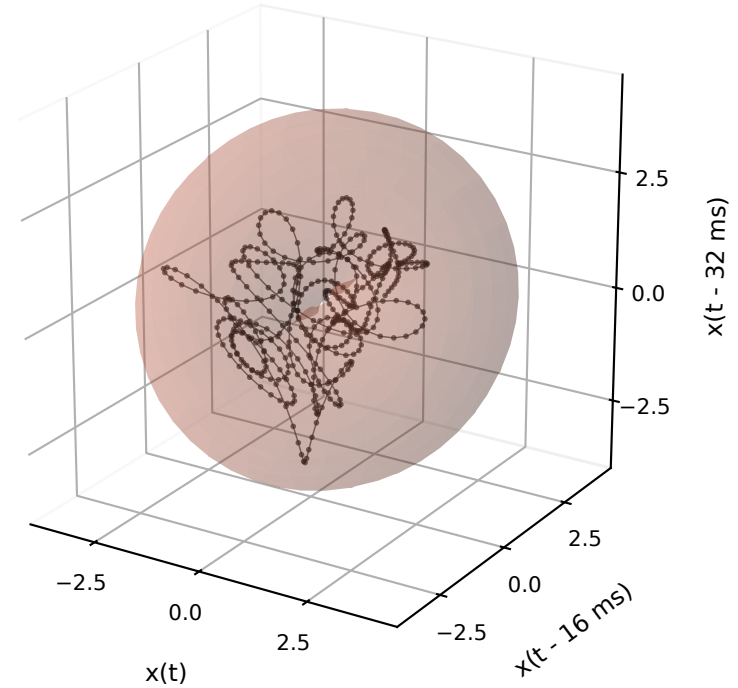
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2503  
Native coverage: 0.912 | nfev: 94  
Flags: nearly circular orientation

2-torus: donut volume



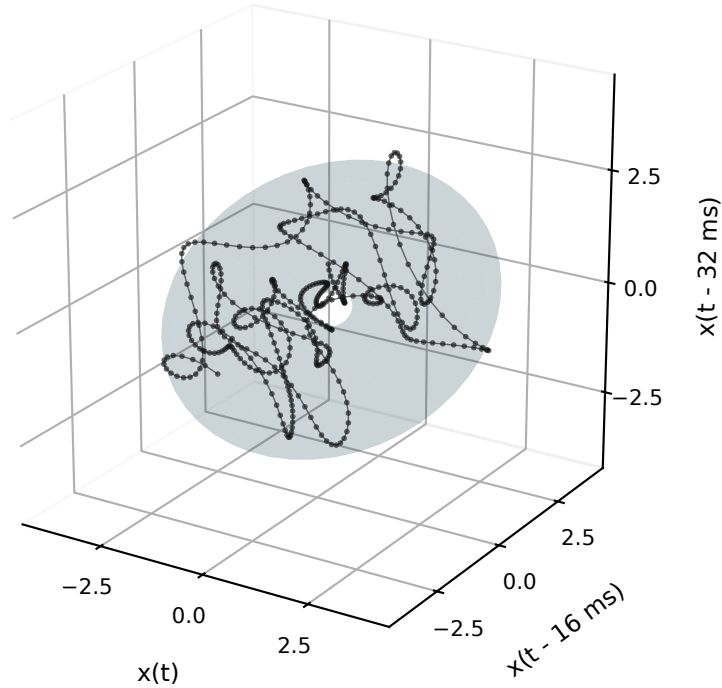
Normalized 3D set error: 0.0011  
Native coverage: 0.910 | nfev: 38  
Flags: none

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 6 | original trial 77

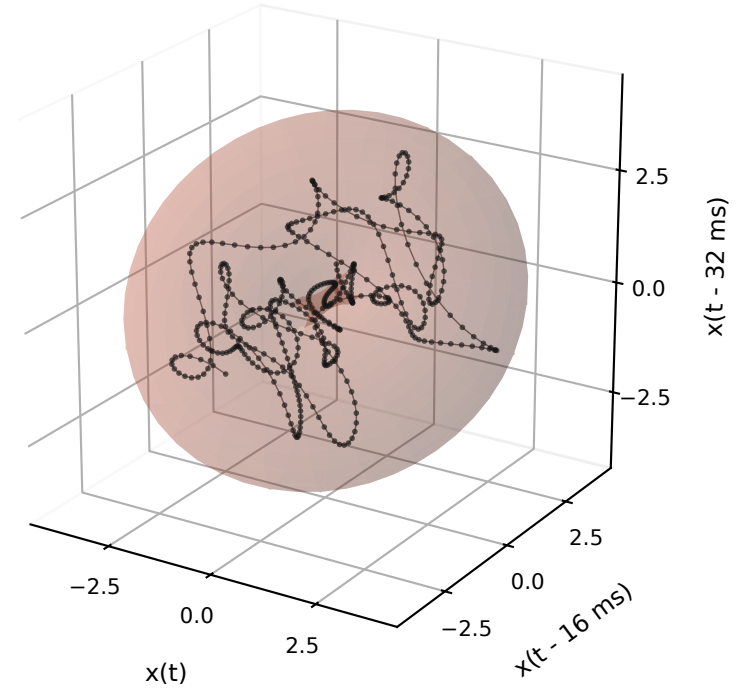
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2216  
Native coverage: 0.910 | nfev: 52  
Flags: none

2-torus: donut volume



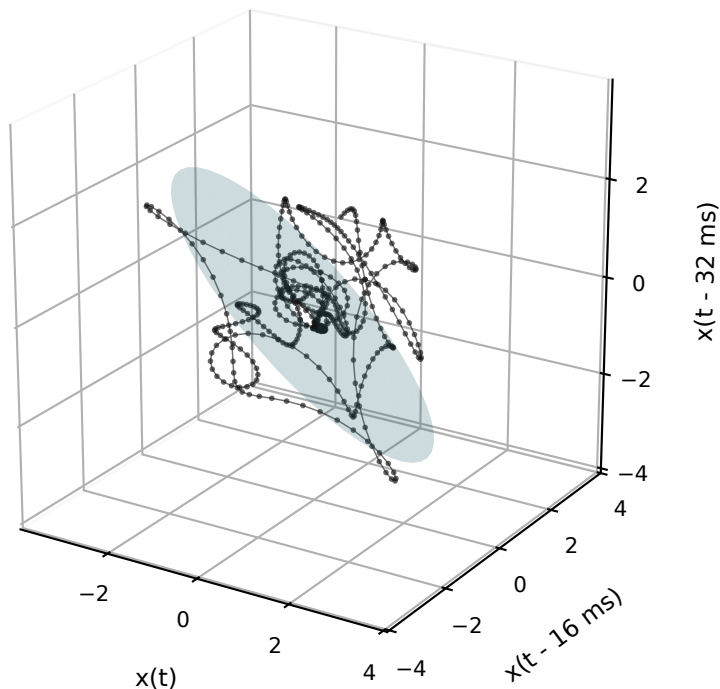
Normalized 3D set error: 0.0007  
Native coverage: 0.906 | nfev: 39  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 6 | original trial 292

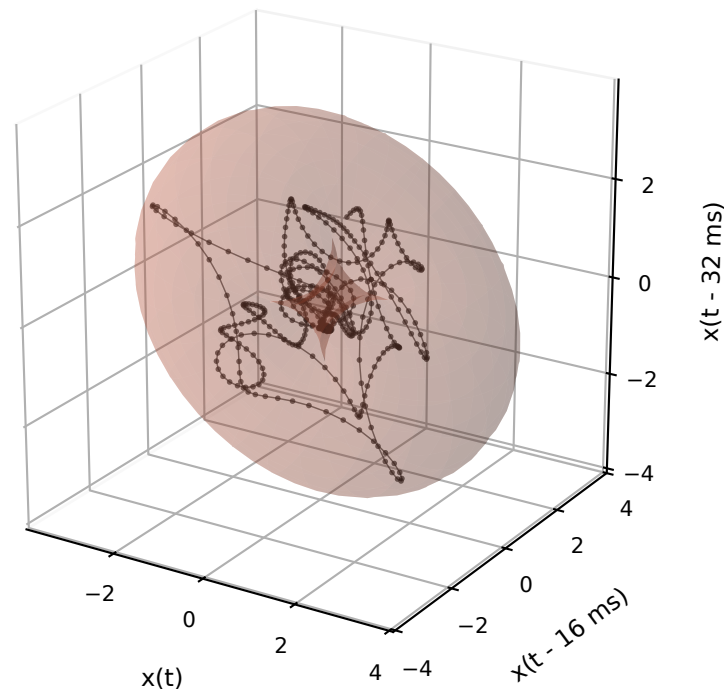
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2858  
Native coverage: 0.895 | nfev: 38  
Flags: none

2-torus: donut volume



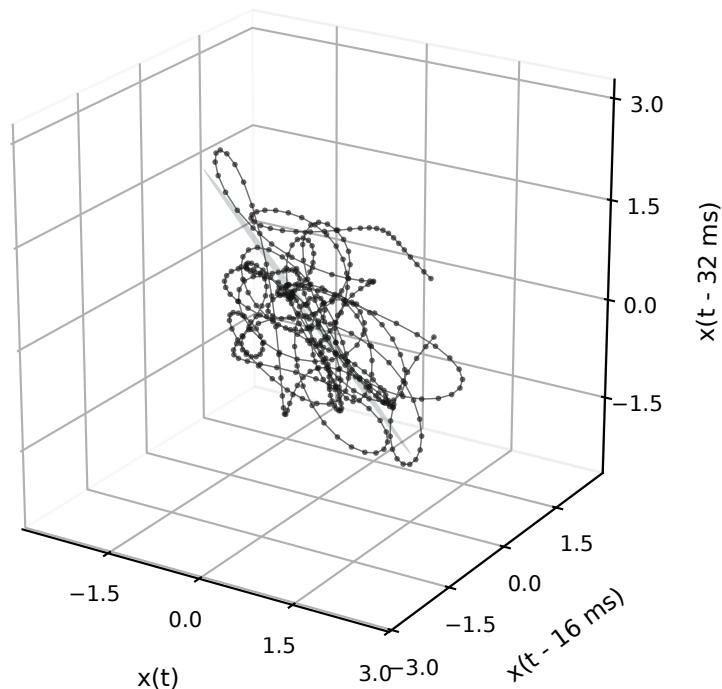
Normalized 3D set error: 0.0015  
Native coverage: 0.885 | nfev: 40  
Flags: collapsed hole; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 6 | original trial 649

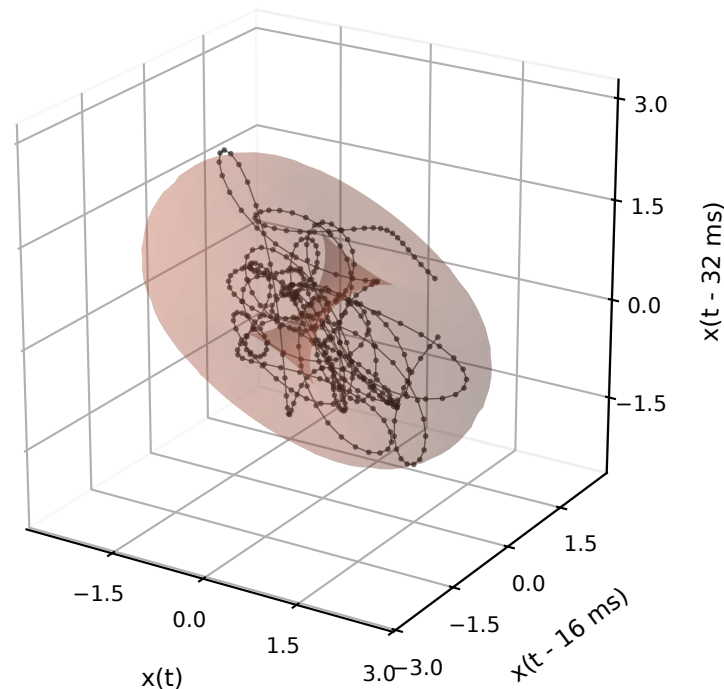
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1386  
Native coverage: 0.904 | nfev: 47  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

2-torus: donut volume



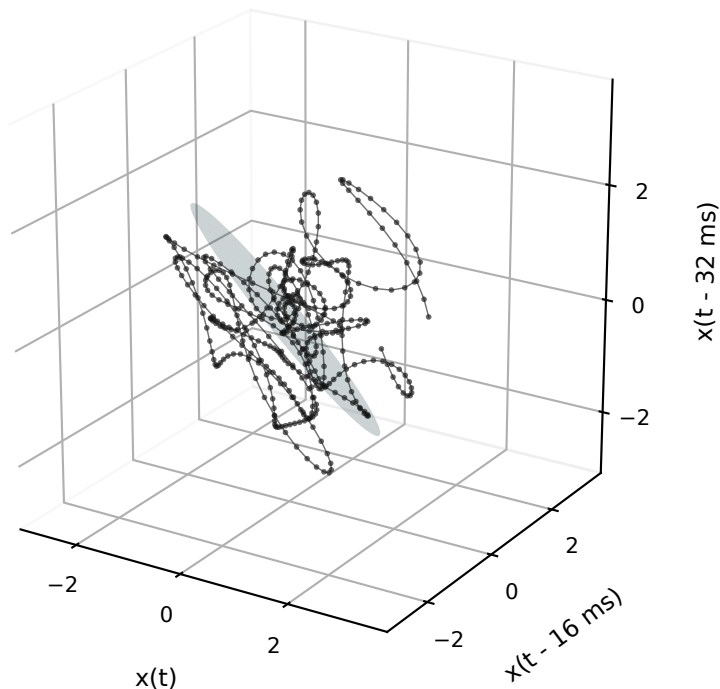
Normalized 3D set error: 0.0014  
Native coverage: 0.863 | nfev: 18  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 6 | original trial 524

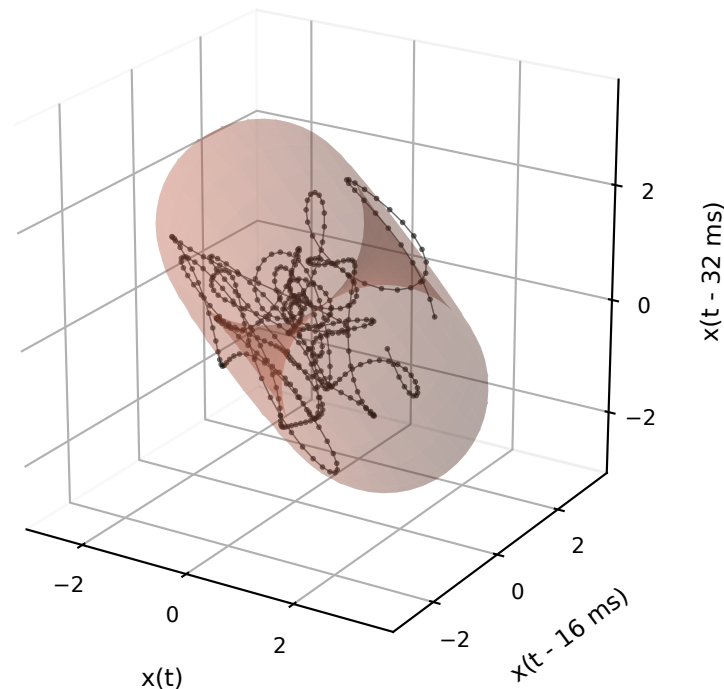
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2539  
Native coverage: 0.891 | nfev: 124  
Flags: nearly circular orientation

2-torus: donut volume



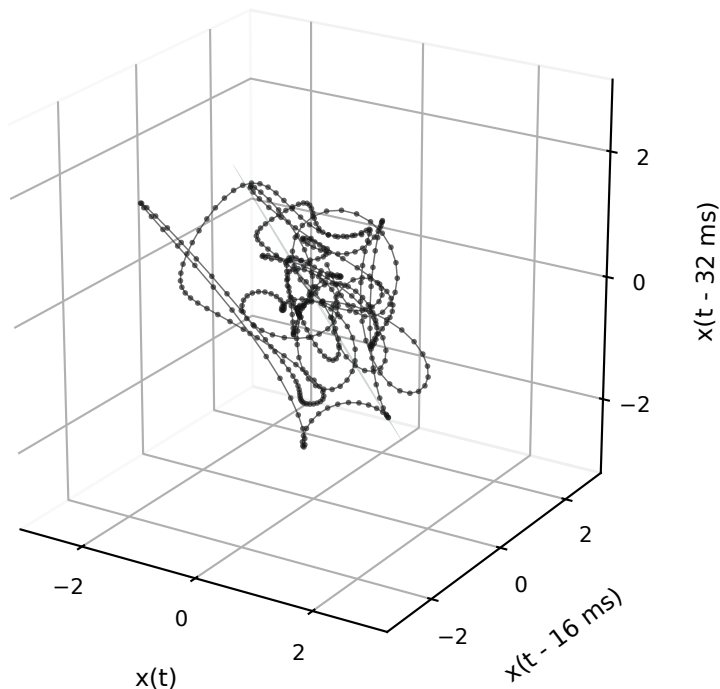
Normalized 3D set error: 0.0013  
Native coverage: 0.895 | nfev: 16  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 6 | original trial 105

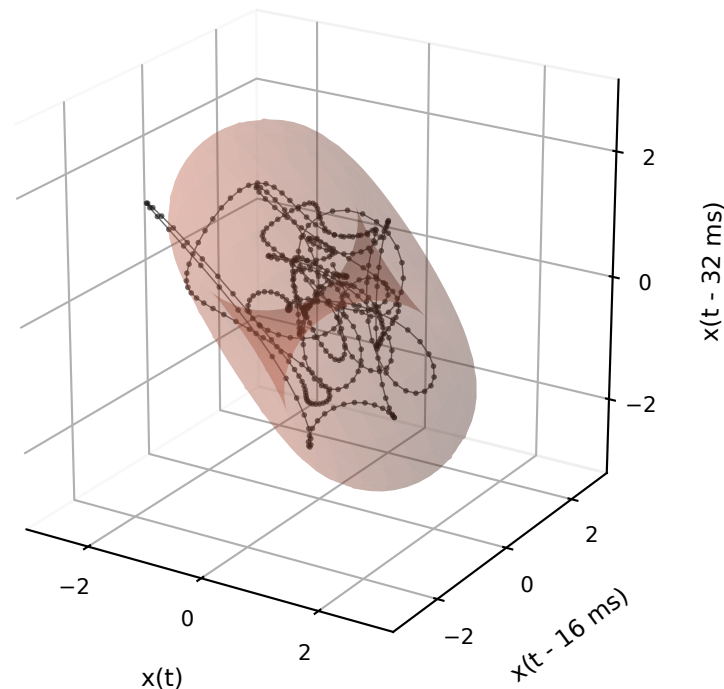
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2025  
Native coverage: 0.900 | nfev: 58  
Flags: nearly circular orientation

2-torus: donut volume



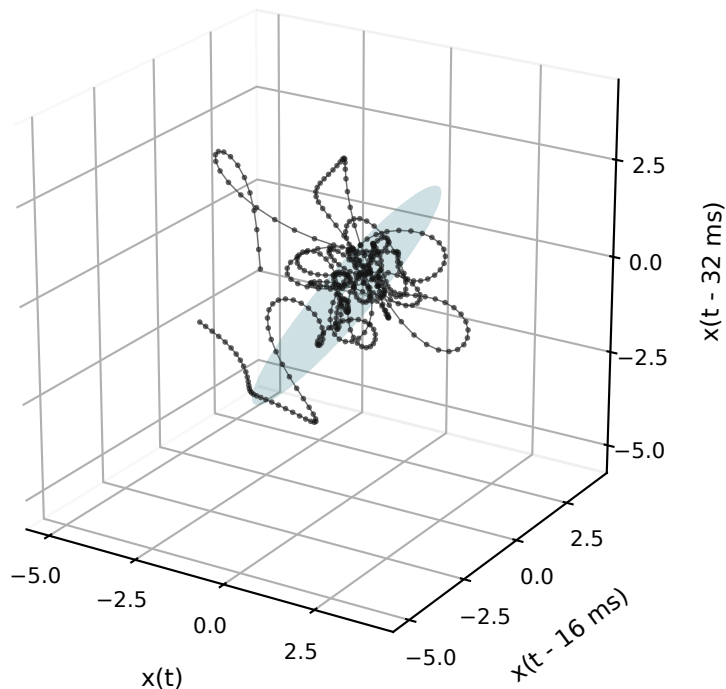
Normalized 3D set error: 0.0012  
Native coverage: 0.887 | nfev: 11  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 6 | original trial 372

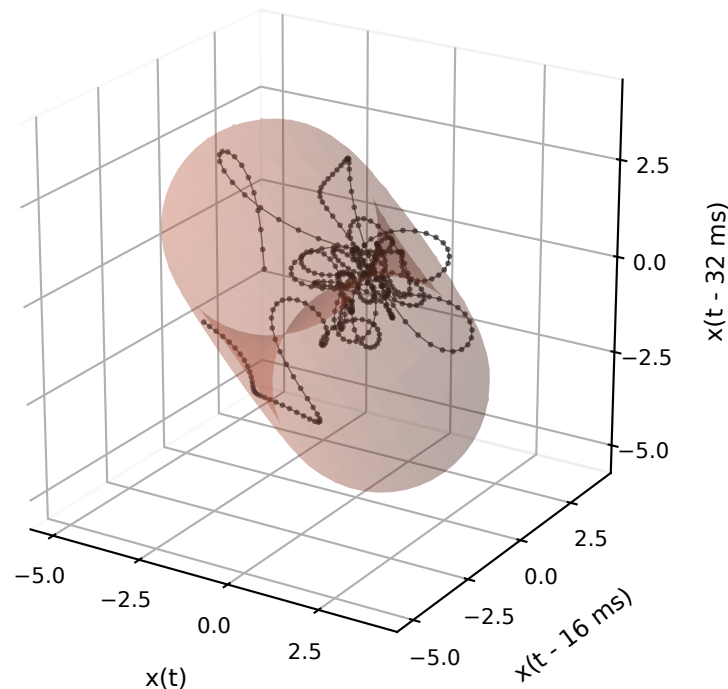
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2998  
Native coverage: 0.919 | nfev: 59  
Flags: nearly circular orientation

2-torus: donut volume



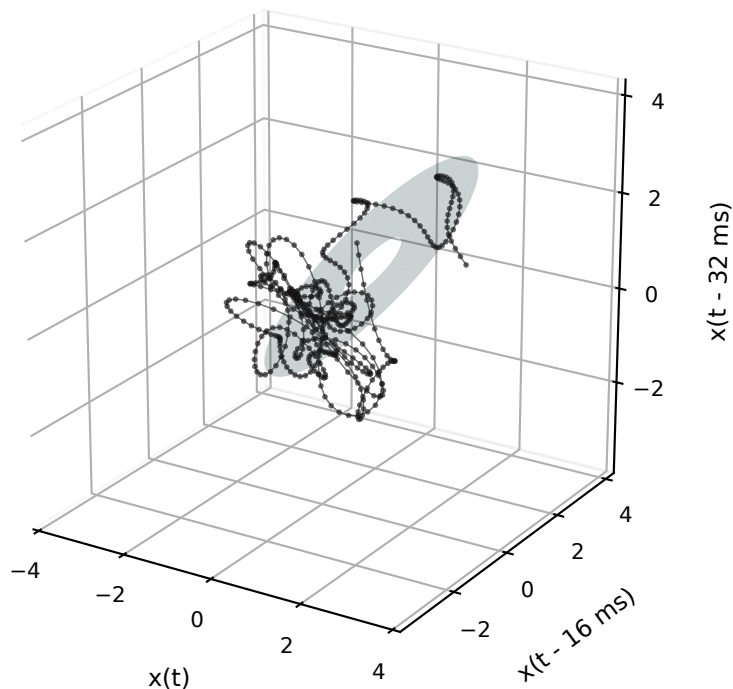
Normalized 3D set error: 0.0005  
Native coverage: 0.915 | nfev: 29  
Flags: collapsed hole; nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 6 | original trial 617

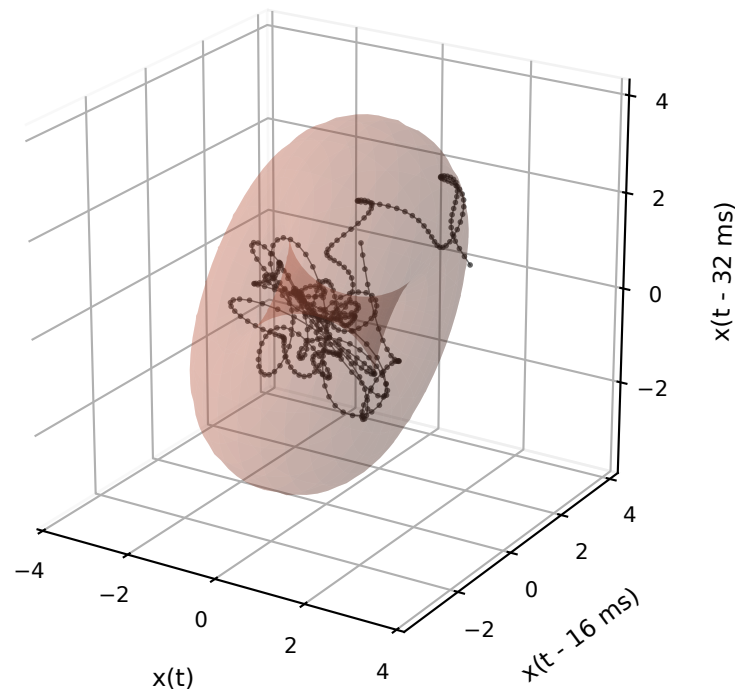
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2462  
Native coverage: 0.934 | nfev: 50  
Flags: none

2-torus: donut volume



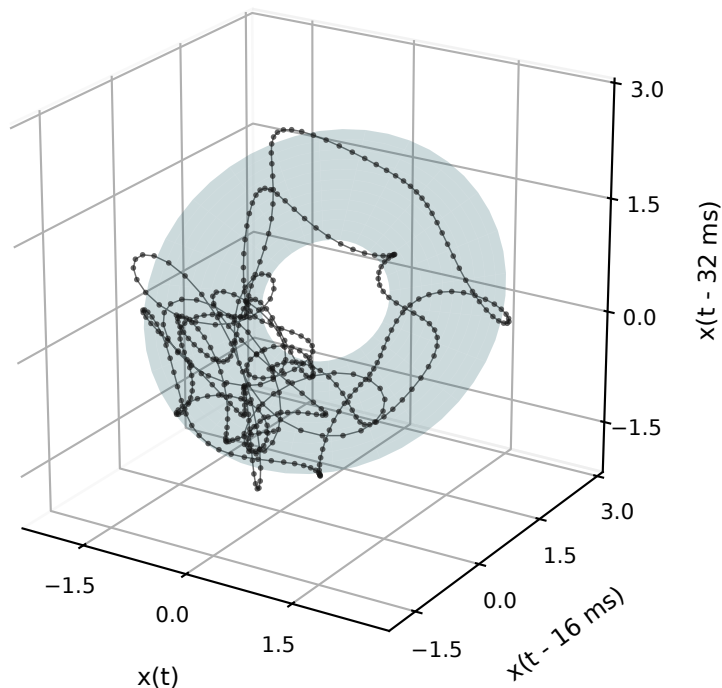
Normalized 3D set error: 0.0013  
Native coverage: 0.865 | nfev: 42  
Flags: active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 6 | original trial 527

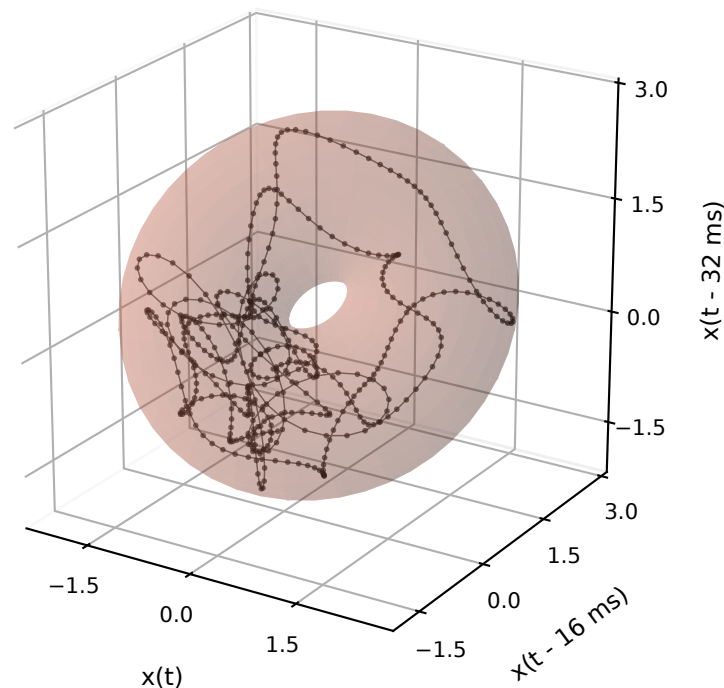
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1691  
Native coverage: 0.870 | nfev: 90  
Flags: none

2-torus: donut volume



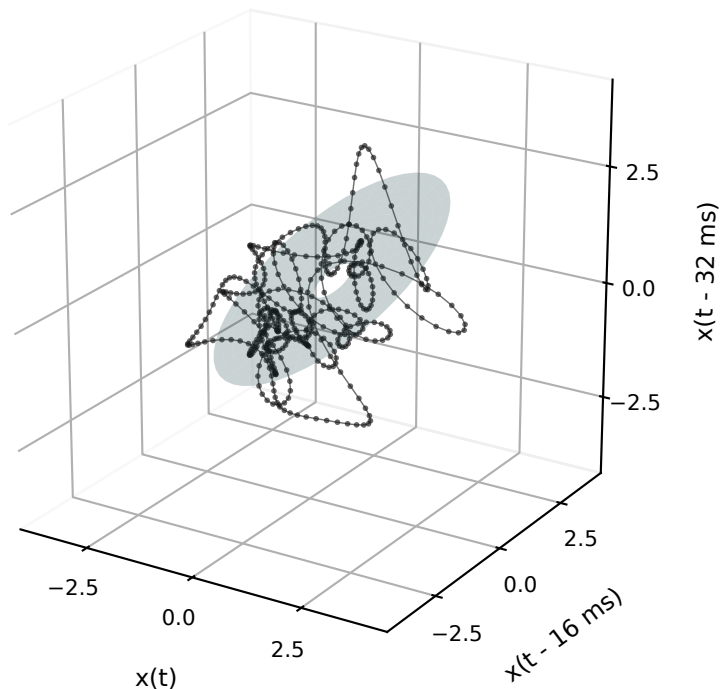
Normalized 3D set error: 0.0012  
Native coverage: 0.825 | nfev: 10  
Flags: active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 6 | original trial 555

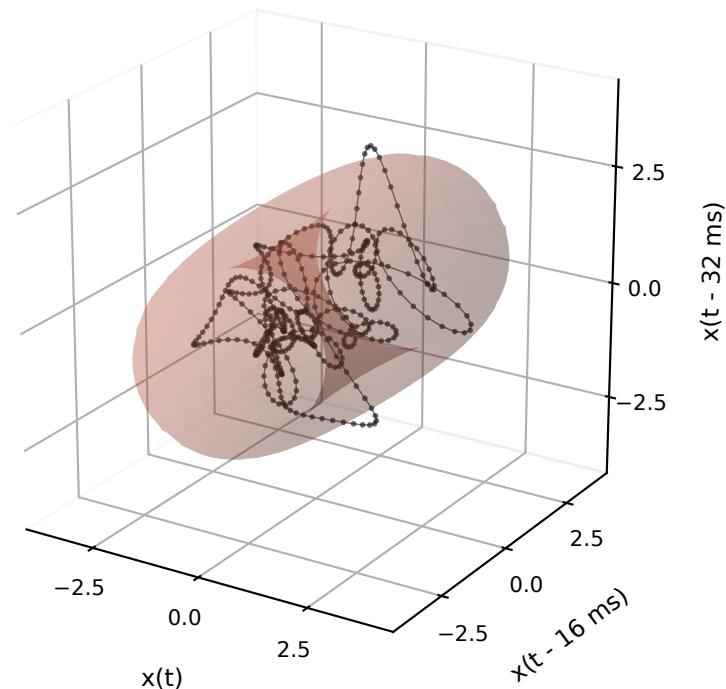
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2660  
Native coverage: 0.929 | nfev: 69  
Flags: none

2-torus: donut volume



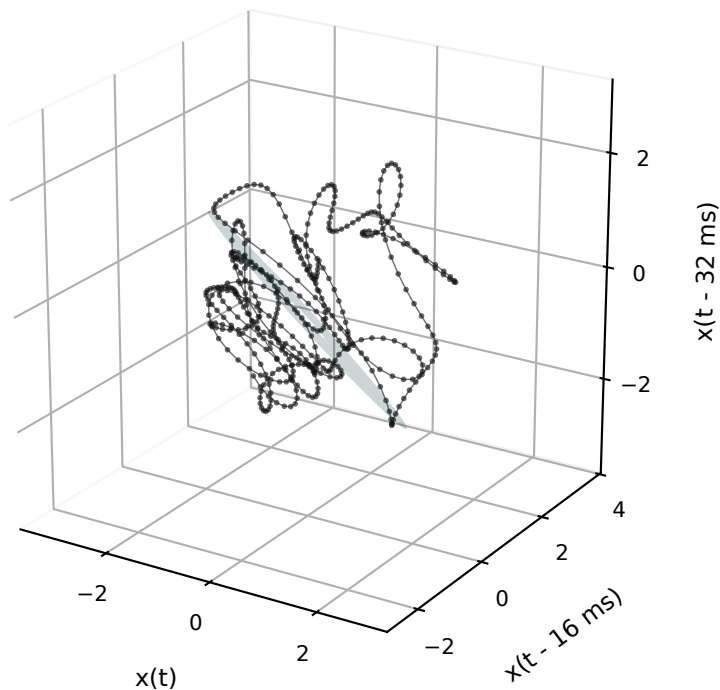
Normalized 3D set error: 0.0010  
Native coverage: 0.893 | nfev: 21  
Flags: collapsed hole

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 6 | original trial 342

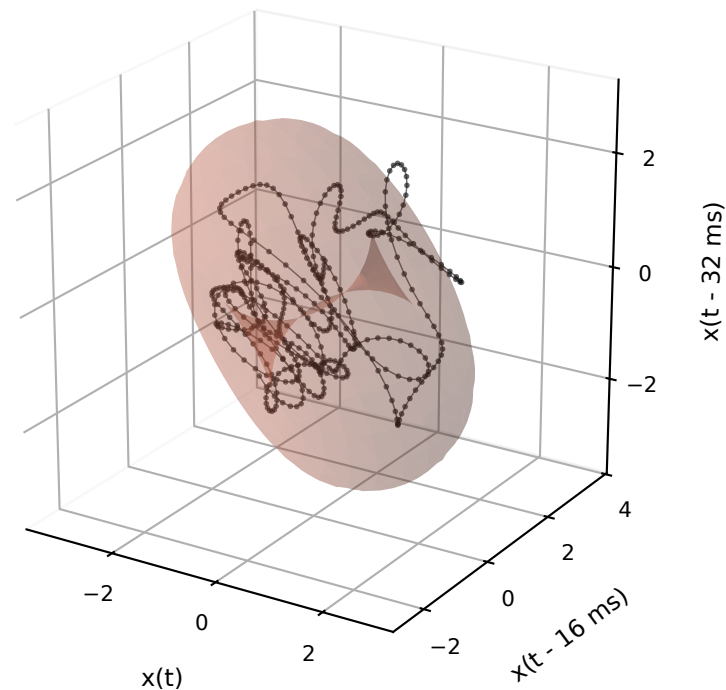
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.3301  
Native coverage: 0.889 | nfev: 88  
Flags: nearly circular orientation

2-torus: donut volume



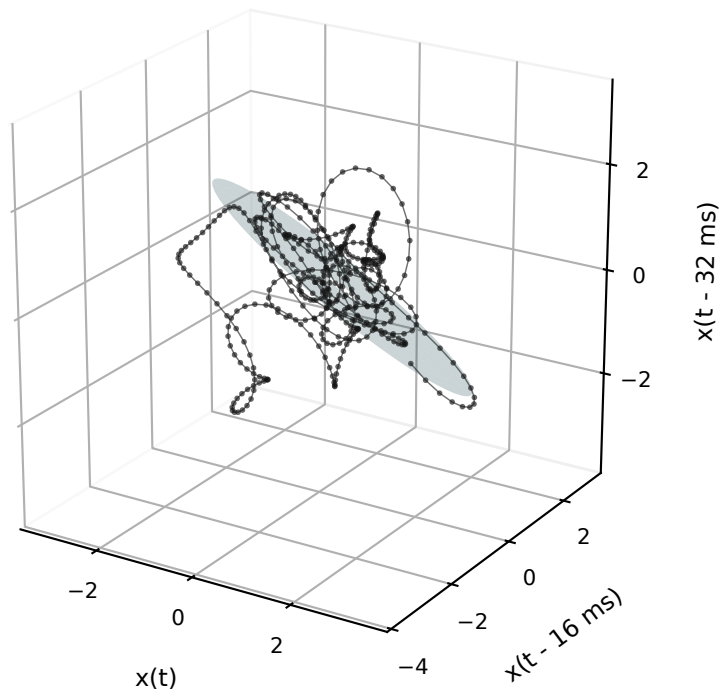
Normalized 3D set error: 0.0011  
Native coverage: 0.855 | nfev: 29  
Flags: collapsed hole; active optimizer bound; near optimizer bound; nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 6 | original trial 211

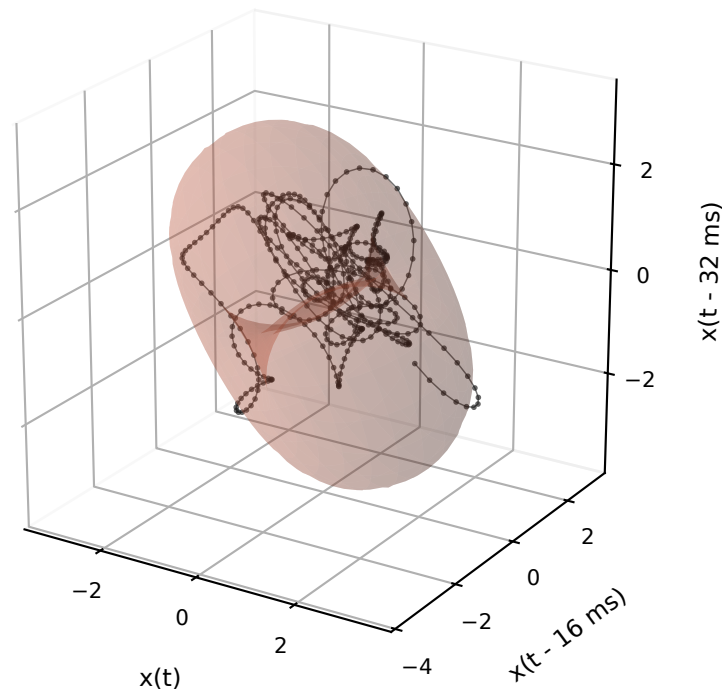
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

**1-torus: planar band**



Normalized 3D set error: 0.2442  
Native coverage: 0.923 | nfev: 22  
Flags: active optimizer bound; near optimizer bound

**2-torus: donut volume**



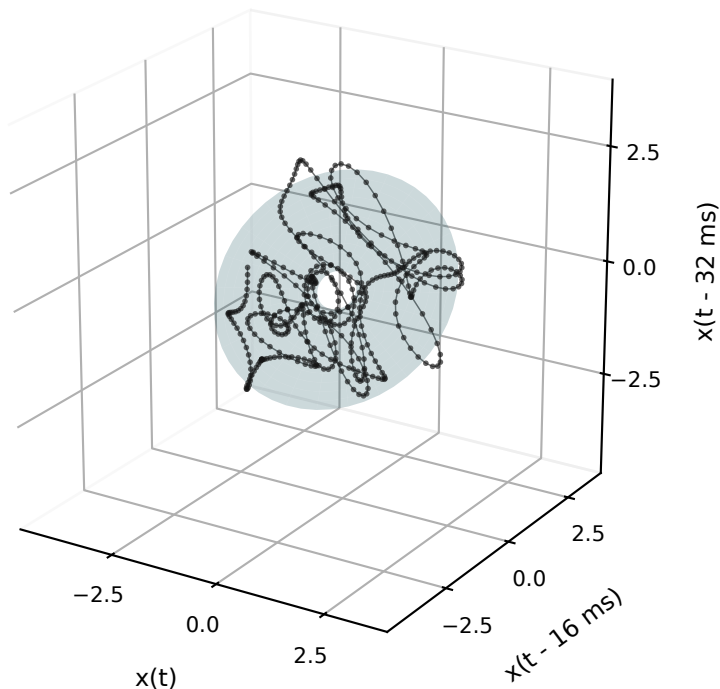
Normalized 3D set error: 0.0015  
Native coverage: 0.874 | nfev: 31  
Flags: collapsed hole; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 6 | original trial 539

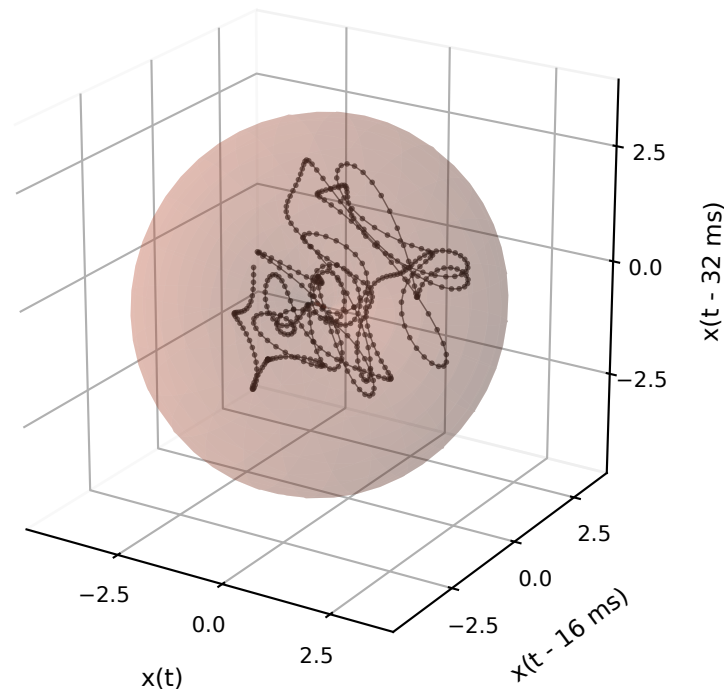
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2763  
Native coverage: 0.891 | nfev: 92  
Flags: none

2-torus: donut volume



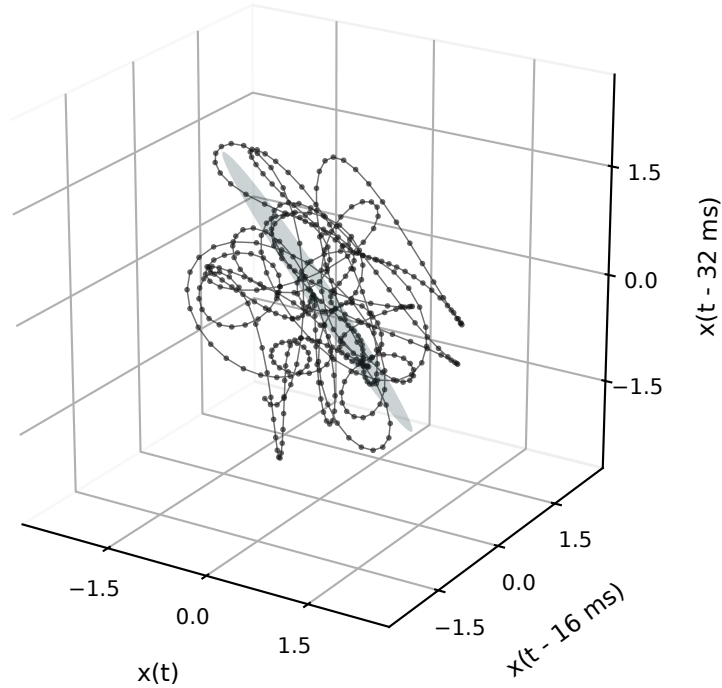
Normalized 3D set error: 0.0007  
Native coverage: 0.893 | nfev: 23  
Flags: collapsed hole; active optimizer bound; near optimizer bound; nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 6 | original trial 24

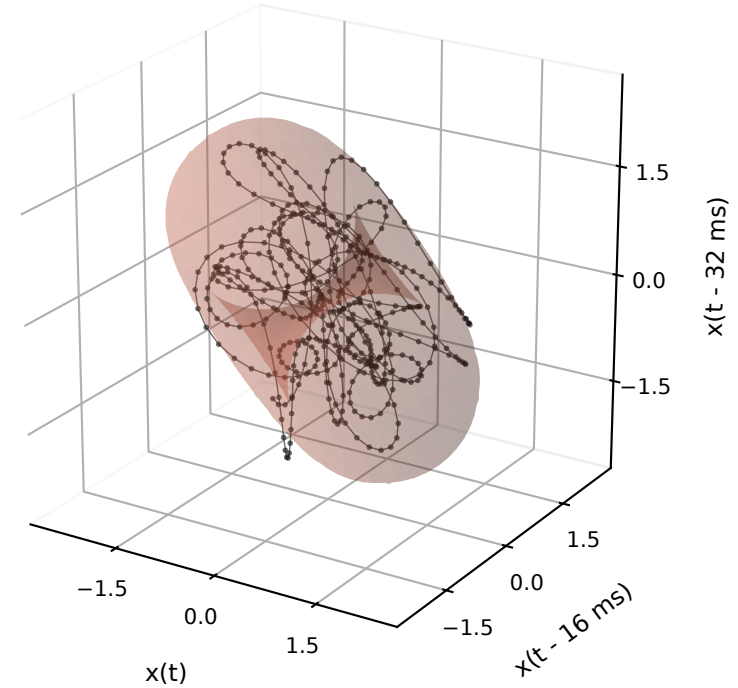
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1860  
Native coverage: 0.906 | nfev: 34  
Flags: none

2-torus: donut volume



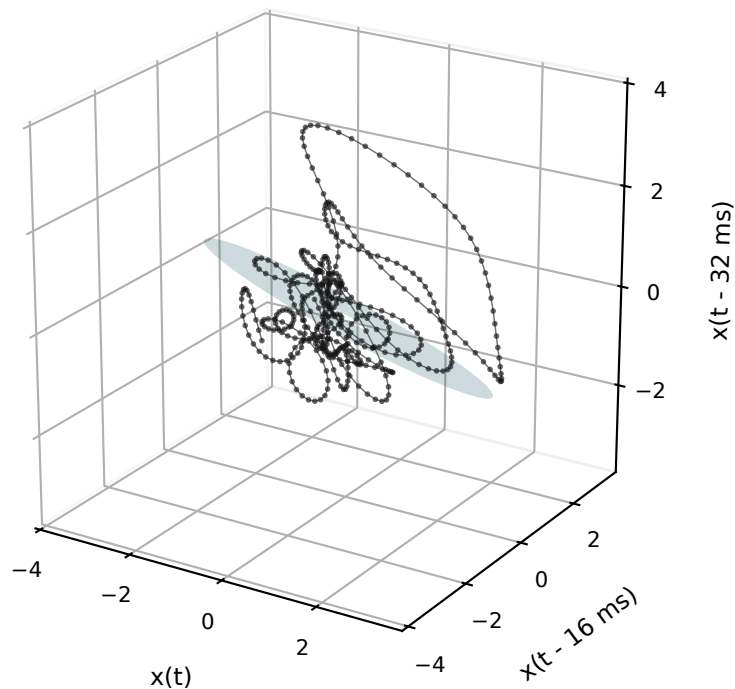
Normalized 3D set error: 0.0011  
Native coverage: 0.840 | nfev: 10  
Flags: active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 6 | original trial 538

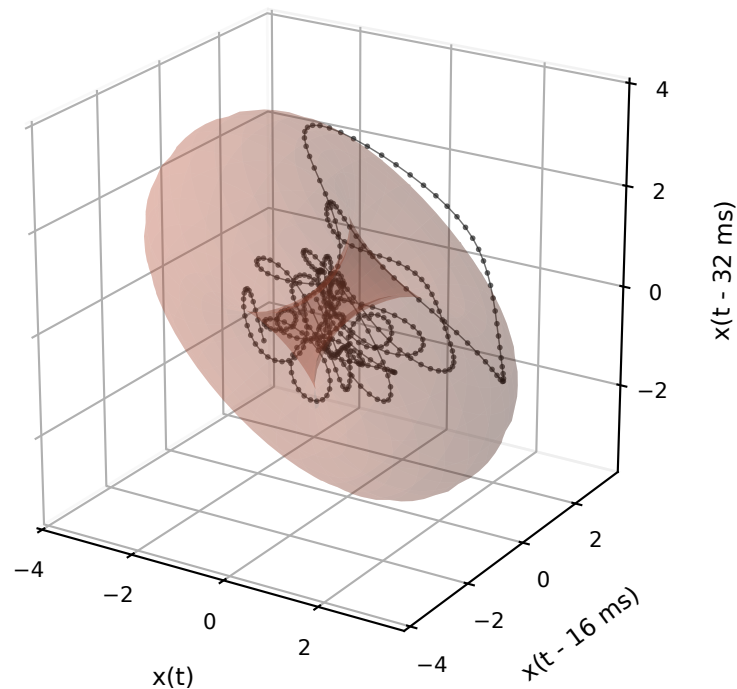
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2965  
Native coverage: 0.902 | nfev: 41  
Flags: active optimizer bound; near optimizer bound

2-torus: donut volume



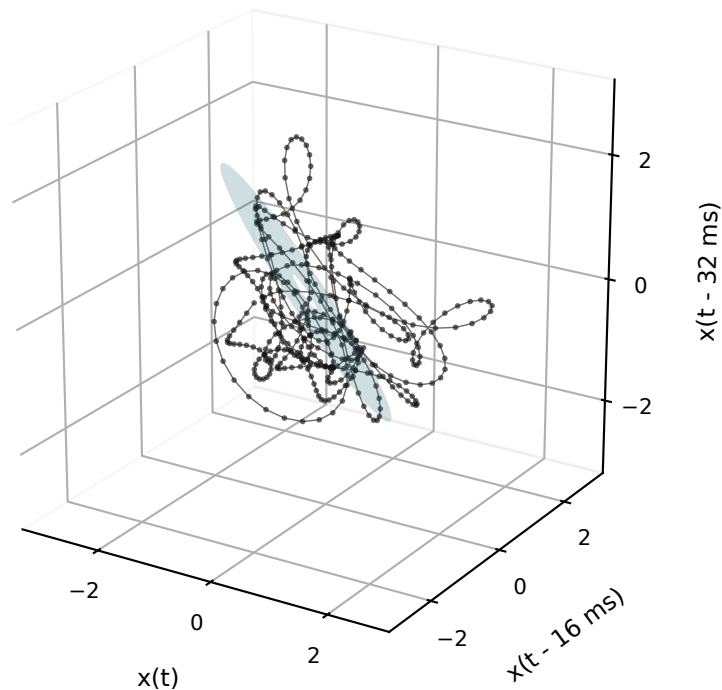
Normalized 3D set error: 0.0017  
Native coverage: 0.893 | nfev: 35  
Flags: collapsed hole; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 6 | original trial 642

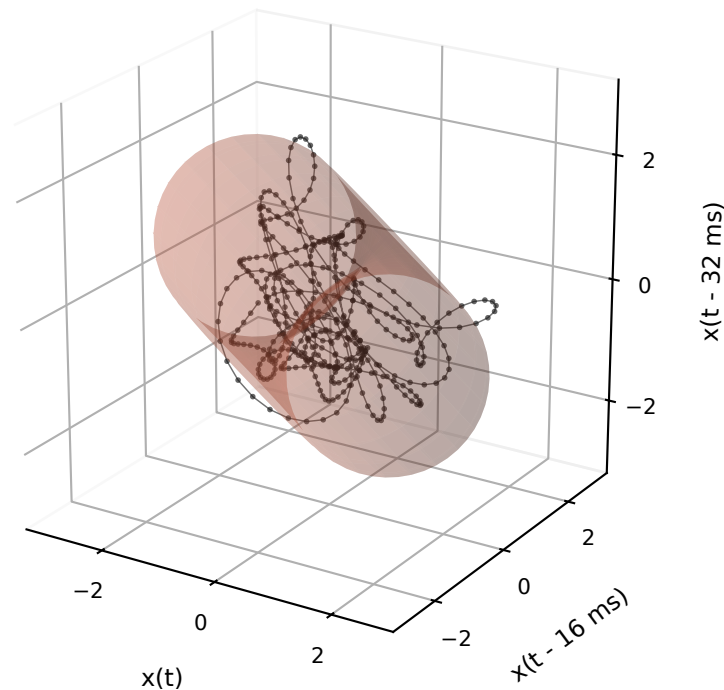
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.1898  
Native coverage: 0.893 | nfev: 54  
Flags: none

2-torus: donut volume



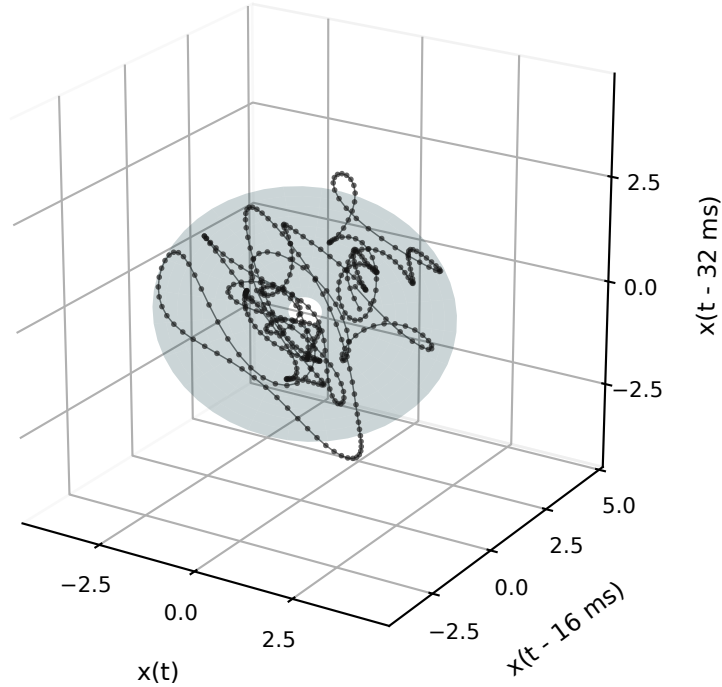
Normalized 3D set error: 0.0017  
Native coverage: 0.876 | nfev: 44  
Flags: collapsed hole

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 6 | original trial 379

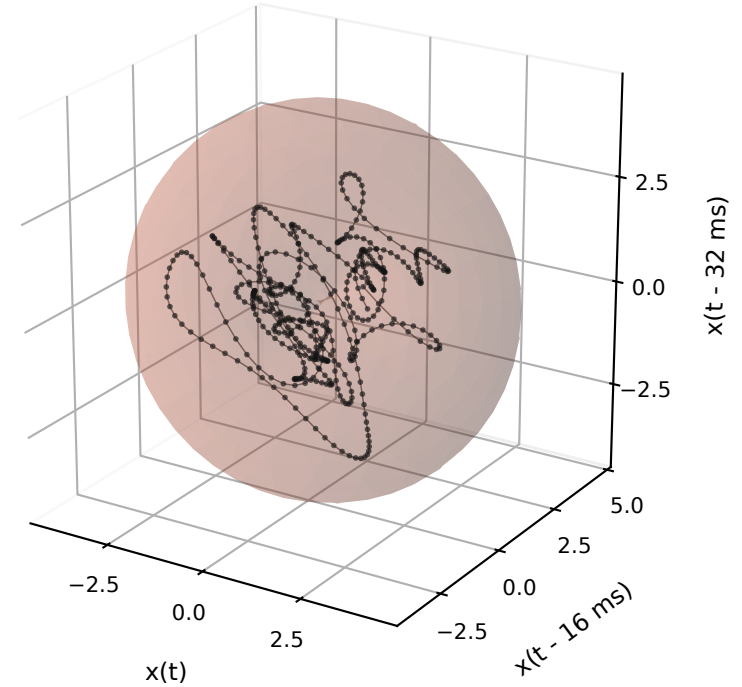
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

**1-torus: planar band**



Normalized 3D set error: 0.3027  
Native coverage: 0.917 | nfev: 71  
Flags: nearly circular orientation

**2-torus: donut volume**



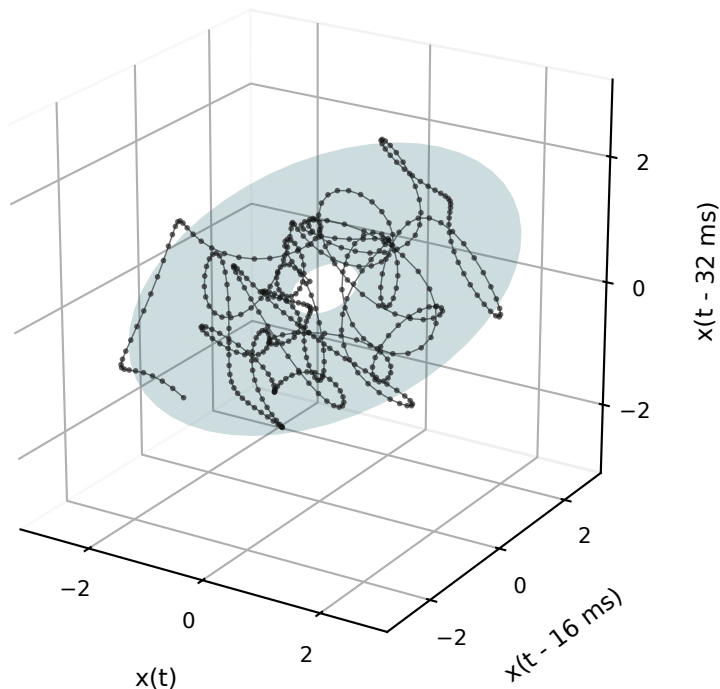
Normalized 3D set error: 0.0008  
Native coverage: 0.910 | nfev: 25  
Flags: active optimizer bound; near optimizer bound;  
nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 6 | original trial 483

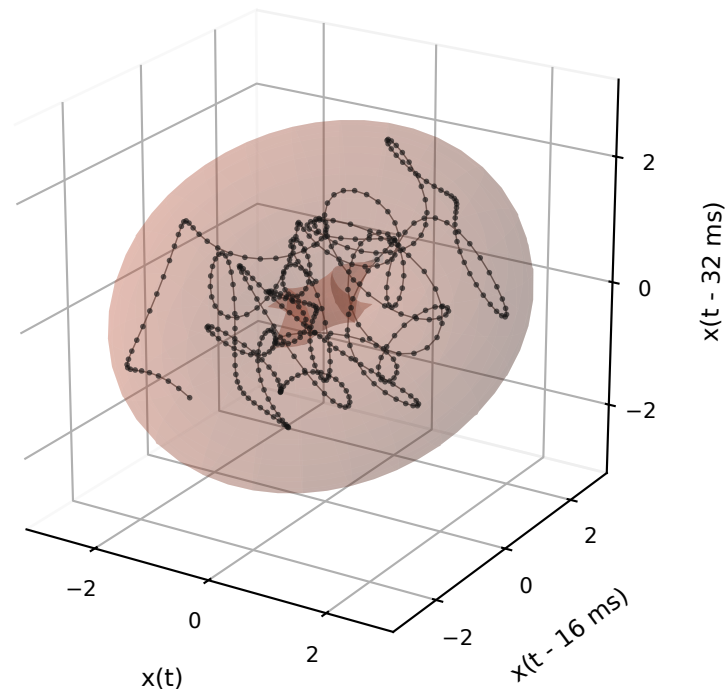
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2117  
Native coverage: 0.897 | nfev: 34  
Flags: none

2-torus: donut volume



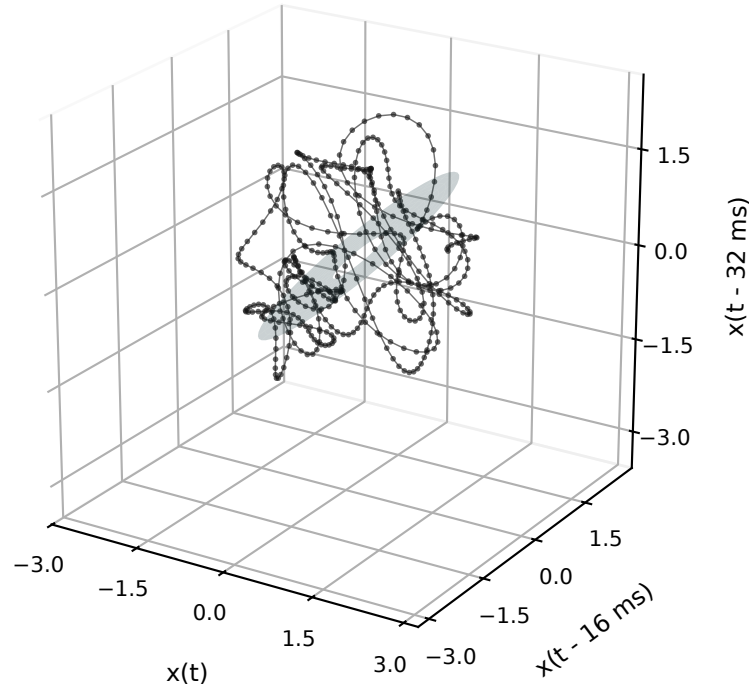
Normalized 3D set error: 0.0018  
Native coverage: 0.891 | nfev: 39  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 6 | original trial 9

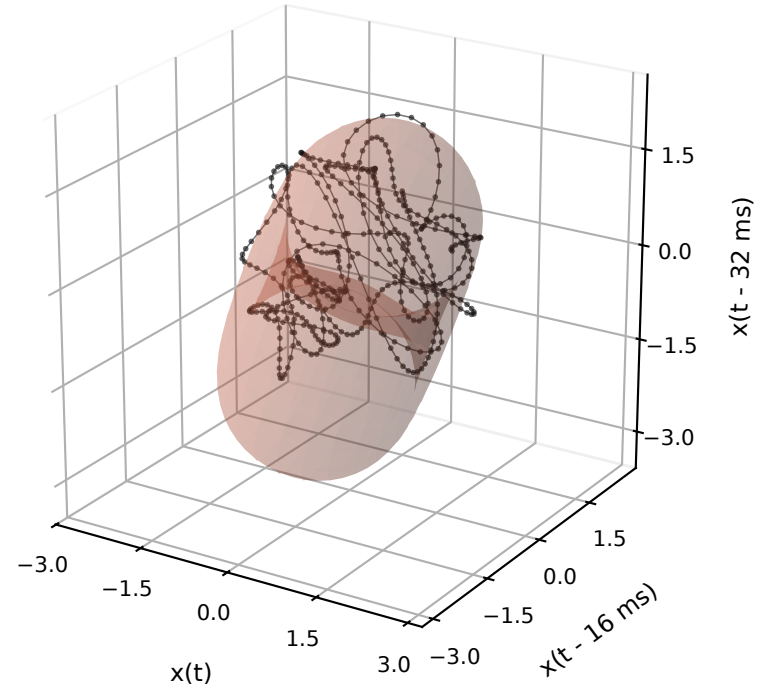
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2799  
Native coverage: 0.889 | nfev: 94  
Flags: none

2-torus: donut volume



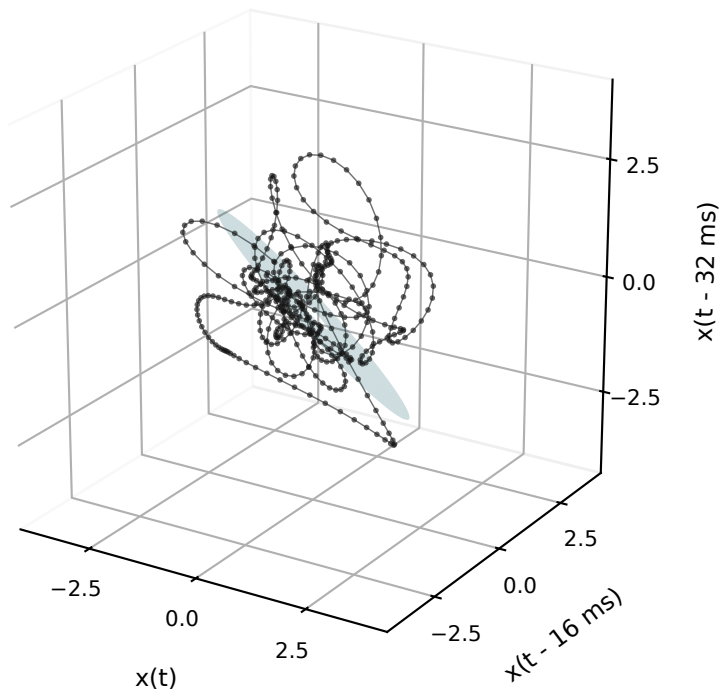
Normalized 3D set error: 0.0019  
Native coverage: 0.844 | nfev: 31  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 6 | original trial 113

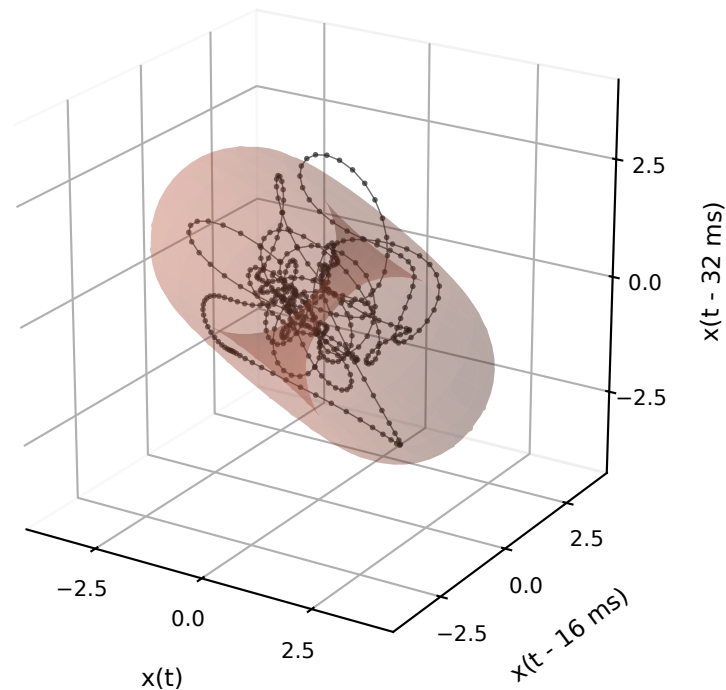
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2553  
Native coverage: 0.900 | nfev: 52  
Flags: none

2-torus: donut volume



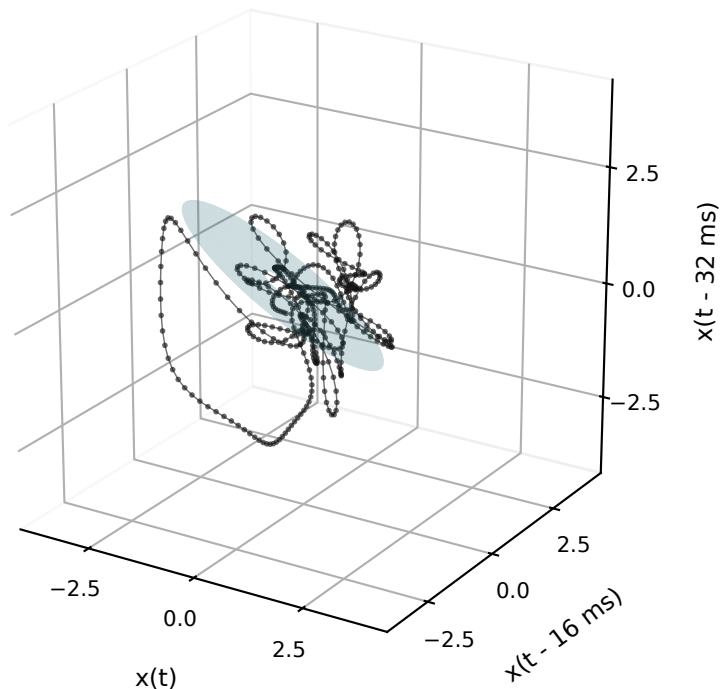
Normalized 3D set error: 0.0013  
Native coverage: 0.900 | nfev: 1334  
Flags: collapsed hole; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 6 | original trial 187

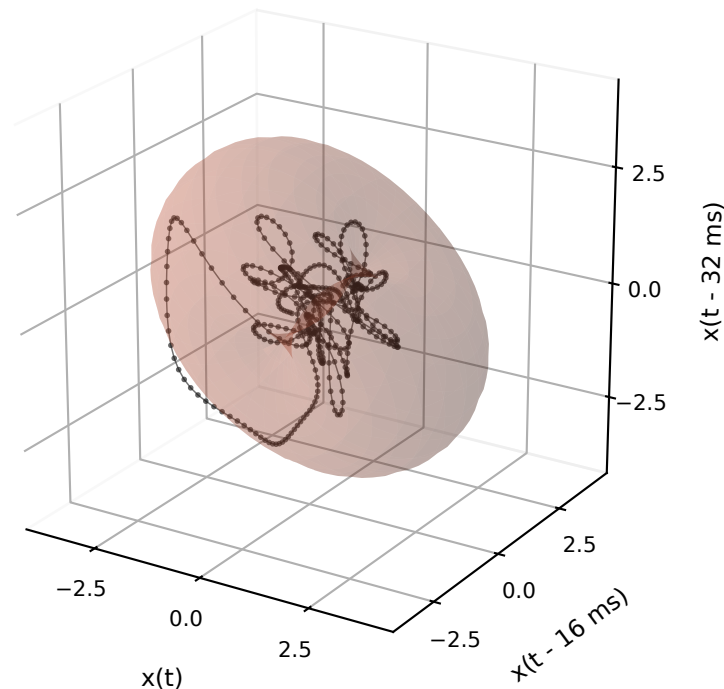
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.3064  
Native coverage: 0.910 | nfev: 121  
Flags: none

2-torus: donut volume



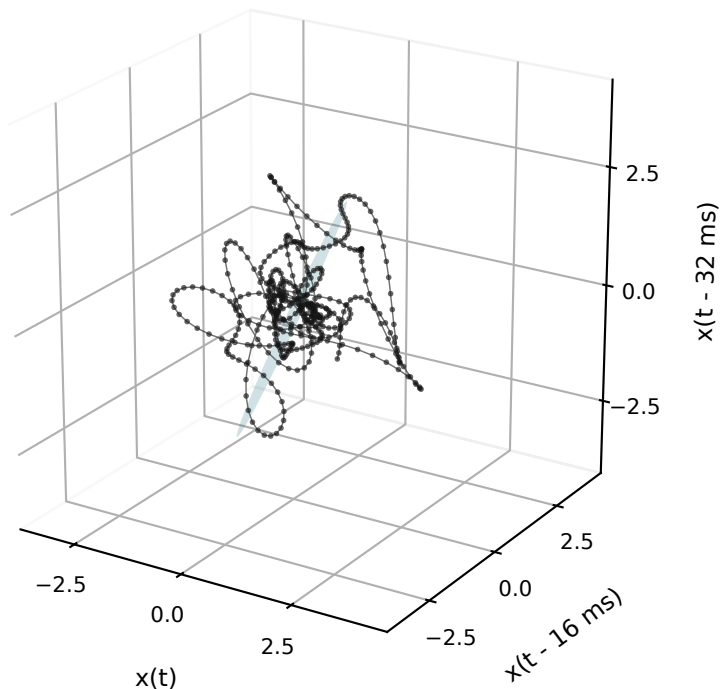
Normalized 3D set error: 0.0011  
Native coverage: 0.902 | nfev: 20  
Flags: collapsed hole; active optimizer bound; near optimizer bound; nearly circular orientation

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.

## Direction 6 | original trial 376

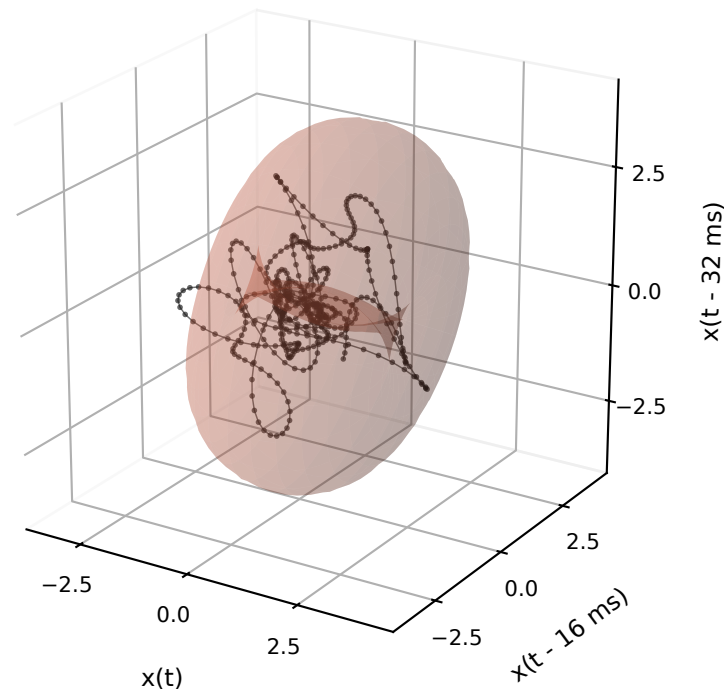
T, channel 7 | final 500 ms before GO |  $m = 3$  |  $\tau = 16$  ms | 468 points

1-torus: planar band



Normalized 3D set error: 0.2883  
Native coverage: 0.887 | nfev: 78  
Flags: active optimizer bound; near optimizer bound

2-torus: donut volume



Normalized 3D set error: 0.0017  
Native coverage: 0.897 | nfev: 53  
Flags: collapsed hole; active optimizer bound; near optimizer bound

Same observed cloud and scale. Band versus solid-tube containment; not a topology test.  
Tube overlay is a parametric sweep, not a volume boundary; thick-tube sweeps can self-intersect.